MATH EXCELLENCE SERIES

SAT MATH

& Complete Mastery

Algebra • Advanced Math • Data Analysis • Geometry

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Contents

PART 1

Algebra

Lesson 1: Linear Equations in One Variable

A linear equation in one variable is a statement that two expressions built from a single unknown — using only addition, subtraction, and multiplication by numbers — are equal. Almost every algebra question on the Digital SAT either is one of these or collapses into one after a line of work, so the routine in this topic is the routine you will reuse in systems, inequalities, functions, and word problems for the rest of the test. The goal is not just to get x; it is to get x quickly, in a way that leaves nowhere for a sign to hide.

What a linear equation is

Any equation that can be rewritten in the form

$$ax + b = 0, \qquad a \neq 0$$

is linear in x. The unknown appears to the first power only: no x^2, no $\sqrt{x}$, no x in a denominator. Equations such as $3x + 12 = 30$, $5x - 8 = 3x + 14$, and $\frac{x+5}{3} - \frac{x-1}{2} = 1$ all look different, but each becomes $ax + b = 0$ after expanding and combining, and each therefore has exactly one solution, $x = -\frac{b}{a}$.

That last sentence is worth reading twice. As long as the coefficient a that survives the simplification is not zero, there is one and only one answer. The special cases that the test loves — no solution, infinitely many solutions — are precisely the cases where a turns out to be zero.

THE BALANCE PRINCIPLE

An equation is a balanced scale. Whatever you do to one side you must do to the other, and if you do, the balance is preserved: the new equation has exactly the same solutions as the old one. You may add or subtract the same quantity on both sides, and you may multiply or divide both sides by the same *nonzero* number.

This is the reason every legal step is reversible, and it is why you can check an answer by substitution. If $x = 7$ solves $5x - 8 = 3x + 14$, then substituting must make both sides read 27. On a timed test, one substitution is often cheaper than re-doing the algebra.

The solving routine

Work in this order and the arithmetic stays clean.

1. Clear fractions and decimals. Multiply every term on both sides by the least common denominator (or by a power of 10). $\frac{3}{4}x + \frac{1}{2} = \frac{5}{8}x + 3$ becomes $6x + 4 = 5x + 24$.

2. Expand. Distribute across every parenthesis, remembering that a factor multiplies *each* term inside: $-4(x - 7) = -4x + 28$.

3. Combine like terms on each side separately, so each side reads $ax + b$.

4. Collect the variable terms on one side and the constants on the other. Move the smaller variable term so the coefficient you keep is positive.

5. Divide by the coefficient of the variable — once, at the very end.

6. Answer the question that was asked. If the question wants $x + 2$, or the length rather than the width, one more line of work is still owed.

HOW MANY SOLUTIONS AN EQUATION HAS

Simplify both sides until the equation reads

$$ax + b = cx + d.$$

Then
One solution: $a \neq c$. The variable survives and $x = \dfrac{d-b}{a-c}$.
No solution: $a = c$ and $b \neq d$. The variable cancels and leaves a false statement such as $15 = 12$.
Infinitely many solutions: $a = c$ and $b = d$. The two sides are the same expression, and the equation reduces to $0 = 0$, which is true for every x.
So when a question hands you a constant and asks for no solution or for infinitely many, you are being asked to match coefficients: set the coefficients equal, then check the constants to see which of the two cases you are in.

FOUR TRAPS THE TEST SETS

Distributing to only the first term. $5(2x-5) = 10x - 25$, not $10x - 5$. The wrong answers on these items are built from exactly this slip.
A minus sign in front of a group. In $\frac{x+5}{3} - \frac{x-1}{2}$, the subtraction applies to the whole second numerator; after clearing denominators it becomes $-3(x-1) = -3x + 3$.
Dividing by a negative. From $-2x = -8$ the answer is $x = 4$; from $-2x = 8$ it is $x = -4$. Do the division once, deliberately.
Solving for the wrong quantity. The equation gives the width, but the question asked for the length. Underline the final phrase of the question before you start.

STRATEGY: SUBSTITUTION AND CLEAN SETUP

If an answer-choice question resists you, put the choices back into the equation. Four substitutions are slow, but one is fast, and the choices are usually ordered, so starting with choice B or C tells you which direction to move.
In word problems, define the variable in writing before you write the equation: *let w be the width in meters*. Everything else in the problem then gets expressed in terms of that one letter — the length becomes $2w+3$, the number of dimes becomes $2n$ — and the sentence turns into a single equation instead of a guess.
Keep units in one system. A value problem with nickels and dimes is easier entirely in cents: $5n+10(2n) = 250$.

WORKED EXAMPLE 1: VARIABLES ON BOTH SIDES

Solve $2(3x-5) - 4(x-7) = 3x + 6$.
Expand first, watching the -4:

$$6x - 10 - 4x + 28 = 3x + 6.$$

Combine like terms on the left:

$$2x + 18 = 3x + 6.$$

The smaller variable term is $2x$, so subtract $2x$ from both sides and subtract 6 from both sides:

$$12 = x.$$

Check by substitution: the left side is $2(31) - 4(5) = 62 - 20 = 42$, and the right side is $36 + 6 = 42$. The two sides agree, so $x = 12$.

WORKED EXAMPLE 2: A NO-SOLUTION QUESTION

In $ax + 15 = 4(3x+5) - 8$ **,** a **is a constant. If the equation has no solution, what is** a **?**
Simplify the side that contains no unknown constant:

$$4(3x+5) - 8 = 12x + 20 - 8 = 12x + 12.$$

The equation now reads $ax + 15 = 12x + 12$. No solution requires the variable to disappear while the constants disagree, so the coefficients must be equal: $a = 12$. Substituting back gives $12x + 15 = 12x + 12$, and subtracting $12x$ leaves $15 = 12$ — false for every x, which is exactly what *no solution* means. If instead the constants had matched, the same value of a would have given infinitely many solutions, so the constant check is not optional.

WORKED EXAMPLE 3: A WORD PROBLEM

Tank A holds 240 liters and drains at 8 liters per minute. Tank B holds 120 liters and fills at 4 liters per minute. After how many minutes do they hold the same amount?

Let t be the number of minutes. Draining makes the rate negative and filling makes it positive, so the two amounts are $240 - 8t$ and $120 + 4t$. *The same amount* means the two expressions are equal:

$$240 - 8t = 120 + 4t.$$

Add $8t$ to both sides and subtract 120:

$$120 = 12t, \qquad t = 10.$$

Notice what the 12 means: the gap between the tanks is 120 liters and it closes at $8 + 4 = 12$ liters per minute. Reading the algebra back into the story like this is the fastest way to catch a setup error.

Module 1

1

If $3x + 12 = 30$, what is the value of x?

A) 6
B) 10
C) 14
D) 18

2

If $7x = 4x + 21$, what is the value of x?

3

If $\frac{x}{4} + 3 = 11$, what is the value of x?

A) 8
B) 32
C) 44
D) 56

4

If $3(x - 4) = 18$, what is the value of x?

A) 2
B) 6
C) 10
D) 22

5

A gym charges a one-time membership fee of \$30 plus \$15 for each month of use. If a member paid a total of \$180, for how many months did the member use the gym?

A) 6
B) 10
C) 12
D) 14

6

If $4x - 9 = 23$, what is the value of $x + 2$?

7

If $-2x + 9 = 1$, what is the value of x?

A) -5
B) -4
C) 4
D) 5

8

If $5x - 8 = 3x + 14$, what is the value of x?

A) -3

B) 3

C) 11

D) 22

9

If $\frac{5}{2}x - 3 = 7$, what is the value of x?

A) 4

B) 10

C) 20

D) 25

10

If $4(x + 3) = 2(x + 9)$, what is the value of x?

11

If $3x + 2y = 12$, which expression gives x in terms of y?

A) $\frac{12 - 2y}{3}$

B) $\frac{12 + 2y}{3}$

C) $12 - \frac{2y}{3}$

D) $\frac{12 - 2y}{2}$

12

At noon the temperature in a town was 12°F, and it fell at a constant rate of 3°F per hour. After how many hours was the temperature -9°F?

A) -7

B) 1

C) 3

D) 7

13

In the equation $4(x - a) = 4x - 12$, a is a constant. If the equation is true for all values of x, what is the value of a?

A) -3

B) 3

C) 12

D) 48

14

The equation $6x + 12 = 3(2x + k)$, where k is a constant, has infinitely many solutions. What is the value of k?

15

If $0.4x + 1.2 = 3.6$, what is the value of x?

A) 0.6

B) 6

C) 9

D) 12

16

If $\frac{x + 5}{3} - \frac{x - 1}{2} = 1$, what is the value of x?

A) -7

B) 1

C) 7

D) 12

17

The formula $C = \frac{5}{9}(F - 32)$ converts a Fahrenheit temperature F to a Celsius temperature C. Which of the following gives F in terms of C?

A) $\frac{5}{9}C + 32$

B) $\frac{9}{5}(C + 32)$

C) $\frac{9}{5}C - 32$

D) $\frac{9}{5}C + 32$

18

If $2(3x - 5) - 4(x - 7) = 3x + 6$, what is the value of x?

19

Tank A holds 240 liters of water and is draining at a constant rate of 8 liters per minute. Tank B holds 120 liters and is filling at a constant rate of 4 liters per minute. After how many minutes will the two tanks hold the same amount of water?

A) 10

B) 15

C) 20

D) 30

20

In the equation $ax+15=4(3x+5)-8$, a is a constant. If the equation has no solution, what is the value of a?

A) 3

B) 4

C) 5

D) 12

21

If $\dfrac{2x-7}{5}=\dfrac{x+4}{3}$, what is the value of x?

22

The perimeter of a rectangle is 48 centimeters. The length of the rectangle is 4 centimeters more than its width. What is the length, in centimeters, of the rectangle?

A) 10

B) 14

C) 20

D) 24

Module 2

1

If $7 - 2x = 19$, what is the value of x?

A) -12
B) -6
C) 6
D) 13

2

If $\frac{x}{3} - \frac{x}{5} = 4$, what is the value of x?

3

If $5(2x - 5) + 4 = 39$, what is the value of x?

A) 1
B) 4
C) 6
D) 60

4

A phone plan costs \$25 per month plus \$0.10 for each text message sent. In one month the bill was \$43. How many text messages were sent that month?

A) 680
B) 430
C) 180
D) 18

5

If $5m - 2n = 20$, which expression gives n in terms of m?

A) $\frac{20 - 5m}{2}$
B) $5m - 10$
C) $\frac{5m + 20}{2}$
D) $\frac{5m - 20}{2}$

6

If $2(x + 6) = 5x - 3$, what is the value of x?

A) -5
B) 3
C) 5
D) 15

7

If $5(x - 2) = 3x + 4$, what is the value of x?

8

If $2(3x - 1) = 4(x + 5) + 2$, what is the value of x?

A) -12
B) 10
C) 12
D) 24

9

If $\frac{2}{3}(x + 6) = 10$, what is the value of x?

A) 6
B) 9
C) 15
D) 21

10

Ana bought 4 identical notebooks and one pen that cost \$8. The total cost of the items, before tax, was \$32. What was the cost, in dollars, of one notebook?

A) 6
B) 8
C) 10
D) 24

11

If $\frac{x + 2}{4} = \frac{x - 6}{2}$, what is the value of x?

12

If $y = \frac{2x - 5}{3}$, which expression gives x in terms of y?

A) $\frac{3y + 5}{2}$
B) $\frac{3y - 5}{2}$
C) $\frac{2y + 5}{3}$
D) $\frac{3y}{2} + 5$

13

The equation $5(2x+3) = ax + 15$, where a is a constant, has infinitely many solutions. What is the value of a?

A) 2

B) 3

C) 5

D) 10

14

In the equation $3(2x-4)+5 = 6x+c$, c is a constant. For which value of c does the equation have infinitely many solutions?

A) -12

B) -7

C) 5

D) 7

15

The length of a rectangle is 3 meters more than twice its width. The perimeter of the rectangle is 36 meters. What is the width, in meters, of the rectangle?

A) 5

B) 6

C) 11

D) 13

16

The equation $\frac{2}{3}(6x-9) = 4x + k - 10$, where k is a constant, has infinitely many solutions. What is the value of k?

17

A studio charges a one-time registration fee of \$60 plus \$10 for each class. A second studio charges \$25 for each class and no registration fee. For how many classes is the total cost at the two studios the same?

A) 100

B) 6

C) 4

D) -4

18

If $3(x-a) = 2(x+a)$, where a is a nonzero constant, what is x in terms of a?

A) $-a$

B) $\frac{a}{5}$

C) a

D) $5a$

19

The sum of three consecutive even integers is 96. What is the least of the three integers?

A) 30

B) 32

C) 34

D) 36

20

A jar contains only nickels and dimes and has twice as many dimes as nickels. The total value of the coins is \$2.50. How many nickels are in the jar?

A) 30

B) 25

C) 20

D) 10

21

If $\frac{3}{4}x + \frac{1}{2} = \frac{5}{8}x + 3$, what is the value of x?

22

In the equation $2(kx-3) = 6x+5-k$, k is a constant. If the equation has no solution, what is the value of k?

A) -6

B) 3

C) 6

D) 11

Lesson 2: Linear Functions

A linear function is a rule whose output changes by the same amount for every equal step in the input. That single property, a constant rate of change, is what makes lines the backbone of the Digital SAT's Algebra domain: it lets you find a whole function from two points, predict a value you were never given, and say in ordinary English what a coefficient means in a real situation. Roughly one question in six on the test is a linear function in disguise, and most of them reward the same three moves: compute a slope, evaluate or invert $f(x)$, and read a model's numbers as quantities with units.

What makes a function linear

A function f is **linear** when equal steps in x always produce equal steps in y. Written out, every linear function can be put in the form

$$f(x) = mx + b,$$

where m is the **slope** (the change in output per unit change in input) and b is the **y-intercept** (the output when the input is 0). The graph of such a function is a straight line, and the two constants are all the information there is: give a test-taker m and b and every question about the function is already answered.

Two facts follow immediately and are worth memorizing as facts, not derivations. First, $f(0) = m(0) + b = b$, so *the value of the function at zero is the y-intercept.* Second, for any two inputs x_1 and x_2,

$$f(x_2) - f(x_1) = (mx_2 + b) - (mx_1 + b) = m(x_2 - x_1),$$

so the change in output is the slope times the change in input. Every slope computation on the test is just this identity rearranged.

SLOPE IS A RATE, AND RATES CARRY UNITS

Solving $f(x_2) - f(x_1) = m(x_2 - x_1)$ for m gives the slope formula. In a bare xy-plane the slope is a number; in a word problem it is a *rate* whose units are the output units divided by the input units. If $C(h) = 45h + 60$ gives a cost in dollars for h hours, then 45 is measured in dollars per hour and 60 in dollars. Getting the units right converts almost every interpretation question into a one-line answer, because the only choice with the correct units is usually the key.

A positive slope means the output grows as the input grows; a negative slope means it shrinks. A slope of -12 in a draining-tank model does not mean 12 gallons; it means 12 gallons lost *per minute.*

THE FOUR FORMULAS THAT COVER THE TOPIC

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \qquad \text{(slope from two points)}$$

$$y = mx + b \qquad \text{(slope-intercept form; } b = f(0)\text{)}$$

$$y - y_1 = m(x - x_1) \qquad \text{(point-slope form)}$$

$$Ax + By = C \implies y = -\frac{A}{B}x + \frac{C}{B} \qquad \text{(standard form; slope } -\tfrac{A}{B}\text{)}$$

Point-slope is the fastest route when you are handed a slope and a point that is *not* the intercept. For standard form, either quote the slope $-\frac{A}{B}$ or, more safely, solve the equation for y and read the coefficient.

Function notation, forwards and backwards

The symbol $f(4)$ is not multiplication. It is an instruction: *substitute 4 for every x in the rule and simplify.* That is the forwards direction, and it is pure arithmetic.

The backwards direction is where points are won and lost. A question that says $f(x) = 30$ hands you the *output* and asks for the *input*, so you write the rule equal to 30 and solve the resulting linear equation. Read carefully which of the two you have been given: *what is* $f(3)$ and *for what* x *is* $f(x) = 3$ are different questions with different answers.

The input may also be an expression. $f(x+3)$ means the whole quantity $x+3$ goes in wherever x appears, so for $f(x) = 2x - 5$ we get $f(x+3) = 2(x+3) - 5 = 2x + 1$. It does not mean $f(x) + 3$.

READING A MODEL IN CONTEXT

When a linear function models a real situation, write it as

$$(\text{output quantity}) = (\text{rate}) \times (\text{input quantity}) + (\text{starting amount}).$$

The constant term is the value *before the process starts* — the flat fee, the initial population, the height of the unlit candle. The coefficient is what happens *per unit* — per hour, per class, per month, per year. Two follow-up questions appear constantly. *What does the solution of* $f(x) = 0$ *represent?* It is the moment the measured quantity runs out: the candle burns away, the tank empties, the balance hits zero. *What does* $f(0)$ *represent?* It is the starting value. Neither requires computation — only a correct reading of what the variables stand for.

FIVE TRAPS THE TEST SETS ON PURPOSE

1. Inverting the slope. $\frac{\Delta y}{\Delta x}$, never $\frac{\Delta x}{\Delta y}$. A slope of $-\frac{3}{4}$ has the distractor $-\frac{4}{3}$ waiting for you.
2. Mixing the order of subtraction. If the numerator runs from point 1 to point 2, the denominator must too. Reversing one of them flips the sign.
3. Reading the intercept as the slope. In $y = -3x + 8$ the slope is -3, not 8, and not 3.
4. Treating a given point as the intercept. If a line of slope 4 passes through $(3, 5)$, then $b \neq 5$; solving $5 = 4(3) + b$ gives $b = -7$.
5. Forgetting to divide by Δx when the inputs are not one apart. If x goes $2, 5, 8$ and g goes $3, 12, 21$, the rate is $\frac{9}{3} = 3$, not 9.

A THREE-SECOND CHECKLIST

Two points given? Subtract in a consistent order, divide, and sanity-check the sign against the picture: falling left-to-right means negative.
A table given? Check that the x-values are evenly spaced before treating the output differences as the rate. If they are, the common difference in the outputs divided by the common difference in the inputs is the slope.
An equation in standard form? Solve for y; it takes ten seconds and removes every sign trap.
An interpretation question? Attach units to each number in the model first. The number multiplied by the variable is a rate "per something"; the lone constant is a starting amount. On the calculator section, graphing $y_1 =$ the model and reading its intercepts confirms both in one step.

WORKED EXAMPLE 1 — A FUNCTION FROM TWO POINTS

For the linear function f, $f(2) = 5$ and $f(6) = 17$. What is $f(10)$?
The two given facts are the points $(2, 5)$ and $(6, 17)$. The slope is

$$m = \frac{17 - 5}{6 - 2} = \frac{12}{4} = 3.$$

There are now two honest finishes. *Stepping:* from $x = 6$ to $x = 10$ is 4 units, each worth $+3$, so $f(10) = 17 + 3(4) = 29$. *Full rule:* from $5 = 3(2) + b$ we get $b = -1$, so $f(x) = 3x - 1$ and $f(10) = 30 - 1 = 29$. Stepping is faster when only one more value is wanted; building the rule is better when the question will ask for several things, or for b itself.

Answer: $f(10) = 29$.

WORKED EXAMPLE 2 — INTERPRETING A MODEL

A storage tank is being drained, and $V(t) = 180 - 12t$ gives the volume in gallons after t minutes. What does 12 mean, and what does the solution of $V(t) = 0$ mean?

Attach units. V is in gallons and t is in minutes, so the coefficient of t must be in gallons per minute. Its sign is negative, so the volume *falls* by 12 gallons each minute. The constant 180 is $V(0)$, the volume at the moment draining began.

Now $V(t) = 0$ asks when the volume reaches zero gallons — that is, when the tank is empty. Solving, $180 - 12t = 0$ gives $t = 15$, so the tank empties after 15 minutes. Notice that no interpretation question needed the number 15; the meaning came entirely from the units.

Answer: 12 is the number of gallons lost per minute; the solution of $V(t) = 0$ is the time at which the tank is empty.

WORKED EXAMPLE 3 — WORKING BACKWARDS THROUGH f

The function f is defined by $f(x) = ax + 7$, where a is a constant, and $f(4) = -5$. What is a, and for what x is $f(x) = 1$?

The statement $f(4) = -5$ means: input 4, output -5. Substituting both gives one equation in the one unknown constant:

$$a(4) + 7 = -5 \implies 4a = -12 \implies a = -3.$$

Stopping at $4a = -12$ and answering -12 is the single most common error on this question type; the division by 4 is part of the problem.

With $f(x) = -3x + 7$, the second question hands you an output and wants an input: $-3x + 7 = 1$, so $-3x = -6$ and $x = 2$.

Answer: $a = -3$, and $f(x) = 1$ when $x = 2$.

Module 1

1

What is the slope of the line that passes through the points $(2, 3)$ and $(6, 11)$?

A) -2

B) $\frac{1}{2}$

C) 2

D) 4

2

The function f is defined by $f(x) = 3x - 5$. What is the value of $f(4)$?

A) -17

B) 2

C) 7

D) 12

3

The function f is defined by $f(x) = 7x + 2$. If $f(x) = 30$, what is the value of x?

4

What is the slope of the graph of $y = -3x + 8$ in the xy-plane?

A) -8

B) -3

C) 3

D) 8

5

A plumber charges a fixed call-out fee plus an hourly rate. The total charge, in dollars, for a job lasting h hours is given by $C(h) = 45h + 60$. What is the best interpretation of the number 60 in this context?

A) The number of hours the plumber works.

B) The plumber's hourly rate, in dollars per hour.

C) The total cost, in dollars, of a one-hour job.

D) The fixed fee, in dollars, charged before any hours are worked.

6

x	1	2	3	4
$f(x)$	5	9	13	17

The table shows several values of the linear function f. What is the value of $f(0)$?

7

A linear function f has a slope of 5, and $f(0) = -2$. Which of the following defines $f(x)$?

A) $5x - 2$

B) $5x + 2$

C) $\frac{1}{5}x - 2$

D) $-2x + 5$

8

A line in the xy-plane passes through the points $(-4, 9)$ and $(6, 4)$. What is the slope of this line?

A) -2

B) $-\frac{1}{2}$

C) $\frac{1}{2}$

D) 2

9

For the linear function f, $f(2) = 5$ and $f(6) = 17$. What is the value of $f(10)$?

10

A storage tank is being drained. The volume of water in the tank, in gallons, after t minutes of draining is given by $V(t) = 180 - 12t$. Which statement is the best interpretation of the number 12 in this context?

A) The volume of water decreases by 12 gallons each minute.

B) The volume of water increases by 12 gallons each minute.

C) The tank holds 12 gallons when draining begins.

D) The tank is completely empty after 12 minutes.

11

A gym charges a one-time membership fee of \$25 plus \$4 for each fitness class attended. Which expression gives the total cost, in dollars, for a member who attends n classes?

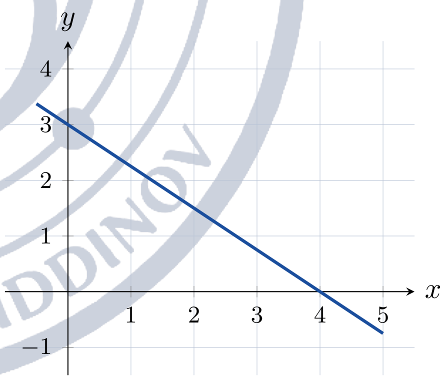
A) $4n + 25$

B) $25n + 4$

C) $29n$

D) $4n - 25$

12



The graph of a linear function is shown in the xy-plane. What is the slope of the graph?

A) $-\frac{4}{3}$

B) $-\frac{3}{4}$

C) $\frac{3}{4}$

D) $\frac{4}{3}$

13

x	0	3	6	9
$f(x)$	14	8	2	-4

The table gives values of the linear function f. For what value of x does $f(x) = 0$?

14

The function f is defined by $f(x) = \frac{3}{4}x + 6$. For what value of x does $f(x) = 0$?

A) -8

B) $-\frac{9}{2}$

C) -6

D) 8

15

At 2:00 p.m. the temperature outside was 68°F, and at 8:00 p.m. it was 56°F. If the temperature changed at a constant rate, what was that rate, in degrees Fahrenheit per hour?

A) -12

B) -6

C) -2

D) 2

16

The function f is defined by $f(x) = ax + b$, where a and b are constants. If $f(1) = 4$ and $f(5) = 16$, what is the value of $f(0)$?

A) -3

B) 1

C) 3

D) 4

17

In the xy-plane, a line with slope 3 passes through the points $(2, k)$ and $(6, 17)$. What is the value of k?

18

The number of subscribers to a streaming service is modeled by $S(m) = 250m + 3200$, where m is the number of months after January 2020. Which statement is the best interpretation of the number 250 in this model?

A) There were 250 subscribers in January 2020.

B) The service reached 250 subscribers after 3200 months.

C) The number of subscribers increases by 250 each year.

D) The number of subscribers increases by 250 each month.

19

What is the slope of the line given by the equation $4x - 6y = 18$ in the xy-plane?

A) $-\frac{3}{2}$

B) $-\frac{2}{3}$

C) $\frac{2}{3}$

D) $\frac{3}{2}$

20

For the linear function g, $g(3) = 10$ and $g(7) = 2$. What is the value of $g(0)$?

21

The function f is defined by $f(x) = 5x - 3$. If $f(c) = 2c + 9$ for some constant c, what is the value of c?

A) -4

B) 2

C) 4

D) 12

22

A candle burns at a constant rate. Its height, in centimeters, after t hours of burning is given by $h(t) = 18 - 1.5t$. Which of the following is the best interpretation of the solution to the equation $h(t) = 0$?

A) The height, in centimeters, of the candle before it is lit.

B) The number of centimeters the candle burns in one hour.

C) The height, in centimeters, of the candle after one hour.

D) The number of hours it takes for the candle to burn down completely.

Module 2

1

The function f is defined by $f(x) = -2x + 11$. What is the value of $f(-3)$?

A) -17
B) 5
C) 6
D) 17

2

A line in the xy-plane passes through the points $(-2, 5)$ and $(4, -7)$. What is the slope of this line?

3

x	2	5	8	11
$g(x)$	3	12	21	30

The table shows several values of the linear function g. What is the rate of change of g with respect to x?

A) $\frac{1}{3}$
B) 3
C) $\frac{9}{2}$
D) 9

4

A car travels at a constant speed along a highway. Its distance from a rest stop, in miles, after t hours of driving is given by $d(t) = 45t + 30$. What is the best interpretation of the number 30 in this model?

A) The car travels 30 miles each hour.
B) The car drives for a total of 30 hours.
C) The car travels a total distance of 30 miles.
D) The car was 30 miles from the rest stop when it started driving.

5

The function f is defined by $f(x) = \frac{1}{2}x - 7$. If $f(x) = -3$, what is the value of x?

A) -20
B) -8
C) 2
D) 8

6

The graph of $2x + 5y = 20$ is drawn in the xy-plane. What is the y-coordinate of the point where the graph crosses the y-axis?

A) -4
B) 4
C) 10
D) 20

7

A linear function f has a slope of 4, and its graph passes through the point $(3, 5)$. Which of the following defines $f(x)$?

A) $4x - 7$
B) $4x + 5$
C) $4x + 17$
D) $\frac{1}{4}x - 7$

8

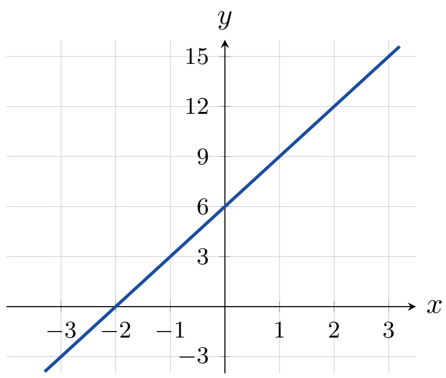
The function f is defined by $f(x) = ax + 7$, where a is a constant. If $f(4) = -5$, what is the value of a?

A) -12
B) -3
C) $-\frac{1}{3}$
D) 3

9

For the linear function f, $f(-2) = 13$ and $f(4) = -5$. What is the value of $f(1)$?

10



The graph of the linear function f is shown in the xy-plane. What is the value of $f(3)$?

A) 9

B) 12

C) 15

D) 18

11

Water is pumped into a pool that already holds some water. The amount of water in the pool, in liters, after t minutes of pumping is given by $W(t) = 25t + 150$. Which statement about the pool must be true?

A) The pool gains 25 liters of water each minute.

B) The pool gains 150 liters of water each minute.

C) The pool is empty when pumping begins.

D) The pool is full after 25 minutes.

12

In the xy-plane, a line with slope $-\frac{3}{2}$ passes through the points $(0, 9)$ and $(t, 0)$. What is the value of t?

13

A candle burns at a constant rate. After 2 hours its height is 21 centimeters, and after 5 hours its height is 15 centimeters. What was the height, in centimeters, of the candle before it was lit?

A) 19

B) 23

C) 25

D) 27

14

The equation $3x - 4y = 24$ defines a line in the xy-plane. What is the value of y when $x = 12$?

A) -3

B) 3

C) 9

D) 12

15

The function f is defined by $f(x) = 2x - 5$. Which of the following is equivalent to $f(x + 3)$?

A) $2x - 15$

B) $2x - 2$

C) $2x + 1$

D) $2x + 6$

16

For the linear function f, $f(3) = 8$ and $f(x + 1) - f(x) = -4$ for every value of x. What is the value of $f(6)$?

A) -4

B) 4

C) 12

D) 20

17

A linear function f satisfies $f(-3) = -1$ and $f(6) = 5$. If $f(x) = mx + b$, what is the value of b?

18

The population of a bird colony is modeled by $P(t) = 1500 - 45t$, where t is the number of years after 2010. Which of the following best describes the meanings of 1500 and 45 in this model?

A) The colony had 1500 birds in 2010, and it gains 45 birds each year.

B) The colony had 45 birds in 2010, and it loses 1500 birds each year.

C) The colony had 1500 birds in 2010, and it disappears after 45 years.

D) The colony had 1500 birds in 2010, and it loses 45 birds each year.

19

The function f is defined by $f(x) = 3x + c$, where c is a constant. If $f(4) = 2f(1)$, what is the value of c?

A) 2

B) 3

C) 6

D) 9

20

The total cost, in dollars, of a gym membership for m months is given by $C(m) = 35m + 80$. After how many months will the total cost be \$1200?

21

x	-4	1	7
$f(x)$	23	8	-10

The table gives three values of the linear function f. What is the value of $f(0)$?

A) 5

B) 8

C) 11

D) 23

22

For the linear function f, $f(0) = k$ and $f(4) = 3k$, where k is a constant. If $f(2) = 10$, what is the value of k?

A) $\frac{5}{2}$

B) 5

C) 10

D) 20

Lesson 3: Linear Equations in Two Variables

A linear equation in two variables describes a straight line in the xy-plane: every point (x, y) that makes the equation true lies on the line, and every point on the line makes it true. The Digital SAT returns to this idea constantly – it asks you to read a slope, to switch between the two standard ways of writing a line, to build a line from two points or from a point and a direction, and to decide when two lines are parallel or perpendicular. Everything in this topic comes back to one number, the slope.

The two forms you must recognise

A line can be written in **slope-intercept form** $y = mx + b$, where m is the slope and $(0, b)$ is the y-intercept, or in **standard form** $Ax + By = C$. Slope-intercept form is the one that tells you everything at a glance, so most SAT work begins by solving for y. Standard form is convenient when a question asks for intercepts, because setting $x = 0$ or $y = 0$ leaves a one-step equation.

Going from $Ax + By = C$ to $y = mx + b$ is always the same two moves: subtract the x-term, then divide every term by the coefficient of y. Doing that in general gives $y = -\frac{A}{B}x + \frac{C}{B}$, so the slope of $Ax + By = C$ is $-\frac{A}{B}$ – worth memorising, because it turns many questions into a single line of work.

THE FACTS OF THIS TOPIC

Slope through two points: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$.
Slope-intercept form: $y = mx + b$, with y-intercept $(0, b)$.
Point-slope form: $y - y_1 = m(x - x_1)$ for a line of slope m through (x_1, y_1).
Standard form: $Ax + By = C$ has slope $-\dfrac{A}{B}$, x-intercept $\left(\dfrac{C}{A}, 0\right)$ and y-intercept $\left(0, \dfrac{C}{B}\right)$.

Parallel: $m_1 = m_2$. **Perpendicular:** $m_1 m_2 = -1$, that is $m_2 = -\dfrac{1}{m_1}$.

WHAT THE SLOPE ACTUALLY SAYS

The slope is a rate: $m = \frac{\text{rise}}{\text{run}}$ says that increasing x by 1 changes y by m. That single sentence lets you move along a line without ever finding its equation. If a line has slope $-\frac{5}{3}$ and passes through $(3, 4)$, then running 3 to the right drops you 5, landing on $(6, -1)$.
It also tells you what the picture looks like before you compute anything: positive slope rises left to right, negative slope falls, slope 0 is horizontal ($y = b$), and a vertical line $x = a$ has no slope at all. Use this as a sanity check – an answer with the wrong sign on the slope is visibly wrong.

Building a line from what you are given

Nearly every "find the equation" question hands you one of three packages, and each has a fixed route.

A point and a slope. Substitute the point into $y = mx + b$ and solve for b. If the point is already on the y-axis, as in $(0, -2)$, then b is handed to you.

Two points. Compute m first, then use either point to find b. Using the second point afterwards is a free check.

Two intercepts. Write them as the points $(x_{\text{int}}, 0)$ and $(0, y_{\text{int}})$ and you are back in the two-point case. Intercepts are points, not loose numbers, and treating them as points prevents most mistakes here.

PARALLEL AND PERPENDICULAR

Two distinct lines are parallel exactly when their slopes are equal; the intercepts must differ, or the two equations describe the same line. Two lines are perpendicular exactly when their slopes multiply to -1, which means each slope is the *negative reciprocal* of the other: flip the fraction *and* change the sign. The negative reciprocal of $\frac{3}{4}$ is $-\frac{4}{3}$; of $-\frac{1}{2}$ it is 2; of 5 it is $-\frac{1}{5}$.
When a question puts an unknown constant inside a coefficient, as in $kx + 6y = 12$, express the slope in terms of that constant, $-\frac{k}{6}$ here, and then impose the parallel or perpendicular condition as an equation in k.

THE THREE MISTAKES THAT COST THE MOST POINTS

1. Half a negative reciprocal. Turning $\frac{2}{3}$ into $\frac{3}{2}$ (flipped, sign forgotten) or into $-\frac{2}{3}$ (sign changed, not flipped). Both appear as answer choices on purpose. Say the phrase in full every time: flip *and* negate.
2. Swapped intercepts. For $3x + 4y = 12$ the x-intercept is 4 and the y-intercept is 3. Set $y = 0$ for the x-intercept – the variable you set to zero is the one you are *not* looking for.
3. Sign errors while rearranging. Moving $2x$ across the equals sign makes it $-2x$, and dividing by a negative coefficient of y changes the sign of *every* term on the other side. Rewrite the whole line rather than editing it in place.

WORKING FASTER ON TEST DAY

Convert to $y = mx + b$ first, almost always: once a line is in that form, slope, intercept, graph and comparison questions all become reading exercises.
When the answer choices are equations, do not solve the whole problem – filter. Find the slope you need and cross out every choice with the wrong one; usually two or three vanish at once, and the survivors are separated by their intercept, which one substitution settles.
On a graph question, read the y-intercept off the picture, then count rise over run between two lattice points the line clearly passes through. On the Bluebook app you can also enter a candidate equation in the graphing calculator and see whether it lands on the given points; that is often faster than algebra for a two-point question.

WORKED EXAMPLE 1: POINT AND PERPENDICULARITY

Line n passes through $(-3, 5)$ and is perpendicular to the line $y = \frac{3}{4}x + 2$. Write the equation of line n.
Step 1 – get the slope. The given line has slope $\frac{3}{4}$. Flip and negate: line n has slope $-\frac{4}{3}$.
Step 2 – use the point. Substitute $(-3, 5)$ into $y = -\frac{4}{3}x + b$: $5 = -\frac{4}{3}(-3) + b = 4 + b$, so $b = 1$.
Step 3 – write and check. Line n is $y = -\frac{4}{3}x + 1$. Check by putting $x = -3$ back in: $-\frac{4}{3}(-3) + 1 = 4 + 1 = 5$. ✓
The step that breaks most often is the sign in $-\frac{4}{3}(-3)$: a negative slope times a negative x is *positive*. Getting $b = 9$ instead of $b = 1$ is exactly what that slip produces.

WORKED EXAMPLE 2: AN UNKNOWN COEFFICIENT

In the xy-plane, the line $kx + 6y = 12$ is perpendicular to the line $2x - 3y = 5$. Find k.
Step 1 – slopes in terms of the letters. Using $m = -\frac{A}{B}$: the first line has slope $-\frac{k}{6}$, and the second has slope $-\frac{2}{-3} = \frac{2}{3}$.
Step 2 – impose the condition. Perpendicular means the product of the slopes is -1: $-\frac{k}{6} \cdot \frac{2}{3} = -1$, so $-\frac{2k}{18} = -1$ and $-\frac{k}{9} = -1$.
Step 3 – solve and sanity-check. $k = 9$. Then the first line is $9x + 6y = 12$ with slope $-\frac{3}{2}$, and $-\frac{3}{2} \cdot \frac{2}{3} = -1$. ✓
Notice that no point, graph or intercept was ever needed. A parallel or perpendicular condition is a statement about slopes alone, so translate it into one equation in the unknown constant and solve.

Module 1

1

The equation of line ℓ in the xy-plane is $y = -3x + 7$. What is the slope of line ℓ?

A) -3

B) $\frac{1}{3}$

C) 3

D) 7

2

Line m is given by $y = \frac{1}{2}x - 6$. What is the y-coordinate of the point where line m crosses the y-axis?

A) $\frac{1}{2}$

B) -6

C) 6

D) 12

3

A line in the xy-plane passes through $(0, 4)$ and has slope 3. What is the value of y when $x = 5$?

4

Which equation is equivalent to $2x + y = 8$?

A) $y = 2x + 8$

B) $y = 2x - 8$

C) $y = -2x + 8$

D) $y = -2x - 8$

5

What is the x-coordinate of the x-intercept of the graph of $3x + 4y = 12$?

A) -4

B) 3

C) $\frac{4}{3}$

D) 4

6

What is the slope of the line that passes through the points $(-1, 4)$ and $(3, 16)$?

7

Which equation represents a line parallel to the line $y = 4x - 1$ in the xy-plane?

A) $y = -4x + 5$

B) $y = 4x + 5$

C) $y = -\frac{1}{4}x + 5$

D) $y = \frac{1}{4}x + 5$

8

What is the slope of the line $3x - 5y = 15$ in the xy-plane?

A) $\frac{3}{5}$

B) $-\frac{3}{5}$

C) $\frac{5}{3}$

D) 3

9

A line in the xy-plane has slope $\frac{5}{2}$ and passes through the point $(0, -2)$. Which equation represents the line?

A) $y = \frac{5}{2}x + 2$

B) $y = -\frac{5}{2}x - 2$

C) $y = \frac{5}{2}x - 2$

D) $y = \frac{2}{5}x - 2$

10

What is the slope of a line that is perpendicular to the line $2x + 3y = 9$?

A) $-\frac{2}{3}$

B) $\frac{2}{3}$

C) $\frac{3}{2}$

D) $-\frac{3}{2}$

11

Which equation represents the line that passes through the points $(1, 4)$ and $(3, 10)$?

A) $y = 3x + 1$

B) $y = 3x - 1$

C) $y = \frac{1}{3}x + 1$

D) $y = 3x + 4$

12

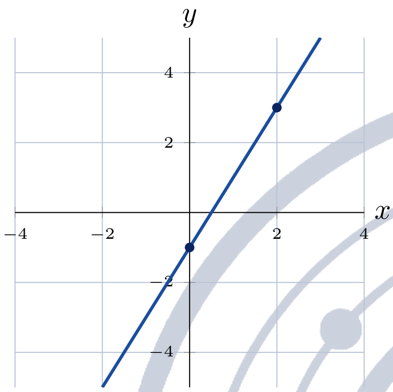
What is the x-coordinate of the x-intercept of the graph of $y = \frac{2}{3}x - 8$?

13

Which equation is equivalent to $y = -\frac{3}{4}x + 2$?

A) $3x - 4y = 8$
B) $4x + 3y = 8$
C) $-3x + 4y = 8$
D) $3x + 4y = 8$

14



The graph of a line in the xy-plane is shown above. Which equation represents this line?

A) $y = -2x - 1$
B) $y = 2x - 1$
C) $y = 2x + 1$
D) $y = \frac{1}{2}x - 1$

15

Line k is parallel to the line $y = -2x + 1$ and passes through the point $(4, 7)$. What is the y-coordinate of the y-intercept of line k?

16

Line n passes through the point $(-3, 5)$ and is perpendicular to the line $y = \frac{3}{4}x + 2$. Which equation represents line n?

A) $y = \frac{3}{4}x + 1$
B) $y = -\frac{4}{3}x + 9$
C) $y = \frac{4}{3}x + 9$
D) $y = -\frac{4}{3}x + 1$

17

In the xy-plane, the line $kx + 6y = 12$ is perpendicular to the line $2x - 3y = 5$. What is the value of k?

A) 9
B) -9
C) 4
D) -4

18

The graph of $4x + ky = 20$ in the xy-plane has the same slope as the graph of $y = -2x + 7$. What is the value of k?

19

A line in the xy-plane has x-intercept 6 and y-intercept -4. Which equation represents this line?

A) $2x + 3y = 12$
B) $3x - 2y = 12$
C) $2x - 3y = 12$
D) $2x - 3y = -12$

20

Line ℓ passes through the point $(2, -1)$ and is perpendicular to the line that passes through $(0, 3)$ and $(4, 1)$. What is the x-coordinate of the x-intercept of line ℓ?

A) -5
B) $\frac{5}{2}$
C) 5
D) $-\frac{2}{5}$

21

In the xy-plane, the line $y = mx + b$ passes through the points $(2, 7)$ and $(5, 16)$. What is the value of $m + b$?

A) 4
B) $\frac{20}{3}$
C) 2
D) 10

22

Line ℓ has equation $y = \frac{2}{5}x - 3$. Line m is perpendicular to line ℓ and passes through the point $(4, 1)$. What is the y-coordinate of the point where line m crosses the y-axis?

Module 2

1

What is the slope of the line that passes through the points $(-1, 4)$ and $(3, -4)$?

A) 2
B) -2
C) $-\frac{1}{2}$
D) $\frac{1}{2}$

2

What is the y-coordinate of the y-intercept of the graph of $2x - 5y = 20$ in the xy-plane?

3

Which equation represents a line perpendicular to the line $y = -\frac{1}{2}x + 3$?

A) $y = -2x + 1$
B) $y = -\frac{1}{2}x - 1$
C) $y = 2x - 1$
D) $y = \frac{1}{2}x + 3$

4

Which equation is equivalent to $6x - 2y = 18$?

A) $y = 3x + 9$
B) $y = -3x + 9$
C) $y = -3x - 9$
D) $y = 3x - 9$

5

A line in the xy-plane passes through the point $(0, 5)$ and has slope $-\frac{3}{4}$. Which equation represents the line?

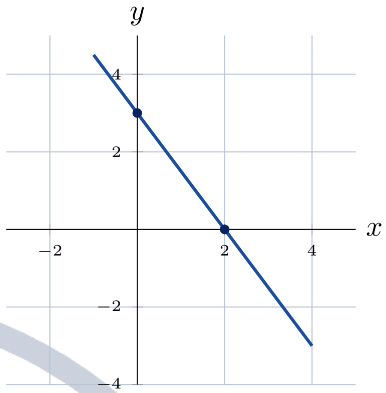
A) $y = -\frac{3}{4}x + 5$
B) $y = \frac{3}{4}x + 5$
C) $y = -\frac{3}{4}x - 5$
D) $y = \frac{4}{3}x + 5$

6

The graph of $y = 4x + c$, where c is a constant, passes through the point $(2, 3)$. What is the value of c?

A) 11
B) -5
C) 5
D) -11

7



A line in the xy-plane is graphed above. Which of the following is an equation of that line?

A) $y = \frac{3}{2}x + 3$
B) $y = -\frac{2}{3}x + 3$
C) $y = -\frac{3}{2}x + 3$
D) $y = -\frac{3}{2}x - 3$

8

A line in the xy-plane has x-intercept 8 and passes through the point $(2, 9)$. What is the y-coordinate of its y-intercept?

A) 8
B) -12
C) $\frac{3}{2}$
D) 12

9

In the xy-plane, the graphs of $2x + 3y = 12$ and $y = mx - 4$ are parallel, where m is a constant. What is the value of m?

10

Line ℓ passes through the point $(-4, 2)$ and has slope $\frac{1}{2}$. What is the x-coordinate of the x-intercept of line ℓ?

A) -8
B) 8
C) 4
D) -4

11

The total cost C, in dollars, of a gym membership for m months is given by $C = 25m + 60$, where 60 dollars is a one-time joining fee. Which equation gives m in terms of C?

A) $m = \frac{C+60}{25}$

B) $m = 25C - 60$

C) $m = \frac{C-60}{25}$

D) $m = \frac{C}{25} - 60$

12

The line that passes through the points $(1, -3)$ and $(k, 9)$ has slope 4, where k is a constant. What is the value of k?

A) -2

B) 4

C) 49

D) 2

13

Which equation represents a line perpendicular to the line $4x - 6y = 15$?

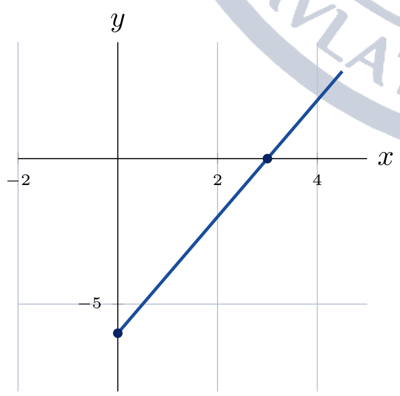
A) $y = \frac{3}{2}x + 1$

B) $y = -\frac{2}{3}x + 1$

C) $y = \frac{2}{3}x + 1$

D) $y = -\frac{3}{2}x + 1$

14



The line shown in the xy-plane above passes through the two marked points. Which equation represents this line in the form $Ax + By = C$?

A) $2x + y = 6$

B) $x - 2y = 6$

C) $2x - y = 6$

D) $2x - y = -6$

15

What is the x-coordinate of the x-intercept of the graph of $y = -\frac{5}{3}x + 10$?

16

In the xy-plane, the graph of $ax - 4y = 12$ is parallel to the line that passes through the points $(1, 2)$ and $(5, 8)$, where a is a constant. What is the value of a?

A) 6

B) -6

C) $\frac{3}{2}$

D) 8

17

Line m is perpendicular to the line $y = \frac{3}{5}x - 2$ and passes through the point $(3, 4)$. What is the value of y on line m when $x = 6$?

18

Line ℓ passes through the points $(-2, 7)$ and $(4, -5)$. Line m is perpendicular to line ℓ and passes through the midpoint of the segment joining those two points. Which equation represents line m?

A) $y = -2x + 3$

B) $y = \frac{1}{2}x - \frac{1}{2}$

C) $y = 2x - 1$

D) $y = \frac{1}{2}x + \frac{1}{2}$

19

In the xy-plane, the graph of $y = mx + 4$ passes through the point at which the graph of $2x + y = 16$ crosses the x-axis. What is the value of m?

A) 2

B) $-\frac{1}{2}$

C) $\frac{1}{2}$

D) -2

20

The graph of a line in the xy-plane has slope -3 and passes through the points $(a, 0)$ and $(0, 12)$, where a is a constant. What is the value of a?

A) -4

B) 36

C) 4

D) -36

21

In the xy-plane, the graph of $3x + ky = 12$ is perpendicular to the graph of $y = \frac{3}{4}x - 1$, where k is a constant. What is the value of k?

22

A technician charges a one-time service fee plus a constant hourly rate. The total charge is 145 dollars for a 2-hour job and 280 dollars for a 5-hour job. If the total charge C, in dollars, is a linear function of the number of hours h, what is the one-time service fee, in dollars?

A) 45

B) 35

C) 100

D) 55

Lesson 4: Systems of Two Linear Equations in Two Variables

A system of two linear equations asks a single question: is there a point (x, y) that satisfies both equations at once? Graphically that point is where two lines cross. The Digital SAT tests three things about such systems – solving them quickly by substitution or elimination, recognising when a system has no solution or infinitely many, and building a system from a short story about tickets, prices, speeds or mixtures. Every one of those tasks is worth practising until it is automatic, because systems appear on both modules of every test.

What a solution is

A solution to a system such as

$$\begin{cases} 2x + y = 7 \\ x - y = 2 \end{cases}$$

is an ordered pair (x, y) that makes *both* equations true. Here $(3, 1)$ works: $2(3) + 1 = 7$ and $3 - 1 = 2$. Because each equation graphs as a line, the solutions of the system are exactly the points the two lines share. Two distinct lines either cross once, or never cross (they are parallel), or coincide completely – which is why a linear system has one solution, no solution, or infinitely many, and never exactly two.

Checking a candidate pair is always legitimate: substitute it into both equations. If it satisfies only one, it is not a solution.

Method 1 – substitution

Substitution is the right tool when one equation is already solved for a variable, or when a variable has coefficient 1 so isolating it costs nothing.

1. Solve one equation for one variable.
2. Substitute that expression into the *other* equation. You now have one equation in one variable.
3. Solve it, then substitute back to get the second variable.

Substituting back into the equation you just came from returns a true but useless statement such as $7 = 7$; always go back to the other equation.

WORKED EXAMPLE – SUBSTITUTION

Solve

$$\begin{cases} y = 3x - 4 \\ 2x + y = 11 \end{cases}$$

The first equation is already solved for y, so replace y in the second equation:

$$2x + (3x - 4) = 11 \implies 5x - 4 = 11 \implies 5x = 15 \implies x = 3.$$

Now substitute $x = 3$ into $y = 3x - 4$: $y = 9 - 4 = 5$. The solution is $(3, 5)$.
Check it in the equation that was not used for the back-substitution: $2(3) + 5 = 11$. True, so the pair is correct.

Method 2 – elimination

Elimination is faster when both equations are in the form $ax + by = c$. Add or subtract multiples of the equations so that one variable cancels.

1. Multiply one or both equations so that the coefficients of one variable are equal in size and opposite in sign.
2. Add the equations; that variable disappears.
3. Solve for the survivor, then substitute back.

When you multiply an equation, multiply *every* term, including the constant on the right.

WORKED EXAMPLE – ELIMINATION

Solve

$$\begin{cases} 4x + 3y = 18 \\ 5x - 2y = 11 \end{cases}$$

To cancel y, scale the first equation by 2 and the second by 3:

$$\begin{aligned} 8x + 6y &= 36 \\ 15x - 6y &= 33 \end{aligned}$$

Adding gives $23x = 69$, so $x = 3$. Substituting into $4x+3y = 18$ gives $12+3y = 18$, so $y = 2$. The solution is $(3, 2)$.
Notice that the choice to eliminate y was deliberate: its coefficients 3 and -2 already had opposite signs, so the scaled equations could simply be added.

HOW MANY SOLUTIONS?

Write both equations as $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$ and compare the three ratios.

- $\dfrac{a_1}{a_2} \neq \dfrac{b_1}{b_2}$: the slopes differ, the lines cross once – **exactly one solution**.
- $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} \neq \dfrac{c_1}{c_2}$: same slope, different intercept – **no solution**.
- $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$: one equation is a multiple of the other – **infinitely many solutions**.

In slope-intercept form the same rules read: different slopes give one solution; equal slopes with different y-intercepts give none; identical equations give infinitely many.

THE FACTS TO MEMORISE

For the system $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$:

$$\text{one solution} \iff a_1b_2 - a_2b_1 \neq 0.$$

When $a_1b_2 = a_2b_1$ the lines are parallel; they coincide only if the constants scale by the same factor.

$$\text{no solution} \iff \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}, \qquad \text{infinitely many} \iff \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}.$$

ANSWERING THE WRONG QUESTION

The single most common way to lose a system question is to solve it perfectly and then report the wrong number. If you find $x = 3$ and $y = 5$, the test may ask for y, for $x + y$, for $2x - y$, or for the price of one shirt *and* one hat. Every one of 3, 5, 8 and 1 will be sitting in the answer choices.

Underline what is being asked before you start, and write both coordinates down before you look at the choices. A second trap: when a question asks for $x+y$ or $x-y$, adding or subtracting the two equations often produces it in one line, so check for that shortcut before grinding out both variables.

CHOOSING A METHOD, AND USING THE CALCULATOR

Scan the system first.

- A variable already isolated, or a coefficient of 1: **substitute**.
- Matching or opposite coefficients: **eliminate**.
- Coefficients swapped between the equations, as in $4x+7y$ and $7x+4y$: add for $x+y$, subtract for $x-y$.
- Fractions or decimals: multiply through to clear them *before* doing anything else.

On the Bluebook Desmos calculator you can type both equations and read the intersection point straight off the graph – fast and reliable for messy numbers. For questions with a parameter such as k, algebra is still faster than graphing, because the ratio test answers them in one line.

Building a system from words

Nearly every SAT system word problem has two unknowns and two facts. One fact usually counts objects and the other counts money, distance or amount.

- *Tickets*: $a+c=$ total tickets and $12a+8c=$ total dollars.
- *Mixture*: $x+y=$ total volume and $0.10x+0.30y=$ total acid.
- *Two plans*: set the two cost expressions equal to find when they match.
- *Upstream and downstream*: the effective speeds are $b+c$ and $b-c$.

Define your variables in writing, with units. "$a=$ number of adult tickets" prevents the wrong-variable error at the end.

WORKED EXAMPLE – A WORD PROBLEM

A shop sells notebooks for \$4 and folders for \$3. A customer buys 10 items and pays \$34. How many notebooks did the customer buy?
Let n be the number of notebooks and f the number of folders. Counting items gives $n+f=10$; counting money gives $4n+3f=34$. Substituting $f=10-n$:

$$4n+3(10-n)=34 \implies n+30=34 \implies n=4.$$

So the customer bought 4 notebooks and 6 folders. Check: $4(4)+3(6)=16+18=34$.
The structure is worth remembering: if all 10 items were folders the bill would be \$30, and each notebook adds \$1 to that baseline, so the \$4 excess means 4 notebooks.

Module 1

1

In the system shown, what is the value of x?

$$\begin{cases} y = 2x \\ 3x + y = 20 \end{cases}$$

A) 4
B) 5
C) 8
D) 20

2

What is the value of x in the system below?

$$\begin{cases} x + y = 10 \\ x - y = 4 \end{cases}$$

A) 3
B) 5
C) 7
D) 14

3

If $x + y = 12$ and $2x + y = 17$, what is the value of y?

4

The system below has exactly one solution (x, y). What is the value of y?

$$\begin{cases} y = x + 3 \\ y = 2x - 1 \end{cases}$$

A) 4
B) 5
C) 7
D) 11

5

For the solution (x, y) of the system shown, what is the value of $x + y$?

$$\begin{cases} 4x + y = 17 \\ y = x + 2 \end{cases}$$

A) 2
B) 3
C) 5
D) 8

6

What is the value of y in the system below?

$$\begin{cases} 2x + y = 7 \\ 3x - y = 8 \end{cases}$$

A) -1
B) 1
C) 2
D) 3

7

A theater sells adult tickets for \$12 each and child tickets for \$8 each. On one evening the theater sold 24 tickets and collected \$232. How many adult tickets were sold?

8

What is the value of y in the system below?

$$\begin{cases} 3x + 2y = 16 \\ 5x - 4y = -10 \end{cases}$$

A) 2
B) 5
C) 7
D) 11

9

If $2x - 3y = 5$ and $4x + 3y = 19$, what is the value of $x + y$?

10

The ordered pair (x, y) is the solution to the system shown. What is the value of $2x - y$?

$$\begin{cases} 3x + 2y = 19 \\ x - 2y = -7 \end{cases}$$

A) 1
B) 3
C) 5
D) 11

11

For what value of k does the system below have infinitely many solutions?

$$\begin{cases} kx + 6y = 10 \\ 2x + 3y = 5 \end{cases}$$

A) 2
B) 3
C) 4
D) 6

12

The system below has no solution. What is the value of k?

$$\begin{cases} 6x + ky = 9 \\ 2x + 5y = 4 \end{cases}$$

A) $\frac{5}{3}$
B) 3
C) 10
D) 15

13

At a school store a pen costs \$2 and a notebook costs \$5. Maria buys 12 items in total and pays \$39. How many notebooks does she buy?

A) 3
B) 5
C) 7
D) 12

14

Two cars start at the same time from points 300 miles apart and drive toward each other along a straight road. They meet after 3 hours. One car travels 20 miles per hour faster than the other. What is the speed, in miles per hour, of the faster car?

A) 40
B) 50
C) 60
D) 80

15

If $5x + 2y = 2$ and $3x - 2y = 14$, what is the value of $x - y$?

16

What is the value of y in the system below?

$$\begin{cases} \frac{x}{2} + \frac{y}{3} = 4 \\ x - y = 3 \end{cases}$$

A) 3
B) 6
C) 9
D) 12

17

If $4x + 7y = 17$ and $7x + 4y = 27$, what is the value of $x + y$?

A) 2
B) 4
C) 11
D) 44

18

The system $ax + 4y = 10$ and $3x + 6y = 12$ has no solution. What is the value of a?

19

A chemist mixes a 10% acid solution with a 30% acid solution to make 20 liters of a 25% acid solution. How many liters of the 30% solution are used?

A) 5
B) 10
C) 12
D) 15

20

Which of the following statements about the system $2x - 5y = 8$ and $-4x + 10y = -16$ is true?

A) The system has no solution.

B) The system has exactly one solution.

C) The system has exactly two solutions.

D) The system has infinitely many solutions.

21

If $5x + 3y = 29$ and $3x + 5y = 27$, what is the value of $x - y$?

22

For what value of c does the system below have infinitely many solutions?

$$\begin{cases} 4x - 6y = 10 \\ -2x + 3y = c \end{cases}$$

A) -10

B) -5

C) 5

D) 10

Module 2

1

The system shown has solution (x, y). What is the value of $x + y$?

$$\begin{cases} y = 4x - 1 \\ y = 2x + 7 \end{cases}$$

A) 4
B) 15
C) 19
D) 23

2

What is the value of y in the system below?

$$\begin{cases} 2x + 5y = 19 \\ 4x - 5y = -7 \end{cases}$$

A) 2
B) 3
C) 5
D) 19

3

A concert sold student tickets for \$9 each and adult tickets for \$15 each. A total of 40 tickets were sold for \$456. How many adult tickets were sold?

4

If (x, y) satisfies $7x + 3y = 27$ and $2x + 3y = 12$, what is the value of $2x - y$?

A) 2
B) 3
C) 4
D) 8

5

The system $8x + ky = 12$ and $2x + 5y = 7$ has no solution. What is the value of k?

A) -20
B) -5
C) 5
D) 20

6

The graphs of $y = -2x + 9$ and $y = x - 3$ intersect at the point (x, y). What is the value of $x + 2y$?

A) 4
B) 5
C) 6
D) 9

7

A bakery sells muffins for \$3 each and scones for \$4 each. One morning it sold 25 of these items for a total of \$87. How many muffins did it sell?

A) 10
B) 11
C) 12
D) 13

8

What is the value of y in the system below?

$$\begin{cases} 0.5x + 0.2y = 2.6 \\ x - y = 1 \end{cases}$$

A) 1
B) 3
C) 4
D) 7

9

If $5x + 4y = 6$ and $3x + 2y = 4$, what is the value of $x - y$?

10

In the system $2x + 3y = 7$ and $6x + 9y = c$, c is a constant. For what value of c does the system have infinitely many solutions?

A) 7
B) 14
C) 21
D) 63

11

A museum charges \$8 for an adult ticket and \$5 for a child ticket. On Saturday it sold 90 tickets and collected \$600. How many child tickets were sold?

A) 20
B) 30
C) 40
D) 50

12

If $4x + y = 13$ and $2x + 3y = 9$, what is the value of $3x + 2y$?

13

Gym A charges a \$40 registration fee plus \$15 per month. Gym B charges a \$10 registration fee plus \$20 per month. After how many months will the two gyms have cost the same total amount?

A) 2
B) 5
C) 6
D) 10

14

If $3(x + y) = 15$ and $2x - y = 4$, what is the value of $2y - x$?

A) 1
B) 3
C) 5
D) 7

15

A boat travels 24 miles downstream in 2 hours and the same 24 miles upstream in 3 hours. What is the speed of the current, in miles per hour?

16

The system $ax + 5y = 12$ and $4x + 10y = 7$ has no solution, where a is a constant. What is the value of a?

A) 2
B) $\frac{5}{2}$
C) 8
D) 20

17

The system $3x + ky = 8$ and $6x + 10y = 16$ has infinitely many solutions, where k is a constant. What is the value of k?

A) 2
B) 5
C) 10
D) 20

18

The sum of two numbers is 34. Three times the larger number exceeds twice the smaller number by 62. What is the larger number?

A) 8
B) 18
C) 26
D) 34

19

If $7x + 5y = 31$ and $5x + 7y = 17$, what is the value of $x - y$?

A) 4
B) 7
C) 12
D) 14

20

The system $y = 5x + 2$ and $y = (k - 1)x - 3$ has no solution, where k is a constant. What is the value of k?

21

In the system $2x - y = 4$ and $6x - 3y = c$, c is a constant. For what value of c does the system have at least one solution?

A) 2
B) 4
C) 8
D) 12

22

At a shop, 3 shirts and 2 hats cost \$85, and 2 shirts and 5 hats cost \$75. What is the total cost, in dollars, of one shirt and one hat?

A) 5

B) 20

C) 25

D) 30

Lesson 5: Linear Inequalities in One or Two Variables

An inequality is an equation that has been loosened: instead of pinning x to a single value it names a whole range of values. The Digital SAT uses inequalities for every situation with a limit in it – a budget, a weight capacity, a minimum score, a maximum number of items. You will be asked to solve a one-variable inequality, to translate a sentence such as "no more than" into symbols, to decide whether a point lies in a shaded half-plane, and to pick the ordered pair that satisfies a whole system. Everything you already know about solving linear equations carries over, with exactly one new rule to remember.

Solving a linear inequality

Solve $3x + 7 \leq 25$ the same way you would solve $3x + 7 = 25$: subtract 7 from both sides to get $3x \leq 18$, then divide both sides by 3 to get $x \leq 6$. Adding or subtracting the same number on both sides, and multiplying or dividing by a *positive* number, never disturb the direction of the inequality sign.

The answer $x \leq 6$ is not one number; it is every number up to and including 6. A quick way to check a solution set is to test one value inside it and one value outside it. Here $x = 0$ gives $7 \leq 25$, true, and $x = 10$ gives $37 \leq 25$, false – so $x \leq 6$ is pointing the right way.

MULTIPLY OR DIVIDE BY A NEGATIVE: FLIP THE SIGN

This is the single most tested idea in the topic. If you multiply or divide both sides of an inequality by a *negative* number, you must reverse the inequality symbol.

$$-2x < 8 \quad \Longrightarrow \quad x > -4 \qquad \text{(divide by } -2\text{, and } < \text{ becomes } >\text{)}$$

Why? Because $3 < 5$ is true but $-3 < -5$ is false: multiplying by a negative reflects both numbers across 0 and swaps their order. Note that subtracting a negative, as in $x - (-4) \leq 9$, does *not* trigger the rule – only multiplying or dividing does. A safe dodge if you distrust yourself: move the variable term to the side that keeps its coefficient positive. From $-2x < 8$, add $2x$ and subtract 8 to get $-8 < 2x$, then divide by 2 to get $-4 < x$, which says $x > -4$ with no flipping at all.

TRANSLATING THE WORDS INTO SYMBOLS

at most, no more than, maximum of, up to	$\leq$
at least, no less than, minimum of, or more	$\geq$
less than, under, fewer than	$<$
more than, over, exceeds	$>$

"At most \$40" means $c \leq 40$: spending exactly \$40 is still allowed. "Under \$40" means $c < 40$: exactly \$40 is not allowed. The test writes distractors that differ from the key only in that one detail, so read the phrase before you read the numbers.

WORKED EXAMPLE 1: A BUDGET

A caterer charges a \$75 setup fee plus \$12 per guest. A club can spend no more than \$500. What is the greatest number of guests the club can have?

Let g be the number of guests. "No more than \$500" is ≤ 500, so the cost $75 + 12g$ must satisfy

$$75 + 12g \leq 500.$$

Subtract 75: $12g \leq 425$. Divide by 12: $g \leq 35.41\ldots$

Now interpret. Guests come in whole numbers, and g must be at most $35.41\ldots$, so the greatest whole number that works is $g = 35$. Check: $75 + 12(35) = 495 \leq 500$, while 36 guests would cost 507. When a context forces whole numbers, always round *toward* the allowed side – down for a maximum, up for a minimum.

Compound inequalities

A compound inequality such as $-5 \le 2x - 1 < 9$ is two statements at once: $-5 \le 2x - 1$ and $2x - 1 < 9$. Solve it by doing the same operation to all three parts. Add 1 everywhere: $-4 \le 2x < 10$. Divide everything by 2: $-2 \le x < 5$.

The solution is every number from -2 up to but not including 5. The least integer in the set is -2 and the greatest is 4. If you ever multiply all three parts by a negative number, both symbols flip and the two ends trade places.

A LINEAR INEQUALITY IN TWO VARIABLES IS A HALF-PLANE

The graph of $y \le 2x + 1$ is the line $y = 2x + 1$ together with everything below it. Two features carry all the information:

The boundary line. Graph the related equation. Draw it *solid* for $\le$ or $\ge$, because points on the line satisfy the inequality, and *dashed* for $<$ or $>$, because they do not.

Which side to shade. When the inequality is solved for y, $y >$ shades above the line and $y <$ shades below it. If the inequality is not solved for y – say $3x - 2y \ge 6$ – do not guess: pick a test point not on the line, usually $(0, 0)$, and substitute. Here $0 \ge 6$ is false, so shade the side that does *not* contain the origin.

SYSTEMS: TEST THE POINT, DO NOT GRAPH

A system of inequalities is solved by the region where all the shaded half-planes overlap. On the SAT you are almost never asked to draw that region – you are asked which ordered pair lies in it, or which system matches a story.

For "which ordered pair is a solution," substitute each choice into the *simpler* inequality first and cross off everything that fails; usually only one or two choices survive to face the second inequality. This is far faster than sketching, and it cannot go wrong. For "which system matches the story," check the direction of each symbol first ($\le$ for a budget or a capacity, $\ge$ for a requirement or a minimum); the choices usually differ there before they differ anywhere else.

WORKED EXAMPLE 2: A SYSTEM IN CONTEXT

A store sells notebooks for $\$4$ and binders for $\$6$. A teacher must buy at least 20 items in total and can spend at most $\$96$. Which system models this, and is $(12, 8)$ a solution?

Let n be the number of notebooks and b the number of binders. "At least 20 items" counts items, not dollars, so it is $n + b \ge 20$. "At most \$96" counts dollars, so it is $4n + 6b \le 96$. The system is

$$n + b \ge 20, \qquad 4n + 6b \le 96.$$

Now test $(12, 8)$: $12 + 8 = 20 \ge 20$ is true (equality is allowed by $\ge$), and $4(12) + 6(8) = 48 + 48 = 96 \le 96$ is true as well. So $(12, 8)$ does satisfy the system – it sits exactly on the corner where both boundary lines meet, which is precisely the sort of point the test uses to catch students who wrote $>$ and $<$ instead of $\ge$ and $\le$.

THREE WAYS STUDENTS LOSE THE POINT

1. Answering the boundary instead of the question. If $g \le 35.4$ and the question asks for the greatest number of guests, the answer is 35, not 35.4 and not 36.

2. Reversing a subtraction. "5 less than x is at most 12" is $x - 5 \le 12$, not $5 - x \le 12$.

3. Attaching the constraint to the wrong quantity. If shirts cost \$4 and hats cost \$7, the money inequality is $4s + 7h \leq 100$; writing $7s + 4h \leq 100$ swaps the prices, and writing $s + h \leq 100$ confuses a count with a cost.

Module 1

1

Which of the following is equivalent to the inequality $3x + 5 \leq 20$?

A) $x \leq 15$

B) $x \leq 5$

C) $x \geq 5$

D) $x \leq \frac{25}{3}$

2

A delivery van can carry a total package weight of at most 1200 pounds. If w represents the total weight, in pounds, of the packages in the van, which inequality represents this situation?

A) $w \geq 1200$

B) $w > 1200$

C) $w \leq 1200$

D) $w < 1200$

3

What is the least integer value of x that satisfies $2x - 7 > 5$?

4

Which of the following ordered pairs (x, y) is a solution to the inequality $y > 2x - 1$?

A) $(0, 0)$

B) $(1, 1)$

C) $(2, 3)$

D) $(3, 4)$

5

A student buys x notebooks that cost \$3 each and y pens that cost \$2 each. The student spends no more than \$30. Which inequality represents this situation?

A) $2x + 3y \leq 30$

B) $3x + 2y \geq 30$

C) $3x + 2y < 30$

D) $3x + 2y \leq 30$

6

What is the greatest value of x that satisfies $4x + 3 \leq 27$?

A) 5

B) 6

C) 7

D) 24

7

If $5 - x \geq 2$, which of the following must be true?

A) $x \geq 3$

B) $x \leq 7$

C) $x \leq 3$

D) $x \geq -3$

8

Which of the following values of x satisfies $-3 < 2x + 1 \leq 7$?

A) -3

B) 3

C) -2

D) 4

9

What is the least integer value of n that satisfies $\dfrac{n}{3} + 4 > 10$?

10

Movie tickets cost \$9 each, and an online order also carries a one-time service fee of \$6. Maria has \$60 to spend. What is the greatest number of tickets she can order?

A) 4

B) 5

C) 6

D) 7

11

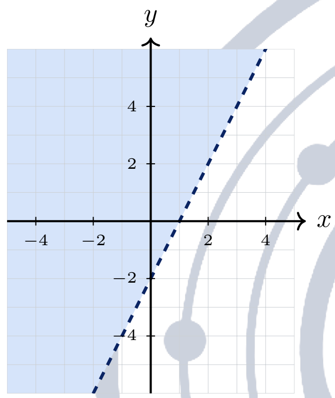
Which ordered pair (x, y) is a solution to the system $y \geq 2x - 3$ and $x + y < 5$?

A) $(2, 4)$
B) $(3, 3)$
C) $(4, 1)$
D) $(0, 0)$

12

The solution to the inequality $12 - 5x \geq -3$ is $x \leq k$. What is the value of k?

13



The shaded region in the xy-plane above is the graph of a linear inequality whose boundary line passes through $(0, -2)$ and $(1, 0)$. Which inequality is graphed?

A) $y \geq 2x - 2$
B) $y > 2x - 2$
C) $y < 2x - 2$
D) $y \leq 2x - 2$

14

A student's grade is the average (arithmetic mean) of four test scores. The student scored 76, 85, and 82 on the first three tests and wants an average of at least 80. Which inequality gives all possible scores s on the fourth test?

A) $s \geq 77$
B) $s > 77$
C) $s \geq 80$
D) $s \leq 77$

15

What is the greatest integer value of x that satisfies $-4x + 9 > -11$?

16

If $-2 \leq 3 - 2x < 7$, which of the following describes all possible values of x?

A) $-2 \leq x < \frac{5}{2}$
B) $-\frac{5}{2} \leq x < 2$
C) $-2 < x \leq \frac{5}{2}$
D) $-\frac{1}{2} < x \leq 2$

17

A caterer needs at least 40 pounds of fruit and will buy a pounds of apples at \$2 per pound and p pounds of pears at \$3 per pound. The caterer can spend at most \$90. Which system of inequalities represents this situation?

A) $a + p \leq 40$ and $2a + 3p \geq 90$
B) $a + p \geq 40$ and $2a + 3p \leq 90$
C) $a + p \geq 40$ and $3a + 2p \leq 90$
D) $a + p \geq 40$ and $2a + 3p \geq 90$

18

A phone plan costs \$25 per month plus \$0.10 for each text message sent. A customer wants the monthly bill to be at most \$40. What is the greatest number of text messages the customer can send in a month?

A) 15
B) 150
C) 250
D) 400

19

How many ordered pairs (x, y) satisfy both $y \leq -x + 4$ and $y \geq -x + 7$?

A) 0
B) 1
C) 2
D) Infinitely many

20

In the xy-plane, the boundary line of the inequality $y \geq mx + 1$ passes through the point $(4, 9)$. What is the value of m?

21

Which of the following describes all values of x that satisfy $\frac{2x-1}{3} \geq \frac{x+4}{2}$?

A) $x \geq \frac{13}{7}$

B) $x \geq 10$

C) $x \geq 14$

D) $x \leq 14$

22

Which of the following describes all values of x that satisfy $-\frac{1}{2}x + 3 > x - 6$?

A) $x < 6$

B) $x > 6$

C) $x < \frac{9}{2}$

D) $x < -6$

Module 2

1

Which of the following is equivalent to $-3x \geq 12$?

A) $x \geq -4$

B) $x \leq -4$

C) $x \leq 4$

D) $x \geq 4$

2

What is the greatest integer value of x that satisfies $7 - 2x \geq 1$?

3

An elevator can hold at most 8 people. There are already 3 people inside. If p represents the number of additional people who can enter, which inequality represents all possible values of p?

A) $p \geq 5$

B) $p \leq 8$

C) $p < 5$

D) $p \leq 5$

4

Which of the following ordered pairs (x, y) is NOT a solution to $2x + 3y \leq 12$?

A) $(0, 0)$

B) $(1, 1)$

C) $(3, 2)$

D) $(6, 1)$

5

Which of the following describes all values of x that satisfy $5(x - 2) < 3x$?

A) $x < 5$

B) $x > 5$

C) $x < \frac{5}{2}$

D) $x < -5$

6

What is the least integer value of x that satisfies $6x + 5 \geq 41$?

7

To qualify for a tournament, a player needs a total of at least 240 points over three games. After two games the player has 152 points. Which inequality gives all possible scores p in the third game that qualify the player?

A) $p \geq 88$

B) $p > 88$

C) $p \leq 88$

D) $p \geq 392$

8

If $4 \leq 3x - 5 \leq 16$, what is the greatest possible value of x?

A) 3

B) 6

C) 7

D) 9

9

An empty truck weighs 4200 pounds. It is loaded with boxes that weigh 65 pounds each. A bridge allows a total weight of no more than 7000 pounds. What is the greatest number of boxes the truck can carry across the bridge?

10

A club buys x shirts at \$4 each and y hats at \$7 each. The club needs at least 20 items in all and can spend at most \$100. Which system represents this situation?

A) $x + y \leq 20$ and $4x + 7y \geq 100$

B) $x + y \geq 20$ and $7x + 4y \leq 100$

C) $x + y \geq 20$ and $4x + 7y \leq 100$

D) $x + y > 20$ and $4x + 7y < 100$

11

Which of the following describes all values of x that satisfy $\dfrac{3}{4}x - 2 \leq 7$?

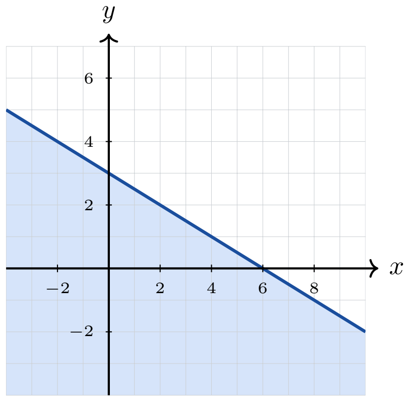
A) $x \leq \frac{20}{3}$

B) $x \leq 9$

C) $x \geq 12$

D) $x \leq 12$

12



The shaded region above is the graph of a linear inequality whose boundary line passes through $(0, 3)$ and $(6, 0)$. Which inequality is graphed?

A) $y \geq -\frac{1}{2}x + 3$
B) $y \leq -\frac{1}{2}x + 3$
C) $y < -\frac{1}{2}x + 3$
D) $y \leq -2x + 3$

13

Which of the following describes all values of x that satisfy $-\dfrac{x}{4} + 1 > 3$?

A) $x < -8$
B) $x > -8$
C) $x < 8$
D) $x > 8$

14

How many integer values of x satisfy $-7 < 2x+1 \leq 9$?

A) 5
B) 7
C) 8
D) 9

15

The point $(3, k)$ is a solution to the inequality $2x+y \geq 11$. What is the least possible value of k?

16

The solution to the inequality $3 - kx \geq 12$ is $x \leq -3$, where k is a constant. What is the value of k?

A) -3
B) $-\frac{1}{3}$
C) $\frac{1}{3}$
D) 3

17

A student has at most 12 hours this weekend to study for a math exam and a science exam, and must spend at least 4 hours on science. If m is the number of hours spent on math and s is the number of hours spent on science, which system represents this situation?

A) $m + s \leq 12$ and $s \geq 4$
B) $m + s \geq 12$ and $s \geq 4$
C) $m + s \leq 12$ and $s \leq 4$
D) $m + s \leq 12$ and $m \geq 4$

18

What is the greatest integer value of x that satisfies $\dfrac{2x+1}{5} < \dfrac{x+4}{3}$?

19

Which of the following has the same solution set as $2(3 - x) \leq 4x - 6$?

A) $x \leq 2$
B) $x \geq \frac{1}{2}$
C) $x \geq 2$
D) $x \leq \frac{2}{3}$

20

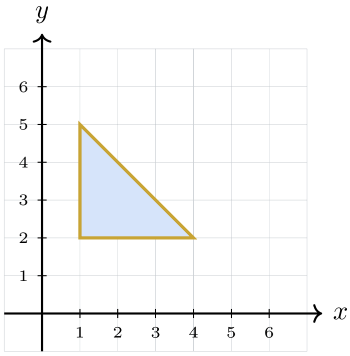
At a school fair a vendor sells hot dogs for \$3 each and drinks for \$2 each, and must take in at least \$180 in total. The vendor has already sold 40 drinks. What is the least number of hot dogs the vendor must sell to reach the goal?

21

Which ordered pair (x, y) is a solution to the system $3x - y > 6$ and $x + 2y \leq 4$?

A) $(0, 2)$
B) $(2, 0)$
C) $(3, -2)$
D) $(4, 1)$

22



The shaded triangular region above is the graph of the system $y \leq -x + 6$, $y \geq 2$, and $x \geq 1$. Which of the following points lies in the shaded region?

A) $(2, 3)$

B) $(0, 3)$

C) $(2, 1)$

D) $(3, 4)$

PART 2

Advanced Math

Lesson 6: Equivalent Expressions

Every *equivalent expressions* question asks the same thing in different clothing: rewrite an expression in a new form without changing its value for any allowed input. The Digital SAT tests this constantly — expanding and multiplying polynomials, factoring, simplifying and combining rational expressions, and rewriting powers and radicals. Nothing here requires clever tricks; it requires knowing which moves are legal and applying them in a sensible order.

What equivalent means

Two expressions are equivalent when they produce the same value for every value of the variable for which both are defined. That definition gives you a free checking tool: substitute a convenient number such as $x = 2$ into the original expression and into each answer choice, and the choices that disagree are gone. It also explains the phrase "for $x \neq 3$" that so often appears in the stem — the two forms agree everywhere except where the original was undefined.

Multiplying polynomials

Multiplication distributes over addition, so every term of the first factor must multiply every term of the second. For two binomials that means four products; for a binomial times a trinomial it means six. After multiplying, collect like terms — terms with the identical power of the variable. x^2 and x are never like terms, and neither are x and a constant.

THE THREE PRODUCTS WORTH MEMORISING

$$(a+b)^2 = a^2 + 2ab + b^2$$
$$(a-b)^2 = a^2 - 2ab + b^2$$
$$(a+b)(a-b) = a^2 - b^2$$

Read them left to right to expand and right to left to factor. The middle term $2ab$ in the first two is the one students lose; the missing middle term in the third is the reason a difference of squares factors so cleanly.

$(a+b)^2$ IS NOT a^2+b^2

Squaring a sum is not squaring each piece. $(2x-5)^2 = 4x^2-20x+25$, not $4x^2+25$. The same warning covers radicals and reciprocals: $\sqrt{a+b} \neq \sqrt{a}+\sqrt{b}$ and $\dfrac{1}{a+b} \neq \dfrac{1}{a}+\dfrac{1}{b}$. Every one of these test writers' favourite distractors comes from applying an operation term by term when the operation does not distribute.

Factoring, in the order you should try it

First pull out the greatest common factor: $3x^2 - 12 = 3\left(x^2-4\right)$. Then look at what is left. If it is a difference of squares, split it: $x^2 - 4 = (x+2)(x-2)$. If it is a trinomial x^2+bx+c, find two numbers with product c and sum b. If the leading coefficient is not 1, as in $6x^2 - x - 15$, find two numbers with product $ac = -90$ and sum $b = -1$ — here 9 and -10 — split the middle term and group. Factoring is finished only when no factor can be factored further.

CANCELLING IS DIVIDING BY A COMMON FACTOR

In a fraction you may cancel only a factor that multiplies the *entire* numerator and the *entire* denominator. So $\dfrac{(x+2)(x-3)}{(x+2)} = x-3$ is legal, but $\dfrac{x+2}{x+4}$ cannot become $\dfrac{2}{4}$ — those are terms of a sum. This is why the first move on any rational expression is to factor top and bottom completely; until you have factored, you have nothing to cancel.

Adding and subtracting rational expressions

Rewrite each fraction over a common denominator, then combine numerators only. For $\dfrac{2}{x+1} + \dfrac{3}{x}$ the common denominator is $x(x+1)$, so the fractions become $\dfrac{2x}{x(x+1)}$ and $\dfrac{3(x+1)}{x(x+1)}$ and the sum is $\dfrac{5x+3}{x^2+x}$. When subtracting, put the second numerator in parentheses first: the minus sign applies to all of it.

EXPONENT AND RADICAL RULES

$$x^a \cdot x^b = x^{a+b} \qquad \frac{x^a}{x^b} = x^{a-b} \qquad (x^a)^b = x^{ab}$$

$$(xy)^a = x^a y^a \qquad x^{-a} = \frac{1}{x^a} \qquad x^{m/n} = \sqrt[n]{x^m} = \left(\sqrt[n]{x}\right)^m$$

For radicals, factor out the largest perfect square: $\sqrt{72x^2} = \sqrt{36x^2}\cdot\sqrt{2} = 6x\sqrt{2}$ for $x > 0$. Radical terms add only when the radicands match: $4\sqrt{3} - 2\sqrt{3} = 2\sqrt{3}$.

EXPONENTS ON COEFFICIENTS AND THE ADD/MULTIPLY MIX-UP

An outside exponent lands on the coefficient too: $(2x^3)^3 = 8x^9$, not $6x^9$, and $(8x^6)^{2/3} = 4x^4$, not $\frac{16}{3}x^4$. And keep the two rules apart — multiplying powers *adds* exponents while a power of a power *multiplies* them. $x^5 \cdot x^3 = x^8$, but $(x^5)^3 = x^{15}$.

PLUG IN A NUMBER WHEN THE ALGEBRA GETS LONG

If the choices are all expressions in x, pick a small number that is legal in every expression (avoid 0, 1 and anything that makes a denominator zero — $x = 2$ or $x = 3$ usually works), evaluate the original, then evaluate the choices. Only the equivalent one matches. This costs about thirty seconds and never requires you to factor anything; it is the fastest route through a hard rational-expression item, and it is a perfect check when you have done the algebra and are not sure.

WORKED EXAMPLE 1: A RATIONAL EXPRESSION

For $x \neq 2$ and $x \neq -2$, which expression is equivalent to $\dfrac{x^2-x-6}{x^2-4}$?

Step 1 — factor the numerator. Two numbers with product -6 and sum -1 are -3 and 2, so $x^2 - x - 6 = (x-3)(x+2)$.

Step 2 — factor the denominator. It is a difference of squares: $x^2 - 4 = (x+2)(x-2)$.

Step 3 — cancel the common factor. The expression is $\dfrac{(x-3)(x+2)}{(x+2)(x-2)}$, and $x+2$ is a factor of both, so it cancels: $\dfrac{x-3}{x-2}$.

Step 4 — check. Put $x = 3$ into the original: $\dfrac{9-3-6}{9-4} = 0$. Put $x = 3$ into $\dfrac{x-3}{x-2}$: $\dfrac{0}{1} = 0$. They agree.

Note what was *not* legal: cancelling x^2 against x^2, since those are terms, not factors.

WORKED EXAMPLE 2: MATCHING COEFFICIENTS

The equation $(3x + a)(2x + b) = 6x^2 + 19x + 15$ is true for all x, where a and b are positive integers. What is $a + b$?

Step 1 — expand the left side. $(3x + a)(2x + b) = 6x^2 + 3bx + 2ax + ab = 6x^2 + (2a + 3b)x + ab$.

Step 2 — match coefficients. "True for all x" means the two sides are the same polynomial, so corresponding coefficients are equal: $2a + 3b = 19$ and $ab = 15$.

Step 3 — test the factor pairs of 15. The positive pairs are $(1, 15)$, $(3, 5)$, $(5, 3)$, $(15, 1)$. Only $a = 5$, $b = 3$ gives $2(5) + 3(3) = 19$.

Step 4 — answer the question asked. It asks for $a + b$, not for a: the answer is 8. Notice that getting the constant term right is not enough — the middle term is what selects the pair.

Module 1

1

Which expression is equivalent to $(x + 3)(x + 5)$?

A) $x^2 + 8x + 15$
B) $x^2 + 15$
C) $x^2 + 8x + 8$
D) $x^2 + 15x + 8$

2

Which expression is equivalent to $3x(2x^2 - 4x + 5)$?

A) $6x^3 - 12x^2 + 15$
B) $6x^3 - 12x^2 + 15x$
C) $6x^3 - 4x + 5$
D) $5x^3 - x^2 + 8x$

3

The expression $(2x + 3)(x - 1)$ can be written in the form $2x^2 + bx - 3$, where b is a constant. What is the value of b?

4

Which of the following is equivalent to $x^2 - 9$?

A) $(x - 3)^2$
B) $(x + 9)(x - 1)$
C) $x(x - 9)$
D) $(x + 3)(x - 3)$

5

Which expression is equivalent to $\dfrac{x^5 \cdot x^3}{x^2}$ for $x \neq 0$?

A) x^6
B) x^8
C) x^{10}
D) x^{13}

6

Which expression is equivalent to $\dfrac{12x^5}{3x^2}$ for $x \neq 0$?

A) $9x^3$
B) $4x^7$
C) $4x^3$
D) $36x^3$

7

Which expression is equivalent to $6x^2 + 9x$?

A) $3x(2x + 9)$
B) $6x(x + 3)$
C) $3x(3x + 2)$
D) $3x(2x + 3)$

8

Which of the following is equivalent to $x^2 + 7x + 12$?

A) $(x + 2)(x + 6)$
B) $(x + 1)(x + 12)$
C) $(x - 3)(x - 4)$
D) $(x + 3)(x + 4)$

9

Which expression is equivalent to $(2x - 5)^2$?

A) $4x^2 + 25$
B) $4x^2 - 25$
C) $4x^2 - 20x + 25$
D) $4x^2 - 10x + 25$

10

For $x \neq 4$, the expression $\dfrac{x^2 - 16}{x - 4}$ is equivalent to $x + c$, where c is a constant. What is the value of c?

11

For $x \neq -2$, which expression is equivalent to $\dfrac{x^2+5x+6}{x+2}$?

A) $x+6$

B) $\dfrac{5x+6}{2}$

C) $x+4$

D) $x+3$

12

For $x \neq 0$, which expression is equivalent to $(3x^{-2})^{-1}$?

A) $\dfrac{x^2}{3}$

B) $3x^2$

C) $\dfrac{1}{3x^2}$

D) $-3x^2$

13

What is the value of $16^{3/4}$?

14

For $x > 0$, which expression is equivalent to $\sqrt{72x^2}$?

A) $36x\sqrt{2}$

B) $6x^2\sqrt{2}$

C) $6x\sqrt{2}$

D) $8x\sqrt{3}$

15

For $x \neq 0$, which expression is equivalent to $\dfrac{1}{x}+\dfrac{1}{2x}$?

A) $\dfrac{1}{3x}$

B) $\dfrac{2}{3x}$

C) $\dfrac{3}{2x}$

D) $\dfrac{1}{2x^2}$

16

Which of the following is equivalent to $2x^2+7x+3$?

A) $(2x+1)(x+3)$

B) $(2x+3)(x+1)$

C) $(2x-1)(x-3)$

D) $(x+1)(x+3)$

17

For $x \neq 2$ and $x \neq -2$, which expression is equivalent to $\dfrac{x+2}{x^2-4}$?

A) $\dfrac{1}{x^2-2}$

B) $\dfrac{1}{x+2}$

C) $x-2$

D) $\dfrac{1}{x-2}$

18

For $x \neq 2$ and $x \neq -3$, the expression $\dfrac{x^2-9}{x^2+x-6}$ is equivalent to $\dfrac{x-3}{x-c}$, where c is a constant. What is the value of c?

19

For $x \neq 1$ and $x \neq -1$, which expression is equivalent to $\dfrac{1}{x-1}-\dfrac{1}{x+1}$?

A) 0

B) $\dfrac{-2}{x^2-1}$

C) $\dfrac{2}{x^2-1}$

D) $\dfrac{2x}{x^2-1}$

20

For $x > 0$, which expression is equivalent to $(8x^6)^{2/3}$?

A) $4x^4$

B) $4x^9$

C) $16x^4$

D) $64x^4$

21

The equation $(3x+a)(2x+b) = 6x^2+19x+15$ is true for all x, where a and b are positive integers. What is the value of $a+b$?

22

For $x \neq -5$, which expression is equivalent to $\dfrac{2x^2+7x-15}{x+5}$?

A) $2x+3$

B) $2x-3$

C) $2x-15$

D) $x-3$

Module 2

1

Which expression is equivalent to $(x-4)(x+4)$?

A) x^2-16

B) x^2+16

C) $x^2-8x-16$

D) x^2-16x

2

Which expression is equivalent to $2x(3x-7)+5x$?

A) $6x^2-2x$

B) $6x^2+5x-7$

C) $6x^2-9x$

D) $6x^2-19x$

3

The expression $x^2+10x+25$ is equivalent to $(x+k)^2$, where k is a positive constant. What is the value of k?

4

Which of the following is equivalent to $4x^2-25$?

A) $(2x-5)^2$

B) $(4x+5)(x-5)$

C) $(2x+5)(2x-5)$

D) $(2x+5)^2$

5

For $x \neq 0$ and $y \neq 0$, which expression is equivalent to $\dfrac{x^4y^3}{xy^5}$?

A) x^3y^2

B) $\dfrac{x^4}{y^2}$

C) $\dfrac{x^3}{y^8}$

D) $\dfrac{x^3}{y^2}$

6

What is the value of $27^{2/3}$?

7

Which expression shows $3x^2-12$ factored completely?

A) $3(x^2-4)$

B) $3(x+2)(x-2)$

C) $(x+2)(x-2)$

D) $3(x-2)^2$

8

Which expression is equivalent to $(x+3)(x^2-2x+4)$?

A) $x^3+x^2+2x+12$

B) $x^3+x^2-2x+12$

C) $x^3-5x^2+10x+12$

D) x^3+27

9

The equation $(2x-3)(3x+k)=6x^2+x-15$ is true for all values of x, where k is a constant. What is the value of k?

10

For $x \neq -3$, which expression is equivalent to $\dfrac{3x^2-27}{x+3}$?

A) $3x-9$

B) $3x+9$

C) $3x-27$

D) $x-9$

11

For $x \neq 0$ and $x \neq -1$, which expression is equivalent to $\dfrac{2}{x+1}+\dfrac{3}{x}$?

A) $\dfrac{5}{x^2+x}$

B) $\dfrac{5x+3}{x^2+x}$

C) $\dfrac{5}{2x+1}$

D) $\dfrac{5x+2}{x^2+x}$

12

For $x>0$, the expression $x^{1/2}\cdot x^{3/2}$ is equivalent to x^k, where k is a constant. What is the value of k?

13

Which expression is equivalent to $\sqrt{48} - \sqrt{12}$?

A) 6
B) $2\sqrt{3}$
C) $6\sqrt{3}$
D) $2\sqrt{6}$

14

For $x \neq 0$ and $y \neq 0$, which expression is equivalent to $(2x^3y^{-2})^3$?

A) $\dfrac{8x^9}{y^6}$
B) $\dfrac{6x^9}{y^6}$
C) $\dfrac{8x^6}{y^5}$
D) $8x^9y^6$

15

For $x \neq 3$, which expression is equivalent to $\dfrac{x^2-5x+6}{x-3}$?

A) $x+2$
B) $x-6$
C) $x-2$
D) $x-3$

16

Which of the following is equivalent to $6x^2 - x - 15$?

A) $(2x-3)(3x+5)$
B) $(2x+3)(3x-5)$
C) $(6x+5)(x-3)$
D) $(2x+5)(3x-3)$

17

For $x \neq 1$ and $x \neq -3$, which expression is equivalent to $\dfrac{x^2-1}{x+3} \cdot \dfrac{x^2+3x}{x-1}$?

A) $x^2 - x$
B) $x+1$
C) $\dfrac{x^2+x}{x+3}$
D) x^2+x

18

For $x \neq \frac{1}{2}$, the expression $\dfrac{2x^2+5x-3}{2x-1}$ is equivalent to $x+c$, where c is a constant. What is the value of c?

19

For nonzero values of x and y, which expression is equivalent to $\dfrac{1}{x^{-1}+y^{-1}}$?

A) $x+y$
B) $\dfrac{xy}{x+y}$
C) $\dfrac{x+y}{xy}$
D) $\dfrac{1}{x+y}$

20

The expression $(2x+3)^2 - (2x-3)^2$ is equivalent to kx, where k is a constant. What is the value of k?

21

For $x \neq 2$ and $x \neq -2$, which expression is equivalent to $\dfrac{x^2-x-6}{x^2-4}$?

A) $\dfrac{x+3}{x+2}$
B) $\dfrac{x+6}{4}$
C) $\dfrac{x-3}{x+2}$
D) $\dfrac{x-3}{x-2}$

22

For $x \neq 1$ and $x \neq -1$, which expression is equivalent to $\dfrac{x}{x-1} - \dfrac{1}{x+1}$?

A) $\dfrac{x-1}{x^2-1}$
B) $\dfrac{x^2+x-1}{x^2-1}$
C) $\dfrac{x^2+1}{x^2-1}$
D) $\dfrac{x^2-x+1}{x^2-1}$

Lesson 7: Quadratic Equations

A quadratic equation is any equation that can be put in the form $ax^2 + bx + c = 0$ with $a \neq 0$. The Digital SAT asks about quadratics constantly: solve one, count how many real solutions it has, find the value of a parameter that makes it have exactly one solution, or read the answer out of a short story about a falling ball or a rectangular garden. There are only four tools — factoring, square roots, completing the square and the quadratic formula — and the whole game is choosing the fastest one for the equation in front of you.

The zero-product property and solving by factoring

Everything about quadratics rests on one fact: if a product is zero, one of the factors is zero. So if $(x-2)(x+5) = 0$, then $x - 2 = 0$ or $x + 5 = 0$, giving $x = 2$ or $x = -5$. To use this, the equation must first be arranged with 0 on one side. Given $x^2 + 3x = 10$, do not divide by x and do not guess — move everything left, $x^2 + 3x - 10 = 0$, then factor into $(x + 5)(x - 2) = 0$.

When $a = 1$, factoring $x^2 + bx + c$ means finding two numbers whose product is c and whose sum is b. For $x^2 - 7x + 10$ the pair is -2 and -5: their product is 10 and their sum is -7, so the factorisation is $(x-2)(x-5)$ and the roots are 2 and 5. Notice that the roots are the negatives of the numbers inside the parentheses; that sign flip is where most careless errors live.

ROOTS, ZEROS AND x-INTERCEPTS ARE THE SAME THING

The solutions of $ax^2 + bx + c = 0$, the zeros of the function $f(x) = ax^2 + bx + c$, and the x-coordinates of the points where the parabola $y = ax^2 + bx + c$ crosses the x-axis are three names for one set of numbers. When a question says "the graph of $y = x^2 - 6x + 8$ intersects the x-axis at $(r, 0)$ and $(s, 0)$", it is simply asking you to solve $x^2 - 6x + 8 = 0$.

Square roots and completing the square

If the equation contains no plain x term — only a perfect square and a constant — take square roots directly and keep both signs: $(x-3)^2 = 49$ gives $x - 3 = \pm 7$, so $x = 10$ or $x = -4$. Any quadratic can be forced into this shape by completing the square. Starting from $x^2 + 8x - 3 = 0$, move the constant across to get $x^2 + 8x = 3$, then add the square of half the coefficient of x, that is $(8/2)^2 = 16$, to both sides: $x^2 + 8x + 16 = 19$, i.e. $(x + 4)^2 = 19$, so $x = -4 \pm \sqrt{19}$.

When the leading coefficient is not 1, factor it out of the first two terms first. From $2x^2 + 12x + 5 = 0$ write $2(x^2 + 6x) + 5 = 0$, complete the square inside, $2[(x + 3)^2 - 9] + 5 = 0$, and tidy up to $2(x + 3)^2 = 13$.

FORMULAS YOU MUST KNOW

For $ax^2 + bx + c = 0$ with $a \neq 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \qquad D = b^2 - 4ac.$$

The quantity D is the discriminant, and it alone decides how many real solutions there are:

$$D > 0 \ \Rightarrow \ \text{two distinct real solutions}, \qquad D = 0 \ \Rightarrow \ \text{exactly one}, \qquad D < 0 \ \Rightarrow \ \text{none}.$$

Vieta's formulas give the sum and product of the two solutions without solving at all:

$$r_1 + r_2 = -\frac{b}{a}, \qquad r_1 r_2 = \frac{c}{a}.$$

Finally, the vertex form $y = a(x - h)^2 + k$ produced by completing the square has its vertex at (h, k) and axis of symmetry $x = h = -\dfrac{b}{2a}$.

CHOOSING A METHOD IN FIVE SECONDS

Scan the equation before you write anything. No x term? Take square roots. Factors jump out (small integer coefficients, $a = 1$)? Factor. Asked only for the sum or product of the solutions? Use Vieta and stop. Asked only how many solutions, or for a parameter that forces one solution? Use the discriminant and never solve. Asked to rewrite in the form $(x + h)^2 = k$ or to find a vertex? Complete the square. Nothing else works? Quadratic formula.

COMMON TRAPS

Forgetting the $\pm$. $x^2 = 25$ has two solutions, 5 and -5; a question that says "the positive solution" is warning you that there is another one.
Sign of the root. From $(x + 6)(x - 1) = 0$ the roots are -6 and 1, not 6 and -1.
Dividing by x. Solving $x^2 = 5x$ by cancelling x loses the solution $x = 0$; factor as $x(x - 5) = 0$ instead.
Dividing wrongly by $2a$. In the quadratic formula the whole numerator, $-b$ included, sits over $2a$: $x = \dfrac{6 \pm \sqrt{8}}{4}$ is not $6 \pm \dfrac{\sqrt{8}}{4}$.
The discriminant is $b^2 - 4ac$, not $b^2 + 4ac$, and a negative c makes $-4ac$ positive.
Answering the wrong question. In a word problem the algebra usually produces two roots and only one of them is a length or a time; a negative time is not an answer.

CALCULATOR / STRATEGY

On the calculator section you may graph $y = ax^2 + bx + c$ in Desmos and read the x-intercepts straight off the curve; that also settles "how many real solutions" instantly. Even better, type the equation itself with a parameter and add a slider when a question asks for the value of k that makes the graph touch the axis exactly once. If a question gives you four numerical choices, testing them is often faster than solving: substitute and see which one makes the equation true. And when the answer must be a single grid-in value, re-read the question — the SAT usually wants the positive solution, the sum of the solutions, or a coefficient, precisely because those are single numbers.

WORKED EXAMPLE 1 — FACTORING VERSUS VIETA

The equation $2x^2 - 7x + 3 = 0$ has solutions p and q with $p < q$. What is the value of $q - p$?
Factor: look for a split of $-7x$ whose parts multiply to $2 \cdot 3 = 6$. The pair -1 and -6 works, so $2x^2 - x - 6x + 3 = x(2x - 1) - 3(2x - 1) = (2x - 1)(x - 3) = 0$. Hence $x = \frac{1}{2}$ or $x = 3$, so $p = \frac{1}{2}$, $q = 3$ and $q - p = \frac{5}{2}$.
Check with Vieta: the sum of the roots is $-b/a = 7/2$ and the product is $c/a = 3/2$, and indeed $\frac{1}{2} + 3 = \frac{7}{2}$ and $\frac{1}{2} \cdot 3 = \frac{3}{2}$. If the question had asked only for $p + q$, Vieta would have answered it in one step with no factoring at all.

WORKED EXAMPLE 2 — A PARAMETER AND THE DISCRIMINANT

In the xy-plane, the line $y = 2x + 3$ intersects the parabola $y = x^2 + k$ at exactly one point. What is the value of k?
Set the expressions equal: $x^2 + k = 2x + 3$, so $x^2 - 2x + (k - 3) = 0$. "Exactly one point of intersection" means this quadratic has exactly one real solution, so its discriminant is zero:

$$D = (-2)^2 - 4(1)(k - 3) = 4 - 4k + 12 = 16 - 4k = 0 \implies k = 4.$$

Check: with $k = 4$ the equation becomes $x^2 - 2x + 1 = (x - 1)^2 = 0$, a single root $x = 1$, and the point of tangency is $(1, 5)$. Note that you never had to solve anything — the discriminant answered a question about the number of solutions directly, which is exactly what it is for.

WORKED EXAMPLE 3 — A PROJECTILE

A ball is launched from a 24-foot platform and its height in feet after t seconds is $h = -16t^2 + 40t + 24$. After how many seconds does it hit the ground?

Hitting the ground means $h = 0$: $-16t^2 + 40t + 24 = 0$. Divide by -8 to keep the numbers clean: $2t^2 - 5t - 3 = 0$, which factors as $(2t + 1)(t - 3) = 0$, so $t = -\frac{1}{2}$ or $t = 3$. Time cannot be negative, so the ball lands after 3 seconds. The rejected root is not a mistake — it is the algebra's way of describing where the parabola would have crossed the axis before the launch.

Module 1

1

What is the larger of the two solutions to $x^2-7x+10 = 0$?

A) 2
B) 5
C) 7
D) 10

2

If $(x-3)^2 = 49$, what is the greatest possible value of x?

A) -4
B) 7
C) 10
D) 52

3

If $2x^2 = 50$, what is the positive value of x?

4

What is the sum of the solutions to $x^2 - 9x + 14 = 0$?

A) -9
B) 2
C) 7
D) 9

5

What is the product of the solutions to $(x+4)(x-6) = 0$?

A) -24
B) -2
C) 2
D) 24

6

What is the value of the discriminant of the equation $x^2 + 6x + 5 = 0$?

A) 4
B) 16
C) 36
D) 56

7

Which of the following is a solution to $x^2+x-12 = 0$?

A) 3
B) 4
C) 6
D) 12

8

The solutions to $x^2 - 6x + 4 = 0$ can be written in the form $3 \pm \sqrt{k}$, where k is a positive constant. What is the value of k?

9

The equation $x^2 + 8x - 3 = 0$ can be rewritten in the form $(x + 4)^2 = b$, where b is a constant. What is the value of b?

A) -19
B) 3
C) 13
D) 19

10

What is the sum of the solutions to $2x^2-10x+3 = 0$?

A) -5
B) $\frac{3}{2}$
C) 5
D) 10

11

A ball is kicked from ground level, and its height in feet after t seconds is given by $h = -16t^2 + 48t$. After how many seconds does the ball return to the ground?

12

In the equation $x^2 + kx + 9 = 0$, k is a positive constant and the equation has exactly one real solution. What is the value of k?

A) 3
B) 6
C) 9
D) 12

13

What is the smaller of the two solutions to $2x^2 - 7x + 3 = 0$?

A) $\frac{1}{2}$
B) 1
C) $\frac{3}{2}$
D) 3

14

If $(2x - 1)^2 = 25$, what is the sum of the possible values of x?

A) -2
B) 1
C) 3
D) 5

15

The equation $3x^2 + bx + 12 = 0$ has exactly one real solution, and b is positive. What is the value of b?

16

The equation $2x^2 + 12x + 5 = 0$ can be written in the form $2(x + 3)^2 = k$, where k is a constant. What is the value of k?

A) -13
B) 5
C) 13
D) 23

17

What is the product of the solutions to $3x^2 - 5x - 2 = 0$?

A) $-\frac{2}{3}$
B) $-\frac{1}{3}$
C) $\frac{2}{3}$
D) $\frac{5}{3}$

18

In the xy-plane, the graph of $y = x^2 + 6x + c$ intersects the x-axis at exactly one point. What is the value of c?

19

A rectangular garden is 3 meters longer than it is wide, and its area is 88 square meters. What is the width of the garden, in meters?

A) 5
B) 8
C) 11
D) 16

20

The solutions to $x^2 - 4x - 8 = 0$ can be written as $a \pm b\sqrt{3}$, where a and b are positive integers. What is the value of $a + b$?

A) 2
B) 3
C) 4
D) 6

21

The solutions to $x^2 + px + 40 = 0$ are $x = 4$ and $x = r$, where p and r are constants. What is the value of $p + r$?

A) -14
B) -4
C) 4
D) 14

22

The equation $2x^2 + 8x + k = 0$ has no real solutions.
Which of the following could be the value of k?

A) 2

B) 4

C) 8

D) 10

Module 2

1

What is the positive solution to $x^2 - 2x - 15 = 0$?

A) -5

B) -3

C) 3

D) 5

2

If $3x^2 - 27 = 0$, what is the positive value of x?

3

How many distinct real solutions does the equation $x^2 + 4x + 7 = 0$ have?

A) 0

B) 1

C) 2

D) 3

4

The equation $x^2 + 10x + 18 = 0$ can be rewritten as $(x+5)^2 = c$, where c is a constant. What is the value of c?

A) -18

B) 7

C) 18

D) 43

5

What is the product of the solutions to $x^2 + 7x - 18 = 0$?

A) -18

B) -7

C) 7

D) 18

6

A stone is thrown from the top of a cliff, and its height in feet after t seconds is $h = -16t^2 + 64t + 80$. After how many seconds does the stone hit the ground?

7

What are the solutions to $x^2 - 4x + 1 = 0$?

A) $2 \pm \sqrt{3}$

B) $2 \pm \sqrt{5}$

C) $-2 \pm \sqrt{3}$

D) $4 \pm 2\sqrt{3}$

8

The length of a rectangular rug is 5 feet greater than its width, and the rug covers 84 square feet. What is the width of the rug, in feet?

A) -12

B) 6

C) 7

D) 12

9

The solutions to $x^2 + bx + c = 0$ are -3 and 5, where b and c are constants. What is the value of c?

10

The equation $3x^2 - 12x + 5 = 0$ can be written in the form $3(x-2)^2 = k$, where k is a constant. What is the value of k?

A) -7

B) 5

C) 7

D) 17

11

In the equation $kx^2 + 12x + 9 = 0$, k is a nonzero constant and the equation has exactly one real solution. What is the value of k?

A) 3

B) 4

C) 6

D) 9

12

A rocket's height in feet after t seconds is $h = -16t^2 + 96t$. What is the smaller of the two values of t at which the height is 128 feet?

13

What is the positive difference between the two solutions to $x^2 - 6x + 2 = 0$?

A) $\sqrt{7}$

B) $2\sqrt{7}$

C) 6

D) 7

14

For which of the following values of k does the equation $x^2 + kx + 4 = 0$ have two distinct real solutions?

A) 0

B) 2

C) 4

D) 5

15

What is the sum of the solutions to $4x^2 - 20x + 9 = 0$?

A) $\frac{9}{4}$

B) $\frac{5}{2}$

C) 5

D) 20

16

If $x^2 + 6x = 16$, what is the sum of all possible values of x?

A) -16

B) -8

C) -6

D) 16

17

If $\dfrac{12}{x} = x - 1$ and $x \neq 0$, what is the greatest possible value of x?

A) -3

B) 3

C) 4

D) 12

18

In the xy-plane, the line $y = 2x + 3$ intersects the parabola $y = x^2 + k$ at exactly one point, where k is a constant. What is the value of k?

19

The equation $x^2 - 14x + k = 0$ has two positive integer solutions that differ by 4. What is the value of k?

20

A ball is launched from a platform, and its height in feet after t seconds is $h = -16t^2 + 40t + 24$. After how many seconds does the ball hit the ground?

A) $\frac{3}{2}$

B) 2

C) 3

D) 5

21

The solutions to $2x^2 - 8x + 3 = 0$ can be written as $\dfrac{a \pm \sqrt{b}}{2}$, where a and b are positive integers. What is the value of $a + b$?

A) 14

B) 18

C) 44

D) 48

22

The equation $x^2 + kx + k + 3 = 0$ has exactly one real solution. What is the sum of all possible values of k?

A) -4

B) 2

C) 4

D) 6

Lesson 8: Quadratic and Polynomial Functions

Topic 7 asked you to *solve* a quadratic equation. This topic asks something different: given a quadratic or polynomial *function*, what does its graph look like, and which algebraic form of the rule shows you that? The Digital SAT tests this constantly, and almost always in the same three ways: read a feature (vertex, intercept, axis of symmetry) off a form that already displays it; rewrite the rule into the form that displays the feature you want; or reason about a higher-degree polynomial from its factors. A question such as "which of the following equivalent forms displays the minimum value as a constant or coefficient?" is not asking you to compute anything at all – it is asking whether you know what each form shows.

The three forms of a parabola

Every quadratic function can be written in three ways, and all three describe the same curve. What changes is which feature is visible without doing any work.

Standard form $f(x) = ax^2 + bx + c$ shows the y-intercept: putting $x = 0$ leaves $f(0) = c$, so the graph crosses the y-axis at $(0, c)$.

Vertex form $f(x) = a(x - h)^2 + k$ shows the vertex (h, k) and therefore the axis of symmetry $x = h$ and the maximum or minimum value k.

Factored form $f(x) = a(x - r)(x - s)$ shows the x-intercepts $(r, 0)$ and $(s, 0)$, the values of x that make the output zero.

In all three, the sign of a decides the direction: $a > 0$ opens upward and the vertex is a minimum, $a < 0$ opens downward and the vertex is a maximum.

THE FACTS TO HAVE AUTOMATIC

Axis of symmetry / vertex x-coordinate from standard form:

$$x = -\frac{b}{2a}$$

Vertex y-coordinate: substitute that x back into the function – $f\left(-\frac{b}{2a}\right)$ is the maximum or minimum value.

Midpoint shortcut: if the graph crosses the x-axis at $x = r$ and $x = s$, the axis of symmetry is exactly halfway between them, $x = \dfrac{r+s}{2}$.

Vertex form: $f(x) = a(x - h)^2 + k$ has vertex (h, k).

y-intercept, always: evaluate $f(0)$, whatever the form.

MATCH THE QUESTION TO THE FORM

When a question names a feature, it is telling you which form to move toward.

- "vertex", "maximum", "minimum", "axis of symmetry" $\Rightarrow$ vertex form, or $x = -\frac{b}{2a}$.
- "x-intercepts", "zeros", "solutions", "where the graph meets the x-axis" $\Rightarrow$ factored form.
- "y-intercept", "initial value", "value when $x = 0$" $\Rightarrow$ standard form, or just substitute 0.

Nothing about the parabola changes when you rewrite it. Only your view of it does.

Completing the square: standard form to vertex form

To rewrite $ax^2 + bx + c$ in vertex form, factor a out of the first two terms, add and subtract the square of half the resulting linear coefficient, and tidy up. On the SAT you can almost always skip the algebra: compute

$h = -\frac{b}{2a}$, compute $k = f(h)$, and write $a(x-h)^2 + k$. That takes two substitutions and is far harder to botch than completing the square under time pressure.

WORKED EXAMPLE 1 – REWRITING TO SHOW THE MINIMUM

Which of the following is equivalent to $f(x) = 2x^2 - 12x + 13$ and displays the minimum value of f as a constant or coefficient?

The phrase "minimum value ... as a constant" means vertex form, so we need $2(x-h)^2 + k$.

Step 1. $h = -\dfrac{b}{2a} = -\dfrac{-12}{2(2)} = 3$.

Step 2. $k = f(3) = 2(9) - 12(3) + 13 = 18 - 36 + 13 = -5$.

Step 3. So $f(x) = 2(x-3)^2 - 5$, and a quick expansion confirms it: $2(x^2 - 6x + 9) - 5 = 2x^2 - 12x + 18 - 5 = 2x^2 - 12x + 13$. ✓

The minimum value is -5, attained at $x = 3$. Notice that -5 is now sitting in the expression as a constant, which is exactly what the question asked for; in the original form it was invisible.

THREE WAYS STUDENTS LOSE THIS POINT

1. The sign of h. In $f(x) = (x+6)^2 - 11$ the vertex is $(-6, -11)$, not $(6, -11)$. Vertex form is $a(x-h)^2 + k$, so $x + 6$ means $h = -6$. Ask yourself: what value of x makes the squared part zero?

2. Answering the wrong coordinate. "What is the maximum height?" wants the y-value; "after how many seconds is the height greatest?" wants the x-value. Both numbers appear in the choices, every time.

3. Reading a zero as its negative. From $g(x) = (x-2)(x+5)$ the x-intercepts are $x = 2$ and $x = -5$ – the values that make each factor zero, not the numbers you see written.

Polynomials: zeros, multiplicity and end behaviour

For a polynomial in factored form, each factor $(x - r)$ contributes a zero at $x = r$. The exponent on that factor is its **multiplicity**, and multiplicity controls what the graph does there: an odd multiplicity means the graph *crosses* the x-axis, an even multiplicity means it *touches* and turns back.

End behaviour depends on only two things, the degree and the sign of the leading coefficient. Odd degree sends the two ends in opposite directions; even degree sends them the same way. A positive leading coefficient sends the right-hand end up, a negative one sends it down. So $p(x) = -2x^3 + 5x - 1$ (odd degree, negative lead) falls to $-\infty$ on the right and rises to $+\infty$ on the left.

THE REMAINDER AND FACTOR THEOREMS

Remainder theorem. When a polynomial $p(x)$ is divided by $x - a$, the remainder is $p(a)$. You never have to do long division on the SAT – substitute.

Factor theorem. $x - a$ is a factor of $p(x)$ exactly when $p(a) = 0$; equivalently, a is a zero of p and $(a, 0)$ is an x-intercept of the graph.

Watch the sign: dividing by $x + 3$ means $a = -3$, so evaluate $p(-3)$.

WORKED EXAMPLE 2 – THE FACTOR THEOREM WITH AN UNKNOWN

The polynomial p is defined by $p(x) = 3x^3 - 5x^2 + kx + 8$, where k is a constant. If $x - 2$ is a factor of $p(x)$, what is the value of k?

Step 1. "$x - 2$ is a factor" translates, by the factor theorem, into $p(2) = 0$. That single translation is the whole question.

Step 2. $p(2) = 3(8) - 5(4) + 2k + 8 = 24 - 20 + 2k + 8 = 2k + 12$.

Step 3. Set it equal to zero: $2k + 12 = 0$, so $k = -6$.

Check: with $k = -6$, $p(2) = 24 - 20 - 12 + 8 = 0$. ✓ If the question had instead said "the remainder is 4 when $p(x)$ is divided by $x - 2$", the only change would be $2k + 12 = 4$.

ON TEST DAY

Read the answer choices first on "which form displays..." questions. If every choice is factored, the question is about intercepts; if every choice is $a(x-h)^2+k$, it is about the vertex. You then only have to check which one is genuinely equivalent, and expanding a choice is faster than completing the square.
Use the graphing calculator for confirmation, not for solving. Typing the function in and reading the vertex takes about as long as $-\frac{b}{2a}$, but it is an excellent check when a question is worth a full re-read.
Underline what is asked – "x-coordinate", "value of f", "how many seconds", "how high". The distractors on this topic are built almost entirely from students answering the right question's neighbour.

Module 1

1

The function f is defined by $f(x) = (x-4)^2+7$. The vertex of the graph of $y = f(x)$ in the xy-plane is the point (h, k). What is the value of h?

A) -7
B) -4
C) 4
D) 7

2

The function f is defined by $f(x) = x^2 - 6x + 11$. The axis of symmetry of the graph of $y = f(x)$ in the xy-plane is the line $x = c$. What is the value of c?

3

The function g is defined by $g(x) = (x-2)(x+5)$. What are the x-intercepts of the graph of $y = g(x)$ in the xy-plane?

A) $x = -2$ and $x = 5$
B) $x = 2$ and $x = 5$
C) $x = -2$ and $x = -5$
D) $x = 2$ and $x = -5$

4

What is the y-coordinate of the y-intercept of the graph of $y = (x-3)(x+8)$ in the xy-plane?

A) -24
B) -11
C) 5
D) 24

5

The function f is defined by $f(x) = -(x-1)^2+6$. What is the maximum value of $f(x)$?

A) -6
B) -1
C) 1
D) 6

6

The remainder when the polynomial $p(x) = x^2+3x-5$ is divided by $x-2$ is r. What is the value of r?

7

In the xy-plane, which of the following equations defines a parabola with vertex $(5, -2)$?

A) $y = (x+5)^2 - 2$
B) $y = (x-5)^2 + 2$
C) $y = (x-5)^2 - 2$
D) $y = (x-2)^2 + 5$

8

Which of the following expressions is equivalent to $x^2 + 8x + 3$ and displays the minimum value of the function $f(x) = x^2 + 8x + 3$ as a constant or coefficient?

A) $(x+4)^2 + 3$
B) $(x-4)^2 - 13$
C) $(x+8)^2 - 13$
D) $(x+4)^2 - 13$

9

The function f is defined by $f(x) = 3x^2 - 12x + 7$. What is the x-coordinate of the vertex of the graph of $y = f(x)$ in the xy-plane?

10

A ball is thrown upward from a platform. Its height above the ground, in feet, t seconds after it is thrown is given by $h(t) = -16t^2 + 64t + 5$. What is the maximum height, in feet, reached by the ball?

A) 2

B) 5

C) 64

D) 69

11

Which of the following expressions is equivalent to $x^2 - 2x - 15$ and displays the x-intercepts of the graph of $y = x^2 - 2x - 15$ as constants or coefficients?

A) $(x+5)(x-3)$

B) $(x-5)(x+3)$

C) $(x-5)(x-3)$

D) $(x-15)(x+1)$

12

The polynomial function p is defined by $p(x) = x(x-3)^2(x+4)$. Which zero of p has multiplicity 2?

A) -4

B) 0

C) 2

D) 3

13

The polynomial function p is defined by $p(x) = -2x^3 + 5x - 1$. Which of the following describes the end behaviour of the graph of $y = p(x)$ in the xy-plane?

A) As x increases without bound, $p(x)$ increases without bound; as x decreases without bound, $p(x)$ decreases without bound.

B) As x increases without bound, $p(x)$ decreases without bound; as x decreases without bound, $p(x)$ increases without bound.

C) As x increases or decreases without bound, $p(x)$ increases without bound.

D) As x increases or decreases without bound, $p(x)$ decreases without bound.

14

Which of the following is a factor of the polynomial $p(x) = x^3 - 7x + 6$?

A) $x+1$

B) $x+2$

C) $x-1$

D) $x-3$

15

In the xy-plane, the graph of the quadratic function g crosses the x-axis at $(1, 0)$ and at $(7, 0)$. What is the x-coordinate of the vertex of the graph of g?

A) 3

B) 4

C) 6

D) 8

16

In the xy-plane, the graph of the quadratic function f has vertex $(2, -3)$ and passes through the point $(4, 5)$. The function can be written as $f(x) = a(x-2)^2 - 3$, where a is a constant. What is the value of a?

A) $\frac{1}{2}$

B) 2

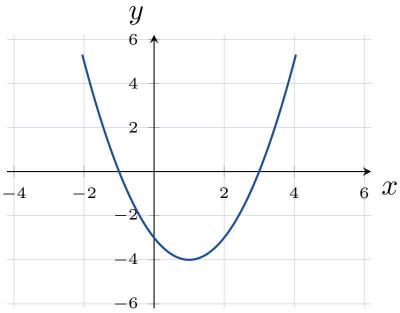
C) 4

D) 8

17

The polynomial p is defined by $p(x) = 2x^3 + kx - 6$, where k is a constant. When $p(x)$ is divided by $x - 2$, the remainder is 4. What is the value of k?

18



The graph of the quadratic function f is shown in the xy-plane above. Which equation could define f?

A) $f(x) = (x + 1)^2 - 4$

B) $f(x) = (x - 1)^2 + 4$

C) $f(x) = -(x - 1)^2 - 4$

D) $f(x) = (x - 1)^2 - 4$

19

Which of the following expressions is equivalent to $(x - 3)^2 - 16$ and displays the y-coordinate of the y-intercept of the graph of $y = (x-3)^2-16$ as a constant or coefficient?

A) $x^2 - 6x + 9$

B) $x^2 + 6x - 7$

C) $x^2 - 6x - 7$

D) $x^2 - 9x - 7$

20

The polynomial p is defined by $p(x) = x^3 + ax^2 - 9x - 18$, where a is a constant. If $x + 2$ is a factor of $p(x)$, what is the value of a?

21

A store's daily revenue, in dollars, from selling an item at a price of p dollars is modelled by $R(p) = p(120 - 4p)$. What is the maximum daily revenue, in dollars?

A) 15

B) 60

C) 900

D) 1800

22

The function f is defined by $f(x) = a(x - 3)^2 + 5$, where a is a negative constant. Which of the following must be true?

A) The graph of $y = f(x)$ has a minimum at the point $(3, 5)$.

B) The maximum value of f is 5.

C) The graph of $y = f(x)$ has no x-intercepts.

D) The graph of $y = f(x)$ is symmetric about the line $x = 5$.

Module 2

1

The function f is defined by $f(x) = -(x+6)^2 + 11$. For what value of x does f attain its maximum value?

A) -11

B) -6

C) 6

D) 11

2

The function g is defined by $g(x) = 2x^2 - 16x + 5$. The graph of $y = g(x)$ in the xy-plane is symmetric about the line $x = c$. What is the value of c?

3

The polynomial function p is defined by $p(x) = x(x-7)(x+2)$. What are the x-intercepts of the graph of $y = p(x)$ in the xy-plane?

A) $x = 0$, $x = 7$, and $x = -2$

B) $x = 0$, $x = -7$, and $x = 2$

C) $x = 7$ and $x = -2$

D) $x = 0$, $x = -7$, and $x = -2$

4

What is the remainder when the polynomial $p(x) = x^3 - 2x^2 + 5$ is divided by $x + 1$?

A) -2

B) 2

C) 4

D) 8

5

The function h is defined by $h(x) = 3(x-2)^2 - 5$. What is the y-coordinate of the y-intercept of the graph of $y = h(x)$ in the xy-plane?

6

The polynomial function q is defined by $q(x) = 4x^4 - x^3 + 2$. Which of the following describes the end behaviour of the graph of $y = q(x)$ in the xy-plane?

A) As x increases or decreases without bound, $q(x)$ increases without bound.

B) As x increases or decreases without bound, $q(x)$ decreases without bound.

C) As x increases without bound, $q(x)$ increases without bound; as x decreases without bound, $q(x)$ decreases without bound.

D) As x increases without bound, $q(x)$ decreases without bound; as x decreases without bound, $q(x)$ increases without bound.

7

In the xy-plane, which of the following equations defines a parabola with x-intercepts at $(6, 0)$ and $(-1, 0)$?

A) $y = (x+6)(x-1)$

B) $y = (x-6)(x-1)$

C) $y = (x-6)(x+1)$

D) $y = (x+6)(x+1)$

8

The function f is defined by $f(x) = x^2 - 10x + 21$. When $f(x)$ is written in the form $(x-h)^2 + k$, where h and k are constants, what is the value of $h + k$?

9

A rocket is launched from a platform. Its height above the ground, in metres, t seconds after launch is given by $h(t) = -5t^2 + 40t + 2$. How many seconds after launch does the rocket reach its greatest height?

A) 2

B) 4

C) 8

D) 82

10

Which of the following expressions is equivalent to $2x^2 - 12x + 13$ and displays the minimum value of the function $f(x) = 2x^2 - 12x + 13$ as a constant or coefficient?

A) $2(x-3)^2 + 5$

B) $2(x-6)^2 - 5$

C) $2(x-3)^2 - 5$

D) $2(x+3)^2 - 5$

11

The polynomial function p is defined by $p(x) = (x-1)(x+3)(x+4)$. What is the value of $p(2)$?

A) -30

B) 6

C) 24

D) 30

12

The polynomial function p is defined by $p(x) = (x+1)^3(x-2)^2$. Which of the following describes the behaviour of the graph of $y = p(x)$ at its x-intercepts?

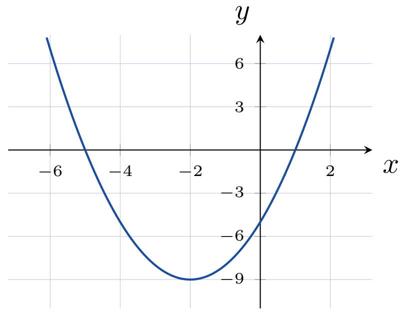
A) The graph crosses the x-axis at $x = -1$ and touches the x-axis at $x = 2$.

B) The graph touches the x-axis at $x = -1$ and crosses the x-axis at $x = 2$.

C) The graph crosses the x-axis at both $x = -1$ and $x = 2$.

D) The graph touches the x-axis at both $x = -1$ and $x = 2$.

13



A quadratic function f has the graph shown in the xy-plane above. Which of the following could be an equation of that graph?

A) $y = (x+2)^2 - 9$

B) $y = (x-2)^2 - 9$

C) $y = (x+2)^2 + 9$

D) $y = -(x+2)^2 - 9$

14

In the xy-plane, the graph of the quadratic function f crosses the x-axis at $(1, 0)$ and $(9, 0)$, and the minimum value of f is -32. The function is defined by $f(x) = a(x-1)(x-9)$, where a is a constant. What is the value of a?

15

Which of the following is a factor of the polynomial $p(x) = x^3 + 2x^2 - 11x - 12$?

A) $x - 2$

B) $x - 1$

C) $x + 3$

D) $x - 3$

16

Which of the following expressions is equivalent to $2x^2 - 2x - 12$ and displays the x-intercepts of the graph of $y = 2x^2 - 2x - 12$ as constants or coefficients?

A) $(x-3)(x+2)$

B) $2(x-3)(x+2)$

C) $2(x+3)(x-2)$

D) $2(x-6)(x+1)$

17

The polynomial p is defined by $p(x) = 3x^3 - 5x^2 + kx + 8$, where k is a constant. If $x - 2$ is a factor of $p(x)$, what is the value of k?

18

The function f is defined by $f(x) = x^2 + 5x + 3$. What is the minimum value of $f(x)$?

A) $-\frac{13}{2}$

B) $-\frac{13}{4}$

C) $-\frac{5}{2}$

D) $\frac{13}{4}$

19

In the xy-plane, the graph of $y = x^2 + bx + c$ has its vertex at $(3, -4)$, where b and c are constants. What is the value of c?

A) -6

B) -4

C) 5

D) 13

20

A company's daily profit, in dollars, from selling x units of a product is modelled by $P(x) = -2x^2 + 56x - 180$. What is the maximum daily profit, in dollars?

21

The polynomial function p is defined by $p(x) = -(x-1)^2(x+3)$. Which of the following statements about the graph of $y = p(x)$ in the xy-plane is true?

A) The graph touches the x-axis at $x = 1$, and $p(x)$ decreases without bound as x increases without bound.

B) The graph crosses the x-axis at $x = 1$, and $p(x)$ increases without bound as x increases without bound.

C) The graph touches the x-axis at $x = -3$, and $p(x)$ decreases without bound as x increases without bound.

D) The graph touches the x-axis at $x = 1$, and $p(x)$ increases without bound as x increases without bound.

22

The polynomial p is defined by $p(x) = x^3 - 6x^2 + 11x - 6$. If $p(x) = (x-3)q(x)$ for some polynomial q, which of the following is the expression $q(x)$?

A) $x^2 - 3x + 2$

B) $x^2 - 3x - 2$

C) $x^2 + 3x + 2$

D) $x^2 - 9x + 2$

Lesson 9: Exponential Functions, Radicals and Rational Expressions

Exponential models, radical equations and rational equations look like three different topics, but the Digital SAT groups them because they share one habit of mind: you must know what the algebra is allowed to do. An exponential model is built by multiplying, not adding; a radical equation is solved by squaring, an operation that can invent solutions; and a rational equation is solved by clearing denominators, an operation that can hide a value the equation was never allowed to use. Roughly one question in eight on the Advanced Math section comes from this cluster, and the majority of the wrong answers on them are traceable to two mistakes: using 0.08 where 1.08 belongs, and keeping a solution that fails the check.

Building an exponential model

Every exponential model on the SAT has the form

$$f(t) = a \cdot b^t,$$

where a is the value at $t = 0$ (the initial amount) and b is the factor the quantity is multiplied by during one time period. A linear model adds the same amount each period; an exponential model multiplies by the same factor each period. When a problem says "increases by 8% each year" it is describing multiplication: next year's value is this year's value plus 8% of it, which is 1.08 times this year's value. When it says "decreases by 8%", what remains is 92%, so the factor is 0.92.

THE FOUR NUMBERS YOU MUST READ OFF

For a percent rate r written as a decimal:

$$\text{growth: } f(t) = a(1+r)^t, \qquad \text{decay: } f(t) = a(1-r)^t.$$

For a quantity that is multiplied by b once every k time units,

$$f(t) = a \cdot b^{t/k},$$

so doubling every 9 years is $a(2)^{t/9}$ and halving every 12 years is $a\left(\frac{1}{2}\right)^{t/12}$. To change the time unit, change the exponent, never the base: if t is in years, then m months means $t = \frac{m}{12}$, giving $a \cdot b^{m/12}$. Interest compounded n times a year at annual rate r is $a\left(1 + \frac{r}{n}\right)^{nt}$.

0.08 IS NOT 1.08

The most common wrong answer in this whole topic is writing $a(0.08)^t$ for a quantity that grows by 8% a year. That model keeps only 8% of the quantity each year — it is a catastrophic decay, not growth. Ask yourself which way the quantity is moving: a growth factor is greater than 1, a decay factor is between 0 and 1. A second version of the same trap is reading the base as the percent: in $200(1.15)^x$ the percent increase is 15%, not 115% and not 1.15%.

Radical equations and the check

To solve an equation containing $\sqrt{\ }$, isolate the radical, then square both sides. Squaring is not reversible: -2 and 2 have the same square, so squaring can turn a false statement into a true one and hand you a number that solves the squared equation but not the original. Such a number is called an *extraneous solution*. Because $\sqrt{A}$ is never negative, any equation of the form $\sqrt{A} = B$ requires $B \geq 0$; that one observation catches most extraneous roots before you even substitute. When two radicals appear, isolate one, square, isolate the remaining radical, and square again.

Rational equations and the domain

To solve an equation with variables in denominators, multiply every term by the least common denominator and solve the polynomial equation that results. Before you do, note which values make any denominator zero: those values are excluded from the domain forever. If clearing denominators produces one of them, it is extraneous. This is why $\frac{1}{x-2} + \frac{1}{x+2} = \frac{4}{x^2-4}$ leads to $x = 2$ and yet has *no* solution.

EXTRANEOUS SOLUTIONS IN ONE SENTENCE

Squaring and multiplying by a variable expression are the two moves that can add solutions, so any answer produced by either move must be substituted back into the *original* equation before it is accepted. A radical equation and a rational equation fail the check for different reasons — a radical because a square root cannot be negative, a rational because a denominator cannot be zero — but the discipline is identical: solve, then check, then discard.

WORKED EXAMPLE 1 — BUILDING AND USING A MODEL

A colony of 4000 bacteria decreases by 15% every 3 hours. Write a model for the population t hours later, and find the population after 12 hours.

Start with the factor. A 15% decrease leaves $100\% - 15\% = 85\%$, so the factor for one 3-hour period is 0.85. That factor belongs to a 3-hour period, so the exponent must count 3-hour periods, and t hours contain $t/3$ of them:

$$P(t) = 4000(0.85)^{t/3}.$$

After 12 hours there are $12/3 = 4$ periods, so

$$P(12) = 4000(0.85)^4 = 4000(0.52200625) \approx 2088.$$

Two checks are worth building into your habits. First, the answer should be smaller than 4000, since the colony is shrinking — if you had used 1.15 you would have got about 6996 and should have caught it. Second, 0.85^4 is a little more than one half, and 2088 is a little more than half of 4000.

WORKED EXAMPLE 2 — A RADICAL AND A RATIONAL EQUATION

(a) Solve $\sqrt{x+9} = x + 3$. Square both sides: $x + 9 = x^2 + 6x + 9$, so $x^2 + 5x = 0$ and $x(x+5) = 0$, giving $x = 0$ or $x = -5$. Now check in the original. For $x = 0$: $\sqrt{9} = 3$ and $0 + 3 = 3$, true. For $x = -5$: $\sqrt{4} = 2$ but $-5 + 3 = -2$, and $2 \neq -2$, so $x = -5$ is extraneous. The only solution is $x = 0$. Notice that the requirement $x + 3 \geq 0$ would have ruled out -5 immediately.

(b) Solve $\frac{x}{x-4} = \frac{4}{x-4} + 3$. The value $x = 4$ is excluded before any work is done. Multiplying every term by $x - 4$ gives $x = 4 + 3(x-4)$, so $x = 3x - 8$, $-2x = -8$ and $x = 4$. That is precisely the excluded value, so it is extraneous and the equation has no solution. A student who skipped the domain check would have confidently gridded in 4.

ON TEST DAY

For "which function models this" questions, do not build the whole model — identify the initial value and the factor, and eliminate. Any choice whose base is the raw percent (0.08 rather than 1.08) is wrong; any choice that puts the period length as a multiplier in the exponent instead of a divisor is wrong. For radical and rational equations, the built-in graphing calculator is faster than algebra: graph each side of the equation and read the intersection, or graph left − right and find the zero. The graph shows only true solutions, so it silently performs the extraneous check for you. If the algebra gives you two candidates and the graph shows one intersection, trust the graph.

Module 1

1

The function $P(x) = 750(1.06)^x$ gives the population of a town x years after 2015. What was the population of the town in 2015?

A) 1.06
B) 750
C) 795
D) 45

2

A car purchased for \$24,000 loses 15% of its value each year. Which function gives the value $V(t)$, in dollars, of the car t years after it was purchased?

A) $V(t) = 24000(0.15)^t$
B) $V(t) = 24000(1.15)^t$
C) $V(t) = 24000 - 0.15t$
D) $V(t) = 24000(0.85)^t$

3

What is the solution to $\sqrt{2x+1} = 7$?

4

What value of x satisfies $\dfrac{12}{x-2} = 4$?

A) 1
B) 3
C) 5
D) 7

5

For an exponential function f, $f(0) = 5$, $f(1) = 15$, and $f(2) = 45$. What is the value of $f(4)$?

A) 75
B) 135
C) 405
D) 1215

6

The function $A(x) = 80(0.5)^x$ gives the number of grams of a sample that remain after x half-lives. How many grams remain after 3 half-lives?

7

A quantity increases by 7% each year. Which of the following is the growth factor for one year?

A) 0.07
B) 0.93
C) 1.07
D) 7

8

What is the solution to $\sqrt{x+7} = x + 1$?

A) -3
B) 2
C) 3
D) -3 and 2

9

What is the solution to $\dfrac{1}{x} + \dfrac{1}{2x} = \dfrac{3}{4}$?

10

The function $P(t) = 1200(1.03)^t$ models the population of a town t years after 2010. Which of the following is the best interpretation of the number 1.03 in this context?

A) The population increases by 1.03 people each year.
B) The population increases by 103% each year.
C) Each year the population is 1.03 times the population the year before.
D) The population in 2010 was 1.03 times 1200.

11

A savings account with an initial balance of $1500 earns 6% annual interest, compounded monthly. Which function gives the balance $B(t)$, in dollars, after t years?

A) $B(t) = 1500(1.06)^{12t}$

B) $B(t) = 1500(1.5)^{t}$

C) $B(t) = 1500(1.005)^{t}$

D) $B(t) = 1500(1.005)^{12t}$

12

What is the solution to $\frac{5}{x+1} = \frac{2}{x-1}$?

A) $\frac{7}{3}$

B) $\frac{3}{7}$

C) $-\frac{7}{3}$

D) 1

13

A machine purchased for $30,000 depreciates by 20% of its value each year. What is the value of the machine, in dollars, after 3 years?

A) 12,000

B) 15,360

C) 18,000

D) 51,840

14

Squaring both sides of $\sqrt{3x+4} = x$ produces an equation with two solutions. Which of the following is an extraneous solution of the original equation?

A) -1

B) 1

C) 4

D) -4

15

A bacteria population doubles every 5 hours. If there are 300 bacteria at time $t = 0$, how many bacteria are there after 20 hours?

16

The value of a rare coin is $5000 today and increases by 12% every 4 years. Which function gives the value $V(t)$, in dollars, of the coin t years from today?

A) $V(t) = 5000(1.12)^{4t}$

B) $V(t) = 5000(1.12)^{t/4}$

C) $V(t) = 5000(0.12)^{t/4}$

D) $V(t) = 5000(1.03)^{t}$

17

What is the solution to $\frac{x^2}{x-3} = \frac{9}{x-3}$?

A) -3

B) 3

C) -3 and 3

D) There is no solution.

18

What is the solution to $\sqrt{x+5} - \sqrt{x} = 1$?

19

The value of an investment doubles every 9 years. If the investment is worth $2500 today, which function gives its value $V(t)$, in dollars, t years from today?

A) $V(t) = 2500(2)^{9t}$

B) $V(t) = 2500(9)^{t/2}$

C) $V(t) = 2500 + 2^{t/9}$

D) $V(t) = 2500(2)^{t/9}$

20

The expression $\frac{8^{x+1}}{2^x}$ can be written in the form 2^{ax+b}, where a and b are constants. What is the value of $a+b$?

A) 2

B) 3

C) 5

D) 6

21

What is the solution to $\sqrt{2x+3} = x - 6$?

22

The population of a town was 8000 in 2000 and 12,000 in 2010, and the population grows exponentially. Which function gives the population $P(t)$ of the town t years after 2000?

A) $P(t) = 8000(1.5)^{t/10}$

B) $P(t) = 8000(1.5)^{10t}$

C) $P(t) = 8000(0.5)^{t/10}$

D) $P(t) = 8000 + 400t$

Module 2

1

The function $f(x) = 200(1.15)^x$ increases by $p\%$ for each increase of 1 in the value of x. What is the value of p?

A) 0.15
B) 15
C) 115
D) 1.15

2

What is the solution to $\frac{8}{x+3} = 2$?

3

What is the solution to $\sqrt{x-1} + 4 = 9$?

A) 6
B) 10
C) 24
D) 26

4

The value of a piece of equipment decreases by 35% each year. Which of the following is the decay factor for one year?

A) 0.35
B) 1.35
C) 35
D) 0.65

5

The function g is defined by $g(x) = 6(3)^x$. What is the value of $g(-1)$?

A) -18
B) $\frac{1}{2}$
C) 2
D) 18

6

A 400-milligram sample of a radioactive substance has a half-life of 12 years. Which function gives the amount $A(t)$, in milligrams, of the sample remaining after t years?

A) $A(t) = 400(0.5)^{12t}$
B) $A(t) = 400(0.5)^{t/12}$
C) $A(t) = 400(1.5)^{t/12}$
D) $A(t) = 400 - \frac{t}{12}$

7

The value of an investment, in dollars, is modeled by $V(t) = 800(1.05)^t$, where t is the number of years since the investment was made. What is the value, in dollars, of the investment after 2 years?

8

What is the solution to $\frac{x}{x-1} = \frac{2}{x-1} + 3$?

A) $\frac{1}{2}$
B) $-\frac{1}{2}$
C) 1
D) 2

9

What is the sum of all solutions to $\sqrt{5x-4} = x$?

10

The function $P(t) = 3000(1.08)^t$ gives the value, in dollars, of an account t years after it was opened. Which function gives the value of the account m months after it was opened?

A) $P(m) = 3000(1.08)^{12m}$
B) $P(m) = 3000\left(\frac{1.08}{12}\right)^m$
C) $P(m) = 250(1.08)^m$
D) $P(m) = 3000(1.08)^{m/12}$

11

The number of bacteria in a culture triples every 4 hours. There are 90 bacteria in the culture at time $t = 0$. How many bacteria are in the culture after 12 hours?

A) 270

B) 810

C) 1080

D) 2430

12

What is the positive solution to $\frac{1}{x} + \frac{1}{x+2} = \frac{3}{4}$?

A) $\frac{2}{3}$

B) 2

C) $-\frac{4}{3}$

D) 4

13

The value of a car t years after it was purchased is modeled by $V(t) = 18000(0.88)^t$. Which of the following is the best interpretation of 0.88 in this model?

A) The car loses 88% of its value each year.

B) The car is worth \$0.88 less each year.

C) Each year the car is worth 0.88 times its value the previous year.

D) The car was worth \$18,000 times 0.88 when it was purchased.

14

Which of the following is equivalent to $\frac{x^2-9}{x+3}$ for $x \neq -3$?

A) $x - 3$

B) $x + 3$

C) $x^2 - 3$

D) $\frac{1}{x-3}$

15

What is the solution to $2\sqrt{x} = x - 3$?

A) 1

B) 3

C) 9

D) 1 and 9

16

The function $N(t) = 1600\left(\frac{1}{2}\right)^{t/8}$ gives the number of grams of a sample remaining t years after the sample was formed. After how many years will 100 grams remain?

A) 16

B) 24

C) 32

D) 64

17

What is the sum of all solutions to $\sqrt{2x+7} = 1 + \sqrt{x+3}$?

A) -3

B) -2

C) 1

D) 4

18

A population grows exponentially. The population is 500 at time $t = 0$ and 720 when $t = 2$, where t is measured in years. If the population increases by $p\%$ each year, what is the value of p?

A) 20

B) 22

C) 44

D) 144

19

What is the solution to $\frac{3}{x-1} + \frac{2}{x+1} = \frac{10}{x^2-1}$?

20

What value of x satisfies $3^{2x-1} = 27^{x-2}$?

21

The function $P(d) = 8(2)^d$ gives the number of thousands of people living in a city d decades after 1990. Which function gives the number of thousands of people living in the city y years after 1990?

A) $P(y) = 8(2)^{10y}$

B) $P(y) = 8(2)^{y/10}$

C) $P(y) = 8(20)^y$

D) $P(y) = 80(2)^y$

22

How many solutions does the equation $\frac{1}{x-2}+\frac{1}{x+2}=\frac{4}{x^2-4}$ have?

A) 0

B) 1

C) 2

D) Infinitely many

Lesson 10: Nonlinear Systems and Function Transformations

Two kinds of question live in this topic, and the test likes to put them side by side. The first is a *nonlinear system*: a line meeting a parabola or a circle, where the work is to substitute one equation into the other and solve the quadratic that appears. The second is *function transformation*: reading $f(x-3)$, $-f(x)$ or $2f(x)$ and knowing exactly how the picture moves. Both reward the same habit — decide what is happening to the input and what is happening to the output, then do the algebra one step at a time.

Solving a nonlinear system by substitution

A system is a pair of conditions that must hold at the same point. When one equation is already solved for y, put that expression into the other equation. Every y disappears and a single equation in x is left — almost always a quadratic.

For $y = x^2 - 3x + 5$ and $y = x + 2$: substitute to get $x^2 - 3x + 5 = x + 2$, collect everything on one side, $x^2 - 4x + 3 = 0$, and factor, $(x-1)(x-3) = 0$. The x-coordinates are 1 and 3. *Finish the job*: put each back into the line to get $y = 3$ and $y = 5$, so the solutions are $(1, 3)$ and $(3, 5)$. A system with two intersection points has two solutions, and a question asking for a sum of coordinates wants both of them.

TWO EQUATIONS BECOME ONE EQUATION

Whatever the curves are, the goal never changes: eliminate a variable. With a circle $x^2 + y^2 = r^2$ and a line, substitute the line's y into the circle. With a circle and a parabola $y = x^2 + k$, it is often faster to eliminate x^2 instead, since $x^2 = y - k$ turns the circle equation into a quadratic in y. Then remember that one y-value can correspond to two x-values: if $x^2 = 7$, that is two points, but if $x^2 = 0$, it is only one.

How many times do the graphs meet?

Once the system has been reduced to $ax^2 + bx + c = 0$, the number of intersection points is the number of *real* solutions, and the discriminant $b^2 - 4ac$ answers that without any solving:

$b^2 - 4ac > 0$: two intersection points. $b^2 - 4ac = 0$: one point, and the line is *tangent* to the curve. $b^2 - 4ac < 0$: the graphs never meet.

Every "for what value of c is the line tangent to the curve" question is this identity read backwards: set the discriminant equal to zero and solve for the parameter.

THE TRANSFORMATION TABLE

Let $k > 0$ and let f be any function.

$f(x) + k$ — shift **up** k. $f(x) - k$ — shift **down** k.
$f(x + k)$ — shift **left** k. $f(x - k)$ — shift **right** k.
$-f(x)$ — reflect across the ***x*-axis**. $f(-x)$ — reflect across the ***y*-axis**.
$af(x)$ with $a > 1$ — vertical **stretch** by a; with $0 < a < 1$ — vertical **shrink**.
$f(bx)$ with $b > 1$ — horizontal **compression** by b; with $0 < b < 1$ — horizontal **stretch**.

One rule organises the whole table: changes *outside* the parentheses act on y and behave as written; changes *inside* act on x and behave in reverse.

THE SHIFT INSIDE THE PARENTHESES GOES THE WRONG WAY

$f(x-3)$ shifts the graph 3 units **right**, not left, and $f(x+6)$ shifts it 6 units **left**. The minus sign looks like "go negative", which is why this is the single most missed transformation on the test.

Never argue from the sign — test a point. If $g(x) = f(x-3)$, then $g(3) = f(0)$: the output f produced at

$x = 0$ now appears at $x = 3$, so the picture moved to the right. The same check settles $f(2x)$: if f has an x-intercept at 6, then $f(2x) = 0$ needs $2x = 6$, so the new intercept is at $x = 3$ — the graph was squeezed toward the y-axis, not stretched away from it.

STRETCHES ACT ON ONE VARIABLE ONLY

$3f(x)$ multiplies every *output* by 3: the point $(2, 5)$ becomes $(2, 15)$, and the x-coordinate does not budge. $f(3x)$ multiplies every *input* by 3: the point $(6, 5)$ becomes $(2, 5)$, and the y-coordinate does not budge. Mixing these up — or adding 3 where the rule says multiply — accounts for most wrong answers on stretch questions.

WORK OUTSIDE-IN, AND USE A SINGLE POINT

For a compound transformation such as $y = -2f(x + 1) - 3$, name the pieces in order: inside first (1 unit left), then the stretch and reflection ($\times(-2)$: flip across the x-axis and stretch vertically by 2), then the vertical shift (3 down).

When a graph or a table is given, do not transform the whole picture — transform one point you can read exactly, usually a vertex or an intercept, and see which choice agrees. And when a question gives a table, remember that the inside change tells you *which row to read* and the outside change tells you *what to do with the number you read.*

WORKED EXAMPLE 1 — A LINE AND A PARABOLA

The graphs of $y = x^2 - 4x + 7$ and $y = 2x - 1$ intersect at two points. What is the sum of the y-coordinates of those points?

Substitute the line into the parabola: $x^2 - 4x + 7 = 2x - 1$. Bring everything to one side: $x^2 - 6x + 8 = 0$, which factors as $(x - 2)(x - 4) = 0$, so $x = 2$ and $x = 4$.

The question asks for y-coordinates, so use the line (it is the easier equation): $y = 2(2) - 1 = 3$ and $y = 2(4) - 1 = 7$. The sum is $3 + 7 = 10$.

Why this order: substituting the *linear* expression is what keeps the algebra to a single quadratic, and returning to the *linear* equation at the end keeps the arithmetic to one multiplication. Stopping at $x = 2, 4$ and answering 6 is the classic incomplete answer.

WORKED EXAMPLE 2 — TANGENCY AND THE DISCRIMINANT

For what value of c is the line $y = 6x + c$ tangent to the parabola $y = x^2 + 2x + 5$?

Substitute: $x^2 + 2x + 5 = 6x + c$, so $x^2 - 4x + (5 - c) = 0$. Tangent means the two intersection points have merged into one, so this quadratic has exactly one real root and its discriminant is zero:

$(-4)^2 - 4(1)(5 - c) = 0 \Rightarrow 16 - 20 + 4c = 0 \Rightarrow c = 1.$

Check: with $c = 1$ the equation is $x^2 - 4x + 4 = (x - 2)^2 = 0$, a repeated root at $x = 2$ — a single point of contact, as tangency requires. Note the sign work in $5 - c$: carrying c across the equals sign incorrectly is where this problem is normally lost.

WORKED EXAMPLE 3 — READING A TRANSFORMED FUNCTION FROM A TABLE

The table below gives values of f. If $g(x) = -2f(x - 1) + 4$, what is $g(3)$?

x	1	2	3	4
$f(x)$	0	5	-2	7

Inside first: $g(3)$ requires $f(3-1) = f(2)$, and the table gives $f(2) = 5$. Now apply the outside operations in order: multiply by -2 to get -10, then add 4 to get -6. So $g(3) = -6$.

Why the inside comes first: $x - 1$ decides which input f receives, and that has to be settled before f can produce a number at all. Reading $f(3) = -2$ (ignoring the shift) gives 8, and reading $f(4) = 7$ (shifting the wrong way) gives -10 — both appear as answer choices on the real test.

Module 1

1

The graph of $y = x^2$ is shifted 5 units up in the xy-plane. Which equation represents the resulting graph?

A) $y = x^2 + 5$
B) $y = x^2 - 5$
C) $y = (x+5)^2$
D) $y = (x-5)^2$

2

The function g is defined by $g(x) = f(x-3)$. Which of the following describes the graph of $y = g(x)$ compared with the graph of $y = f(x)$?

A) The graph of f shifted 3 units to the left.
B) The graph of f shifted 3 units to the right.
C) The graph of f shifted 3 units up.
D) The graph of f shifted 3 units down.

3

In the xy-plane, the graphs of $y = 2x$ and $y = x^2 - 3$ intersect at two points. What is the greatest x-coordinate of an intersection point?

4

The graph of $y = f(x)$ is reflected across the x-axis. Which equation represents the reflected graph?

A) $y = -f(x)$
B) $y = f(-x)$
C) $y = -f(-x)$
D) $y = |f(x)|$

5

The graph of $y = 3f(x)$ is obtained from the graph of $y = f(x)$ by which transformation?

A) A shift of 3 units up.
B) A horizontal stretch by a factor of 3.
C) A vertical stretch by a factor of 3.
D) A horizontal compression by a factor of 3.

6

The graphs of $y = x^2$ and $y = x + 6$ intersect at two points in the xy-plane. What is the sum of the y-coordinates of those two points?

A) 1
B) 5
C) 9
D) 13

7

How many points of intersection do the graphs of $y = x^2 + 1$ and $y = x - 2$ have in the xy-plane?

A) 0
B) 1
C) 2
D) 4

8

$$y = x^2 - 4x + 7$$
$$y = 2x - 1$$

Which ordered pair (x, y) is a solution to the system of equations above?

A) $(1, 1)$
B) $(2, 3)$
C) $(3, 5)$
D) $(4, 9)$

9

In the xy-plane, the line $y = 4x + c$ is tangent to the parabola $y = x^2$, where c is a constant. What is the value of c?

10

The system of equations $x^2 + y^2 = 25$ and $y = x + 1$ has two solutions in the xy-plane. What is the greatest y-coordinate of a solution?

A) 3
B) 4
C) 5
D) 7

11

The table gives five values of the function f.

x	-1	0	1	2	3
$f(x)$	5	2	6	-1	8

The function g is defined by $g(x) = f(x + 2)$. What is the value of $g(1)$?

A) 2
B) 5
C) 6
D) 8

12

The graph of $y = x^2$ is transformed to produce the graph of $y = (x + 4)^2 - 3$. Which of the following describes the transformation?

A) A shift 4 units to the right and 3 units down.
B) A shift 4 units to the right and 3 units up.
C) A shift 4 units to the left and 3 units down.
D) A shift 4 units to the left and 3 units up.

13

The point $(2, 5)$ lies on the graph of $y = f(x)$. The graph of $y = 3f(x)$ passes through the point $(2, k)$. What is the value of k?

A) $\frac{5}{3}$
B) 8
C) 15
D) 45

14

The function f is defined by $f(x) = x^2 - 6x$, and $g(x) = f(-x)$. Which expression is equivalent to $g(x)$?

A) $x^2 - 6x$
B) $x^2 + 6x$
C) $-x^2 + 6x$
D) $-x^2 - 6x$

15

In the xy-plane, the graph of $y = -x^2 + 6x$ intersects the line $y = 5$ at two points. What is the positive difference between the x-coordinates of those two points?

16

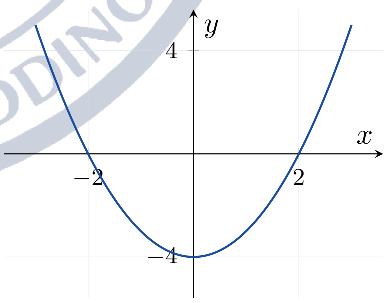
In the xy-plane, the line $y = mx - 4$ is tangent to the graph of $y = x^2$, where m is a positive constant. What is the value of m?

A) 2
B) 4
C) 8
D) 16

17

The solutions to the system $y = x^2 - 2x - 3$ and $y = 2x + 2$ are (x_1, y_1) and (x_2, y_2). What is the value of $y_1 + y_2$?

18



The graph of $y = f(x)$ is shown, where $f(x) = x^2 - 4$. The function g is defined by $g(x) = f(x + 3)$. What is the vertex of the graph of $y = g(x)$?

A) $(-3, -4)$
B) $(3, -4)$
C) $(-3, -1)$
D) $(0, -1)$

19

The minimum value of the function f is -3, and it occurs at $x = 1$. The function g is defined by $g(x) = -2f(x)$. What is the maximum value of g?

A) -6

B) -3

C) 3

D) 6

20

In the xy-plane, the line $y = 2x + c$ is tangent to the circle $x^2 + y^2 = 45$, where c is a positive constant. What is the value of c?

21

The graph of $y = f(x)$ is first shifted 2 units to the right and then reflected across the x-axis. Which equation represents the resulting graph?

A) $y = -f(x + 2)$

B) $y = f(-x - 2)$

C) $y = -f(x - 2)$

D) $y = -f(x) - 2$

22

The graph of $y = f(x)$ has an x-intercept at $x = 6$. The function g is defined by $g(x) = f(2x)$. At what value of x does the graph of $y = g(x)$ have an x-intercept?

A) 3

B) 4

C) 8

D) 12

Module 2

1

The function h is defined by $h(x) = f(x+6)$. Which of the following describes the graph of $y = h(x)$ compared with the graph of $y = f(x)$?

A) The graph of f shifted 6 units to the left.

B) The graph of f shifted 6 units to the right.

C) The graph of f shifted 6 units up.

D) The graph of f shifted 6 units down.

2

The graphs of $y = x^2 + 3x$ and $y = x + 8$ intersect at two points in the xy-plane. What is the greater of the two x-coordinates?

3

In the xy-plane, how many points does the graph of $y = x^2 - 4$ have in common with the line $y = -4$?

A) 0

B) 1

C) 2

D) 3

4

How does the graph of $y = f(x) - 7$ compare with the graph of $y = f(x)$?

A) It is shifted 7 units to the left.

B) It is shifted 7 units to the right.

C) It is shifted 7 units up.

D) It is shifted 7 units down.

5

The point $(-3, 8)$ lies on the graph of $y = f(x)$. The graph of $y = f(x)+5$ passes through the point $(-3, k)$. What is the value of k?

6

The function f is defined by $f(x) = x^3 - 2x$, and $h(x) = f(-x)$. Which expression is equivalent to $h(x)$?

A) $x^3 - 2x$

B) $x^3 + 2x$

C) $-x^3 + 2x$

D) $-x^3 - 2x$

7

The graph of $y = \frac{1}{2}f(x)$ is obtained from the graph of $y = f(x)$ by which transformation?

A) A horizontal compression by a factor of 2.

B) A horizontal stretch by a factor of 2.

C) A shift down $\frac{1}{2}$ unit.

D) A vertical shrink by a factor of $\frac{1}{2}$.

8

In the xy-plane, the line $y = 2x + c$ intersects the parabola $y = x^2 + 6x + 11$ at exactly one point, where c is a constant. What is the value of c?

9

How many solutions does the system of equations $y = x^2 - 6x + 9$ and $y = x - 3$ have?

A) 0

B) 1

C) 2

D) 3

10

The table gives four values of the function f.

x	1	3	5	7
$f(x)$	2	6	-1	4

The function g is defined by $g(x) = f(x-2)+3$. What is the value of $g(5)$?

A) 2

B) 6

C) 7

D) 9

11

Which transformation takes the graph of $y = x^2$ to the graph of $y = -(x-2)^2$?

A) A reflection across the x-axis and a shift 2 units to the left.

B) A reflection across the x-axis and a shift 2 units to the right.

C) A reflection across the y-axis and a shift 2 units to the right.

D) A reflection across the y-axis and a shift 2 units to the left.

12

The function f is defined by $f(x) = (x+1)^2$. The graph of $y = f(x)$ is shifted 3 units to the right. Which equation represents the resulting graph?

A) $y = (x+4)^2$

B) $y = (x+1)^2 + 3$

C) $y = (x-2)^2$

D) $y = (x-3)^2 + 1$

13

The function f is defined by $f(x) = x^2 - 16$, and g is defined by $g(x) = f\left(\frac{x}{2}\right)$. What is the positive x-intercept of the graph of $y = g(x)$?

14

$$y = x^2 + 5x + 4$$
$$y = 3x + 7$$

The system above has two solutions. What is the sum of the y-coordinates of the two solutions?

A) -2

B) 8

C) 10

D) 12

15

The function g is defined by $g(x) = -f(-x)$. Which of the following describes the graph of $y = g(x)$ in terms of the graph of $y = f(x)$?

A) The graph of f reflected across both the x-axis and the y-axis.

B) The graph of f reflected across the x-axis only.

C) The graph of f reflected across the y-axis only.

D) The graph of f reflected across the line $y = x$.

16

In the xy-plane, at how many points does the graph of $x^2 + y^2 = 16$ intersect the graph of $y = x^2 - 4$?

A) 1

B) 2

C) 3

D) 4

17

In the xy-plane, the line $y = kx$ is tangent to the graph of $y = x^2 - 2x + 4$, where k is a positive constant. What is the value of k?

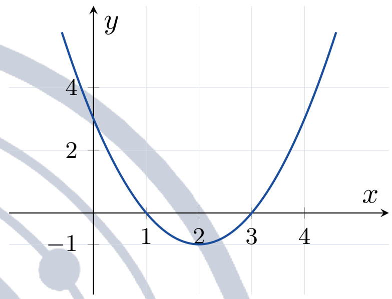
A) 2

B) 4

C) 6

D) 8

18



The graph of $y = f(x)$ is shown, where $f(x) = (x-2)^2 - 1$. What is the vertex of the graph of $y = f(x+1) - 2$?

A) $(3, -3)$

B) $(1, 1)$

C) $(3, 1)$

D) $(1, -3)$

19

The table gives five values of the function f.

x	0	1	2	3	4
$f(x)$	5	-3	8	6	1

The function h is defined by $h(x) = 2f(x+1)$. What is the value of $h(2)$?

20

Which sequence of transformations takes the graph of $y = x^2$ to the graph of $y = 2(x+3)^2$?

A) A vertical stretch by a factor of 2 and a shift 3 units to the right.

B) A vertical stretch by a factor of 2 and a shift 3 units to the left.

C) A horizontal stretch by a factor of 2 and a shift 3 units to the left.

D) A shift 3 units to the left and 2 units up.

21

In the xy-plane, the graph of $x^2 + y^2 = 25$ and the graph of $y = x^2 - 13$ intersect at how many points?

22

$$y = x^2 - 2x + c$$

$$y = 2x - 5$$

In the system of equations above, c is a constant. For what value of c does the system have exactly one real solution?

A) -1

B) 1

C) 5

D) 9

PART 3

Problem-Solving and Data Analysis

Lesson 11: Ratios, Rates, Proportional Relationships and Units

A ratio compares two quantities, a rate compares two quantities with different units, and a proportional relationship is what you get when that comparison never changes. The Digital SAT leans on this trio harder than on almost anything else in Problem-Solving and Data Analysis: recipes, fuel economy, currency, map scales, population per square kilometre and density all reduce to the same move — find the constant multiplier, then multiply. This topic also carries the test's unit-conversion questions, where the arithmetic is easy but the bookkeeping is not.

Ratios: parts, the whole and the multiplier

A ratio $a : b$ does not say how many of anything there are; it says how the quantities are built out of equal *parts*. If red and blue marbles are in the ratio $3 : 4$, then the reds come in 3 parts and the blues in 4 parts, and every part has the same size k. So the counts are $3k$ and $4k$ and the total is $7k$.

That single letter k is the whole topic. Every ratio question hands you one piece of information — one of the quantities, the total, or the difference between the quantities — and asks for another. Turn the given information into an equation in k, solve for k, then read off whatever was asked.

For example, with red : blue $= 3 : 4$ and 84 marbles in all: $3k + 4k = 84$, so $7k = 84$ and $k = 12$. There are $3(12) = 36$ red marbles and $4(12) = 48$ blue ones.

PART-TO-PART IS NOT PART-TO-WHOLE

The ratio $3 : 4$ says the reds are $\frac{3}{4}$ of the *blues* — it does *not* say the reds are $\frac{3}{4}$ of the bag. As a fraction of the whole, the reds are $\dfrac{3}{3+4} = \dfrac{3}{7}$.

Whenever a question gives you a *total*, add the parts of the ratio first and work with the part-to-whole fractions $\frac{3}{7}$ and $\frac{4}{7}$. Whenever it gives you *one quantity*, use the part-to-part fraction. Choosing the wrong one of these two is the single most common error on the topic, and the test always supplies the answer it produces.

Rates, unit rates and proportional relationships

A rate is a ratio between quantities measured in different units: 245 kilometres per 35 litres, \$6 per 500 sheets, 250 people per square kilometre. A *unit rate* is the same information rewritten with a denominator of 1: 7 kilometres per litre, \$0.012 per sheet, and so on. Dividing to get the unit rate first almost always makes the rest of the question one multiplication.

Two quantities are *directly proportional* when $y = kx$ for a constant k; equivalently, $\dfrac{y_1}{x_1} = \dfrac{y_2}{x_2}$. That equality is the proportion you cross-multiply. Proportional quantities scale multiplicatively, never additively: if x goes up by a factor of $\frac{5}{3}$, so does y. Adding the same amount to both is not a proportional relationship.

THE RELATIONSHIPS WORTH MEMORISING

$$\text{proportion: } \frac{y_1}{x_1} = \frac{y_2}{x_2} \quad \Longrightarrow \quad y_1 x_2 = y_2 x_1$$

$$\text{unit rate} = \frac{\text{total amount}}{\text{number of units}} \qquad \text{density} = \frac{\text{mass}}{\text{volume}} \qquad \text{population density} = \frac{\text{population}}{\text{area}}$$

$$\text{actual distance} = (\text{drawing distance}) \times (\text{scale factor})$$

For a scale factor r applied to a figure: lengths multiply by r, areas multiply by r^2, volumes multiply by r^3.

DIMENSIONAL ANALYSIS: MAKE THE UNITS CANCEL

The hardest items on this topic are conversion chains, and there is a mechanical method that never fails. Write the starting quantity as a fraction, then multiply by conversion factors — fractions equal to 1, such as $\frac{1000\text{ m}}{1\text{ km}}$ — chosen so that every unit you do not want cancels.
Speed of 72 kilometres per hour into metres per second:

$$\frac{72\text{ km}}{1\text{ h}} \times \frac{1000\text{ m}}{1\text{ km}} \times \frac{1\text{ h}}{3600\text{ s}} = \frac{72\,000}{3600}\,\frac{\text{m}}{\text{s}} = 20\text{ m/s}$$

Two rules make this reliable. First, never decide whether to multiply or divide by feel — write the factor with the unwanted unit on the side where it cancels, and the arithmetic decides for you. Second, chain as many factors as you need in one line: hours to seconds and kilometres to metres and litres to cubic metres can all sit in the same product, so you only round once, at the end.

SQUARED AND CUBED UNITS, AND AREAS UNDER A SCALE

$1\text{ m} = 100\text{ cm}$, but $1\text{ m}^2 = 100^2 = 10\,000\text{ cm}^2$ and $1\text{ m}^3 = 100^3 = 1\,000\,000\text{ cm}^3$. The conversion factor gets squared or cubed along with the unit, and the answer choice that used the plain factor 100 is always on the list.
The same trap wears a second costume. If a map has 1 cm representing 4 km, then 1 cm^2 represents $4^2 = 16\text{ km}^2$, so a lake covering 12 cm^2 on the map really covers $12 \times 16 = 192\text{ km}^2$, not $12 \times 4 = 48$. Whenever the quantity is an area, square the scale factor; whenever it is a volume, cube it.

SCALE DRAWINGS, MAPS AND SIMILAR FIGURES

A scale written $1 : 25\,000$ has no units: one unit on the drawing is 25 000 of the same units in reality, so 8 cm on the map is 200 000 cm, which is 2 km. A scale written "1 cm represents 25 km" already carries its units and needs no conversion.
Similar figures are the same idea without a map: corresponding lengths are in a fixed ratio r. Enlarging a 4×6 photograph so that its width becomes 10 gives $r = \frac{10}{4} = 2.5$, so the length becomes $6(2.5) = 15$. Perimeters scale by r too, because a perimeter is a sum of lengths — but areas scale by r^2.

WORKED EXAMPLE 1 — A RATIO GIVEN AS A DIFFERENCE

In a survey, the ratio of people who prefer tea to people who prefer coffee is $5 : 7$. There are 96 more people who prefer coffee. How many people were surveyed?
Write the two groups as $5k$ and $7k$. The information given is a difference, so turn it into an equation in k:

$$7k - 5k = 96 \quad\Longrightarrow\quad 2k = 96 \quad\Longrightarrow\quad k = 48.$$

The question asks for the total, which is $5k + 7k = 12k = 12(48) = 576$ people.
Notice what went wrong for anyone who divided 96 by 12 instead of by 2: the 96 is not the whole, it is the gap of 2 parts. Always match the number you are given to the number of parts it represents before you solve for k.

WORKED EXAMPLE 2 — A FULL CONVERSION CHAIN

A bottling machine fills 250 bottles per hour and each bottle holds 0.75 litres. Given that 1 cubic metre = 1000 litres, how many cubic metres does the machine bottle in 8 hours?
Set out one product in which every unwanted unit cancels:

$$\frac{250\text{ bottles}}{1\text{ h}} \times 8\text{ h} \times \frac{0.75\text{ L}}{1\text{ bottle}} \times \frac{1\text{ m}^3}{1000\text{ L}} = \frac{250 \cdot 8 \cdot 0.75}{1000}\text{ m}^3 = 1.5\text{ m}^3.$$

The intermediate value $250 \cdot 8 \cdot 0.75 = 1500$ litres is worth writing down: it is the sanity check. A machine bottling 1500 litres in a working day should land near a cubic metre and a half, so 0.15 or 15 would both be visibly wrong — and both are on the answer list, one for each direction of dividing by 1000.

ESTIMATE FIRST, THEN COMPUTE

Almost every wrong answer on this topic is off by a factor of 10, of 100, or of the ratio you inverted. A one-second estimate catches all three. Before computing, ask whether the answer should be bigger or smaller than the number you started with: metres per second are smaller than kilometres per hour, grams are bigger than kilograms, the actual field is bigger than the map.
Then compute exactly, keeping fractions rather than decimals wherever you can — $\frac{3}{24} \cdot 60$ is cleaner by hand than 0.125×60 — and finish by rereading the last line of the question. Rate problems routinely ask for the quantity you did *not* just solve for: the blue marbles, the total, the cost rather than the amount.

Module 1

1

In a school choir, the ratio of boys to girls is $2 : 3$. If there are 18 boys in the choir, how many girls are in the choir?

A) 12
B) 15
C) 18
D) 27

2

A car travels 245 kilometres using 35 litres of fuel. At this rate, how many kilometres does the car travel per litre of fuel?

3

A bag of rice has a mass of 3.5 kilograms. Given that 1 kilogram $= 1000$ grams, what is the mass of the bag, in grams?

A) 350
B) 3500
C) 35 000
D) 350 000

4

Two business partners divide a profit of \$60 in the ratio $2 : 3$. How many dollars does the partner with the larger share receive?

A) 12
B) 24
C) 30
D) 36

5

Four identical notebooks cost \$6.00. At this rate, what is the cost, in dollars, of 10 of these notebooks?

A) 15
B) 20
C) 24
D) 60

6

On a map, 1 centimetre represents 25 kilometres. Two cities are 6 centimetres apart on the map. What is the actual distance, in kilometres, between the two cities?

7

A metal block has a mass of 480 grams and a volume of 60 cubic centimetres. What is the density of the block, in grams per cubic centimetre?

A) 0.125
B) 8
C) 80
D) 540

8

A bag contains only red marbles and blue marbles. The ratio of red marbles to blue marbles is $3 : 4$, and there are 84 marbles in the bag. How many of the marbles are red?

A) 12
B) 36
C) 48
D) 63

9

A train travels at a constant speed of 72 kilometres per hour. What is the speed of the train, in metres per second?

10

A recipe that uses 3 cups of flour makes 24 cookies. At the same rate, how many cups of flour are needed to make 60 cookies?

A) 4.5
B) 7.5
C) 8
D) 20

11

A bank exchanges currency at the rate 1 euro = 1.25 dollars. How many euros does a traveller receive in exchange for 400 dollars?

A) 320
B) 400
C) 425
D) 500

12

A photograph is 4 inches wide and 6 inches long. The photograph is enlarged proportionally so that the new width is 10 inches. What is the new length, in inches?

A) 2.4
B) 12
C) 15
D) 18

13

A tabletop has an area of 2 square metres. Given that 1 metre = 100 centimetres, what is the area of the tabletop, in square centimetres?

A) 20
B) 200
C) 2000
D) 20 000

14

Aluminium has a density of 2.7 grams per cubic centimetre. What is the mass, in grams, of a solid aluminium part with a volume of 50 cubic centimetres?

A) 27
B) 52.7
C) 135
D) 270

15

The quantity y is directly proportional to the quantity x. When $x = 12$, the value of y is 45. What is the value of y when $x = 20$?

A) 53
B) 75
C) 80
D) 135

16

A bottling machine fills 250 bottles per hour, and each bottle holds 0.75 litres. Given that 1 cubic metre = 1000 litres, how many cubic metres of liquid does the machine bottle in 8 hours?

A) 0.0015
B) 0.015
C) 0.15
D) 1.5

17

A pastry recipe mixes flour, sugar and butter by mass in the ratio $5 : 2 : 3$. One batch of the recipe uses 400 grams of sugar. What is the total mass, in grams, of the three ingredients in that batch?

18

A map is drawn to a scale of $1 : 25\,000$. Two landmarks are 8 centimetres apart on the map. What is the actual distance between the landmarks, in kilometres?

A) 0.02
B) 0.2
C) 2
D) 20

19

On a map, 1 centimetre represents 4 kilometres. A lake covers an area of 12 square centimetres on the map. What is the actual area of the lake, in square kilometres?

A) 48

B) 96

C) 192

D) 768

20

A car uses 8 litres of fuel for every 100 kilometres driven, and fuel costs \$1.60 per litre. What is the cost, in dollars, of the fuel used on a trip of 250 kilometres?

A) 32

B) 40

C) 50

D) 320

21

A solid metal cube has edges of length 5 centimetres and is made of a metal with a density of 8.4 grams per cubic centimetre. What is the mass of the cube, in kilograms?

22

A printer prints 15 pages per minute, and each page uses one sheet of paper. Paper costs \$6.00 per package of 500 sheets. What is the cost, in dollars, of the paper used if the printer runs continuously for 3 hours?

A) 16.20

B) 27

C) 32.40

D) 54

Module 2

1

A paint mixture is made by combining red paint and blue paint in the ratio $4 : 7$. A batch of this mixture uses 28 litres of red paint. How many litres of blue paint does it use?

A) 16
B) 25
C) 35
D) 49

2

A printer produces 84 pages in 6 minutes. At this constant rate, how many pages does the printer produce per minute?

3

An experiment ran for 5400 seconds. Given that 1 hour $= 3600$ seconds, how many hours did the experiment run?

A) 0.9
B) 1.5
C) 15
D) 90

4

A prize of \$96 is divided among three students in the ratio $1 : 2 : 5$. How many dollars does the student with the largest share receive?

A) 12
B) 24
C) 60
D) 80

5

Seven identical textbooks have a total mass of 2.8 kilograms. What is the total mass, in kilograms, of 12 of these textbooks?

A) 4.8
B) 5.6
C) 7.2
D) 8.4

6

A model car is built to a scale of $1 : 18$. The model is 25 centimetres long. What is the length, in centimetres, of the actual car?

7

A rural district has a population density of 250 people per square kilometre and an area of 36 square kilometres. What is the population of the district?

A) 286
B) 900
C) 9000
D) 90 000

8

In a survey, the ratio of the number of people who prefer tea to the number who prefer coffee is $5 : 7$. There are 96 more people who prefer coffee than prefer tea. How many people were surveyed?

A) 288
B) 576
C) 672
D) 1152

9

A drone flies at a constant speed of 15 metres per second. What is this speed, in kilometres per hour?

10

A storage tank has a volume of 2.5 cubic metres. Given that 1 metre $= 100$ centimetres, what is the volume of the tank, in cubic centimetres?

A) 2500
B) 25 000
C) 250 000
D) 2 500 000

11

A triangle has sides of length 6 centimetres, 8 centimetres and 10 centimetres. A second triangle is similar to the first and has a longest side of length 25 centimetres. What is the perimeter, in centimetres, of the second triangle?

A) 30
B) 48
C) 60
D) 75

12

In a certain country, petrol costs 1.40 euros per litre, and 1 euro is worth 1.10 dollars. What is the cost, in dollars, of 40 litres of petrol?

A) 44
B) 50.91
C) 56
D) 61.60

13

A map has a scale of $1 : 50\,000$. Two towns are 12 kilometres apart. How far apart, in centimetres, are the two towns on the map?

14

A 40-litre mixture of water and syrup contains water and syrup in the ratio $7 : 3$. How many litres of syrup must be added so that the mixture contains equal amounts of water and syrup?

A) 16
B) 20
C) 24
D) 28

15

Cooking oil has a density of 0.8 grams per cubic centimetre. What is the volume, in cubic centimetres, of 1.2 kilograms of this oil?

A) 150
B) 960
C) 1500
D) 1600

16

A rectangular field measures 250 metres by 160 metres. Fertiliser is applied at a rate of 3 kilograms per hectare, where 1 hectare $= 10\,000$ square metres. How many kilograms of fertiliser are needed to cover the field?

A) 1.2
B) 12
C) 120
D) 12 000

17

A car travels at a constant speed of 90 kilometres per hour. How many metres does the car travel in 12 seconds?

18

Two rectangles are similar, and the ratio of the length of a side of the smaller rectangle to the length of the corresponding side of the larger rectangle is $3 : 5$. The area of the smaller rectangle is 27 square centimetres. What is the area, in square centimetres, of the larger rectangle?

A) 45
B) 75
C) 125
D) 225

19

A solid rectangular block measures 20 centimetres by 15 centimetres by 8 centimetres and has a mass of 6.72 kilograms. What is the density of the block, in grams per cubic centimetre?

20

A car travels 30 miles on each gallon of fuel. Using 1 mile $= 1.6$ kilometres and 1 gallon $= 4$ litres, how many kilometres does the car travel on each litre of fuel?

A) 12
B) 18.75
C) 48
D) 75

21

A floor plan is drawn to a scale of 1 : 200. On the plan, a rectangular room measures 4.5 centimetres by 3 centimetres. What is the actual floor area of the room, in square metres?

A) 13.5
B) 27
C) 54
D) 108

22

A water tank with a capacity of 4.5 cubic metres is filled by a hose that delivers 15 litres every 10 seconds. Given that 1 cubic metre = 1000 litres, how many minutes does it take to fill the tank?

A) 30
B) 50
C) 75
D) 300

Lesson 12: Percentages

A percent is nothing more than a fraction whose denominator is 100: 37% means $\frac{37}{100} = 0.37$. Every percent question on the Digital SAT is one of four questions in disguise — *what is p% of a number*, *what percent is one number of another*, *by what percent did a quantity change*, and *what was the quantity before it changed*. The last of these, together with two changes applied one after the other, is where most of the points are won and lost, because both reward the multiplier method and punish the instinct to add percents together.

Percent of a number, and one number as a percent of another

Translate the English literally. The word *of* means multiply, the word *is* means equals, and *what* means the unknown. So $p\%$ **of** N is $\frac{p}{100} \cdot N$: 30% of 250 is $0.30 \times 250 = 75$.

Running the sentence backwards gives the other two questions. **What percent is** A **of** B**?** asks for the number p with $\frac{p}{100} \cdot B = A$, so $p = \frac{A}{B} \times 100$. And $p\%$ **of what number is** A**?** asks for N with $\frac{p}{100} \cdot N = A$, so $N = \frac{A}{p/100}$. Notice that in *every* version there is one equation, $\frac{p}{100} \cdot N = A$, and you are told two of the three letters.

THE FOUR FORMULAS

$$p\% \text{ of } N = \frac{p}{100} \cdot N$$

$$A \text{ as a percent of } B = \frac{A}{B} \times 100\%$$

$$\text{percent change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100\%$$

$$\text{percent error} = \frac{|\text{measured} - \text{actual}|}{\text{actual}} \times 100\%$$

In the last two the denominator is always the *starting* or *true* value — the thing the change is being measured against.

THE MULTIPLIER METHOD

Do not compute the change and then add or subtract it. Turn the change into a single number you multiply by:

$$\text{an increase of } p\% \;\longrightarrow\; \times\left(1 + \tfrac{p}{100}\right), \qquad \text{a decrease of } p\% \;\longrightarrow\; \times\left(1 - \tfrac{p}{100}\right).$$

A 15% increase is $\times 1.15$; a 15% decrease is $\times 0.85$; adding an 8% tax is $\times 1.08$; taking 40% off is $\times 0.60$. This one habit pays for itself three times over. It makes a single change a one-step calculation, it makes two changes in a row a product, and — because multiplying is reversible — it turns every *reverse* percent question into a division.

WORKED EXAMPLE 1 — A DISCOUNT, THEN THE PERCENT CHANGE

A sweater priced at \$85 is marked down to \$68. By what percent was the price reduced, and what single number could the store have multiplied the old price by?

Percent change. The change is $68 - 85 = -17$ dollars, and the change is measured against the *old* price, 85:

$$\frac{68 - 85}{85} \times 100\% = \frac{-17}{85} \times 100\% = -0.20 \times 100\% = -20\%,$$

a 20% decrease.

Multiplier. A 20% decrease is $\times(1 - 0.20) = \times 0.80$, and indeed $85 \times 0.80 = 68$.

The check is worth doing every time: if your percent change is right, the old value times the multiplier must land exactly on the new value. Had you divided by 68 instead of 85 you would have gotten 25%, and $85 \times 0.75 = 63.75 \neq 68$ — the check catches the error immediately.

Successive change: percents do not add

When one change follows another, the second percent is taken of the *new* amount, not of the original. So you multiply the multipliers:

$$\text{final} = \text{original} \times m_1 \times m_2.$$

A 20% rise followed by a 20% fall gives $1.20 \times 0.80 = 0.96$: the quantity ends at 96% of where it started, a 4% *net loss*, not a return to the beginning. A 10% rise then another 10% rise gives $1.10 \times 1.10 = 1.21$, a 21% increase — not 20%.

Two consequences worth memorising. A rise of $p\%$ and a fall of $p\%$, in either order, always leaves you *below* the start. And to undo a $p\%$ fall you need a rise of *more* than $p\%$: after a 20% drop the multiplier is 0.8, so the recovery multiplier is $\frac{1}{0.8} = 1.25$, a 25% rise.

THE THREE MISTAKES THE TEST IS FISHING FOR

1. Adding the percents. "30% off, then 10% more off" is not 40% off. It is $\times 0.70 \times 0.90 = \times 0.63$, i.e. 37% off. The choice built from 40% is always sitting there.
2. Dividing by the wrong value. Percent change and percent error use the *original* (or *actual*) value as the denominator. Going from 200 to 250 is a $\frac{50}{200} = 25\%$ increase, not $\frac{50}{250} = 20\%$.
3. Multiplying when you should divide. If a price is \$68 *after* a 15% discount, the original is $\frac{68}{0.85} = 80$, not $68 \times 0.85 = 57.80$ and not $68 \times 1.15 = 78.20$. The \$68 is the output of the multiplication, so you must undo it.

REVERSE PERCENT: WORK BACKWARDS THROUGH THE MULTIPLIER

Whenever a question gives you the value *after* a change and asks for the value *before*, name the original x, write the forward sentence, and solve:

$$x \cdot m = \text{known value} \quad \Longrightarrow \quad x = \frac{\text{known value}}{m}.$$

The skill is entirely in writing m correctly: "after a 15% discount" gives $m = 0.85$; "including a 6% tax" gives $m = 1.06$; "after a 20% increase and then a 10% decrease" gives $m = 1.20 \times 0.90 = 1.08$.
A fast sanity check: after an increase the original must be *smaller* than the number you were given; after a decrease it must be *larger*. If your answer is on the wrong side, you multiplied instead of dividing.

WORKED EXAMPLE 2 — REVERSE AND SUCCESSIVE TOGETHER

A retailer raises the price of a bicycle by 25%, then puts the raised price on sale at 20% off. The sale price is \$150. What was the price before the increase?
Step 1: build the multipliers. An increase of 25% is $\times 1.25$; a discount of 20% is $\times 0.80$.
Step 2: write the forward equation. With x the original price,

$$x \times 1.25 \times 0.80 = 150.$$

Step 3: combine and divide. $1.25 \times 0.80 = 1.00$, so $x = 150$. The two changes cancelled exactly — which happens here only because $\frac{1}{1.25} = 0.80$, i.e. because 20% is exactly the discount that undoes a 25% increase. That coincidence is itself a favourite SAT question: *after raising a price by 25%, what percent discount returns it to the original?* The answer is not 25%. The needed multiplier is $\frac{1}{1.25} = 0.80$, so the discount is 20%.

STRATEGY AND CALCULATOR USE

Pick 100. When a question is entirely in percents with no numbers attached ("the final value is what percent of the original?"), let the original be 100, run the multipliers, and read the answer straight off. A 20% rise then a 25% fall: $100 \to 120 \to 90$, so the final value is 90% of the original, a 10% decrease.
Keep everything as one product. Type $120 \times 0.7 \times 0.9$ in one go rather than rounding an intermediate price; percent answers on the SAT are exact, so a rounded middle step is what turns 75.60 into a wrong choice.
Read what is being asked for. Questions alternate between "what is the final amount", "what is the percent change", and "the final is what percent of the original". Those are three different numbers — 90, -10% and 90% in the example above — and all three are usually among the choices.

Module 1

1
What is 30% of 250?

A) 7.5
B) 75
C) 175
D) 750

2
45 is what percent of 180? (Disregard the percent sign when entering your answer.)

3
A jacket regularly priced at \$80 is on sale for 25% off. What is the sale price of the jacket?

A) \$20
B) \$55
C) \$60
D) \$75

4
The number of members of a chess club increased from 200 to 250. What was the percent increase in the number of members?

A) 20%
B) 25%
C) 50%
D) 80%

5
A dinner bill is \$60. A 15% tip is added to the bill. What is the total amount paid?

A) \$9
B) \$63
C) \$69
D) \$75

6
In a shipment of 150 bicycles, 24 of the bicycles are electric. What percent of the bicycles in the shipment are electric?

A) 16%
B) 24%
C) 76%
D) 84%

7
On a 40-question test, Maria answered 34 questions correctly. What percent of the questions did Maria answer correctly?

A) 6%
B) 15%
C) 34%
D) 85%

8

Increasing a quantity by 8% gives the same result as multiplying that quantity by which of the following?

A) 0.08

B) 0.8

C) 0.92

D) 1.08

9

The price of a laptop was reduced from \$850 to \$680. By what percent was the price of the laptop reduced? (Disregard the percent sign when entering your answer.)

10

40% of what number is 26?

A) 10.4

B) 65

C) 66

D) 104

11

In a survey of 800 residents, 45% of the residents said they own a car. Of the residents who own a car, 25% said they also own a bicycle. How many of the residents surveyed own both a car and a bicycle?

A) 90

B) 200

C) 360

D) 560

12

Favorite subject	Number of students
Mathematics	96
Science	84
History	72
English	148
Total	400

The table shows the favorite subject reported by each of the 400 students at a school. What percent of these students reported that their favorite subject is mathematics or science?

A) 21%

B) 24%

C) 45%

D) 55%

13

A carpenter estimated that a board was 480 centimeters long, but the board was actually 500 centimeters long. The percent error of an estimate is the positive difference between the estimate and the actual value, expressed as a percent of the actual value. What is the percent error of the carpenter's estimate?

A) 4%

B) 4.2%

C) 20%

D) 25%

14

After a 20% decrease, the monthly rent for an apartment is \$960. What was the monthly rent, in dollars, before the decrease?

A) \$768

B) \$1000

C) \$1152

D) \$1200

15

The price of a bicycle is \$200. The price is first increased by 10%, and then the increased price is decreased by 10%. What is the final price of the bicycle?

A) \$160

B) \$180

C) \$198

D) \$200

16

A coat is priced at \$120. A store takes 30% off the price of the coat and then takes an additional 10% off the discounted price. What is the final price of the coat?

A) \$72
B) \$75.60
C) \$78
D) \$84

17

Including a 6% sales tax, the total cost of a television is \$265. What was the price of the television, in dollars, before tax?

18

The value of an investment increased by 20% during its first year and then decreased by 25% during its second year. At the end of the second year, the value of the investment was what percent of its original value?

A) 90%
B) 95%
C) 100%
D) 105%

19

A number is increased by 30%, and the result is then decreased by 30%. The final value is 91. What was the original number?

20

60% of a number x is equal to 45% of 80. What is the value of x?

A) 36
B) 45
C) 48
D) 60

21

A store increases the price of a lamp by 25%. During a sale, the store then reduces the increased price by 25%. If the sale price of the lamp is \$60, what was the price of the lamp before the increase?

A) \$48
B) \$60
C) \$64
D) \$80

22

A shop raises the price of every bicycle by 25%. The shop then wants to advertise a discount that returns the price of a bicycle to its value before the increase. What percent discount must the shop advertise? (Disregard the percent sign when entering your answer.)

Module 2

1

What is 65% of 400?

A) 26
B) 235
C) 260
D) 465

2

18 is what percent of 45? (Disregard the percent sign when entering your answer.)

3

A store sold 240 phones last month, and 35% of those phones were refurbished. How many of the phones sold last month were refurbished?

A) 35
B) 84
C) 156
D) 205

4

A store's daily sales fell from \$250 on Monday to \$200 on Tuesday. What was the percent decrease in daily sales?

A) 20%
B) 25%
C) 50%
D) 80%

5

30% of what number is 72?

A) 21.6
B) 102
C) 216
D) 240

6

Decreasing a quantity by 35% gives the same result as multiplying that quantity by which of the following?

A) 0.35
B) 0.65
C) 1.35
D) 1.65

7

A student's score on a practice test rose from 60 points to 75 points. What was the percent increase in the student's score?

A) 15%
B) 20%
C) 25%
D) 80%

8

After a 25% discount, the price of a jacket is \$96. What was the price of the jacket, in dollars, before the discount?

9

The population of a town was 500 at the start of 2021. The population increased by 10% during 2021 and then increased by 20% during 2022. What was the population of the town at the end of 2022?

A) 620
B) 650
C) 660
D) 680

10

A recipe calls for 250 grams of flour, but a baker used 265 grams. The amount of flour the baker used was what percent greater than the amount the recipe calls for?

A) 5.7%
B) 6%
C) 15%
D) 106%

11

Ticket type	Number sold
Adult	180
Student	260
Senior	60
Total	500

The table shows the number of tickets of each type sold for a concert. What percent of the tickets sold were not student tickets? (Disregard the percent sign when entering your answer.)

12

96 is what percent of 60?

A) 36%

B) 60%

C) 62.5%

D) 160%

13

A quantity is decreased by 40%, and the result is then increased by 40%. The final value is what percent of the original value?

A) 84%

B) 96%

C) 100%

D) 116%

14

A pair of headphones is on sale for $52 after a 20% discount. What was the price of the headphones before the discount?

A) $41.60

B) $62.40

C) $65

D) $72

15

40% of a number is 26 more than 15% of the same number. What is the number?

16

A town's population was 8000 at the start of 2022. During 2022 the population decreased by 15%, and during 2023 it increased by 20%. What was the town's population at the end of 2023?

A) 8000

B) 8160

C) 8400

D) 8560

17

The price of a share of a stock increased by 20% during January and then decreased by 10% during February. If the price of a share at the end of February was $216, what was the price of a share at the start of January?

A) $194.40

B) $200

C) $233.28

D) $240

18

A store buys a jacket from a supplier and sets a listed price that is 40% greater than what the store paid. The store later sells the jacket at a 25% discount off the listed price. The amount the store received for the jacket is what percent of the amount the store paid for it?

A) 15%

B) 95%

C) 100%

D) 105%

19

The price of a share of a stock fell by 20%. By what percent must the reduced price rise in order to return the share to its original price? (Disregard the percent sign when entering your answer.)

20

Year	Revenue (thousands of dollars)
2021	250
2022	300
2023	270
2024	324

The table shows a company's annual revenue, in thousands of dollars, for four years. According to the table, the company's revenue in 2024 was what percent greater than its revenue in 2021?

A) 22.8%

B) 24%

C) 29.6%

D) 74%

21

A printer is sold for \$180 after two successive discounts off its original price: first a discount of 25%, and then a discount of 20% off the reduced price. What was the original price of the printer?

A) \$240

B) \$270

C) \$288

D) \$300

22

In 2023 a company's revenue was 25% greater than its revenue in 2022. In 2024 the company's revenue was 8% less than its revenue in 2023. If the company's revenue in 2024 was \$115,000, what was its revenue, in dollars, in 2022?

Lesson 13: One-Variable Data: Distributions and Measures of Center and Spread

One-variable data questions hand you a small pile of numbers — as a list, as a frequency table, as a dot plot, as a histogram, or as a box plot — and ask what is typical about them (a measure of *centre*) or how scattered they are (a measure of *spread*). Every one of these questions is answered with arithmetic you already know; the work is in reading the display correctly and in knowing which of the mean, median, mode and range the question actually wants. Expect two or three of them on a Digital SAT.

The four measures

A **data set** is just a list of values. Four numbers summarise it.

The **mean** is the total of the values divided by how many values there are. The **median** is the middle value once the list is written in order — and if there is an even number of values, it is the mean of the two middle ones. The **mode** is the value that appears most often. The **range** is the largest value minus the smallest value.

For the list 4, 9, 6, 4, 12: the mean is $\frac{4+9+6+4+12}{5} = \frac{35}{5} = 7$; ordered the list is $4, 4, 6, 9, 12$, so the median is 6; the mode is 4; the range is $12 - 4 = 8$.

FORMULAS YOU MUST KNOW

$$\text{mean} = \frac{\text{sum of the values}}{\text{number of values}} \iff \text{sum} = \text{mean} \times \text{number of values}$$

$$\text{range} = \text{maximum} - \text{minimum} \qquad \text{IQR} = Q_3 - Q_1$$

The right-hand form of the mean formula is the one that solves "working backwards" questions: if you know a mean and a count, you know the total.

READING A FREQUENCY TABLE, DOT PLOT OR HISTOGRAM

All three displays say the same thing: *this value occurred this many times.* In a dot plot each dot is one data value; in a histogram or frequency table the height (or the tabulated frequency) tells you how many times that value occurred.

To find the **mean**, multiply each value by its frequency, add the products, and divide by the *total frequency*: mean $= \dfrac{\sum(\text{value} \times \text{frequency})}{\sum \text{frequency}}$.

To find the **median**, first find the total frequency N. The median sits at position $\frac{N+1}{2}$ when N is odd, and is the mean of positions $\frac{N}{2}$ and $\frac{N}{2}+1$ when N is even. Run a running total of the frequencies from the left until you reach that position.

THREE MISTAKES THAT ARE BUILT INTO THE ANSWER CHOICES

1. Taking the median without sorting. The middle entry of the list as printed is almost never the median. Write the list in order first — every time.

2. Averaging the frequencies instead of the values. In a table with values $0, 1, 2, 3, 4$ and frequencies $4, 7, 5, 3, 1$, the mean is *not* $\frac{20}{5} = 4$ and it is *not* $\frac{0+1+2+3+4}{5} = 2$. It is $\frac{0(4)+1(7)+2(5)+3(3)+4(1)}{20} = \frac{30}{20} = 1.5$.

3. Confusing range with interquartile range. The range uses the two whisker ends of a box plot; the IQR uses the two ends of the *box*. They are different numbers and both appear among the choices.

OUTLIERS: WHAT MOVES AND WHAT DOES NOT

An **outlier** is a value far from the rest of the data. Because the mean is built from the total, one extreme value drags it a long way. The median only cares about position, so it barely shifts — usually by one place in the ordered list.

So for the list $12, 14, 15, 16, 18$ (mean 15, median 15), adding the value 90 gives a mean of $\frac{165}{6} = 27.5$ but a median of only 15.5. **Add a high outlier:** the mean jumps up, the median creeps up. **Remove a high outlier:** the mean drops sharply, the median drops slightly. The rule reverses in sign for a low outlier. This is why a strongly skewed data set has a mean pulled toward the long tail and a median that stays with the bulk of the data.

STANDARD DEVIATION — COMPARED, NEVER COMPUTED

Standard deviation measures how far a typical value sits from the mean. **The SAT never asks you to compute one.** Every standard-deviation question is a comparison, and it is decided by a single question: *in which data set are the values packed more tightly around the mean?*

Tightly clustered values $\Rightarrow$ **smaller** standard deviation. Values spread far from the mean, or piled up at the two ends, $\Rightarrow$ **larger** standard deviation. Identical values give a standard deviation of exactly 0.

Two facts worth memorising: equal means tell you *nothing* about standard deviation, and adding a far-away outlier always increases it.

CALCULATOR / STRATEGY

Sort first, always. Ten seconds of sorting removes the single most common error on this topic.

Work in totals. “The mean of 12 numbers is 7” is really the statement “the numbers add to 84”. Convert every mean in the question into a total, do the arithmetic on the totals, and convert back at the end.

Don’t compute what you can see. For a spread comparison, look at the picture. For a box plot, read the five marks off the axis — minimum, Q_1, median, Q_3, maximum — and answer from those five numbers alone.

WORKED EXAMPLE 1 — A FREQUENCY TABLE

The table shows the number of pets owned by each of 20 households.

Pets	0	1	2	3	4
Households	4	7	5	3	1

(a) What is the mean number of pets? (b) What is the median?

(a) Multiply each value by its frequency and add: $0(4)+1(7)+2(5)+3(3)+4(1) = 0+7+10+9+4 = 30$ pets in all. There are $4+7+5+3+1 = 20$ households, so the mean is $\frac{30}{20} = 1.5$ pets.

(b) With $N = 20$ values the median is the mean of the 10th and 11th values in order. Running totals: the first 4 values are 0; values 5 through 11 are 1. So both the 10th and the 11th value are 1, and the median is 1.

Notice that $\frac{20}{5} = 4$ (the mean of the frequencies) and $\frac{0+1+2+3+4}{5} = 2$ (the mean of the listed values) are both wrong — and both will be sitting in the answer choices.

WORKED EXAMPLE 2 — WORKING BACKWARDS, AND AN OUTLIER

The mean of nine numbers is 32. A tenth number is added and the mean of all ten numbers becomes 38. What is the tenth number?

Turn both means into totals. Nine numbers with mean 32 add to $9 \times 32 = 288$. Ten numbers with mean 38 add to $10 \times 38 = 380$. The tenth number is therefore $380 - 288 = 92$.

That 92 is an outlier: it sits far above a group averaging 32. Consistent with the rule above, it pushed the mean up by 6 — while the median would have moved by at most half a place. If a follow-up question asked what happened to the standard deviation, the answer is that it increased, because a value that far from the mean makes the typical distance from the mean larger.

Module 1

1

The high temperatures, in degrees Fahrenheit, recorded in a town on five days were 61, 70, 72, 78, and 84. What is the mean of these five temperatures?

A) 71

B) 72

C) 73

D) 74

2

The numbers of books read last month by seven students were 5, 2, 9, 4, 7, 3, and 6. What is the median number of books read?

A) 4

B) 5

C) 6

D) 9

3

The masses, in grams, of six mineral samples were 12, 19, 7, 15, 22, and 10. What is the range of these masses, in grams?

4

A shoe store recorded the sizes of the eight pairs of shoes it sold in one hour: 8, 9, 7, 9, 10, 8, 9, and 11. What is the mode of these sizes?

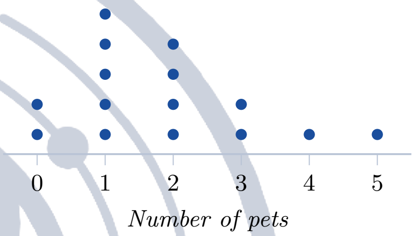
A) 3

B) 8

C) 9

D) 10

5



The dot plot shows the number of pets owned by each of 15 families. What is the median number of pets?

A) 1

B) 2

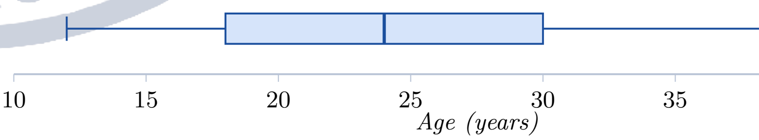
C) 2.5

D) 3

6

Over eight weeks, a runner ran 14, 18, 12, 20, 16, 22, 19, and 15 miles. What was the mean number of miles the runner ran per week?

7



The box plot summarizes the ages, in years, of the trees in a small orchard. What is the median age of the trees?

A) 12

B) 18

C) 24

D) 30

8

Number of siblings	Number of students
0	4
1	7
2	5
3	3
4	1

The frequency table shows the number of siblings reported by each of 20 students. What is the mean number of siblings per student?

A) 1
B) 1.5
C) 2
D) 4

9

A biologist recorded the heights, in centimetres, of ten seedlings: 31, 45, 28, 52, 39, 47, 33, 41, 36, and 50. What is the median height?

A) 39
B) 40
C) 41
D) 43

10

The mean of five test scores is 84. Four of the scores are 78, 91, 85, and 80. What is the fifth score?

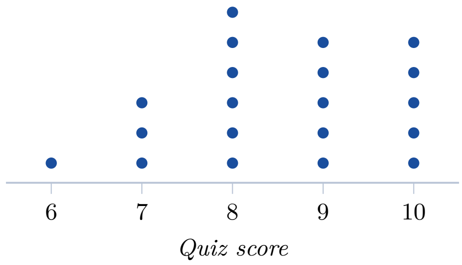
A) 83.5
B) 84
C) 86
D) 88

11

A data set consists of the nine values 12, 14, 15, 16, 18, 19, 21, 23, and 24. The value 90 is then added to the data set. Which of the following best describes the effect of adding 90?

A) The mean and the median increase by the same amount.
B) The median increases by more than the mean does.
C) The mean increases and the median decreases.
D) The mean increases by much more than the median does.

12



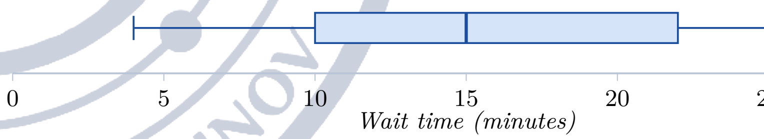
The dot plot shows the scores of 20 students on a 10-point quiz. What is the mean score?

13

Data set X consists of the values 20, 21, 22, 23, and 24. Data set Y consists of the values 12, 17, 22, 27, and 32. Each data set has a mean of 22. Which of the following is true about the standard deviations of the two data sets?

A) Data set X has the larger standard deviation, because its values are closer to the mean.
B) The standard deviations are equal, because the two data sets have the same mean.
C) The standard deviations are equal, because the two data sets have the same number of values.
D) Data set Y has the larger standard deviation, because its values are farther from the mean.

14



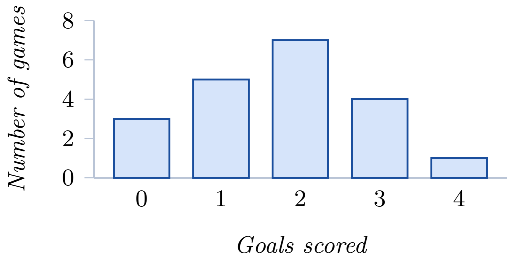
The box plot summarizes the number of minutes that each of 40 commuters waited for a train. What is the interquartile range of the wait times, in minutes?

A) 12
B) 15
C) 18
D) 26

15

The mean of six numbers is 15. When one of the numbers is removed, the mean of the remaining five numbers is 16. What is the value of the number that was removed?

16



The histogram shows the number of goals scored in each of 20 soccer games. What is the mean number of goals per game?

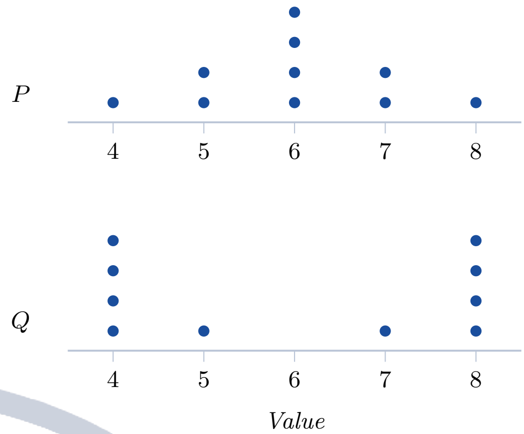
A) 1.75
B) 2
C) 4
D) 7

17

Value	Frequency
2	5
3	8
4	x
5	3

In the frequency table shown, the value 4 occurs x times. The mean of all the data is 3.4. What is the value of x?

18



The dot plots show two data sets, each consisting of 10 values with a mean of 6. Which data set has the larger standard deviation, and why?

A) Data set P, because more of its values are far from the mean.
B) Data set P, because its values are more tightly clustered.
C) Neither, because the two data sets have the same mean.
D) Data set Q, because more of its values are far from the mean.

19

The yearly salaries, in thousands of dollars, of the eight employees at a small company are 32, 35, 36, 38, 40, 42, 45, and 180. The salary of 180 is removed from the data set. Which of the following describes the change in the mean and in the median?

A) The mean decreases by about 18 and the median decreases by 1.
B) The mean decreases by 1 and the median decreases by about 18.
C) The mean and the median each decrease by about 18.
D) The mean decreases and the median is unchanged.

20

In one class of 24 students, the mean score on a test was 78. In a second class of 16 students, the mean score on the same test was 88. What is the mean score of all 40 students?

21

Days absent	Number of students
0	6
1	9
2	8
3	7
4	5

The table shows the number of days absent for each of the 35 students in a class. What is the median number of days absent?

A) 0

B) 1

C) 2

D) 7

22

The mean of 12 numbers is 7. A thirteenth number is added to the data set, and the mean of all 13 numbers is 8. What is the thirteenth number?

A) 8

B) 15

C) 20

D) 21

Module 2

1

The prices, in dollars, of eight items sold at a shop are 14, 9, 21, 16, 12, 18, 25, and 11. What is the median price, in dollars?

A) 14

B) 15

C) 15.75

D) 16

2

Number of children	Number of households
0	6
1	14
2	19
3	8
4	3

The table shows the number of children in each of 50 households in a neighbourhood. What is the mode of the number of children per household?

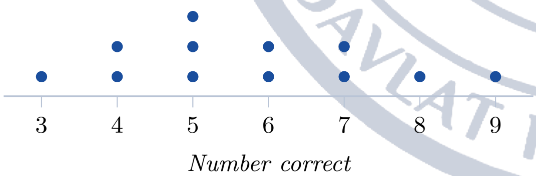
A) 2

B) 3

C) 14

D) 19

3



The dot plot shows the number of questions answered correctly by each of 12 students on a short test. What is the range of the number of correct answers?

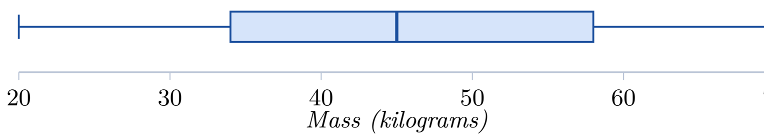
A) 6

B) 7

C) 9

D) 12

4



The box plot summarizes the masses, in kilograms, of the dogs treated at a clinic in one week. What is the value of the third quartile, Q_3?

A) 34

B) 45

C) 46

D) 58

5

The rainfall, in inches, recorded in a town in each of six months was 2.4, 3.1, 1.8, 2.9, 3.6, and 4.2. What was the mean monthly rainfall, in inches?

6

Data set A consists of the five values 50, 50, 50, 50, and 50. Data set B consists of the five values 46, 48, 50, 52, and 54. Which data set has the greater standard deviation, and why?

A) Data set A, because all of its values are equal to the mean.

B) Data set A, because it has the greater mean.

C) They have the same standard deviation, because they have the same mean.

D) Data set B, because its values vary while those of data set A do not.

7

Number correct	Number of students
1	2
2	3
3	4
4	6

The table shows the number of correct answers, out of 4, for each of 15 students on a short quiz. What is the median number of correct answers?

A) 3

B) 4

C) 6

D) 15

8

Number of cars	Number of families
0	5
1	8
2	6
3	4
4	2

The table shows the number of cars owned by each of 25 families. What is the mean number of cars per family?

A) 1

B) 1.6

C) 2

D) 5

9

A data set consists of the six values 4, 9, 11, x, 17, and 20, listed in increasing order. The median of the data set is 13. What is the value of x?

10

A data set of 10 numbers has a mean of 46. One value of 23 is replaced by 63, and no other value is changed. What is the mean of the new data set?

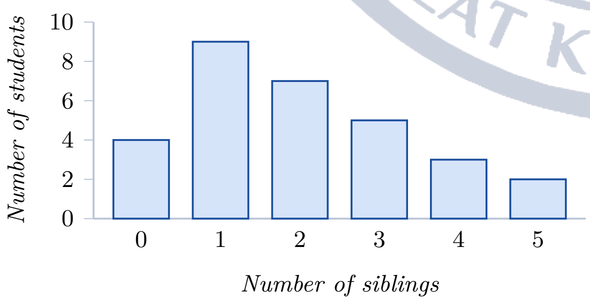
A) 46

B) 48

C) 50

D) 86

11



The histogram shows the number of siblings of each of 30 students. What is the median number of siblings?

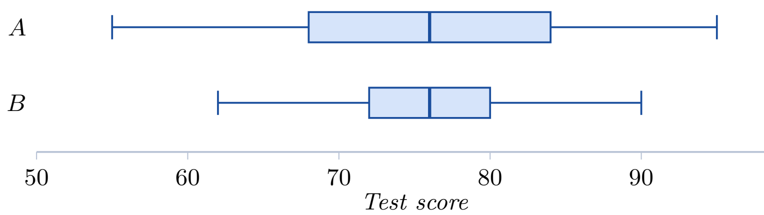
A) 1

B) 2

C) 2.5

D) 5

12



The box plots summarize the test scores of the students in class A and class B. Which of the following statements is true?

A) The two classes have the same median, and class A has the larger interquartile range.

B) Class B has the larger median and the larger interquartile range.

C) The two classes have the same median, and class B has the larger range.

D) Class A has the larger median, and the two classes have the same range.

13

The 18 students in a class have a mean height of 62 inches. Two new students, of heights 70 inches and 74 inches, join the class. What is the mean height, in inches, of the 20 students?

A) 62.4

B) 63

C) 64

D) 67

14

The numbers of customers a cashier served during each of eight hours were 3, 7, 8, 12, 13, 15, 18, and 25. What is the interquartile range of these numbers?

15

A data set consists of the seven values 4, 6, 7, 9, 10, 12, and 40. The value 40 is removed from the data set. Which of the following describes the change in the mean and in the median?

A) The median decreases by more than the mean does.

B) The mean is unchanged and the median decreases.

C) Both the mean and the median increase.

D) The mean decreases by more than the median does.

16

The mean of seven numbers is 24. Two of the numbers, 15 and 28, are removed from the data set. What is the mean of the remaining five numbers?

17

Two data sets each consist of 20 values with a mean of 50. In data set R, 16 of the values are between 48 and 52. In data set S, 16 of the values are less than 40 or greater than 60. Which of the following statements is true?

A) Data set R has the larger standard deviation, because more of its values are near 50.

B) The standard deviations are equal, because the means are equal.

C) The standard deviations cannot be compared without calculating each one.

D) Data set S has the larger standard deviation, because more of its values are far from 50.

18

The mean of nine numbers is 15. Each of the nine numbers is increased by 4, and each result is then doubled. What is the mean of the nine new numbers?

A) 30

B) 34

C) 38

D) 46

19

Score	Number of students
70	6
75	9
80	11
85	9
90	5

The table shows the scores of the 40 students who took a quiz. What is the median score?

A) 75

B) 79.75

C) 80

D) 85

20

A data set consists of 15 values, each between 40 and 60. A new value of 200 is then added to the data set. Compared with the original data set, which of the following must be true of the new data set?

A) Both the mean and the standard deviation are larger.

B) The mean is larger and the standard deviation is smaller.

C) The mean is unchanged and the standard deviation is larger.

D) The median increases by more than the mean does.

21

A class of 30 students took a test. The 12 students in group A had a mean score of 85, and the class as a whole had a mean score of 79. What was the mean score of the other 18 students?

22

A data set of 20 values has a mean of 42. When two of the values are removed, the mean of the remaining 18 values is 40. What is the mean of the two values that were removed?

A) 41

B) 44

C) 46

D) 60

Lesson 14: Two-Variable Data: Models and Scatterplots

Two-variable data means every subject in the study carries two numbers, and those pairs are plotted as points in the xy-plane. The Digital SAT asks a small, very predictable set of questions about such a picture: which way does it slope, what line or curve fits it, what do the slope and the y-intercept of that line *mean* in the words of the problem, what does the model predict, and how far off is the model at a point we actually observed. Almost none of it is heavy computation; nearly all of it is careful reading.

Reading a scatterplot

A **scatterplot** plots one point (x, y) for each subject. Describe it with three words.

Direction. If y tends to rise as x rises the association is *positive*; if y tends to fall as x rises it is *negative*; if there is no tendency at all there is *no association.*

Form. If the cloud of points hugs a straight line the form is *linear.* If it bends — rising slowly and then explosively, or falling fast and then flattening out — the form is *nonlinear*, and on the SAT the nonlinear choice is almost always *exponential.*

Strength. Tightly packed points mean a strong association; a wide scattered cloud means a weak one. A strong association is still only an association: the SAT never lets you conclude that x *causes* y from a scatterplot alone.

THE LINE OF BEST FIT

The **line of best fit** (or *least-squares regression line*) is the single straight line that passes as close as possible to all of the points at once. You are never asked to compute it by hand. You are asked to *use* it:

- read a predicted y off the line for a given x;
- read the x that produces a given predicted y;
- find its slope from two points the line passes through;
- say what its slope and its y-intercept mean in context.

Everything the line says is a *prediction*, never a fact about an individual subject. The word “predicted” belongs in every correct interpretation.

THE MODEL AND WHAT EACH PIECE MEANS

For a line of best fit written as

$$\hat{y} = mx + b,$$

slope m = the predicted change in y for each *increase of one unit* in x. Its units are (units of y) per (unit of x).
y-intercept b = the predicted value of y when $x = 0$. Its units are the units of y.
From two points (x_1, y_1) and (x_2, y_2) *on the line*,

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

For an exponential model $\hat{y} = a(b)^x$: a is the predicted value at $x = 0$, and b is the factor y is multiplied by for each unit increase in x. If $b = 1 + r$ the quantity grows by r (as a percent) per unit; if $b = 1 - r$ it decays by r per unit.

Linear or exponential?

Look at how the y-values change as x goes up by 1.

Constant difference $\Rightarrow$ linear. 5, 9, 13, 17 adds 4 every time, so $y = 4x + 5$.

Constant ratio $\Rightarrow$ exponential. 5, 15, 45, 135 multiplies by 3 every time, so $y = 5(3)^x$.

In words: "increases by $40 per year" is linear; "increases by 4% per year" is exponential, because a percent of a growing amount is a growing amount. On a graph, a linear scatter rises at a steady rate, while an exponential scatter is nearly flat and then shoots up (growth), or drops steeply and then levels toward zero (decay).

RESIDUALS

A **residual** measures how wrong the model is at one observed point:

$$\text{residual} = \text{actual} - \text{predicted}.$$

- Residual **positive** $\Rightarrow$ the point sits *above* the line $\Rightarrow$ the model **underpredicts**.
- Residual **negative** $\Rightarrow$ the point sits *below* the line $\Rightarrow$ the model **overpredicts**.

Rearranged, actual = predicted + residual, which is how you recover a missing data value. If the residuals of a linear fit run positive, then negative, then positive again, the data curve and a nonlinear model would fit better.

THE SLOPE AND THE INTERCEPT GET SWAPPED

In an interpretation question the wrong choices are not random. The favourite distractor is the correct sentence with the two numbers traded: for $\hat{y} = 24x + 150$, the key says "the predicted number of members increases by 24 each month," and the trap says "... increases by 150 each month." Two more to watch for:

- **Variables swapped**: "for each additional member, 24 more months," which reverses what is predicted from what.
- **Units swapped**: if y is measured in thousands of people or in dollars, a slope of 1.6 may mean 1600 people, and a slope of 0.19 dollars may mean 19 cents.

And when a question asks for the *residual*, the predicted value is always sitting there among the choices.

HOW TO ATTACK THESE QUESTIONS

1. Label the axes in words before you read the choices. Write "x = years since 2000, y = population in thousands." Almost every distractor dies immediately.
2. For a slope question say the sentence yourself first: "for each extra *[one unit of x]*, the predicted *[y]* changes by *[m]* *[units of y]*." Then find the choice that matches.
3. For an intercept question set $x = 0$ and ask what that means in the story — the day the shop opened, the moment the candle was lit, the year 2000.
4. Never extrapolate far. A model built on x from 1 to 12 says nothing reliable at $x = 90$; if a choice raises that objection, it is usually the answer.
5. Watch the sign convention. Actual minus predicted — in that order, always.

WORKED EXAMPLE 1 — SLOPE, INTERCEPT, PREDICTION

A city recorded the number of electric-vehicle charging stations y in the year x years after 2015. The line of best fit is $\hat{y} = 18x + 42$.
(a) What does 18 mean? The slope is the predicted change in y per 1-unit increase in x, and one unit

of x is one year. So the number of charging stations is predicted to increase by 18 per year. (The trap choice says "increases by 42 per year.")
(b) What does 42 mean? Set $x = 0$, which is the year 2015: the model predicts 42 charging stations in 2015. Not "42 stations per year," and not "18 stations in 2015."
(c) Predict the count for 2024. 2024 is $x = 9$, so $\hat{y} = 18(9) + 42 = 162 + 42 = 204$ stations.
(d) In which year does the model first predict 300? Solve $18x + 42 = 300$: $18x = 258$, $x = 14.33\ldots$, so during the 15th year after 2015, i.e. 2030. Notice that 2030 is well past the data, so this prediction deserves a shrug.

WORKED EXAMPLE 2 — RESIDUALS AND CHOOSING A MODEL

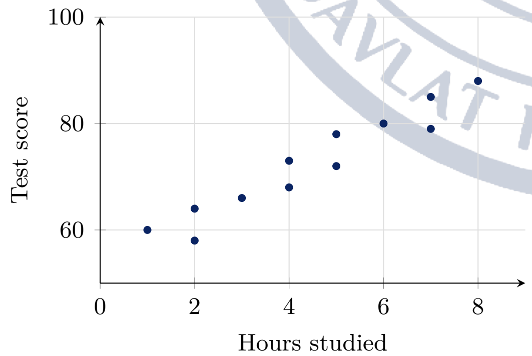
The table gives a shop's actual weekly sales, in hundreds of dollars, for four weeks; the line of best fit is $\hat{y} = 6x + 20$.

Week x	1	2	3	4
Actual y	28	30	36	46

(a) Residual in week 3. Predicted: $6(3) + 20 = 38$. Residual: $36 - 38 = -2$. It is negative, so the point lies below the line and the model *overpredicts* week 3's sales by 2 hundred dollars.
(b) Actual sales in a week where the prediction was 50 and the residual was +3. actual = predicted + residual = $50 + 3 = 53$ hundred dollars. If you answer 50, you answered a question that was not asked.
(c) Would an exponential model be better? The actual values change by $+2, +6, +10$: neither a constant difference nor a constant ratio ($30/28 \approx 1.07$ but $46/36 \approx 1.28$). The increases are growing, so the data bend upward and an exponential model is the better of the two candidates — exactly the reasoning the SAT wants: constant difference means linear, constant ratio means exponential.

Module 1

1



The scatterplot shows the number of hours each of 12 students spent studying for an exam and the score that student earned. Which of the following best describes the association between hours studied and score?

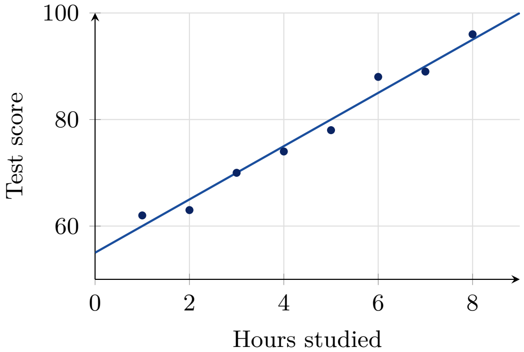
A) The association is positive and approximately linear.

B) The association is negative and approximately linear.

C) The association is positive and exponential, with the score doubling each hour.

D) There is no association between hours studied and score.

2



The scatterplot shows test scores against hours studied for 8 students, along with the line of best fit. According to the line of best fit, which of the following is closest to the predicted score of a student who studied for 5 hours?

A) 70
B) 75
C) 80
D) 85

3

A line of best fit passes through the points $(2, 11)$ and $(6, 27)$. What is the slope of this line of best fit?

4

A gym recorded its membership each month after opening. For the scatterplot of x, the number of months since the gym opened, and y, the number of members, the line of best fit is $\hat{y} = 24x + 150$. Which of the following is the best interpretation of the slope of this line?

A) The predicted number of members increases by 24 each month.
B) The predicted number of members increases by 150 each month.
C) The gym had 24 members when it opened.
D) It takes 24 months for the gym to gain one member.

5

For a scatterplot of x, the number of weeks since a seedling was potted, and y, the height of the seedling in centimeters, the line of best fit is $\hat{y} = 1.8x + 4$. Which of the following is the best interpretation of the y-intercept of this line?

A) The predicted height increases by 4 centimeters each week.
B) The predicted height of the seedling when it was potted was 1.8 centimeters.
C) The seedling reaches a height of 4 centimeters after 1.8 weeks.
D) The predicted height of the seedling when it was potted was 4 centimeters.

6

For a group of cafés, the line of best fit relating x, the number of tables, to y, the number of chairs, is $\hat{y} = 6x + 13$. How many chairs does this model predict for a café with 12 tables?

7

x	0	1	2	3
y	200	400	800	1600

The table shows four values of a data set. Which of the following best describes a model for these data?

A) An exponential model, because the values increase by a constant factor.
B) A linear model, because the values increase by a constant amount.
C) A linear model, because the values increase by a constant factor.
D) An exponential model, because the values increase by a constant amount.

8

For a scatterplot of the daily high temperature and the number of gallons of ice cream a shop sells, the line of best fit predicts 58 gallons on a certain day. The shop actually sold 64 gallons that day. What is the residual for that day?

A) -6
B) 6
C) 58
D) 122

9

A line of best fit predicts that a repair will cost \$212, but the repair actually costs \$197. Which of the following correctly describes the model at this data point?

A) The model underpredicts the cost by \$15, so the residual is 15 dollars.

B) The model overpredicts the cost by \$15, so the residual is 15 dollars.

C) The model underpredicts the cost by \$15, so the residual is -15 dollars.

D) The model overpredicts the cost by \$15, so the residual is -15 dollars.

10

For a scatterplot of x, the number of years since 2010, and y, the number of landline telephone subscriptions per 100 households in a county, the line of best fit is $\hat{y} = -3.4x + 61$. Which of the following is the best interpretation of the slope of this line?

A) The predicted number of subscriptions per 100 households increases by 3.4 each year.

B) The predicted number of subscriptions per 100 households decreases by 61 each year.

C) In 2010 there were 3.4 subscriptions per 100 households.

D) The predicted number of subscriptions per 100 households decreases by 3.4 each year.

11

The line of best fit for the height of a certain species of tree is $\hat{y} = 4.5x + 12$, where x is the number of years since the tree was planted and $\hat{y}$ is the predicted height in feet. What height, in feet, does the model predict for a tree 16 years after planting?

12

A taxi company plotted x, the length of a ride in miles, against y, the fare in dollars, and found the line of best fit $\hat{y} = 2.4x + 3.5$. Which of the following is the best interpretation of the number 3.5 in this model?

A) The predicted fare for a ride of 0 miles is \$3.50.

B) The predicted fare increases by \$3.50 for each additional mile.

C) The predicted fare for a ride of 0 miles is \$2.40.

D) The predicted length of a ride costing \$0 is 3.5 miles.

13

x	1	2	3	4	5	6
Actual y	28	33	42	47	60	63

The table shows six data points from a study. The line of best fit for these data is $\hat{y} = 7x + 20$. What is the residual for the data point with $x = 5$?

14

t	0	1	2	3
P	80	120	180	270

The table shows the value of P at four times t. Which of the following describes the type of model that fits the data and the constant that governs it?

A) Linear, with a constant increase of 40.

B) Linear, with a constant increase of 1.5.

C) Exponential, with a constant multiplier of 1.5.

D) Exponential, with a constant multiplier of 40.

15

A town's population was modeled by $\hat{P} = 1.6t + 34$, where t is the number of years since 2000 and $\hat{P}$ is the predicted population in thousands. Which of the following is the best interpretation of the slope of this model?

A) The population is predicted to increase by 1.6 people each year.

B) The population is predicted to increase by 34000 people each year.

C) The population in 2000 was predicted to be 1600 people.

D) The population is predicted to increase by 1600 people each year.

16

A reading study produced the line of best fit $\hat{y} = 13x + 45$, where x is the number of minutes spent reading and $\hat{y}$ is the predicted number of words recalled. According to this model, how many more words are predicted for a student who reads for 18 minutes than for a student who reads for 11 minutes?

A) 7

B) 13

C) 91

D) 324

17

A museum uses a line of best fit to predict daily attendance. On one day the model predicted 138 visitors and the residual for that day was -24. How many visitors actually came that day?

A) 24
B) 114
C) 138
D) 162

18

A tutoring company plotted x, the number of practice tests a student completed, against y, that student's score improvement in points, and found the line of best fit $\hat{y} = 17x + 5$. Which of the following is the best interpretation of the slope of this line?

A) Each additional practice test causes a score improvement of exactly 17 points.
B) On average, each additional point of score improvement is associated with 17 more practice tests.
C) A student who takes no practice tests is predicted to improve by 17 points.
D) On average, each additional practice test is associated with a predicted increase of 17 points in score improvement.

19

The line of best fit relating x, the number of weeks of training, to $\hat{y}$, the predicted number of push-ups an athlete can complete, is $\hat{y} = 2.5x + 18$. For what value of x does this model predict 68 push-ups?

20

Day	1	2	3	4	5
Actual	12	24	29	42	48
Predicted	13	22	31	40	49

The table shows the actual number of bicycles a shop rented and the number predicted by the line of best fit on each of five days. On how many of these days does the model overpredict the number of rentals?

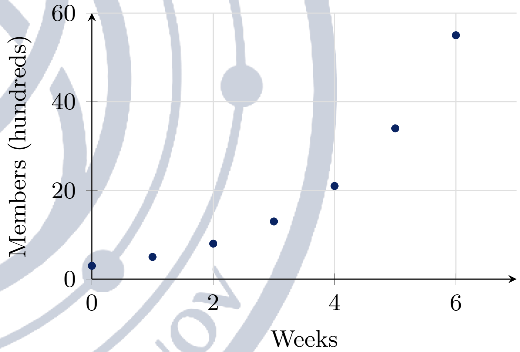
A) 1
B) 2
C) 3
D) 4

21

A biologist restocked a pond and then counted the fish each month. For the scatterplot of x, the number of months since restocking, and y, the number of fish, the line of best fit is $\hat{y} = 34x + 210$. Which of the following statements about this model is true?

A) The predicted number of fish increases by 210 each month, and the pond contained 34 fish at restocking.
B) The predicted number of fish increases by 34 each month, and the pond contained 210 fish at restocking.
C) The predicted number of fish increases by 34 each month, and the pond contained 34 fish at restocking.
D) The predicted number of fish increases by 210 each month, and the pond contained 210 fish at restocking.

22

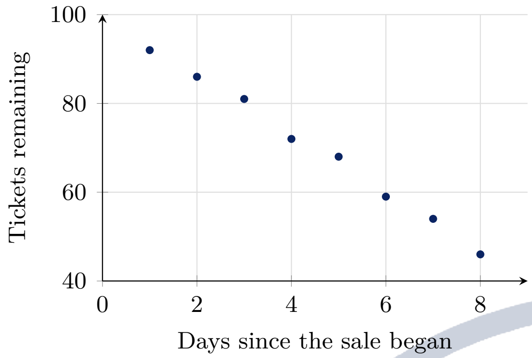


The scatterplot shows the membership of a new club, in hundreds, over seven weeks. Which of the following best describes a model for these data?

A) A linear model, because the points fall close to a straight line.
B) An exponential decay model, because the values decrease at a decreasing rate.
C) An exponential growth model, because the values increase at an increasing rate.
D) A linear model, because the values increase by a constant amount for each additional week.

Module 2

1



The scatterplot shows the number of tickets remaining for a concert on each of eight days after a sale began. Which of the following best describes the association shown?

A) The association is negative and approximately linear.

B) The association is positive and approximately linear.

C) The association is negative and exponential, with the tickets halving each day.

D) There is no association between the day and the number of tickets remaining.

2

A candle's height was measured as it burned. For the scatterplot of x, the number of minutes since the candle was lit, and y, the height of the candle in centimeters, the line of best fit is $\hat{y} = 15 - 0.2x$. Which of the following is the best interpretation of the number 15 in this model?

A) The candle's predicted height decreases by 15 centimeters per minute.

B) The candle's predicted height when it was lit was 0.2 centimeters.

C) The candle's predicted height when it was lit was 15 centimeters.

D) The candle is predicted to burn out after 15 minutes.

3

In a study of 30 precincts, the line of best fit relating x, the number of registered voters in hundreds, to $\hat{y}$, the predicted number of ballots cast in hundreds, is $\hat{y} = 0.8x + 12$. How many hundreds of ballots does the model predict for a precinct with 35 hundred registered voters?

4

For a group of phone customers, the line of best fit relating x, the number of text messages sent in a month, to y, that month's bill in dollars, is $\hat{y} = 0.05x + 30$. Which of the following is the best interpretation of the slope of this line?

A) The predicted bill increases by \$30 for each additional text message.

B) The predicted bill increases by \$0.05 for each additional text message.

C) The predicted bill is \$0.05 when no text messages are sent.

D) For each additional dollar in the bill, 0.05 more text messages are predicted.

5

A line of best fit passes through the points $(4, 30)$ and $(14, 5)$. What is the slope of this line of best fit?

6

Which of the following situations is best modeled by an exponential function?

A) A car's value decreases by 15% of its current value each year.

B) A pool is drained at a constant rate of 12 gallons per minute.

C) A worker earns 18 dollars for each hour worked.

D) A tank gains 5 liters of water every 2 minutes.

7

In a scatterplot with a line of best fit, one data point lies below the line. Which of the following must be true about that data point?

A) The residual for that point is positive, and the model underpredicts the value.

B) The residual for that point is negative, and the model underpredicts the value.

C) The residual for that point is negative, and the model overpredicts the value.

D) The residual for that point is positive, and the model overpredicts the value.

8

The line of best fit for a data set is $\hat{y} = 5.5x + 30$. One data point in the set has $x = 8$ and an actual y-value of 70. What is the residual for that data point?

9

The population of a town is modeled by $P(t) = 4800(0.85)^t$, where t is the number of years since 2015. Which of the following is the best interpretation of this model?

A) The population decreases by 85% each year from an initial value of 4800.

B) The population decreases by 15 people each year from an initial value of 4800.

C) The population increases by 15% each year from an initial value of 4800.

D) The population decreases by 15% each year from an initial value of 4800.

10

A researcher collected data for x between 1 and 12 months and found the line of best fit $\hat{y} = 45x + 120$. She then used the model to predict $\hat{y}$ when $x = 90$. Which of the following is the best reason to doubt that prediction?

A) The slope of the line of best fit is positive, so the model cannot be used to make predictions.

B) A line of best fit can only be used for values of x that appear exactly in the data set.

C) The y-intercept of 120 is too large for the model to be accurate.

D) The value $x = 90$ is far outside the range of the data used to build the model, so the pattern may not continue.

11

The line of best fit for a set of data is $\hat{y} = -0.6x + 45$. For what value of x does this model predict a y-value of 15?

12

For a scatterplot of x, the number of years since 1990, and y, the average price of a movie ticket in dollars, the line of best fit is $\hat{y} = 0.19x + 4.2$. Which of the following is the best interpretation of the slope of this line?

A) The predicted price of a ticket increased by about 19 cents per year.

B) The predicted price of a ticket increased by about 19 dollars per year.

C) The predicted price of a ticket increased by about 4.20 dollars per year.

D) The predicted price of a ticket in 1990 was about 19 cents.

13

A machine purchased for \$25000 loses value according to the model $V(t) = 25000(0.8)^t$, where t is the number of years since purchase. What value does this model predict after 3 years?

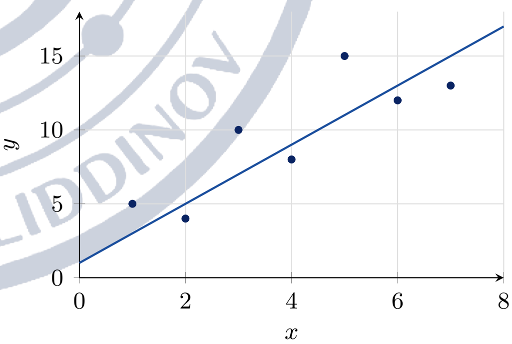
A) \$10000

B) \$12800

C) \$15000

D) \$20000

14



The scatterplot shows seven data points together with a line of best fit. For how many of these data points is the residual positive?

15

On one afternoon a weather station recorded the temperature every hour. For the scatterplot of x, the number of hours after noon, and y, the temperature in degrees Fahrenheit, the line of best fit is $\hat{y} = 2.3x + 68$. Which of the following is the best interpretation of the number 68 in this model?

A) The temperature increased by 68 degrees Fahrenheit each hour.

B) The predicted temperature at noon was 2.3 degrees Fahrenheit.

C) It took 68 hours for the temperature to rise 2.3 degrees Fahrenheit.

D) The predicted temperature at noon was 68 degrees Fahrenheit.

16

t	0	1	2	3
N	6	18	54	162

The table shows the number N of cells in a culture t hours after an experiment began. Which of the following equations best models these data?

A) $N = 3(6)^t$

B) $N = 6t + 3$

C) $N = 6(3)^t$

D) $N = 12t + 6$

17

Day	1	2	3	4
Actual visitors	46	48	71	68
Predicted visitors	40	55	62	80

The table shows the actual number of visitors to a park and the number predicted by a line of best fit on each of four days. Which of the following correctly identifies the day on which the model overpredicts by the greatest amount?

A) On day 3 the model overpredicts by 9 visitors.

B) On day 4 the model overpredicts by 12 visitors.

C) On day 2 the model overpredicts by 7 visitors.

D) On day 1 the model overpredicts by 6 visitors.

18

A study of high school students produced the line of best fit $\hat{y} = 8.4 - 0.006x$, where x is the number of minutes per day a student spends on social media and $\hat{y}$ is the predicted number of hours of sleep that student gets per night. Which of the following is the best interpretation of the number -0.006 in this model?

A) For each additional minute per day on social media, the model predicts 8.4 fewer hours of sleep per night.

B) For each additional hour of sleep per night, the model predicts 0.006 fewer minutes per day on social media.

C) A student who spends no time on social media is predicted to sleep 0.006 hours per night.

D) For each additional minute per day on social media, the model predicts 0.006 fewer hours of sleep per night.

19

For a set of data the line of best fit is $\hat{y} = 7.2x + 15$. According to this model, how much greater is the predicted value of y when $x = 25$ than the predicted value of y when $x = 10$?

20

A linear model is fitted to a data set, and the residuals, listed in order of increasing x, are 6, 2, -3, -5, -2, 3, and 8. Which of the following conclusions is best supported by these residuals?

A) The linear model is appropriate, because the residuals sum to a small number.

B) The linear model is appropriate, because every residual is less than 10 in absolute value.

C) A nonlinear model would fit better, because the residuals move from positive to negative and back in a U-shaped pattern.

D) A nonlinear model would fit better, because every residual is positive.

21

For a data set the line of best fit is $\hat{y} = 3.4x + 12$. One data point in the set has $x = 15$, and its residual is -7. What is the actual y-value of that data point?

A) -7

B) 56

C) 63

D) 70

22

For 40 employees at a firm, the line of best fit relating x, the number of years of experience, to y, the annual salary in thousands of dollars, is $\hat{y} = 2.8x + 41$. Which of the following gives the best interpretation of both the slope and the y-intercept of this model?

A) Each additional year of experience is associated with a predicted increase of \$2800 in salary, and an employee with no experience is predicted to earn \$41000.

B) Each additional year of experience is associated with a predicted increase of \$41000 in salary, and an employee with no experience is predicted to earn \$2800.

C) Each additional year of experience is associated with a predicted increase of \$2.80 in salary, and an employee with no experience is predicted to earn \$41.

D) Each additional \$2800 of salary is associated with one more year of experience, and an employee earning no salary is predicted to have 41 years of experience.

Lesson 15: Probability and Conditional Probability

Every probability question on the Digital SAT is the same fraction: how many outcomes you want, divided by how many outcomes you are choosing from. The whole difficulty of the topic lives in that second number. When a question says "one of the 400 people surveyed is chosen at random" the denominator is 400; when it says "one of the people who oppose the proposal is chosen at random" the denominator is no longer 400 — it is the number of people who oppose. Reading the correct denominator out of a two-way frequency table is the single skill this topic tests, and it is where almost every wrong answer comes from.

Simple probability

If every outcome is equally likely, then

$$P(\text{event}) = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}.$$

A jar holds 5 red, 7 blue and 8 green marbles, so there are $5 + 7 + 8 = 20$ marbles in all and $P(\text{red}) = \frac{5}{20} = \frac{1}{4}$. Notice that the total is built by adding *every* category, including the one you are asking about — writing $\frac{5}{15}$ because you set the red marbles aside is a real and very common slip.

Probabilities are always between 0 and 1. The SAT accepts a fraction or its decimal equivalent in a grid-in box, so $\frac{1}{4}$ and 0.25 are both fine; leave the fraction unreduced only if it fits the answer field.

THE THREE FORMULAS THAT COVER THIS TOPIC

$$P(A) = \frac{n(A)}{n(\text{total})} \qquad P(\text{not } A) = 1 - P(A) \qquad P(A \text{ given } B) = \frac{n(A \text{ and } B)}{n(B)}$$

The first two share the grand total as denominator. The third does not: its denominator is the size of the group you were told to assume, $n(B)$, and its numerator is the overlap of the two categories. For counting how many people are in one category *or* the other, use

$$n(A \text{ or } B) = n(A) + n(B) - n(A \text{ and } B),$$

subtracting the overlap once so it is not counted twice.

Reading a two-way frequency table

A two-way table sorts one group of people by two characteristics at once. The body of the table holds the counts of each combination; the margins hold the row totals, the column totals, and in the bottom-right corner the grand total.

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

Every probability you will be asked for is a ratio of two numbers already printed in (or computable from) this table. The words of the question tell you which two: the numerator is the cell or margin you want, and the denominator is the pool the question draws from.

THREE SENTENCES, THREE DENOMINATORS

Read these three questions about the table above and watch the denominator move.
1. "One of the 100 students is chosen at random. What is the probability the student is a senior in robotics?" — the pool is everybody: $\frac{26}{100} = \frac{13}{50}$.
2. "A student is chosen at random *from the seniors*. What is the probability the student is in robotics?" — the pool is the senior row: $\frac{26}{40} = \frac{13}{20}$.
3. "A student is chosen at random *from the robotics club*. What is the probability the student is a senior?" — the pool is the robotics column: $\frac{26}{50} = \frac{13}{25}$.
Same cell on top all three times. Three different denominators, three different answers.

THE TRAP: THE GRAND TOTAL IS USUALLY THE WRONG DENOMINATOR

The signature error on this topic is reaching for the grand total in a question that has already restricted you to a row or a column. Phrases such as *given that*, *among the*, *of those who*, *a senior is chosen at random* and *if the student plays a sport* all shrink the pool. The moment you meet one, the grand total is out of play — circle the row or column it names and use *that* total.
The second half of the trap is reversing the condition. "The probability that a debate member is a senior" is $\frac{14}{50}$, but "the probability that a senior is a debate member" is $\frac{14}{40}$. Both have 14 on top; the group named right after *given* (or the group you are told the chosen person already belongs to) supplies the denominator, never the group you are asked about.
The third variant: *and* means the overlap cell, *or* means everything in the row plus everything in the column minus that cell. Answering $\frac{50+40}{100}$ for "debate or senior" double-counts the 14 students who are both.

WORKED EXAMPLE 1 — PICKING THE DENOMINATOR

Using the table above: a student in the debate club is selected at random. What is the probability that the student is a junior?
The phrase "a student in the debate club" restricts the pool to the debate column, whose total is 50. Of those 50 students, 36 are juniors, so

$$P(\text{junior} \mid \text{debate}) = \frac{36}{50} = \frac{18}{25}.$$

The two tempting wrong answers are $\frac{36}{100} = \frac{9}{25}$, which ignores the restriction, and $\frac{36}{60} = \frac{3}{5}$, which answers the reversed question "a junior is selected; what is the probability the student is in debate?" Both use the number 36 correctly and the denominator incorrectly.

WORKED EXAMPLE 2 — "HOW MANY MORE MUST BE ADDED"

Using the table above: how many additional students, all of whom join robotics, must enrol so that the probability that a randomly selected student is in robotics becomes $\frac{3}{5}$?
Let x be the number added. Each new student adds 1 to the robotics count *and* 1 to the grand total — forgetting the second is what makes this question hard.

$$\frac{50+x}{100+x} = \frac{3}{5} \implies 5(50+x) = 3(100+x) \implies 250 + 5x = 300 + 3x \implies x = 25.$$

Check: $\frac{75}{125} = \frac{3}{5}$. ✓
If instead the new students were added inside one row only — say x more seniors who all oppose — then only that row's numerator and that row's total move, and the equation is built from the row, not from the grand total.

STRATEGY ON TEST DAY

1. Underline the pool first. Before you look at a single number, decide whether the question draws from everybody, from one row, or from one column, and write that total down as the denominator.
2. Then find the numerator — one cell for an *and* question, one margin for a plain question, row + column − cell for an *or* question.
3. Fill in blanks as you go. A table with a missing cell is asking you to subtract; every row and every column must add to its total.
4. Use the complement when a question says *at least one*, *not*, or *fails to*: $P(\text{not } A) = 1 - P(A)$ is nearly always fewer cells to add.
5. Sanity-check the size. A conditional probability taken inside a small row is often much larger than the overall probability; if your answer is a tiny fraction of the grand total when the question restricted you to 40 people, you used the wrong denominator.

Module 1

1

A bag contains 5 red marbles, 7 blue marbles, and 8 green marbles. If one marble is selected at random from the bag, what is the probability that it is red?

A) $\frac{1}{4}$
B) $\frac{1}{3}$
C) $\frac{2}{5}$
D) $\frac{3}{4}$

2

A basket contains only 12 lemon candies and 18 cherry candies. If one candy is selected at random, what is the probability that it is *not* a lemon candy?

A) $\frac{2}{5}$
B) $\frac{3}{5}$
C) $\frac{2}{3}$
D) $\frac{3}{4}$

3

An integer is selected at random from the integers 1 through 20, inclusive. What is the probability that the integer selected is a multiple of 5?

A) $\frac{1}{20}$
B) $\frac{1}{10}$
C) $\frac{1}{5}$
D) $\frac{1}{4}$

4

A class contains 25 students, 15 of whom are girls. If one student is selected at random from the class, what is the probability that the student is a boy?

5

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. If one of these 100 students is selected at random, what is the probability that the student is in the robotics club?

A) $\frac{2}{5}$
B) $\frac{1}{2}$
C) $\frac{13}{25}$
D) $\frac{13}{20}$

6

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. Given that a randomly selected member is a junior, what is the probability that the member is in the debate club?

A) $\frac{9}{25}$
B) $\frac{2}{5}$
C) $\frac{3}{5}$
D) $\frac{18}{25}$

7

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. If one of these 100 students is selected at random, what is the probability that the student is a junior in the robotics club?

A) $\frac{6}{25}$
B) $\frac{2}{5}$
C) $\frac{12}{25}$
D) $\frac{21}{25}$

8

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. A member of the robotics club is selected at random. What is the probability that this member is a senior?

9

A charity sells 250 raffle tickets and will draw one winning ticket at random. Maria buys 10 of the tickets. What is the probability that Maria's tickets do *not* include the winning ticket?

A) $\frac{1}{25}$
B) $\frac{1}{24}$
C) $\frac{23}{25}$
D) $\frac{24}{25}$

10

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. If one of the 160 students is selected at random, what is the probability that the student walks to school?

A) $\frac{3}{16}$
B) $\frac{5}{16}$
C) $\frac{1}{3}$
D) $\frac{2}{5}$

11

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. Given that a randomly selected student is in grade 10, what is the probability that the student takes the bus?

A) $\frac{7}{32}$
B) $\frac{2}{7}$
C) $\frac{7}{16}$
D) $\frac{1}{2}$

12

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. A student is selected at random from those who take the bus. What is the probability that the student is in grade 9?

13

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. If one of the 160 students is selected at random, what is the probability that the student does *not* travel by car?

A) $\frac{3}{16}$
B) $\frac{3}{14}$
C) $\frac{5}{16}$
D) $\frac{13}{16}$

14

A jar contains only red marbles and green marbles. The probability that a marble selected at random is red is $\frac{3}{8}$, and the jar contains 15 red marbles. How many green marbles are in the jar?

A) 24
B) 25
C) 40
D) 64

15

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. If one of the 160 students is selected at random, what is the probability that the student is in grade 10 and walks to school?

A) $\frac{1}{8}$
B) $\frac{2}{7}$
C) $\frac{5}{16}$
D) $\frac{2}{5}$

16

A bag contains 8 red marbles and 12 blue marbles. How many additional red marbles must be added to the bag so that the probability of selecting a red marble at random becomes $\frac{1}{2}$?

17

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. A member of the debate club is selected at random. What is the probability that this member is a junior?

A) $\frac{18}{25}$
B) $\frac{3}{5}$
C) $\frac{9}{25}$
D) $\frac{7}{25}$

18

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. If one of these 100 students is selected at random, what is the probability that the student is in the debate club or is a senior?

A) $\frac{7}{50}$
B) $\frac{31}{50}$
C) $\frac{19}{25}$
D) $\frac{9}{10}$

19

	Debate	Robotics	Total
Juniors	36	24	60
Seniors	14	26	40
Total	50	50	100

The table shows the 100 members of a school's two after-school clubs, classified by grade. How many additional students, all of whom join the robotics club, must enrol so that the probability that a randomly selected member of the two clubs is in the robotics club becomes $\frac{3}{5}$?

A) 10
B) 15
C) 20
D) 25

20

A store's display holds 30 shirts, 6 of which are blue. How many additional blue shirts must be added to the display so that the probability that a randomly selected shirt from the display is blue becomes $\frac{1}{4}$?

21

	Bus	Walk	Car	Total
Grade 9	45	30	15	90
Grade 10	35	20	15	70
Total	80	50	30	160

The table shows how 160 students at a school travel to school, by grade. Given that a randomly selected student is in grade 9, what is the probability that the student does *not* take the bus?

A) $\frac{9}{32}$
B) $\frac{1}{3}$
C) $\frac{1}{2}$
D) $\frac{9}{16}$

22

At a gathering of 60 people, 35 like tea, 30 like coffee, and 12 like both tea and coffee. If one of the 60 people is selected at random, what is the probability that the person likes neither tea nor coffee?

A) $\frac{1}{12}$
B) $\frac{7}{60}$
C) $\frac{1}{5}$
D) $\frac{53}{60}$

Module 2

1

A spinner is divided into 8 sections of equal area, 3 of which are shaded. If the spinner is spun once and lands on a random section, what is the probability that it lands on a shaded section?

A) $\frac{1}{8}$
B) $\frac{3}{8}$
C) $\frac{3}{5}$
D) $\frac{5}{8}$

2

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. If one of the 400 residents is selected at random, what is the probability that the resident supports the proposal?

A) $\frac{1}{2}$
B) $\frac{3}{5}$
C) $\frac{2}{3}$
D) $\frac{3}{4}$

3

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. Given that a randomly selected resident is under 40, what is the probability that the resident supports the proposal?

A) $\frac{3}{10}$
B) $\frac{1}{2}$
C) $\frac{3}{5}$
D) $\frac{3}{4}$

4

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. If one of the 400 residents is selected at random, what is the probability that the resident is 40 or older and opposes the proposal?

5

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. A resident who supports the proposal is selected at random. What is the probability that this resident is under 40?

A) $\frac{4}{5}$
B) $\frac{3}{4}$
C) $\frac{3}{5}$
D) $\frac{3}{10}$

6

Each of 12 cards is labelled with a different integer from 1 through 12. If one card is selected at random, what is the probability that the number on the card is prime?

A) $\frac{1}{4}$
B) $\frac{1}{3}$
C) $\frac{5}{12}$
D) $\frac{1}{2}$

7

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines. The table shows the results. If one of the 500 items is selected at random, what is the probability that it is defective?

A) $\frac{1}{10}$
B) $\frac{7}{40}$
C) $\frac{3}{10}$
D) $\frac{9}{10}$

8

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. An item produced by machine B is selected at random. What is the probability that the item is defective?

9

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. A defective item is selected at random. What is the probability that it was produced by machine A?

A) $\frac{3}{10}$
B) $\frac{1}{10}$
C) $\frac{1}{20}$
D) $\frac{3}{100}$

10

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. Given that a randomly selected item passed inspection, what is the probability that it was produced by machine B?

A) $\frac{33}{100}$
B) $\frac{11}{30}$
C) $\frac{19}{30}$
D) $\frac{33}{40}$

11

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. A resident who opposes the proposal is selected at random. What is the probability that this resident is 40 or older?

A) $\frac{2}{5}$
B) $\frac{3}{5}$
C) $\frac{2}{3}$
D) $\frac{4}{5}$

12

A bag contains 60 marbles. The probability that a marble selected at random from the bag is green is $\frac{1}{5}$. How many of the marbles are *not* green?

13

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. An item produced by machine A is selected at random. What is the probability that the item passed inspection?

A) $\frac{11}{30}$

B) $\frac{57}{100}$

C) $\frac{9}{10}$

D) $\frac{19}{20}$

14

An integer is selected at random from the integers 1 through 30, inclusive. What is the probability that the integer selected is *not* a multiple of 3?

A) $\frac{1}{3}$

B) $\frac{1}{2}$

C) $\frac{2}{3}$

D) $\frac{3}{4}$

15

	Yes	No	Total
Male		30	70
Female	20	30	50
Total		60	120

The incomplete table shows the results of a survey of 120 people who were asked whether they support a change to the bus timetable. Given that a randomly selected respondent answered Yes, what is the probability that the respondent is female?

A) $\frac{1}{6}$

B) $\frac{1}{3}$

C) $\frac{2}{5}$

D) $\frac{2}{3}$

16

	Passed	Defective	Total
Machine A	285	15	300
Machine B	165	35	200
Total	450	50	500

A factory inspected the 500 items produced in one shift by its two machines, with the results shown. The next day the factory produces some additional items, all of which are defective, and adds them to this group. How many additional defective items would make the probability that a randomly selected item from the enlarged group is defective equal to $\frac{1}{6}$?

17

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. How many additional residents under 40, all of whom support the proposal, would have to be surveyed so that $\frac{5}{9}$ of all the residents surveyed support the proposal?

A) 40

B) 50

C) 60

D) 100

18

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. Of the residents who do *not* support the proposal, what fraction are under 40?

A) $\frac{4}{5}$

B) $\frac{1}{4}$

C) $\frac{1}{5}$

D) $\frac{1}{10}$

19

A bag contains 24 marbles, 6 of which are red. How many red marbles must be added to the bag so that the probability of selecting a red marble at random is $\frac{1}{2}$?

20

In a school of 120 students, 70 play a sport, 45 play a musical instrument, and 25 do both. If one of the 120 students is selected at random, what is the probability that the student does neither?

A) $\frac{3}{4}$
B) $\frac{3}{8}$
C) $\frac{1}{4}$
D) $\frac{5}{24}$

21

A distributor receives lamps from two plants: 60% come from plant X and the rest from plant Y. Of the lamps from plant X, 5% are defective, and of the lamps from plant Y, 10% are defective. If one lamp is selected at random from the distributor's stock, what is the probability that it is defective?

A) $\frac{3}{100}$
B) $\frac{7}{100}$
C) $\frac{3}{40}$
D) $\frac{3}{20}$

22

	Support	Oppose	Total
Under 40	120	40	160
40 or older	80	160	240
Total	200	200	400

The table summarises the responses of 400 residents asked whether they support a proposed bicycle lane, by age group. How many additional residents aged 40 or older, all of whom oppose the proposal, would have to be surveyed so that the probability that a randomly selected respondent aged 40 or older opposes the proposal is $\frac{3}{4}$?

Lesson 16: Inference, Margin of Error and Evaluating Statistical Claims

Every question in this topic starts from the same situation: someone measured a small group and wants to say something about a much larger one. The SAT asks three things about that leap. How do you scale a sample statistic up to a population? What does the margin of error let you claim, and what does it not let you claim? And what does the design of the study — who was chosen, and who was assigned — permit you to conclude? The arithmetic here is easy on purpose; almost all of the difficulty is in reading a claim carefully enough to see whether the study actually supports it.

From a sample to a population

A **population** is the entire group you want to describe. A **sample** is the part of it you actually measure. A number computed from the sample (a sample mean, a sample proportion) is a **statistic**; the matching number for the whole population (the population mean, the population proportion) is a **parameter**. The statistic is known and the parameter is not — estimating the parameter from the statistic is the whole game.

The estimate itself is nothing more than a proportion applied to a bigger group. If 84 of 240 sampled trees are maples, the sample proportion is $\frac{84}{240} = 0.35$, so in a park of 8,000 trees the estimated number of maples is $0.35 \times 8000 = 2800$. Two habits will keep this clean: reduce the sample proportion to a decimal first, and check what the question wants back — a proportion, a percent, or a count.

THE THREE FORMULAS

Estimating a population count.

$$\text{estimated count} = \frac{\text{count in sample}}{\text{sample size}} \times \text{population size}$$

The plausible range. If an estimate is reported with a margin of error E,

$$\text{estimate} - E \ \le\ \text{parameter} \ \le\ \text{estimate} + E$$

so the interval is $[\,\text{estimate} - E,\ \text{estimate} + E\,]$.

Reading an interval backwards. Given endpoints a and b,

$$\text{estimate} = \frac{a+b}{2}, \qquad E = \frac{b-a}{2}.$$

The estimate is the midpoint and the margin of error is half the width. Any question that hands you an interval is asking for one of these two.

WHAT MOVES THE MARGIN OF ERROR

Only two things on this test change the margin of error, and they move it in opposite directions.

Sample size. A larger random sample means a smaller margin of error. The relationship is $E \propto \frac{1}{\sqrt{n}}$: to cut the margin of error in half you need four times the sample; to divide it by 3 you need nine times the sample.

Confidence level. A higher confidence level means a *larger* margin of error, hence a wider interval. Demanding to be right more often, from the same data, forces you to cast a wider net.

What does *not* change the margin of error: the size of the population, and the value of the estimate itself. A poll of 1,000 people is about as precise in a city of fifty thousand as in a country of fifty million.

WHAT A 95% CONFIDENCE INTERVAL DOES NOT SAY

Suppose a 95% confidence interval for a mean is 48 to 62. All four of these are wrong:

- "There is a 95% probability that the population mean is between 48 and 62." The population mean is a fixed number, not a random one. It is either inside this interval or it is not.
- "95% of the population lies between 48 and 62." The interval estimates *one number*, the mean — not the spread of individuals.
- "95% of the sample lies between 48 and 62." Same error, aimed at the sample.
- "The population mean must be 55." No single value is established; the interval offers a range of plausible values.

The correct reading is a statement about the *procedure*: if you drew many random samples and built an interval from each one the same way, about 95% of those intervals would contain the population mean. And the usable consequence on test day is simply this: every value inside the interval is plausible, every value outside it is not.

RANDOM SELECTION VERSUS RANDOM ASSIGNMENT

These two phrases sound alike and do completely different jobs. Keep them in separate boxes.
Random selection is how participants get *into* the study. If subjects were randomly selected from a population, the results generalize to that population — and to no wider one. A random sample of one university describes that university, not every university.
Random assignment is how participants, once in the study, get sorted into *treatment groups*. If the researcher randomly assigned who receives what, the groups start out alike, so a difference at the end is attributed to the treatment. This — and only this — licenses the word *causes*.
So a survey, in which nobody is assigned anything, can never establish cause, no matter how large or how carefully random it is. And an experiment on 300 volunteers who answered an advertisement can establish cause for those volunteers, but cannot be generalized beyond them.

HOW TO ATTACK THE VERBAL ITEMS

The conclusion questions are all the same question in different clothing. Read the stem and answer two yes/no questions before you look at the choices:

1. **Were subjects randomly selected from a population?** If yes, you may generalize to that population. If no (volunteers, self-selected respondents, one convenient location), the conclusion is confined to the participants.
2. **Were subjects randomly assigned to groups?** If yes, cause is allowed. If no, only *association* is allowed — words like "is associated with," "tend to," "related to."

Then strike out every choice that claims more than your two answers permit. The wrong choices are almost always over-claims: cause from a survey, a whole country from one school, an exact value from an estimate. The right choice is usually the most cautious one that still says something.

WORKED EXAMPLE 1 — ESTIMATE, INTERVAL, AND COUNT

A random sample of 500 of the 6,400 employees at a company was surveyed. Based on the sample, the proportion of all employees who bike to work is estimated to be 0.36, with a margin of error of 0.04. What is the greatest number of employees who bike to work that is consistent with this estimate?

Step 1 — build the plausible range for the proportion. The estimate is 0.36 and the margin of error is 0.04, so plausible proportions run from $0.36 - 0.04 = 0.32$ to $0.36 + 0.04 = 0.40$.
Step 2 — take the end the question asks for. "Greatest number" means the largest proportion, 0.40.
Step 3 — scale up to the population. $0.40 \times 6400 = 2560$ employees.
Why this order. The margin of error attaches to the *proportion*, so widen first and multiply second. Notice also that the sample size 500 never appears in the arithmetic — it has already done its work by determining how large the margin of error is. A number in the stem that is not needed is normal on this topic.

WORKED EXAMPLE 2 — READING A STUDY'S DESIGN

A researcher selected a random sample of 250 students from the 3,000 students at one university, then recorded whether each student ate breakfast and how much energy the student reported. Students in the sample who ate breakfast reported higher energy levels. Which conclusion is supported?
Step 1 — random selection? Yes: 250 students drawn at random from all 3,000 at this university. So the finding may be generalized to the students at *this university* — not to university students in general.
Step 2 — random assignment? No. The researcher recorded what students already did; nobody was assigned to eat breakfast. Students who eat breakfast may also sleep more, live on campus, or have earlier classes, and any of those could explain the energy difference. So no causal claim is available.
Step 3 — combine. Generalize, but only to an association: *among the students at this university, eating breakfast is associated with higher reported energy levels.*
The fix. If the researcher wanted to say breakfast *causes* higher energy, the design would have to change: randomly assign students to eat or skip breakfast, then compare. Collecting more survey data would never be enough.

Module 1

1

A researcher selected a random sample of 200 of the 1,500 students at a high school and found that 60 of the students in the sample walk to school. Based on this sample, which of the following is the best estimate of the total number of students at the school who walk to school?

A) 60
B) 300
C) 450
D) 750

2

A poll of a random sample of voters estimated that 42% of all voters in a city support a new transit plan, with an associated margin of error of 3%. What is the least percent of all voters that is consistent with this estimate? (Ignore the percent sign when entering your answer.)

3

A city library has 2,000 registered members. A librarian surveyed 150 members selected at random from the list of all registered members. Which of the following is the population that this survey is designed to describe?

A) The 150 members who were surveyed.
B) All 2,000 registered members of the library.
C) All residents of the city.
D) All people who visited the library on the day of the survey.

4

A researcher will use a random sample to estimate the mean number of hours per week that adults in a town exercise. If the researcher increases the size of the random sample but keeps the confidence level the same, what is the most likely effect on the margin of error?

A) It will decrease.

B) It will increase.

C) It will stay exactly the same.

D) It will become negative.

5

In a random sample of 80 of the 1,200 households in a town, 28 households own a pet. Based on this sample, which of the following is the best estimate of the number of households in the town that own a pet?

A) 336

B) 420

C) 480

D) 560

6

Based on a random sample, a biologist estimates that the mean mass of a certain species of frog is between 12.4 grams and 13.6 grams. What is the margin of error associated with this estimate?

A) 0.6

B) 1.2

C) 12.4

D) 13.0

7

A store manager wants to estimate the mean number of minutes that all of the store's customers wait in line. Which of the following sampling methods is most likely to produce a biased estimate?

A) Selecting a random sample of customers from a list of all purchases made during the week.

B) Selecting a random sample of customers from each day of the week.

C) Selecting only customers who shop during the single busiest hour of the week.

D) Selecting a random sample of customers throughout the entire week.

8

A survey of a random sample of the students at a college found that the mean number of hours per week that the students in the sample study is 15.2 hours, with an associated margin of error of 1.3 hours. Which of the following is the most appropriate conclusion?

A) Every student at the college studies between 13.9 and 16.5 hours per week.

B) Exactly 95 percent of the students at the college study between 13.9 and 16.5 hours per week.

C) The mean number of hours per week that all students at the college study is exactly 15.2 hours.

D) It is plausible that the mean number of hours per week that all students at the college study is between 13.9 and 16.5 hours.

9

A random sample of 250 adults in a city was asked how many minutes they commute to work each day. The mean for the sample was 27.5 minutes, with an associated margin of error of 1.2 minutes. What is the greatest number of minutes that is consistent with this estimate?

10

A swim club at one gym has 400 members. All 400 names were listed, 50 members were selected at random, and each was asked about a proposed change to the pool schedule. Of those surveyed, 34 supported the change. Which of the following conclusions is best supported by these results?

A) Most members of the swim club at this gym likely support the change.

B) Most people who swim in the city likely support the change.

C) The proposed schedule change causes members to swim more often.

D) Exactly 34 of the 400 members of the swim club support the change.

11

A random sample of 320 of the 12,000 registered voters in a county was surveyed, and 208 of those surveyed said they plan to vote in an upcoming election. Based on this sample, which of the following is the best estimate of the number of registered voters in the county who plan to vote?

A) 6 000

B) 7 800

C) 8 400

D) 9 600

12

Two polls were conducted to estimate the percent of a city's residents who support a bond measure. Both polls used a random sample of residents and the same confidence level, but Poll A surveyed 400 residents and Poll B surveyed 1,600 residents. Which of the following statements about the margins of error is most likely true?

A) Poll A has the smaller margin of error, because it surveyed fewer residents.

B) The two polls must have the same margin of error, because they used the same confidence level.

C) The margins of error cannot be compared without knowing the results of the two polls.

D) Poll B has the smaller margin of error, because it surveyed more residents.

13

Based on a random sample of the trees in a forest, a plausible range for the proportion of all trees in the forest that are oak trees is 0.38 to 0.46. What is the proportion of oak trees estimated from the sample?

14

A researcher recruited 200 volunteers who experience frequent headaches and randomly assigned each volunteer to take either a new medication or a placebo. After eight weeks, the volunteers taking the new medication reported significantly fewer headaches than those taking the placebo. Which of the following conclusions is best supported?

A) The new medication caused a reduction in headaches for the volunteers in the study.

B) The new medication would reduce headaches for all adults who experience frequent headaches.

C) Adults who take medication are more likely to experience frequent headaches.

D) No conclusion about cause can be drawn, because the volunteers were not selected at random.

15

Based on a random sample of a population of fish, a 95% confidence interval for the mean weight of the fish in the population is 2.1 kilograms to 2.7 kilograms. Which of the following statements is best supported by this interval?

A) Ninety-five percent of the fish in the population weigh between 2.1 kilograms and 2.7 kilograms.

B) Ninety-five percent of the fish in the sample weigh between 2.1 kilograms and 2.7 kilograms.

C) It is plausible that the mean weight of all the fish in the population is 2.5 kilograms.

D) The mean weight of all the fish in the population must be 2.4 kilograms.

16

A nutritionist selected a random sample of 250 students from the 3,000 students at one university and recorded each student's breakfast habits and reported energy level. Students in the sample who ate breakfast reported higher energy levels than students who did not. Which of the following conclusions is best supported by this study?

A) Eating breakfast causes higher energy levels in students at this university.

B) Eating breakfast causes higher energy levels in university students throughout the country.

C) Students at this university who want more energy must eat breakfast.

D) Among the students at this university, eating breakfast is associated with higher reported energy levels.

17

A random sample of 500 of the 6,400 employees at a company was surveyed. Based on the sample, the proportion of all employees at the company who bike to work is estimated to be 0.36, with an associated margin of error of 0.04. What is the greatest number of employees at the company who bike to work that is consistent with this estimate?

18

From one random sample of batteries, a statistician computes both a 90% confidence interval and a 99% confidence interval for the mean lifetime of the batteries produced at a factory. Which of the following correctly compares the two intervals?

A) The 99 percent interval is wider, because a higher confidence level requires a larger margin of error.

B) The 90 percent interval is wider, because a lower confidence level requires a larger margin of error.

C) The two intervals have the same width, because they are computed from the same sample.

D) The 99 percent interval is narrower, because a higher confidence level is more precise.

19

A magazine printed a survey in one issue and asked readers to mail in their responses. Of the 3,000 readers who responded, 82% said they were dissatisfied with a new city policy. Which of the following is the best reason that this result may be a poor estimate of the percent of all city residents who are dissatisfied with the policy?

A) The sample of 3,000 is too small for a city-wide estimate.

B) A percent cannot be estimated unless a mean is also reported.

C) The survey asked about only one policy instead of several.

D) The readers who chose to respond were not randomly selected, so people with strong opinions are likely overrepresented.

20

Based on a random sample of the bags of chips produced at a factory, a 95% confidence interval for the mean mass, in grams, of all bags produced at the factory is 142.9 to 148.3. What is the margin of error for this estimate?

21

Based on a random sample of the students at a school, a 95% confidence interval for the mean number of text messages sent per day by the students at the school is 48 to 62. Which of the following best describes what the confidence level of 95% means?

A) There is a 95 percent chance that the mean for all students at the school is between 48 and 62.

B) If this sampling procedure were repeated many times, about 95 percent of the intervals produced would contain the mean for all students at the school.

C) About 95 percent of the students at the school send between 48 and 62 text messages per day.

D) About 95 percent of the students in the sample send between 48 and 62 text messages per day.

22

A survey of a random sample of adults in a country found that the adults who own a dog walk more minutes per week, on average, than the adults who do not own a dog. A reporter concluded that owning a dog causes adults to walk more. Which of the following changes to the study would be needed to support the reporter's conclusion?

A) Surveying a larger random sample of adults in the country.

B) Reporting a margin of error along with each estimate.

C) Randomly assigning adults to own a dog or not to own a dog and then comparing the two groups.

D) Surveying only adults who already walk regularly.

Module 2

1

A random sample of 150 of the 900 books in a school library was examined, and 45 of the books in the sample were published before 2010. Based on this sample, which of the following is the best estimate of the number of books in the library that were published before 2010?

A) 135
B) 225
C) 270
D) 300

2

Based on a random sample of its residents, a city estimates that the proportion of all residents who recycle is 0.58, with an associated margin of error of 0.03. What is the least proportion of all residents who recycle that is consistent with this estimate?

3

Two independent random samples were taken from the same population to estimate the mean daily rainfall, in millimeters, using the same confidence level. Study A used 100 days and Study B used 900 days. The margin of error reported by Study B was smaller than the one reported by Study A. Which of the following best explains why?

A) A larger random sample gives more information about the population, which reduces the margin of error.
B) A larger random sample always produces a larger mean, which reduces the margin of error.
C) Study B must have used a higher confidence level than Study A.
D) Study B measured rainfall on more days, so its measurements must be more accurate individually.

4

A random sample of 60 employees at one branch of a company was surveyed about the branch's parking policy. To which of the following groups can the results of the survey be generalized?

A) All employees at that branch of the company.
B) All employees at every branch of the company.
C) All people who work in the city where the branch is located.
D) Only the 60 employees who were surveyed.

5

In a random sample of 300 of the 5,500 students at a university, 24 students said that they own a car. Based on this sample, what is the best estimate of the number of students at the university who own a car?

6

Based on a random sample of the students at a school, the mean number of hours of sleep per night for all students at the school is estimated to be 7.4 hours, with an associated margin of error of 0.5 hours. Which of the following is a plausible value for the mean number of hours of sleep per night for all students at the school?

A) 6.8
B) 7.1
C) 8.0
D) 8.4

7

A researcher wants to estimate the mean number of hours per week that all residents of a town spend exercising. Which of the following samples is most likely to produce an estimate that is too high?

A) A random sample of residents chosen from the town census list.
B) A random sample of residents chosen from a list of all households in the town.
C) A random sample of residents chosen from a list of all registered voters in the town.
D) A sample of residents chosen as they leave a fitness center on a weekday evening.

8

Researchers followed a random sample of 1,200 adults for one year and recorded how much coffee each adult drank and how many hours each adult slept. Adults who drank more coffee reported fewer hours of sleep. The researchers did not assign any adult to drink a particular amount of coffee. Which of the following conclusions is best supported?

A) Drinking coffee causes adults to sleep fewer hours.

B) Sleeping fewer hours causes adults to drink more coffee.

C) There is an association between coffee consumption and hours of sleep among adults, but the study cannot establish a cause.

D) The study shows that coffee consumption and hours of sleep are unrelated.

9

Based on a random sample of the packages shipped by a company, a plausible range for the mean weight, in pounds, of all packages shipped by the company is 21.4 to 24.8. What is the mean weight, in pounds, estimated from the sample?

10

A researcher randomly selected 100 of the 2,500 members of a professional organization and asked each selected member whether they attended the organization's annual conference. Which of the following is an appropriate use of the results?

A) Estimating the proportion of all professionals in this field who attended the conference.

B) Concluding that attending the conference causes members to be more satisfied with their jobs.

C) Determining the exact number of members of the organization who attended the conference.

D) Estimating the proportion of all 2,500 members of the organization who attended the conference.

11

A pollster used a random sample of voters to compute a 90% confidence interval for the percent of all voters who support a measure. Using the same sample, the pollster then computes a 99% confidence interval instead. Compared with the first interval, the second interval will be described by which of the following?

A) It is wider, and the estimate at its center is the same.

B) It is narrower, and the estimate at its center is the same.

C) It is wider, and the estimate at its center is larger.

D) It is unchanged, because the sample did not change.

12

In a random sample of 240 of the 8,000 trees in a park, 84 of the trees are maple trees. Based on this sample, which of the following is the best estimate of the number of maple trees in the park?

A) 1 960

B) 2 400

C) 2 600

D) 2 800

13

Based on a random sample of a store's customers, a 95% confidence interval for the mean number of minutes per visit that all of the store's customers spend in the store is 28.6 to 31.2. Which of the following values of the mean for all customers is NOT plausible based on this interval?

A) 28.9

B) 30.0

C) 31.1

D) 32.4

14

A school emailed a survey to all 1,000 of its students, and 120 students responded. Of the students who responded, 90% said they were satisfied with the cafeteria. Which of the following is the best criticism of using this result to estimate the percent of all students at the school who are satisfied with the cafeteria?

A) The sample of 120 students is a random sample, so the estimate applies only to those 120 students.

B) The students who chose to respond may differ systematically from the students who did not respond.

C) The survey should have been emailed to fewer students so that more would respond.

D) A percent cannot be computed from only 120 responses.

15

Based on a random sample of the households in a town, the mean number of hours per day that a television is on in the households of the town is estimated to be 3.6 hours, with an associated margin of error of 0.45 hours. What is the greatest number of hours that is consistent with this estimate?

16

A city randomly selected 500 of its residents and surveyed them about how they commute and how much stress they experience. Residents who reported commuting by bicycle also reported lower stress levels. Which of the following is the strongest conclusion justified by this survey?

A) Commuting by bicycle lowers stress for the residents of the city.

B) Lower stress causes residents of the city to commute by bicycle.

C) Because the sample was randomly selected, the association between bicycle commuting and lower stress can be generalized to the city's residents, but no cause-and-effect relationship is established.

D) Because the sample was randomly selected, the survey establishes that bicycle commuting causes lower stress for the city's residents.

17

Study 1 surveyed a random sample of 800 adults selected from a national list and found an association between hours of screen time and reported sleep quality. Study 2 recruited 200 volunteers and randomly assigned each volunteer to one of two screen-time schedules, then compared the groups' reported sleep quality. Which of the following correctly describes what the two studies can support?

A) Study 1 can establish cause and effect for adults nationally, and Study 2 cannot establish cause and effect at all.

B) Study 2 can establish cause and effect for its volunteers, and Study 1 can generalize an association to adults nationally.

C) Both studies can establish cause and effect for adults nationally.

D) Neither study can support any conclusion about anyone other than the people who took part.

18

A random sample of 400 of the 4,500 students at a college was surveyed. Based on the sample, the proportion of all students at the college who live on campus is estimated to be 0.42, with an associated margin of error of 0.03. What is the greatest number of students at the college who live on campus that is consistent with this estimate?

19

A survey of a random sample of 400 people produced a margin of error of 4 percentage points at a 95% confidence level. The margin of error is inversely proportional to the square root of the sample size. Approximately how many people would need to be surveyed, at the same confidence level, to reduce the margin of error to 2 percentage points?

A) 800

B) 1 200

C) 1 600

D) 3 200

20

A polling organization reports, "Based on our random sample, we are 95% confident that between 46% and 52% of all voters support the measure." Which of the following is the most accurate interpretation of this report?

A) There is a 95 percent probability that the percent of all voters who support the measure is between 46 and 52.

B) Between 46 percent and 52 percent of the voters in the sample support the measure.

C) The interval was produced by a method that, used repeatedly, captures the percent of all voters who support the measure about 95 percent of the time.

D) If 95 more random samples were taken, every one of them would give a percent between 46 and 52.

21

A study using a random sample of 900 adults reported a margin of error of 2.4 percentage points. The margin of error is inversely proportional to the square root of the sample size. How many adults must be included in a random sample, at the same confidence level, for the margin of error to be 0.8 percentage points?

22

Researchers recruited 300 volunteers who responded to an online advertisement and randomly assigned each volunteer to one of two exercise programs. After twelve weeks, the volunteers in Program A had significantly greater improvement in endurance than those in Program B. Which of the following conclusions is best supported?

A) Program A causes greater improvement in endurance than Program B for all adults.

B) Program A caused greater improvement in endurance for the volunteers in this study, but the result may not extend to all adults.

C) No conclusion about cause can be drawn, because the volunteers were not randomly selected.

D) Adults who want to improve their endurance are more likely to choose Program A.

PART 4

Geometry and Trigonometry

Lesson 17: Area and Volume

Every formula this topic needs is printed on the reference sheet the Digital SAT gives you. That is exactly why the questions are never "do you remember the formula?" They are "which quantity in this sentence is the radius, which is the height, and what am I actually being asked for?" The three things the test really measures are: picking the right relationship, running it backwards when the area or volume is the number you are given, and knowing what happens to area and volume when every length in a figure is multiplied by the same factor.

Plane figures: area and perimeter

Perimeter is a length: add up the outside edges, and it is measured in inches, cm, feet. Area is a covering: it is measured in *square* units. A question that asks for fencing, trim, ribbon, or the distance around a track wants perimeter; a question that asks for carpet, paint, sod, or glass wants area. Reading which one is being asked for is worth more points on this topic than any algebra.

THE PLANE-FIGURE FORMULAS

Rectangle: $A = \ell w$, $P = 2\ell + 2w$.
Triangle: $A = \frac{1}{2}bh$.
Parallelogram: $A = bh$.
Trapezoid: $A = \frac{1}{2}(b_1 + b_2)h$.
Circle of radius r: $A = \pi r^2$, $C = 2\pi r = \pi d$.
In every one of these, h means the *perpendicular* height, not the length of a slanted side.

ONE IDEA BEHIND FOUR FORMULAS

A parallelogram is a rectangle with a triangle slid from one end to the other, so its area is still bh. A triangle is half of a parallelogram, so it is $\frac{1}{2}bh$. A trapezoid is a parallelogram whose base is the *average* of the two parallel sides, so it is $\frac{b_1+b_2}{2} \cdot h$. If you can see the figure as base times perpendicular height, you can rebuild any of them.

Solids: volume and surface area

Volume is how much a solid holds, in cubic units. Surface area is how much material wraps around it, in square units. For any prism or cylinder, volume is (area of the base) × (height) — that single sentence covers boxes, triangular prisms and cylinders at once. Cones and pyramids are the same base and height, taken one third of the way: exactly one third of the prism or cylinder that would enclose them.

THE SOLIDS ON THE REFERENCE SHEET

Rectangular box: $V = \ell wh$, $SA = 2(\ell w + \ell h + wh)$.
Prism or cylinder: $V = Bh$; cylinder $V = \pi r^2 h$, lateral area $2\pi rh$, total surface area $2\pi r^2 + 2\pi rh$.
Cone: $V = \frac{1}{3}\pi r^2 h$. Pyramid: $V = \frac{1}{3}Bh$.
Sphere: $V = \frac{4}{3}\pi r^3$, $SA = 4\pi r^2$.

RADIUS, DIAMETER, AND SLANT HEIGHT

Three misreadings account for most lost points here. (1) A problem that hands you a *diameter* of 16 and a formula that wants r: using 16 as the radius multiplies a circle's area by 4. (2) The *slant height* of a cone or pyramid is the distance up the sloped surface, not the vertical height; you must use the Pythagorean theorem to get the real height first. (3) Dropping the $\frac{1}{3}$ in a cone or pyramid triples the answer, and the tripled value is always sitting there among the choices.

WORKING BACKWARDS FROM AN AREA OR A VOLUME

When the question gives you the area or volume and asks for a length, do not hunt for a different formula. Write the same formula, put the unknown in as a letter, set it equal to the number you were given, and solve. *A cylinder holds* 100π *and is* 4 *tall*: write $\pi r^2(4) = 100\pi$, divide by 4π to get $r^2 = 25$, so $r = 5$. Then — and this is where points are lost — reread the question, because it often wants something built from that length, such as the surface area, not the length itself.

WORKED EXAMPLE 1: A COMPOSITE FIGURE

A running-track infield is a 20 m by 10 m rectangle with a semicircle attached to each of the two short ends. What is its area?

Cut it into pieces you have formulas for. The rectangle contributes $20 \cdot 10 = 200$ square metres. Each short end is 10 m long, so each semicircle has *diameter* 10 and therefore radius 5 — this is the step the test is testing. The two semicircles together make one full circle of radius 5, area $\pi(5)^2 = 25\pi$. Total: $200 + 25\pi$ square metres. Note what would have happened had you called the radius 10: you would get $200 + 100\pi$, which is one of the choices every time.

Scaling: the k, k^2, k^3 law

Suppose every length in a figure is multiplied by the same factor k. Then every length — side, perimeter, radius, circumference, diagonal — is multiplied by k; every area — the figure's area, a solid's surface area — is multiplied by k^2; and every volume is multiplied by k^3. This one sentence is the source of most hard questions in this topic.

WHY THE EXPONENTS ARE 1, 2 AND 3

Look at the formulas. A rectangle's area is ℓw: scale both lengths by k and you get $(k\ell)(kw) = k^2\ell w$. A circle's area is πr^2: replacing r with kr gives $\pi k^2 r^2$. A box's volume is ℓwh: three lengths scaled means k^3. Every area formula multiplies exactly two lengths and every volume formula exactly three, so the exponent counts the dimensions.

SCALING BY k WHEN YOU SHOULD SCALE BY k^2

A photo's dimensions are each multiplied by 3*; its area was* 12 *square inches, so the new area is* 36*.* That is the single most common wrong answer on the test, and it is wrong: the new area is $12 \cdot 3^2 = 108$. Run the law backwards too. If a volume grows by a factor of 27, the lengths grew by $\sqrt[3]{27} = 3$, and so the surface area grew by $3^2 = 9$ — not by 27.

WORKED EXAMPLE 2: SCALING IN REVERSE

Two similar cones have volumes 27 cm^3 and 125 cm^3. The smaller cone's base radius is 6 cm. What is the larger cone's base radius?

Similar means one is a scale copy of the other, so some factor k multiplies every length. Volumes are in the ratio k^3, and $\frac{125}{27} = k^3$, so $k = \sqrt[3]{125/27} = \frac{5}{3}$. Radii are lengths, so they scale by k, not by k^3: the larger radius is $6 \cdot \frac{5}{3} = 10$ cm. Cube-rooting the volume ratio first, then applying the plain factor to a length, is the whole method.

A FOUR-LINE CHECKLIST

1. Circle the words *perimeter*, *area*, *volume*, *surface area* in the question, and check the units in the answer choices — square or cubic. 2. Label the figure with every number in the text and ask of each: is that a radius or a diameter, a height or a slant height? 3. If the given number *is* the area or volume, set the

formula equal to it and solve for the letter. 4. If nothing is numeric and lengths are being multiplied, it is the k, k^2, k^3 question.

Module 1

1

A rectangular tabletop is 12 inches long and 7 inches wide. What is the area, in square inches, of the tabletop?

A) 19

B) 38

C) 42

D) 84

2

A triangle has a base of length 10 centimeters and a height of 6 centimeters. What is the area, in square centimeters, of the triangle?

A) 16

B) 30

C) 32

D) 60

3

A circle has a radius of 5 inches. What is the area, in square inches, of the circle?

A) 5π

B) 10π

C) 25π

D) 100π

4

A rectangular box has a length of 9 inches, a width of 5 inches, and a height of 4 inches. What is the volume, in cubic inches, of the box?

5

A circular garden has a diameter of 14 feet. What is the circumference, in feet, of the garden?

A) 7π

B) 14π

C) 28π

D) 49π

6

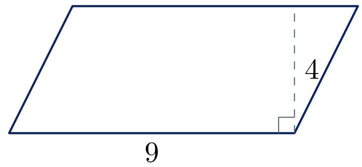


Figure not drawn to scale.

In the parallelogram shown, the base has length 9 and the height is 4. What is the area of the parallelogram?

A) 13

B) 18

C) 26

D) 36

7

A cube has edges of length 4 centimeters. What is the surface area, in square centimeters, of the cube?

8

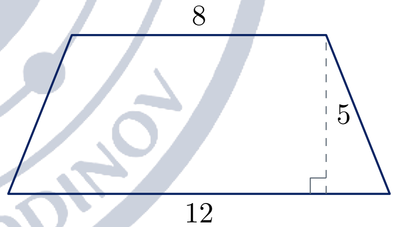


Figure not drawn to scale.

The trapezoid shown has parallel sides of length 8 and 12 and a height of 5. What is the area of the trapezoid?

A) 40

B) 50

C) 60

D) 100

9

A cylindrical can has a radius of 3 inches and a height of 10 inches. What is the volume, in cubic inches, of the can?

A) 30π

B) 60π

C) 90π

D) 360π

10

A right circular cone has a base radius of 6 centimeters and a height of 10 centimeters. What is the volume, in cubic centimeters, of the cone?

A) 120π

B) 180π

C) 360π

D) 480π

11

A triangle has an area of 84 square units and a base of length 24 units. What is the length of the height drawn to that base?

12

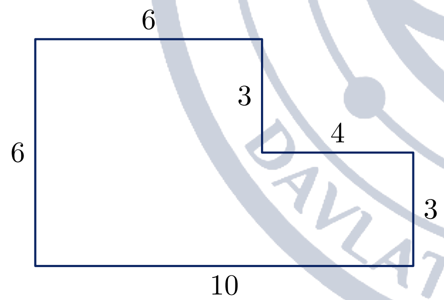


Figure not drawn to scale.

In the figure shown, every angle is a right angle. What is the area of the region?

A) 36

B) 48

C) 60

D) 72

13

Rectangle B is formed by multiplying both the length and the width of rectangle A by 3. The area of rectangle B is how many times the area of rectangle A?

A) 3

B) 6

C) 9

D) 27

14

A sphere has a radius of 3 inches. What is the volume, in cubic inches, of the sphere?

A) 27π

B) 36π

C) 108π

D) 288π

15

A right circular cylinder has a volume of 100π cubic centimeters and a height of 4 centimeters. What is the radius, in centimeters, of the cylinder's base?

16

The radius of a right circular cone is multiplied by 2 and its height is multiplied by 3. The volume of the new cone is how many times the volume of the original cone?

A) 6

B) 12

C) 24

D) 36

17

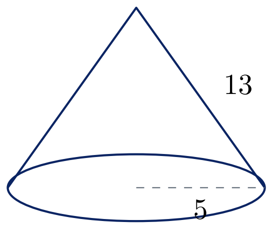


Figure not drawn to scale.

A right circular cone has a base radius of 5 and a slant height of 13, as shown. What is the volume of the cone?

A) 100π
B) $\frac{325\pi}{3}$
C) 300π
D) 325π

18

A rectangular prism with a square base has a volume of 500 cubic inches and a height of 5 inches. What is the surface area, in square inches, of the prism?

A) 200
B) 300
C) 400
D) 500

19

A sculptor builds a scale model of a statue. Every length on the statue is 4 times the corresponding length on the model. The model has a total surface area of 25 square inches. What is the total surface area, in square inches, of the statue?

20

A cylindrical tank with a base radius of 4 feet and a height of 9 feet is filled with water to $\frac{2}{3}$ of its capacity. What is the volume, in cubic feet, of the water in the tank?

A) 48π
B) 96π
C) 144π
D) 384π

21

A trapezoid has an area of 60 square units and a height of 6 units. One of its parallel sides is 3 times as long as the other. What is the length of the longer parallel side?

A) 5
B) 10
C) 15
D) 20

22

The volume of a sphere is 27 times the volume of a smaller sphere. The surface area of the larger sphere is how many times the surface area of the smaller sphere?

A) 3
B) 9
C) 18
D) 27

Module 2

1

A square has an area of 144 square meters. What is the perimeter, in meters, of the square?

A) 12
B) 24
C) 48
D) 576

2

A circle has an area of 49π square centimeters. What is the circumference, in centimeters, of the circle?

A) 7π
B) 14π
C) 49π
D) 98π

3

A rectangular box has a volume of 240 cubic inches. The base of the box measures 8 inches by 6 inches. What is the height, in inches, of the box?

A) 5
B) 10
C) 30
D) 40

4

A cube has a volume of 125 cubic centimeters. What is the surface area, in square centimeters, of the cube?

5

A circular pizza has a diameter of 16 inches. What is the area, in square inches, of the pizza?

A) 16π
B) 32π
C) 64π
D) 256π

6

Each edge of a cube is multiplied by 4 to produce a larger cube. The volume of the larger cube is how many times the volume of the original cube?

A) 4
B) 12
C) 16
D) 64

7

A tabletop is shaped like a trapezoid whose parallel sides measure 6 feet and 10 feet, with a distance of 4 feet between them. What is the area, in square feet, of the tabletop?

A) 20
B) 24
C) 32
D) 64

8

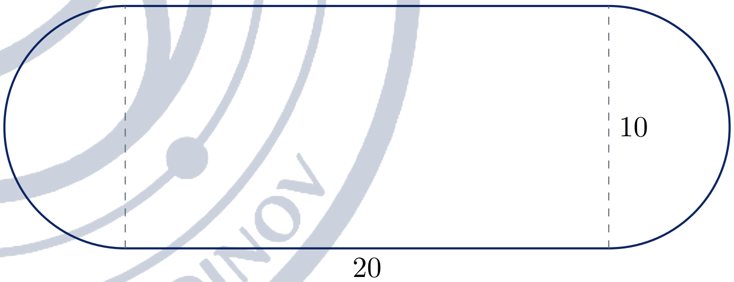


Figure not drawn to scale.

The region shown consists of a rectangle that is 20 meters long and 10 meters wide, with a semicircle attached to each of its two shorter sides. What is the area, in square meters, of the region?

A) $200 + 25\pi$
B) $200 + 50\pi$
C) $400 + 25\pi$
D) $200 + 100\pi$

9

A right circular cone has a volume of 96π cubic inches and a height of 8 inches. What is the radius, in inches, of the cone's base?

10

A closed cylindrical container has a base radius of 5 centimeters and a height of 6 centimeters. What is the total surface area, in square centimeters, of the container?

A) 60π

B) 85π

C) 110π

D) 260π

11

A rectangle has a perimeter of 34 inches, and one of its sides has length 12 inches. What is the area, in square inches, of the rectangle?

A) 5

B) 17

C) 22

D) 60

12

A rectangular photograph with an area of 48 square inches is enlarged so that each of its dimensions is multiplied by 2.5. What is the area, in square inches, of the enlarged photograph?

13

A solid consists of a right circular cylinder of radius 3 inches and height 10 inches with a hemisphere of radius 3 inches attached to its top. What is the volume, in cubic inches, of the solid?

A) 90π

B) 108π

C) 126π

D) 234π

14

A closed rectangular box measures 3 inches by 4 inches by 5 inches. What is the minimum number of square inches of paper needed to cover the outside of the box exactly?

A) 24

B) 47

C) 60

D) 94

15

A rectangular garden measures 15 feet by 8 feet. A walkway of uniform width 2 feet is built around the outside of the garden. What is the area, in square feet, of the walkway alone?

A) 46

B) 92

C) 108

D) 228

16

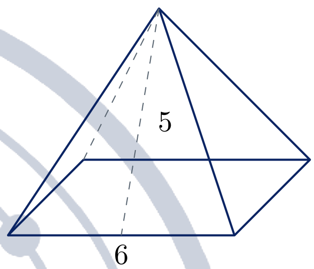


Figure not drawn to scale.

A pyramid has a square base with edges of length 6 centimeters. The slant height, measured from the apex to the midpoint of a base edge, is 5 centimeters. What is the volume, in cubic centimeters, of the pyramid?

A) 48

B) 60

C) 144

D) 180

17

Cylinder A and cylinder B are similar, and every length on cylinder B is 3 times the corresponding length on cylinder A. The volume of cylinder A is 20 cubic centimeters. What is the volume, in cubic centimeters, of cylinder B?

18

A sphere has a volume of 288π cubic units. What is the surface area, in square units, of the sphere?

A) 36π

B) 144π

C) 288π

D) 576π

19

Cylinder P has radius r and height h. Cylinder Q has a radius half as long as cylinder P's and a height twice as long as cylinder P's. The volume of cylinder Q is what fraction of the volume of cylinder P?

A) $\frac{1}{2}$

B) 1

C) 2

D) 4

20

A rectangular prism has a square base, and its height is 4 times the length of an edge of the base. The volume of the prism is 500 cubic centimeters. What is the length, in centimeters, of an edge of the base?

21

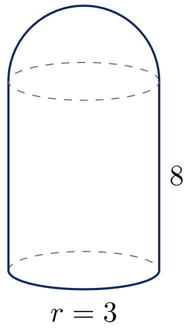


Figure not drawn to scale.

A solid is formed by attaching a hemisphere of radius 3 inches to the top of a right circular cylinder of radius 3 inches and height 8 inches, as shown. The bottom of the cylinder is a flat circular disk. What is the total surface area, in square inches, of the solid?

A) 48π

B) 57π

C) 66π

D) 75π

22

Two similar right circular cones have volumes of 27 cubic centimeters and 125 cubic centimeters. The base radius of the smaller cone is 6 centimeters. What is the base radius, in centimeters, of the larger cone?

Lesson 18: Lines, Angles and Triangles

Almost every Digital SAT test form asks two or three questions built out of nothing but angles, triangles and proportion. The figures look different every time, but the toolkit is small and fixed: angles that add to 180° or to 90°, the equal-angle pairs made by a transversal crossing parallel lines, the fact that the three angles of a triangle add to 180°, the rules that govern the sides of a triangle, and the proportion that comes out of two similar triangles. Learn to name what you are looking at and these questions become bookkeeping.

Angles at a point and on a line

An angle is a measure of turning, and the whole turn around a point is 360°. Two facts follow immediately and account for a large share of the easy geometry questions on the test.

If two angles together form a straight line, they are **supplementary**: their measures add to 180°. If they together form a right angle, they are **complementary**: their measures add to 90°. And when two straight lines cross, the two angles opposite each other – the **vertical** angles – are equal, because each of them is the supplement of the same neighbouring angle.

THREE ANGLE FACTS AT A POINT

- Angles on a straight line add to 180° (supplementary).
- Angles in a right angle add to 90° (complementary).
- Vertical angles are equal.

When a figure gives you one angle, mark every angle you can deduce from it before you read the question again. The answer is usually two marks away.

Parallel lines and a transversal

A line that crosses two parallel lines is called a **transversal**. It creates eight angles, but only two different sizes: a small one and a large one, and any small plus any large makes 180°.

The named pairs are worth knowing because the test uses the names. **Corresponding** angles sit in matching positions at the two crossings and are equal. **Alternate interior** angles sit between the parallel lines on opposite sides of the transversal and are equal. **Co-interior** (same-side interior) angles sit between the parallel lines on the same side of the transversal and are supplementary.

PARALLEL-LINE ANGLE FACTS

If $\ell \parallel m$ and a transversal cuts both:

- corresponding angles are equal;
- alternate interior angles are equal;
- alternate exterior angles are equal;
- co-interior (same-side interior) angles add to 180°.

Practical shortcut: in the whole figure every angle is either equal to the one you were given or is its supplement. Decide which by looking at the picture – angles that *look* the same size are the same size.

EQUAL, OR SUPPLEMENTARY?

The single most common wrong answer in this topic is the supplement of the right answer. A student finds that the co-interior partner of $(3x + 10)°$ is $(2x + 40)°$, then sets the two expressions *equal* instead of adding them to 180° – or does the reverse.

Before you write an equation, look at the two marked angles in the figure. If they look about the same size, set them equal. If one looks acute and the other obtuse, add them to 180°. The figure is drawn accurately unless it says otherwise, and even a figure that is not to scale usually keeps acute angles acute.

The angles of a triangle

The three interior angles of any triangle add to 180°. Extend one side and you get the **exterior angle**: because it is the supplement of the interior angle next to it, and because the three interior angles add to 180°, an exterior angle equals the sum of the two *remote* interior angles – the two it does not touch. That identity saves a step every time it appears.

A triangle with two equal sides is **isosceles**, and the angles opposite those two sides are equal. A triangle with three equal sides is **equilateral**, and all three of its angles measure 60°.

TRIANGLE RULES

- Angle sum: $\angle A + \angle B + \angle C = 180°$.
- Exterior angle: an exterior angle = the sum of the two remote interior angles.
- Isosceles: equal sides face equal angles, so if $AB = AC$ then $\angle B = \angle C$ and $\angle A = 180° - 2\angle B$.
- Equilateral: all angles 60°.
- Triangle inequality: with two sides a and b, the third side c satisfies $|a - b| < c < a + b$.
- The largest angle of a triangle faces the longest side.

THE TRIANGLE INEQUALITY IS STRICT

With sides 7 and 12 the third side must satisfy $5 < c < 19$. The values 5 and 19 are *not* allowed: at $c = 19$ the three segments flatten into a straight line and there is no triangle at all. Both endpoints appear as answer choices on purpose.
So the largest integer third side here is 18, not 19, and the smallest is 6, not 5.

Similar and congruent triangles

Two triangles are **congruent** when they have the same shape and the same size: every pair of corresponding sides and angles is equal. They are **similar** when they have the same shape but possibly different sizes: corresponding angles are equal and corresponding sides are all in the same ratio.

On the SAT you almost never have to prove similarity from scratch – the figure hands it to you. Two triangles that share an angle and have a second pair of equal angles are similar (AA). A segment drawn parallel to one side of a triangle cuts off a small triangle at the top that is similar to the whole triangle. Two segments crossing at a point with the outer ends joined by parallel segments give a pair of similar triangles through vertical angles plus alternate interior angles.

WRITE THE CORRESPONDENCE BEFORE YOU WRITE THE PROPORTION

The statement $\triangle ABC \sim \triangle DEF$ carries information in the *order* of the letters: A matches D, B matches E, C matches F. Therefore

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}.$$

Write that chain of fractions down before substituting numbers. Almost every wrong answer to a similarity question comes from pairing a side of one triangle with the wrong side of the other, and the wrong pairing produces a number that looks perfectly reasonable.

RATIOS OF SIMILAR FIGURES

If two similar figures have corresponding lengths in the ratio k (so every length in the second is k times the matching length in the first), then

- every length – side, perimeter, height, median – is in the ratio k;
- every area is in the ratio k^2;
- (for solids) every volume is in the ratio k^3.

Running it backwards: if the areas are in the ratio $9 : 16$, the lengths are in the ratio $3 : 4$.

A LENGTH RATIO IS NOT AN AREA RATIO

Similar triangles with sides in the ratio $3 : 5$ and the smaller one of area 18: the larger has area $18 \cdot \left(\frac{5}{3}\right)^2 = 50$, not $18 \cdot \frac{5}{3} = 30$.
The test always offers 30. It also offers the reverse error – taking the square root when it was not needed – so decide first whether the quantity you are scaling is a length or an area, and only then choose between k and k^2.

HOW TO ATTACK A GEOMETRY FIGURE

1. Mark on the figure everything you are told: equal sides with ticks, given angles with numbers.
2. Fill in free consequences: vertical angles, supplements on a line, the third angle of any triangle whose other two you know.
3. Name the relationship the question actually needs (co-interior? exterior angle? similarity?) before writing algebra.
4. If two triangles are involved, write the similarity statement with the vertices in matching order, then the proportion, then substitute.
5. Check the answer against the picture. A figure marked "not drawn to scale" still tells you whether an angle is acute or obtuse, and a side that looks shorter usually is.

WORKED EXAMPLE 1 – AN ANGLE CHASE

Lines ℓ and m are parallel. Point A lies on ℓ, points B and C lie on m, the angle at A between $\overline{AB}$ and line ℓ measures $38°$, and $\angle ACB$ measures $74°$. What is the measure of $\angle BAC$?
Step 1 – transfer the angle across the parallel lines. The $38°$ angle at A and $\angle ABC$ are alternate interior angles for $\ell \parallel m$ with transversal $\overline{AB}$, so $\angle ABC = 38°$.
Step 2 – use the triangle. In $\triangle ABC$ the three angles add to $180°$:

$$\angle BAC = 180° - 38° - 74° = 68°.$$

Why this is the right path. The question gives an angle at one line and asks about a triangle sitting on the other; the parallel-line facts are the only bridge between the two. Once the $38°$ has been carried down to B, the problem is a one-line angle sum. Note the trap: $180° - 68° = 112°$, the supplement of the answer, is exactly the sort of value the test offers as a choice.

WORKED EXAMPLE 2 – A NESTED SIMILAR TRIANGLE

In $\triangle ABC$, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE} \parallel \overline{BC}$. If $AD = 8$, $DB = 6$, and $BC = 21$, what is the length of $\overline{DE}$?
Step 1 – name the similar triangles, in order. Because $\overline{DE} \parallel \overline{BC}$, the angles $\angle ADE$ and $\angle ABC$ are corresponding angles and are equal, and the two triangles share $\angle A$. So $\triangle ADE \sim \triangle ABC$, with $A \leftrightarrow A$, $D \leftrightarrow B$, $E \leftrightarrow C$.
Step 2 – write the proportion the correspondence dictates.

$$\frac{AD}{AB} = \frac{DE}{BC}.$$

Step 3 – substitute whole sides, not pieces. $AB = AD + DB = 8 + 6 = 14$, so

$$\frac{8}{14} = \frac{DE}{21} \implies DE = 21 \cdot \frac{8}{14} = 12.$$

Why this is the right path. The similarity statement pairs AD with AB – a side of the small triangle with the *whole* side of the big one. Using $AD/DB = 8/6$ instead gives $DE = 28$, which is the classic wrong answer: DB is a piece of $\overline{AB}$, not a side of either triangle. The ratio $AD : DB$ is only legitimate when it is set against the matching pieces $AE : EC$.

Module 1

1

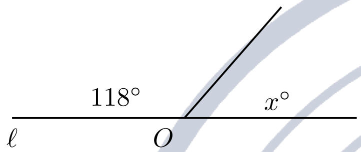


Figure not drawn to scale.

In the figure, ray OA meets line ℓ at point O, forming the two marked angles. What is the value of x?

A) 28

B) 62

C) 118

D) 152

2

Angle A and angle B are complementary. The measure of angle A is $27°$. What is the measure, in degrees, of angle B?

3

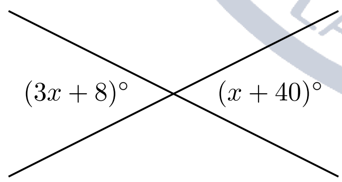


Figure not drawn to scale.

In the figure, two lines intersect and the two marked angles are vertical angles. What is the value of x?

A) 8

B) 16

C) 20

D) 33

4

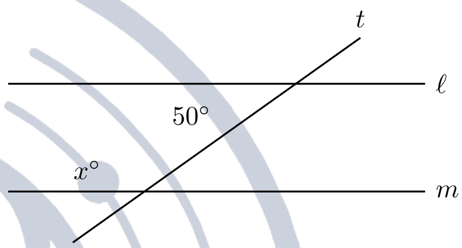


Figure not drawn to scale.

In the figure, line ℓ is parallel to line m and both are crossed by transversal t. The two marked angles are interior angles on the same side of t. What is the value of x?

A) 40

B) 50

C) 65

D) 130

5

Two of the interior angles of a triangle measure $47°$ and $68°$. What is the measure, in degrees, of the third interior angle?

A) 65

B) 112

C) 115

D) 133

6

In isosceles triangle ABC, $AB = AC$ and the measure of $\angle A$ is $40°$. What is the measure, in degrees, of $\angle B$?

7

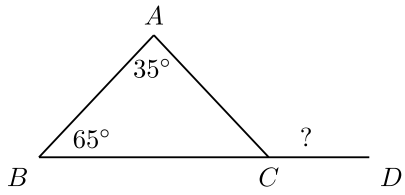


Figure not drawn to scale.

In the figure, side $\overline{BC}$ of triangle ABC is extended through C to point D. If $\angle A$ measures $35°$ and $\angle B$ measures $65°$, what is the measure, in degrees, of $\angle ACD$?

A) 80
B) 100
C) 115
D) 145

8

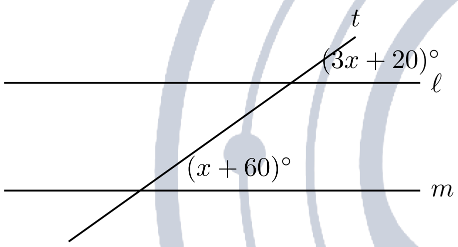


Figure not drawn to scale.

In the figure, $\ell \parallel m$ and the two marked angles are corresponding angles. What is the measure, in degrees, of each marked angle?

A) 20
B) 25
C) 80
D) 100

9

Two sides of a triangle have lengths 7 and 12. Which of the following could be the length of the third side?

A) 4
B) 5
C) 17
D) 19

10

The lengths of two sides of a triangle are 9 and 5. What is the greatest possible integer length of the third side?

11

Triangle ABC is similar to triangle DEF, where A corresponds to D, B corresponds to E, and C corresponds to F. If $AB = 8$, $BC = 12$, and $DE = 6$, what is the length of $\overline{EF}$?

A) 9
B) 10
C) 16
D) 18

12

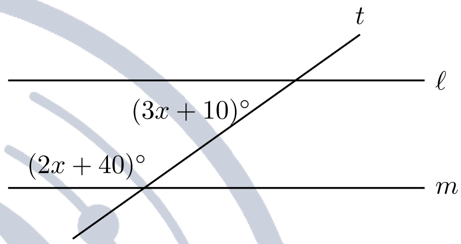


Figure not drawn to scale.

In the figure, $\ell \parallel m$ and the two marked angles are interior angles on the same side of the transversal. What is the value of x?

A) 26
B) 30
C) 88
D) 92

13

In triangle ABC, side $\overline{AB}$ is extended beyond B to point D. The measure of $\angle CBD$ is $(5x)°$, the measure of $\angle A$ is $40°$, and the measure of $\angle C$ is $(3x + 10)°$. What is the value of x?

14

In triangle ABC, $AB = AC$. The measure of $\angle A$ is $(x + 20)°$ and the measure of $\angle B$ is $(2x)°$. What is the value of x?

A) 20
B) 32
C) 52
D) 64

15

Triangle ABC is similar to triangle DEF, and the ratio of AB to DE is 3 to 5. The area of triangle ABC is 18. What is the area of triangle DEF?

A) 10.8
B) 30
C) 50
D) 90

16

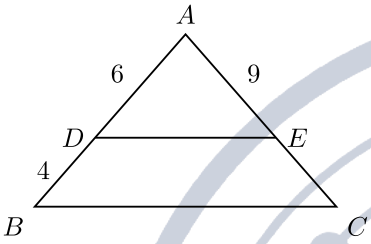


Figure not drawn to scale.

In the figure, D lies on $\overline{AB}$, E lies on $\overline{AC}$, and $\overline{DE} \parallel \overline{BC}$. If $AD = 6$, $DB = 4$, and $AE = 9$, what is the length of $\overline{EC}$?

A) 4
B) 6
C) 13.5
D) 15

17

In triangle PQR, point S lies on $\overline{PQ}$ and point T lies on $\overline{PR}$ so that $\overline{ST} \parallel \overline{QR}$. If $PS = 8$, $SQ = 12$, and $ST = 10$, what is the length of $\overline{QR}$?

18

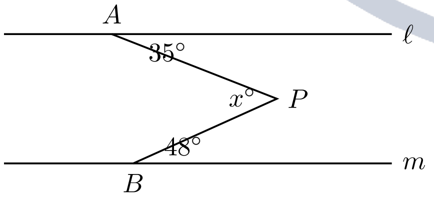


Figure not drawn to scale.

In the figure, $\ell \parallel m$, point A is on ℓ, point B is on m, and point P lies between the two lines. Segment $\overline{PA}$ makes a 35° angle with ℓ and segment $\overline{PB}$ makes a 48° angle with m, as shown. What is the value of x, the measure in degrees of $\angle APB$?

A) 13
B) 48
C) 83
D) 97

19

The side lengths of a triangle are 5, 11, and x, where x is an integer. What is the greatest possible perimeter of the triangle?

A) 21
B) 23
C) 31
D) 32

20

Triangle ABC is similar to triangle DEF, and the ratio of the perimeter of triangle ABC to the perimeter of triangle DEF is 2 to 3. If the area of triangle ABC is 24, what is the area of triangle DEF?

21

At the same time of day, a person who is 6 feet tall casts a shadow 9 feet long, and a nearby flagpole casts a shadow 24 feet long. How tall, in feet, is the flagpole?

A) 16
B) 21
C) 27
D) 36

22

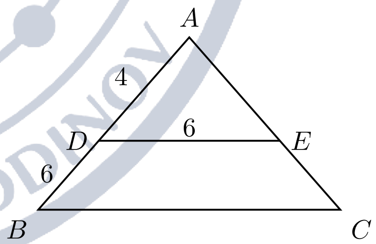


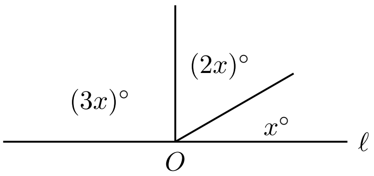
Figure not drawn to scale.

In the figure, D lies on $\overline{AB}$ and E lies on $\overline{AC}$, and $\angle ADE$ is congruent to $\angle ABC$. If $AD = 4$, $DB = 6$, and $DE = 6$, what is the length of $\overline{BC}$?

A) 2.4
B) 9
C) 12
D) 15

Module 2

1



In the figure, two rays drawn from point O on line ℓ divide the straight angle above the line into three angles with measures x°, $(2x)^\circ$, and $(3x)^\circ$. What is the value of x?

A) 20
B) 30
C) 45
D) 60

2

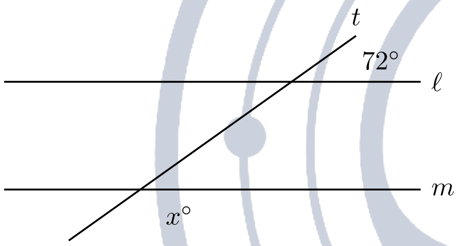


Figure not drawn to scale.

In the figure, line ℓ is parallel to line m and both are crossed by transversal t. What is the value of x?

A) 18
B) 72
C) 108
D) 162

3

The measures of the three interior angles of a triangle are in the ratio 2 : 3 : 4. What is the measure, in degrees, of the largest angle?

4

In isosceles triangle ABC, $AB = AC$ and the measure of $\angle B$ is 37°. What is the measure, in degrees, of $\angle A$?

A) 37
B) 53
C) 74
D) 106

5

In triangle ABC, side $\overline{AB}$ is extended through B to point D, and $\angle CBD$ measures 130°. If $\angle A$ measures 55°, what is the measure, in degrees, of $\angle C$?

A) 50
B) 75
C) 105
D) 125

6

Triangle RST is similar to triangle XYZ, where R corresponds to X, S corresponds to Y, and T corresponds to Z. If $RS = 12$, $ST = 15$, and $XY = 8$, what is the length of $\overline{YZ}$?

A) 8
B) 10
C) 11
D) 22.5

7

Two similar triangles have corresponding side lengths in the ratio 2 to 5. The area of the smaller triangle is 12. What is the area of the larger triangle?

8

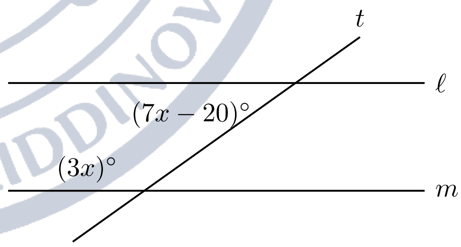


Figure not drawn to scale.

In the figure, $\ell \parallel m$ and the two marked angles are interior angles on the same side of the transversal. What is the measure, in degrees, of the larger of the two marked angles?

A) 20
B) 40
C) 60
D) 120

9

The measures of the three interior angles of a triangle are $(x + 10)^\circ$, $(2x - 20)^\circ$, and $(3x + 10)^\circ$. What is the measure, in degrees, of the largest angle?

10

Two sides of a triangle have lengths 8 and 15. If the length of the third side is an integer, how many different values are possible for that length?

A) 13
B) 14
C) 15
D) 16

11

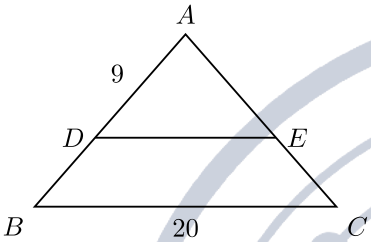


Figure not drawn to scale.

In the figure, D lies on $\overline{AB}$, E lies on $\overline{AC}$, and $\overline{DE} \parallel \overline{BC}$. If $AD = 9$, $AB = 15$, and $BC = 20$, what is the length of $\overline{DE}$?

A) 12
B) 14
C) 20
D) 30

12

In triangle ABC, $AB = AC$. Side $\overline{BC}$ is extended through C to point D, and $\angle ACD$ measures $118°$. What is the measure, in degrees, of $\angle A$?

A) 56
B) 62
C) 118
D) 124

13

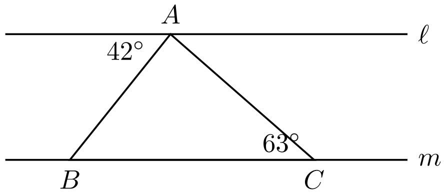


Figure not drawn to scale.

In the figure, $\ell \parallel m$, vertex A of triangle ABC lies on ℓ, and B and C lie on m. Side $\overline{AB}$ makes a $42°$ angle with line ℓ as shown, and $\angle ACB$ measures $63°$. What is the measure, in degrees, of $\angle BAC$?

A) 75
B) 105
C) 117
D) 138

14

Triangle ABC is similar to triangle PQR, where A corresponds to P and B corresponds to Q. The area of triangle ABC is 27 and the area of triangle PQR is 48. If $AB = 9$, what is the length of $\overline{PQ}$?

A) 6.75
B) 9
C) 12
D) 16

15

Triangle ABC is congruent to triangle XYZ, where A corresponds to X, B corresponds to Y, and C corresponds to Z. If $AB = 3x - 4$, $XY = 20$, $BC = 2y + 1$, and $YZ = 15$, what is the value of $x + y$?

A) 7
B) 8
C) 15
D) 16

16

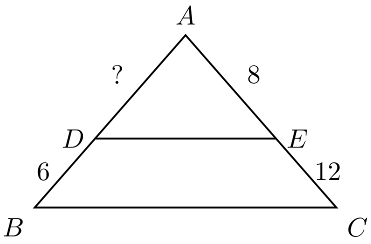


Figure not drawn to scale.

In the figure, D lies on $\overline{AB}$, E lies on $\overline{AC}$, and $\overline{DE} \parallel \overline{BC}$. If $DB = 6$, $AE = 8$, and $EC = 12$, what is the length of $\overline{AD}$?

17

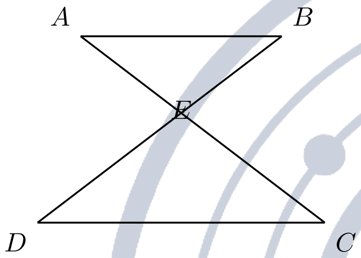


Figure not drawn to scale.

In the figure, $\overline{AC}$ and $\overline{BD}$ intersect at point E, and $\overline{AB} \parallel \overline{CD}$. If $AE = 4$, $EC = 10$, and $BE = 6$, what is the length of $\overline{ED}$?

A) 2.4

B) 6

C) 10

D) 15

18

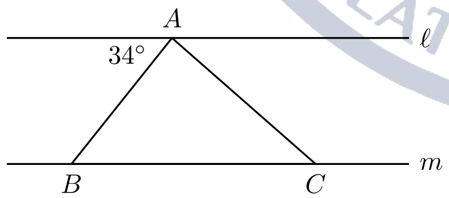


Figure not drawn to scale.

In the figure, $\ell \parallel m$, point A is on ℓ, and points B and C are on m, where $AB = AC$. Side $\overline{AB}$ makes a $34°$ angle with line ℓ as shown. What is the measure, in degrees, of $\angle BAC$?

A) 56

B) 68

C) 112

D) 146

19

Two similar triangular plots of land have corresponding side lengths in the ratio 1 to 1.5. The area of the smaller plot is 40 square meters. What is the area, in square meters, of the larger plot?

A) 60

B) 90

C) 120

D) 160

20

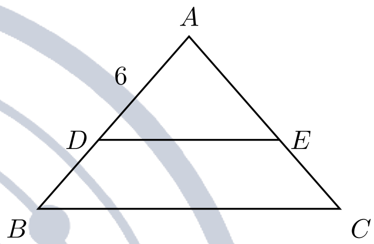


Figure not drawn to scale.

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE} \parallel \overline{BC}$. The area of triangle ADE is 16 and the area of quadrilateral $DBCE$ is 48. If $AD = 6$, what is the length of $\overline{AB}$?

21

An isosceles triangle has two sides of lengths 5 and 12. What is the perimeter of the triangle?

A) 17

B) 22

C) 24

D) 29

22

In triangle ABC, point D lies on $\overline{BC}$ so that $\angle BAD$ is congruent to $\angle BCA$. If $AB = 8$ and $BD = 4$, what is the length of $\overline{BC}$?

A) 4

B) 8

C) 12

D) 16

Lesson 19: Right Triangles and Trigonometry

Every right-triangle question on the Digital SAT comes down to three tools: the Pythagorean theorem, the two special right triangles, and the three trigonometric ratios. The test almost never asks you to evaluate a trigonometric function of a messy angle; instead it hands you a ratio, or a pair of complementary angles, and asks what follows. Learn the triples on sight, keep the ratios straight, and these questions become some of the fastest points on the section.

The Pythagorean theorem

In a right triangle, the two sides that meet at the right angle are the **legs** and the side across from the right angle is the **hypotenuse**. The hypotenuse is always the longest side, which gives you a free sanity check on every answer.

If the legs are a and b and the hypotenuse is c, then

$$a^2 + b^2 = c^2.$$

Used forwards, it produces the hypotenuse from two legs. Used backwards, it produces a leg: $a^2 = c^2 - b^2$. Deciding which side is c before you write anything is the whole game.

TRIPLES WORTH KNOWING ON SIGHT

3-4-5 5-12-13 8-15-17 7-24-25

Any multiple of a triple is also a right triangle: 6-8-10, 9-12-15, 10-24-26, 15-36-39. If two sides of a right triangle match a scaled triple, write the third side down without squaring anything.

The two special right triangles

Two right triangles have side ratios you are expected to know cold, because the test uses them to avoid giving you a calculator-dependent answer.

A 45°-45°-90° triangle is half a square. A 30°-60°-90° triangle is half an equilateral triangle, which is why its short leg is exactly half the hypotenuse.

SPECIAL RIGHT TRIANGLE RATIOS

45°-45°-90°: leg : leg : hypotenuse $= x : x : x\sqrt{2}$

30°-60°-90°: short leg : long leg : hypotenuse $= x : x\sqrt{3} : 2x$

In both, the shortest side faces the smallest angle. Going from a short side up to a long one you *multiply*; going from a long side down to a short one you *divide*.

SOHCAHTOA

For an acute angle θ in a right triangle,

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}, \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Opposite and **adjacent** are read relative to the angle you named, so the same side is opposite one acute angle and adjacent to the other. The hypotenuse never changes roles.

Because a single ratio names two sides, one ratio plus the Pythagorean theorem rebuilds the entire triangle. If $\sin\theta = \frac{3}{5}$, set opposite $= 3k$ and hypotenuse $= 5k$; the third side is $4k$, and every other ratio follows.

THE IDENTITY THE TEST LOVES: $\sin(x^\circ) = \cos((90-x)^\circ)$

The two acute angles of a right triangle add to 90°. The leg opposite one of them is the leg adjacent to the other, so

$$\sin(x^\circ) = \cos((90-x)^\circ) \qquad \text{and} \qquad \cos(x^\circ) = \sin((90-x)^\circ).$$

Two consequences drive almost every SAT question on this:

- If $\sin A = \cos B$ for acute angles, then $A + B = 90^\circ$. A trigonometric equation becomes a *linear* equation.
- In right triangle ABC with the right angle at C, $\sin A = \cos B$ and $\cos A = \sin B$ always—no angle measure required.

Tangent behaves differently: complementary angles have *reciprocal* tangents, $\tan(x^\circ) = \dfrac{1}{\tan((90-x)^\circ)}$.

FOUR MISTAKES THAT COST REAL POINTS

- **Using a leg as the hypotenuse.** In a ladder or wire problem the slanted object is always the hypotenuse. Adding $10^2 + 26^2$ instead of subtracting is the most common error in the topic.
- **Swapping opposite and adjacent.** $\tan A = \frac{3}{4}$ and $\tan A = \frac{4}{3}$ describe different triangles, and both appear among the choices.
- **Inverting a ratio.** $\sin\theta = \frac{8}{17}$ means the hypotenuse is on the bottom; $\frac{17}{8}$ is always a distractor.
- **Swapping the special-triangle multipliers.** $\sqrt{2}$ belongs to 45°-45°-90° and $\sqrt{3}$ to 30°-60°-90°; the hypotenuse of the second is $2x$, not $x\sqrt{3}$.

HOW TO ATTACK THESE FAST

1. **Mark the right angle first,** then label the hypotenuse. Everything else is a leg.
2. **Check for a triple** before reaching for the calculator. Two sides from a scaled triple give the third for free.
3. **When only a ratio is given, draw the triangle** with those two sides and find the third. Never look for an angle measure you were not given.
4. **If a sine equals a cosine, add the angles to** 90**.** Do not evaluate anything.
5. **Figures are not drawn to scale,** so trust the labels, not the picture—but do trust that the longest label belongs to the hypotenuse.

WORKED EXAMPLE 1 — A RATIO, BACKWARDS

In right triangle ABC, angle C is a right angle, $\tan A = \frac{5}{12}$, and $AB = 26$. What is the length of $\overline{BC}$?
$\tan A = \frac{5}{12}$ names the two legs: the leg opposite A is $5k$ and the leg adjacent is $12k$. The Pythagorean theorem gives the hypotenuse $13k$, since 5-12-13 is a triple.
We are told the hypotenuse $AB = 26$, so $13k = 26$ and $k = 2$. The side $\overline{BC}$ is the leg opposite A, so $BC = 5k = \mathbf{10}$.
Notice what we never did: we never found the measure of angle A. The ratio alone was enough.

WORKED EXAMPLE 2 — A SPECIAL TRIANGLE IN DISGUISE

A ramp rises from level ground to a platform 7 feet high, making a 30° angle with the ground. How long is the ramp?

The ramp, the ground and the vertical height form a right triangle. The height is opposite the 30° angle, so it is the *short leg*; the ramp is the hypotenuse.
In a 30°-60°-90° triangle the hypotenuse is twice the short leg, so the ramp is $2 \cdot 7 = \mathbf{14}$ feet.
The trap answer is $7\sqrt{3} \approx 12.1$, which is the horizontal run—the long leg—not the ramp itself. Always ask which side the question named.

WORKED EXAMPLE 3 — SINE EQUALS COSINE

In a right triangle, one acute angle measures $(2x+5)^\circ$ *and* $\sin((2x+5)^\circ) = \cos((3x)^\circ)$. *What is the value of* x*?*
A sine equals a cosine exactly when the angles are complementary, so

$$(2x+5) + 3x = 90.$$

Then $5x + 5 = 90$, so $5x = 85$ and $x = \mathbf{17}$.
Check: the angles are 39° and 51°, which do add to 90°. The entire trigonometric dressing collapsed into a two-line linear equation—which is exactly why this item type is worth practising until it is automatic.

Module 1

1

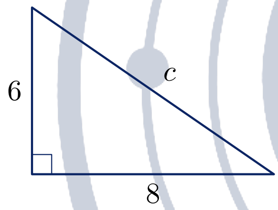


Figure not drawn to scale.

In the right triangle shown, the two legs have lengths 6 and 8. What is the length of the hypotenuse?

A) $\sqrt{28}$

B) 10

C) 14

D) 48

2

A right triangle has a hypotenuse of length 13 and one leg of length 5. What is the length of the other leg?

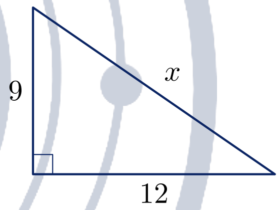
A) 8

B) 12

C) $\sqrt{194}$

D) 18

3



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The legs of a right triangle have lengths 9 and 12. What is the length of the hypotenuse of the triangle?

4

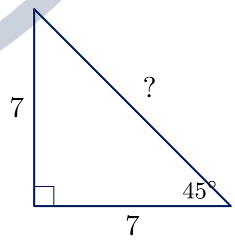

Figure not drawn to scale.

In a right triangle, both acute angles measure 45° and each leg has length 7. What is the length of the hypotenuse?

A) $\frac{7}{2}\sqrt{2}$

B) $7\sqrt{2}$

C) $7\sqrt{3}$

D) 14

5

In a right triangle, the hypotenuse has length 10 and one of the acute angles measures 30°. What is the length of the side opposite the 30° angle?

A) 5
B) $5\sqrt{2}$
C) $5\sqrt{3}$
D) $10\sqrt{3}$

6

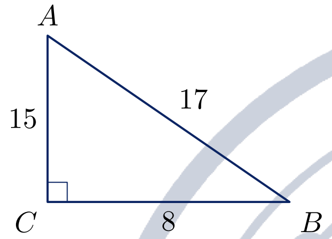

Figure not drawn to scale.

In the right triangle shown, angle A is an acute angle. What is the value of $\sin A$?

A) $\frac{8}{17}$
B) $\frac{8}{15}$
C) $\frac{15}{17}$
D) $\frac{17}{8}$

7

If $\sin(x°) = \frac{3}{5}$, what is the value of $\cos((90 - x)°)$?

A) $\frac{3}{5}$
B) $\frac{3}{4}$
C) $\frac{4}{5}$
D) $\frac{5}{3}$

8

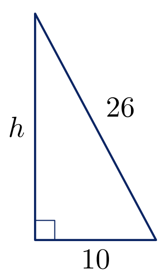

Figure not drawn to scale.

A 26-foot ladder leans against a vertical wall, with the base of the ladder 10 feet from the wall on level ground. How many feet up the wall does the top of the ladder reach?

9

In right triangle ABC, angle C is a right angle, $\tan A = \frac{3}{4}$, and $AC = 20$. What is the length of $\overline{BC}$?

A) 12
B) 15
C) 25
D) $\frac{80}{3}$

10

In a right triangle, one acute angle measures 60° and the side opposite the 30° angle has length 6. What is the length of the side opposite the 60° angle?

A) $3\sqrt{3}$
B) $6\sqrt{2}$
C) $6\sqrt{3}$
D) 12

11

Right triangles ABC and DEF have right angles at C and F, and angle A is congruent to angle D. If $AC = 6$, $BC = 8$, and $DF = 9$, what is the length of $\overline{EF}$?

A) 10
B) 11
C) 12
D) 15

12

In right triangle PQR, angle R is a right angle and $\cos P = \frac{7}{25}$. What is the value of $\sin Q$?

A) $\frac{7}{25}$
B) $\frac{7}{24}$
C) $\frac{24}{25}$
D) $\frac{25}{7}$

13

A vertical pole stands on level ground. The angle of elevation of the sun is 45°, and the pole's shadow is 18 feet long. What is the height, in feet, of the pole?

A) $9\sqrt{2}$
B) $9\sqrt{3}$
C) 18
D) $18\sqrt{2}$

14

The legs of a right triangle have lengths 8 and 15. What is the perimeter of the triangle?

A) 17

B) 23

C) $\sqrt{161} + 23$

D) 40

15

In right triangle ABC, angle C is a right angle and $\sin A = \frac{3}{5}$. What is the value of $\cos A$?

16

In right triangle ABC, angle B is a right angle, $AB = 9$, and $BC = 12$. Point D lies on $\overline{AC}$ so that $\overline{BD}$ is perpendicular to $\overline{AC}$. What is the length of $\overline{BD}$?

A) $\frac{20}{3}$

B) $\frac{36}{5}$

C) $\frac{15}{2}$

D) $\frac{48}{5}$

17

The acute angles of a right triangle measure $a°$ and $b°$. If $\sin(a°) = \frac{5}{13}$, what is the value of $\cos(b°) \cdot \tan(a°)$?

A) $\frac{25}{169}$

B) $\frac{25}{156}$

C) $\frac{60}{169}$

D) $\frac{144}{169}$

18

In a right triangle, one acute angle measures $(3x+10)°$ and $\sin((3x+10)°) = \cos((x+20)°)$. What is the value of x?

19

An equilateral triangle has side length 12. What is the length of an altitude of the triangle?

A) 6

B) $6\sqrt{2}$

C) $6\sqrt{3}$

D) $12\sqrt{3}$

20

In a right triangle, θ is an acute angle and $\tan\theta = \frac{20}{21}$. What is the value of $\sin\theta$?

A) $\frac{20}{29}$

B) $\frac{21}{29}$

C) $\frac{20}{21}$

D) $\frac{29}{20}$

21

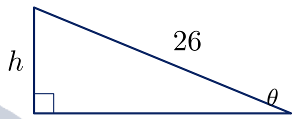

Figure not drawn to scale.

A straight ramp 26 feet long runs from level ground to a loading dock. The ramp makes an angle θ with the ground, where $\sin\theta = \frac{5}{13}$. How many feet above the ground is the top of the ramp?

22

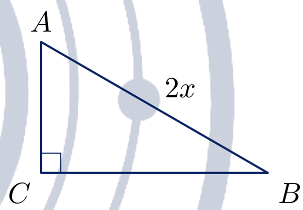

Figure not drawn to scale.

In right triangle ABC, angle C is a right angle and the length of $\overline{AB}$ is twice the length of $\overline{BC}$. What is the value of $\tan A$?

A) $\frac{1}{2}$

B) $\frac{1}{3}\sqrt{3}$

C) $\frac{1}{2}\sqrt{3}$

D) $\sqrt{3}$

Module 2

1

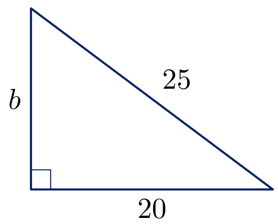

Figure not drawn to scale.

In the right triangle shown, one leg has length 20 and the hypotenuse has length 25. What is the length of the other leg?

A) 5

B) 15

C) $\sqrt{1025}$

D) 45

2

The hypotenuse of a 45°-45°-90° triangle has length $8\sqrt{2}$. What is the length of each leg?

A) 4

B) $\frac{8}{3}\sqrt{3}$

C) $4\sqrt{2}$

D) 8

3

In right triangle ABC, angle B is a right angle, $AB = 24$, and $BC = 7$. What is the value of $\tan A$?

4

Angles A and B are the two acute angles of a right triangle. If $\tan A = \frac{5}{12}$, what is the value of $\tan B$?

A) $\frac{5}{13}$

B) $\frac{5}{12}$

C) $\frac{13}{12}$

D) $\frac{12}{5}$

5

In a 30°-60°-90° triangle, the side opposite the 60° angle has length $9\sqrt{3}$. What is the length of the hypotenuse?

A) $\frac{9}{2}\sqrt{3}$

B) 9

C) $9\sqrt{3}$

D) 18

6

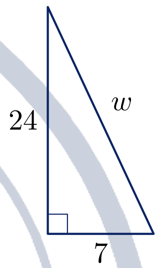

Figure not drawn to scale.

A support wire runs from the top of a vertical 24-foot pole to a point on level ground 7 feet from the base of the pole. What is the length, in feet, of the wire?

A) 17

B) $\sqrt{527}$

C) 25

D) 31

7

In right triangle DEF, angle F is a right angle and $\sin D = \frac{8}{17}$. What is the value of $\cos E$?

8

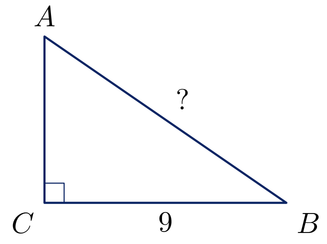

Figure not drawn to scale.

In right triangle ABC, angle C is a right angle, $BC = 9$, and $\sin A = \frac{3}{5}$. What is the length of $\overline{AB}$?

A) $\frac{27}{5}$
B) $\frac{45}{4}$
C) 12
D) 15

9

In a right triangle, one acute angle measures $(4x-6)°$ and $\cos((4x-6)°) = \sin((x+16)°)$. What is the value of x?

10

Right triangles ABC and XYZ are similar, with right angles at B and Y and with angle A corresponding to angle X. If $AB = 8$, $BC = 6$, and $XY = 20$, what is the length of $\overline{XZ}$?

A) 10
B) 15
C) $\sqrt{436}$
D) 25

11

A vertical flagpole 40 feet tall casts a shadow $40\sqrt{3}$ feet long on level ground. What is the measure, in degrees, of the angle of elevation of the sun?

A) 30
B) 45
C) 60
D) 75

12

The legs of a right triangle have lengths 10 and 24. What is the perimeter of the triangle?

13

In right triangle ABC, angle B is a right angle and $\cos A = \frac{5}{13}$. What is the value of $\tan A$?

A) $\frac{5}{12}$
B) $\frac{12}{13}$
C) $\frac{12}{5}$
D) $\frac{13}{5}$

14

A square has a diagonal of length 10. What is the area of the square?

A) 25
B) 50
C) $50\sqrt{2}$
D) 100

15

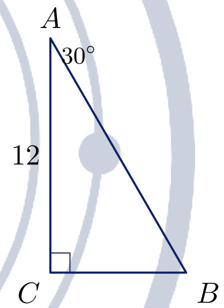

Figure not drawn to scale.

In right triangle ABC, angle C is a right angle, angle A measures 30°, and $AC = 12$. What is the length of $\overline{BC}$?

A) 6
B) $4\sqrt{3}$
C) $6\sqrt{3}$
D) $12\sqrt{3}$

16

In right triangle ABC, angle C is a right angle and $\sin A = \frac{2}{3}$. What is the value of $\cos A$?

A) $\frac{1}{3}$
B) $\frac{2}{3}$
C) $\frac{1}{3}\sqrt{5}$
D) $\frac{1}{3}\sqrt{13}$

17

In right triangle ABC, angle C is a right angle and $\tan A = \frac{3}{4}$. The perimeter of triangle ABC is 36. What is the length of $\overline{AB}$?

18

The acute angles of a right triangle measure $a°$ and $b°$. If $\cos(a°) = \frac{4}{5}$, what is the value of $\dfrac{\sin(b°)}{\cos(b°)}$?

A) $\frac{3}{4}$
B) $\frac{4}{5}$
C) $\frac{4}{3}$
D) $\frac{5}{3}$

19

In right triangle ABC the right angle is at C. Point D lies on $\overline{AB}$ so that $\overline{CD}$ is perpendicular to $\overline{AB}$. If $AD = 4$ and $DB = 9$, what is the length of $\overline{CD}$?

A) 6
B) $\frac{13}{2}$
C) $\sqrt{97}$
D) 13

20

In right triangle ABC, angle C is a right angle, $AB = 20$, and $\sin B = \cos B$. What is the area of triangle ABC?

21

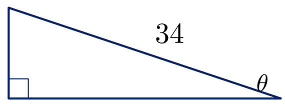

Figure not drawn to scale.

A straight ramp 34 feet long rises from level ground at an angle θ, where $\sin\theta = \frac{8}{17}$. What is the horizontal distance, in feet, covered by the ramp?

A) 16
B) 18
C) 30
D) 32

22

In right triangle ABC, angle C is a right angle and the measure of angle A is twice the measure of angle B. What is the value of $\dfrac{AC}{AB}$?

A) $\frac{1}{2}$
B) $\frac{1}{2}\sqrt{2}$
C) $\frac{1}{2}\sqrt{3}$
D) $\frac{2}{3}\sqrt{3}$

Lesson 20: Circles

A circle is the set of all points a fixed distance r from a fixed point, the centre. Almost everything the Digital SAT asks about circles follows from that one sentence: the two size formulas ($C = 2\pi r$ and $A = \pi r^2$), the fact that an arc or a sector is just a fraction of the whole circle, the two angle rules (central and inscribed), and the equation $(x-h)^2 + (y-k)^2 = r^2$, which is nothing more than the distance formula written without a square root. Answers on this topic are almost always exact — left in terms of π or as a clean integer — so resist the urge to reach for 3.14.

Radius, diameter, circumference and area

The diameter is twice the radius, $d = 2r$. Every other formula is written in terms of r, so the very first thing to do in any circle question is to decide what r is.

The two size formulas run in both directions. If a question hands you the area and asks for the circumference, do not hunt for a formula connecting them: go through the radius. From $A = 49\pi$ you get $r^2 = 49$, so $r = 7$, and then $C = 2\pi(7) = 14\pi$. The radius is the hinge of the whole topic.

FORMULAS YOU MUST KNOW

For a circle of radius r:

$$C = 2\pi r = \pi d \qquad A = \pi r^2$$

For a central angle of θ degrees:

$$\text{arc length} = \frac{\theta}{360} \cdot 2\pi r \qquad \text{sector area} = \frac{\theta}{360} \cdot \pi r^2$$

For a central angle of θ radians:

$$s = r\theta \qquad \text{sector area} = \tfrac{1}{2} r^2 \theta$$

Conversions: multiply by $\dfrac{\pi}{180}$ to go from degrees to radians, and by $\dfrac{180}{\pi}$ to go the other way.

Equation of a circle with centre (h, k) and radius r:

$$(x-h)^2 + (y-k)^2 = r^2$$

AN ARC OR A SECTOR IS A FRACTION OF A CIRCLE

You never need a separate arc formula. A central angle of $\theta°$ cuts off the fraction $\dfrac{\theta}{360}$ of the circle, so it cuts off that same fraction of the circumference (an arc) and that same fraction of the area (a sector). A $90°$ sector is a quarter of the circle; a $60°$ sector is a sixth of it.

In radians the fraction is $\dfrac{\theta}{2\pi}$, and the algebra collapses to $s = r\theta$. That is the whole reason radians exist: they make arc length a single multiplication.

CENTRAL ANGLES AND INSCRIBED ANGLES

A central angle has its vertex at the centre; its measure equals the measure of the arc it cuts off. An inscribed angle has its vertex on the circle; its measure is **half** the arc it cuts off.

So if a central angle and an inscribed angle intercept the same arc, the inscribed angle is half the central angle. Two consequences the test loves:

- An angle inscribed in a semicircle is a right angle. If $\overline{AB}$ is a diameter and C is on the circle, then $\angle ACB = 90°$.

- A tangent line meets the radius drawn to the point of tangency at $90°$, which turns tangent questions into right-triangle questions.

COMMON TRAP

Four mistakes account for most wrong answers on this topic.

- Using the **diameter** where the formula wants the radius. A circle of diameter 10 has area 25π, not 100π.
- Reading r^2 off the equation and reporting it as r. In $(x-3)^2+(y+5)^2=49$ the radius is 7, not 49.
- Getting the **sign** of the centre backwards. The form is $(x-h)^2+(y-k)^2=r^2$, so $(x+4)^2$ means $h=-4$: flip the sign you see.
- Halving the wrong angle. The inscribed angle is the small one. If the central angle is $80°$, the inscribed angle on the same arc is $40°$ — not $160°$.

CALCULATOR / STRATEGY

Keep π as a symbol. If a grid-in answer would contain π, the question will always be phrased so that it does not — it will ask for the radius, the degree measure, or the value of k where the area is $k\pi$. That phrasing is a hint about what to solve for.
When an equation of a circle is given expanded, do not try to test the answer choices by plugging in points; complete the square once and read everything off. And when a question gives you a tangent line, draw the radius to the point of tangency immediately — the right angle it creates is the only reason the tangent was mentioned.

WORKED EXAMPLE 1 — ARC, SECTOR AND AN INSCRIBED ANGLE

In a circle of radius 12, an inscribed angle of $30°$ intercepts arc AB. Find the length of arc AB and t
The inscribed angle is half the arc, so arc AB measures $2(30°)=60°$; equivalently the central angle for that arc is $60°$. That is $\dfrac{60}{360}=\dfrac{1}{6}$ of the circle.
The full circumference is $2\pi(12)=24\pi$, so

$$\text{arc } AB = \tfrac{1}{6}(24\pi) = 4\pi.$$

The full area is $\pi(12)^2=144\pi$, so

$$\text{sector area} = \tfrac{1}{6}(144\pi) = 24\pi.$$

Notice that the $30°$ was never used directly in a formula — it had to be doubled first. Skipping that step gives 2π and 12π, which is exactly why those numbers appear as answer choices.

The equation of a circle

A point (x,y) lies on the circle with centre (h,k) and radius r exactly when its distance from the centre is r. Squaring the distance formula gives

$$(x-h)^2+(y-k)^2=r^2.$$

Reading a circle off this form is mechanical: flip the signs inside the parentheses to get the centre, and take a square root on the right to get the radius. So $(x+4)^2+(y-1)^2=36$ has centre $(-4,1)$ and radius 6.

Going the other way is just as mechanical. Centre $(0,3)$ and radius 5 give $x^2+(y-3)^2=25$. And if a question gives the two endpoints of a diameter, the centre is their midpoint and the radius is half the distance between them.

WORKED EXAMPLE 2 — COMPLETING THE SQUARE

The equation $x^2 + y^2 - 6x + 8y - 11 = 0$ represents a circle. Find its centre and radius.
Group the x terms and the y terms and move the constant across:

$$(x^2 - 6x) + (y^2 + 8y) = 11.$$

Complete each square by halving the linear coefficient and squaring it. Half of -6 is -3, and $(-3)^2 = 9$. Half of 8 is 4, and $4^2 = 16$. Add 9 and 16 to **both** sides:

$$(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16,$$

$$(x - 3)^2 + (y + 4)^2 = 36.$$

The centre is $(3, -4)$ and the radius is $\sqrt{36} = 6$.
Two checks worth making every time. First, the numbers you add on the left must also be added on the right — forgetting that is the single most common error here, and it produces $\sqrt{11}$. Second, the answer to "what is the radius" is 6, not 36; 36 is r^2.

A SHORTCUT FOR THE CENTRE

If a circle is written as $x^2 + y^2 + Dx + Ey + F = 0$, the centre is $\left(-\frac{D}{2}, -\frac{E}{2}\right)$ and $r^2 = \frac{D^2}{4} + \frac{E^2}{4} - F$. For $x^2 + y^2 - 12x + 2y + 12 = 0$ that gives centre $(6, -1)$ and $r^2 = 36 + 1 - 12 = 25$, so $r = 5$ — in one line. One caution: this only works when the coefficients of x^2 and y^2 are both 1. If you see $2x^2 + 2y^2 + \cdots = 0$, divide the whole equation by 2 first.

Module 1

1

A circle has a radius of 7 inches. What is the circumference of this circle, in inches?

A) 7π
B) 14π
C) 28π
D) 49π

2

A circle has a diameter of 10 centimeters. What is the area of this circle, in square centimeters?

A) 5π
B) 10π
C) 25π
D) 100π

3

In the xy-plane, the equation of a circle is $(x - 3)^2 + (y + 5)^2 = 49$. What is the radius of this circle?

4

An angle has a measure of $60°$. What is the measure of this angle, in radians?

A) $\frac{\pi}{6}$
B) $\frac{\pi}{4}$
C) $\frac{\pi}{3}$
D) $\frac{\pi}{2}$

5

In the xy-plane, a circle has the equation $(x + 4)^2 + (y - 1)^2 = 36$. What are the coordinates of the center of this circle?

A) $(-4, 1)$
B) $(-4, -1)$
C) $(4, -1)$
D) $(4, 1)$

6

The area of a circle is 36π square meters. What is the radius of this circle, in meters?

A) 6
B) 18
C) 36
D) 72

7

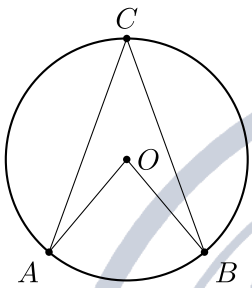

Figure not drawn to scale.

In the circle shown, O is the center and the central angle AOB measures $80°$. Point C lies on the major arc AB. What is the measure of the inscribed angle ACB?

A) $20°$
B) $40°$
C) $80°$
D) $160°$

8

In a circle with radius 9, a central angle measures $40°$. What is the length of the arc intercepted by this central angle?

A) 2π
B) 4π
C) 9π
D) 18π

9

In a circle with radius 6, a sector has a central angle of $60°$. What is the area of this sector?

A) 2π
B) 6π
C) 12π
D) 36π

10

An angle has a measure of $\frac{5\pi}{6}$ radians. What is the measure of this angle, in degrees?

11

In the xy-plane, a circle has center $(0, 3)$ and radius 5. Which of the following is an equation of this circle?

A) $x^2 + (y + 3)^2 = 25$
B) $x^2 + (y - 3)^2 = 5$
C) $(x - 3)^2 + y^2 = 25$
D) $x^2 + (y - 3)^2 = 25$

12

In the xy-plane, the graph of $(x+2)^2+(y-6)^2 = 100$ is a circle with center (h, k) and radius r. What is the value of $h + k + r$?

13

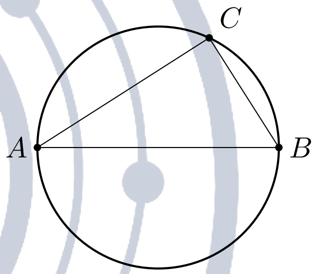

Figure not drawn to scale.

In the figure, $\overline{AB}$ is a diameter of the circle and point C lies on the circle. If the measure of angle CAB is $35°$, what is the measure of angle ABC?

A) $35°$
B) $45°$
C) $55°$
D) $90°$

14

A line is tangent to a circle with center O at point P. The radius of the circle is 5, and Q is a point on the tangent line such that $OQ = 13$. What is the length of $\overline{PQ}$?

A) 5
B) 8
C) 12
D) 18

15

A circle has a radius of 10. A central angle of $\frac{3\pi}{4}$ radians intercepts an arc of this circle. What is the length of that arc?

A) $\frac{15\pi}{8}$

B) $\frac{15\pi}{4}$

C) $\frac{15\pi}{2}$

D) 30π

16

In the xy-plane, the equation $x^2+y^2-6x+8y-11=0$ represents a circle. What is the radius of this circle?

A) $\sqrt{11}$

B) $\sqrt{14}$

C) 6

D) 36

17

In the xy-plane, the graph of $x^2+y^2+10x-4y+13=0$ is a circle. What is the radius of this circle?

18

In the xy-plane, the equation $x^2+y^2-12x+2y+12=0$ represents a circle. What are the coordinates of the center of this circle?

A) $(-6, 1)$

B) $(-6, -1)$

C) $(6, 1)$

D) $(6, -1)$

19

A sector of a circle has a central angle of $120°$ and an area of 27π. What is the radius of the circle?

A) 3

B) 9

C) 27

D) 81

20

In a circle with radius 12, an inscribed angle of $30°$ intercepts arc AB. What is the length of arc AB?

A) π

B) 2π

C) 3π

D) 4π

21

In the xy-plane, a circle has center $(4, -3)$ and passes through the point $(9, 9)$. What is the radius of this circle?

22

Which of the following equations is equivalent to $x^2+y^2+4x-10y+20=0$?

A) $(x+2)^2+(y-5)^2=9$

B) $(x+2)^2+(y-5)^2=29$

C) $(x-2)^2+(y+5)^2=9$

D) $(x+2)^2+(y-5)^2=3$

Module 2

1

The circumference of a circle is 36π centimeters. What is the diameter of this circle, in centimeters?

A) 6
B) 18
C) 36
D) 72

2

The area of a circle is 49π square inches. What is the circumference of this circle, in inches?

A) 7π
B) 14π
C) 28π
D) 98π

3

An angle has a measure of $\frac{3\pi}{4}$ radians. What is the measure of this angle, in degrees?

4

In the xy-plane, the equation of a circle is $(x-7)^2+(y+2)^2=45$. What is the radius of this circle?

A) $3\sqrt{5}$
B) $5\sqrt{3}$
C) $9\sqrt{5}$
D) 45

5

A sector of a circle with radius 8 has a central angle of $45°$. What is the area of this sector?

A) 2π
B) 4π
C) 8π
D) 16π

6

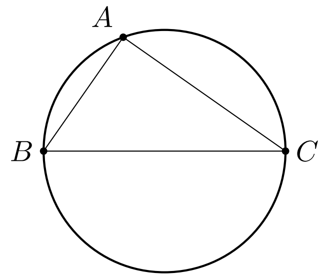

Figure not drawn to scale.

In the figure, $\overline{BC}$ is a diameter of the circle and point A lies on the circle. If the measure of angle ABC is $28°$, what is the measure of angle BCA?

A) $28°$
B) $62°$
C) $90°$
D) $152°$

7

An angle has a measure of $210°$. What is the measure of this angle, in radians?

A) $\frac{7\pi}{12}$
B) $\frac{5\pi}{6}$
C) $\frac{6\pi}{7}$
D) $\frac{7\pi}{6}$

8

In the xy-plane, a circle has center $(2,-2)$ and passes through the point $(6,1)$. What is the radius of this circle?

9

A sector of a circle with radius 12 has an area of 24π. What is the measure, in degrees, of the central angle of this sector?

A) 30
B) 45
C) 60
D) 120

10

In a circle with radius 15, an arc has length 10π. What is the measure, in degrees, of the central angle that intercepts this arc?

11

In the xy-plane, the points $(1, 2)$ and $(7, 10)$ are the endpoints of a diameter of a circle. Which of the following is an equation of this circle?

A) $(x-4)^2+(y-6)^2=25$
B) $(x-4)^2+(y-6)^2=100$
C) $(x+4)^2+(y+6)^2=25$
D) $(x-1)^2+(y-2)^2=100$

12

A central angle of $\frac{2\pi}{3}$ radians intercepts an arc of length 12π. What is the radius of the circle?

A) 6
B) 9
C) 12
D) 18

13

In the xy-plane, the graph of $x^2+y^2-8x+6y-11=0$ is a circle. What is the radius of this circle?

14

In a circle, a central angle and an inscribed angle intercept the same arc. The inscribed angle measures $(3x)^\circ$ and the central angle measures $(x+40)^\circ$. What is the value of x?

A) 8
B) 20
C) 40
D) 80

15

The circumference of a circle is 24π. What is the area of a sector of this circle with a central angle of 30°?

A) 2π
B) 6π
C) 12π
D) 144π

16

A sector of a circle with radius 12 has an arc length of 8π. What is the area of this sector?

A) 4π
B) 24π
C) 48π
D) 96π

17

Which of the following equations is equivalent to $x^2+y^2+6x-14y+22=0$?

A) $(x+3)^2+(y-7)^2=36$
B) $(x-3)^2+(y+7)^2=36$
C) $(x+3)^2+(y-7)^2=6$
D) $(x+3)^2+(y-7)^2=58$

18

In the xy-plane, the equation $2x^2+2y^2-12x+4y-6=0$ represents a circle. What is the radius of this circle?

A) $\sqrt{7}$
B) $\sqrt{13}$
C) $\sqrt{26}$
D) 13

19

In the xy-plane, the graph of $x^2+y^2-10x+4y+20=0$ is a circle with center (h, k) and radius r. What is the value of $h+k+r$?

20

In the xy-plane, a circle has center $(3, 4)$ and is tangent to the x-axis. Which of the following is an equation of this circle?

A) $(x-3)^2+(y-4)^2=9$
B) $(x+3)^2+(y+4)^2=16$
C) $(x-3)^2+(y-4)^2=25$
D) $(x-3)^2+(y-4)^2=16$

21

In the xy-plane, the graph of $x^2+y^2-4x-2y-20=0$ is a circle. The area of this circle is $k\pi$. What is the value of k?

22

In the xy-plane, the equation $x^2+y^2-2x+8y=8$ represents a circle. What is the circumference of this circle?

A) 5π
B) 10π
C) 25π
D) 50π

PART 5

Worked Solutions

Topic 01 — Linear Equations in One Variable

Module 1

1 Answer: A

1. Subtract 12 from both sides: $3x = 18$.
2. Divide both sides by 3: $x = 6$.

Every linear equation is undone in the reverse of the order it was built: the 12 was added last, so it is removed first, and only then is the multiplication by 3 undone. Peeling the constant off before the coefficient is what keeps the arithmetic clean.

2 Answer: 7

1. Subtract $4x$ from both sides: $3x = 21$.
2. Divide both sides by 3: $x = 7$.

When the variable appears on both sides, the first move is to collect it on one side. Subtracting $4x$ is legal because subtracting the same quantity from both sides leaves the two sides equal, and it turns the problem into an ordinary one-step equation.

3 Answer: B

1. Subtract 3 from both sides: $\dfrac{x}{4} = 8$.
2. Multiply both sides by 4: $x = 32$.

The variable is divided by 4, so the last undoing step is a multiplication by 4 — but only after the $+3$ is gone. Multiplying before subtracting would force the 3 to be multiplied as well.

4 Answer: C

1. Divide both sides by 3: $x - 4 = 6$.
2. Add 4 to both sides: $x = 10$.

A product equal to a number can be split immediately: if 3 times a quantity is 18, that quantity is 6. Dividing first avoids distributing at all, which is faster and removes a chance to slip a sign.

5 Answer: B

1. Let m be the number of months: $15m + 30 = 180$.
2. Subtract 30: $15m = 150$.
3. Divide by 15: $m = 10$.

A one-time charge is a constant and a repeated charge is a rate, so the total is $15m + 30$. Recognising which number is attached to the variable and which stands alone is the whole task; the algebra afterwards is two steps.

6 Answer: 10

1. Add 9 to both sides: $4x = 32$, so $x = 8$.
2. The question asks for $x + 2 = 8 + 2 = 10$.

The question asks for $x + 2$, not for x, so the work is not finished when x is found. Reading the last line of the question again before gridding an answer is the habit that prevents this whole family of errors.

7 Answer: C

1. Subtract 9 from both sides: $-2x = -8$.
2. Divide both sides by -2: $x = 4$.

The coefficient of x is -2, and dividing by a negative number is what trips students here. Isolating $-2x$ first and then dividing once by -2 keeps exactly one sign change in the whole problem.

8 Answer: C

1. Subtract $3x$ from both sides: $2x - 8 = 14$.
2. Add 8 to both sides: $2x = 22$.
3. Divide by 2: $x = 11$.

With variables on both sides, move the smaller variable term across so the remaining coefficient stays positive, then move the constants the other way. Each move is the same operation applied to both sides, which is why the equality survives.

9 Answer: A

1. Add 3 to both sides: $\dfrac{5}{2}x = 10$.
2. Multiply both sides by $\dfrac{2}{5}$: $x = 4$.

A fractional coefficient is undone by multiplying by its reciprocal: $\frac{2}{5}$, not $\frac{5}{2}$. Multiplying by the fraction itself is the single most common slip on this item type.

10 Answer: 3

1. Expand: $4x + 12 = 2x + 18$.
2. Subtract $2x$ and 12: $2x = 6$.
3. Divide by 2: $x = 3$.

Both sides are products, so both must be expanded before like terms can be combined. Expanding is just the distributive property, and it is what turns two grouped expressions into a form where the x terms can meet.

11 Answer: A

1. Subtract $2y$ from both sides: $3x = 12 - 2y$.
2. Divide both sides by 3: $x = \dfrac{12 - 2y}{3}$.

Solving a literal equation uses exactly the same moves as solving a numerical one; the only difference is that the answer is an expression. Because the whole quantity $12 - 2y$ is what equals $3x$, the entire numerator must be divided by 3, not just one term of it.

12 Answer: D

1. Write the model: $12 - 3h = -9$.
2. Subtract 12: $-3h = -21$.
3. Divide by -3: $h = 7$.

A falling quantity gets a negative rate, so the model is $12 - 3h$. Setting that equal to -9 and solving carefully keeps both negatives straight: the constant moves left, the negative coefficient divides last.

13 **Answer: B**

1. Expand the left side: $4x - 4a = 4x - 12$.
2. The $4x$ terms already match, so $-4a = -12$.
3. Divide by -4: $a = 3$.

T̈rue for all values of xm̈eans the two sides are the same expression, so their constant terms must match once both are expanded. Expanding gives $4x - 4a$ on the left, and $-4a$ can only equal -12 when $a = 3$.

14 **Answer: 4**

1. Expand the right side: $6x + 12 = 6x + 3k$.
2. The $6x$ terms cancel, leaving $12 = 3k$.
3. Divide by 3: $k = 4$.

Infinitely many solutions means the equation collapses to a true statement such as $0 = 0$, which happens only when the two sides are identical expressions. Expanding the right side gives $6x + 3k$, so the constants must agree: $3k = 12$.

15 **Answer: B**

1. Multiply every term by 10: $4x + 12 = 36$.
2. Subtract 12: $4x = 24$.
3. Divide by 4: $x = 6$.

Decimals obey the same rules as integers, but multiplying every term by 10 first turns $0.4x + 1.2 = 3.6$ into $4x + 12 = 36$, where no decimal point can be misplaced. Clearing decimals is a legal move because both sides are scaled equally.

16 **Answer: C**

1. Multiply every term by 6: $2(x + 5) - 3(x - 1) = 6$.
2. Expand: $2x + 10 - 3x + 3 = 6$.
3. Combine: $-x + 13 = 6$, so $x = 7$.

Multiplying by the least common denominator 6 clears both fractions at once, but every term — including the 1 on the right — must be multiplied. The minus sign in front of the second fraction then applies to the whole numerator, which is where the sign errors live.

17 **Answer: D**

1. Multiply both sides by $\dfrac{9}{5}$: $\dfrac{9}{5}C = F - 32$.
2. Add 32 to both sides: $F = \dfrac{9}{5}C + 32$.

To free F, undo the operations that were applied to it: the multiplication by $\frac{5}{9}$ comes off first (multiply by $\frac{9}{5}$), and only then does the -32 come off. Reversing that order produces $\frac{9}{5}(C + 32)$, which is a different function.

18 **Answer: 12**

1. Expand: $6x - 10 - 4x + 28 = 3x + 6$.
2. Combine on the left: $2x + 18 = 3x + 6$.
3. Subtract $2x$ and 6: $12 = x$.

The factor -4 multiplies both terms inside its parentheses, so $-4(x-7)$ becomes $-4x+28$, not $-4x-28$. Once each side is simplified to the form $ax + b$, one subtraction of variables and one of constants finishes it.

19 **Answer: A**

1. Model the tanks: $240 - 8t = 120 + 4t$.
2. Add $8t$ and subtract 120: $120 = 12t$.
3. Divide by 12: $t = 10$.

Each tank is a linear model in the same variable, so setting the two expressions equal locates the moment they agree. Because one tank falls while the other rises, collecting the t terms adds the rates: the gap of 120 liters closes at 12 liters per minute.

20 **Answer: D**

1. Simplify the right side: $4(3x + 5) - 8 = 12x + 12$.
2. For no solution the x coefficients must be equal: $a = 12$.
3. Check the constants: $15 \neq 12$, so there is indeed no solution.

An equation has no solution exactly when the variable terms match but the constants do not, leaving something like $15 = 12$. Simplifying the right side to $12x + 12$ shows the coefficient must be $a = 12$; the constants 15 and 12 then disagree, which is what makes the equation impossible.

21 **Answer: 41**

1. Cross multiply: $3(2x - 7) = 5(x + 4)$.
2. Expand: $6x - 21 = 5x + 20$.
3. Subtract $5x$ and add 21: $x = 41$.

When a single fraction equals a single fraction, cross multiplication is just multiplying both sides by both denominators at once. Each numerator must be kept in parentheses so the whole numerator gets multiplied, not only its first term.

22 **Answer: B**

1. Let w be the width; the length is $w + 4$.
2. Perimeter: $2(w + w + 4) = 48$, so $4w + 8 = 48$.
3. Then $w = 10$, and the length is $w + 4 = 14$.

Writing both dimensions in terms of one variable is what makes this a single linear equation: if the width is w, the length is $w + 4$. The perimeter formula then gives one equation in w, and the final step is to convert the w that is found into the length that is asked for.

Module 2

1 Answer: B

1. Subtract 7 from both sides: $-2x = 12$.
2. Divide both sides by -2: $x = -6$.

The variable term is $-2x$, so the constant 7 must leave first and the division by -2 must be done in one step at the end. Treating the coefficient as 2 instead of -2 is what produces the positive answer.

2 Answer: 30

1. Write with denominator 15: $\dfrac{5x - 3x}{15} = 4$.
2. Combine: $\dfrac{2x}{15} = 4$, so $2x = 60$.
3. Divide by 2: $x = 30$.

The two fractions are like terms in disguise: over the common denominator 15 they combine into $\frac{2x}{15}$. Combining before solving is faster than clearing denominators here, and it shows why the answer is a multiple of 15.

3 Answer: C

1. Expand: $10x - 25 + 4 = 39$.
2. Combine constants: $10x - 21 = 39$, so $10x = 60$.
3. Divide by 10: $x = 6$.

The 5 outside the parentheses multiplies both terms inside, so the -5 becomes -25. Distributing to only the first term leaves a different equation entirely, which is exactly the mistake the wrong answers here record.

4 Answer: C

1. Write the model: $25 + 0.10t = 43$.
2. Subtract 25: $0.10t = 18$.
3. Divide by 0.10: $t = 180$.

The fixed monthly charge is the constant and the per-text charge is the coefficient, giving $25 + 0.10t = 43$. Dividing the remaining \$18 by \$0.10 — not by 10 — is the step where the decimal must be handled honestly.

5 Answer: D

1. Subtract $5m$ from both sides: $-2n = 20 - 5m$.
2. Divide both sides by -2: $n = \dfrac{5m - 20}{2}$.

Isolating $-2n$ first and then dividing by -2 changes the sign of both remaining terms, which is why the numerator is $5m - 20$ and not $20 - 5m$. Dividing only one term by 2 is the other common slip.

6 Answer: C

1. Expand the left side: $2x + 12 = 5x - 3$.
2. Subtract $2x$ and add 3: $15 = 3x$.
3. Divide by 3: $x = 5$.

Expanding first puts both sides in the form $ax + b$, and then moving the smaller x term to the right keeps the coefficient positive. The constants travel the opposite way, which is the standard two-move routine.

7 **Answer: 7**

1. Expand: $5x - 10 = 3x + 4$.
2. Subtract $3x$ and add 10: $2x = 14$.
3. Divide by 2: $x = 7$.

Because the right side is not a single product, dividing by 5 first would create fractions on the right; expanding is the cleaner opening move. After expanding, the variable terms and the constants each get one move.

8 **Answer: C**

1. Expand both sides: $6x - 2 = 4x + 22$.
2. Subtract $4x$ and add 2: $2x = 24$.
3. Divide by 2: $x = 12$.

Both sides must be fully simplified before anything is moved: the right side is $4x + 20 + 2 = 4x + 22$. Combining the two constants on the right before crossing sides is what keeps the second step to a single subtraction.

9 **Answer: B**

1. Multiply both sides by $\dfrac{3}{2}$: $x + 6 = 15$.
2. Subtract 6: $x = 9$.

The fraction multiplies the whole parenthesis, so multiplying both sides by $\frac{3}{2}$ removes it in one step and leaves $x + 6 = 15$. Attaching the $\frac{2}{3}$ to x alone — as if the equation read $\frac{2}{3}x + 6 = 10$ — changes the problem.

10 **Answer: A**

1. Let n be the price of one notebook: $4n + 8 = 32$.
2. Subtract 8: $4n = 24$.
3. Divide by 4: $n = 6$.

The pen is a single fixed cost and the notebooks are a repeated cost, so the receipt reads $4n + 8 = 32$. Dividing the full \$32 by 4 would charge the pen to the notebooks as well, which is why the constant leaves first.

11 **Answer: 14**

1. Cross multiply: $2(x + 2) = 4(x - 6)$.
2. Expand: $2x + 4 = 4x - 24$.
3. Subtract $2x$ and add 24: $28 = 2x$, so $x = 14$.

Cross multiplying turns a proportion into a plain linear equation, and the parentheses around each numerator are essential so that the -6 is multiplied by 4 as well. The variable then appears on both sides and is collected in one move.

12 **Answer: A**

1. Multiply both sides by 3: $3y = 2x - 5$.
2. Add 5: $3y + 5 = 2x$.
3. Divide by 2: $x = \dfrac{3y + 5}{2}$.

Undo the division by 3 before anything else, because that division was applied to the whole numerator: $3y = 2x - 5$. Only after the $+5$ is restored can both terms be divided by 2.

13 **Answer: D**

1. Expand the left side: $10x + 15 = ax + 15$.
2. The constants match, so the coefficients must match: $a = 10$.

Infinitely many solutions means the two sides are the same expression after simplification. The left side expands to $10x + 15$; the constants already agree, so the coefficients must agree too, forcing $a = 10$.

14 **Answer: B**

1. Expand and combine: $6x - 12 + 5 = 6x - 7$.
2. Compare with $6x + c$: the coefficients agree automatically.
3. Match the constants: $c = -7$.

Simplify the left side to $6x - 7$. The x terms already match, so the equation is an identity precisely when the constants match as well, which means $c = -7$; any other c makes the equation impossible.

15 **Answer: A**

1. Let w be the width; the length is $2w + 3$.
2. Perimeter: $2(w + 2w + 3) = 36$, so $6w + 6 = 36$.
3. Then $6w = 30$ and $w = 5$.

Naming the width w lets the length be written as $2w + 3$, so the perimeter becomes a single linear equation. The factor of 2 in the perimeter formula applies to the sum of a length and a width, not to one of them, which is the step most often dropped.

16 **Answer: 4**

1. Expand the left side: $\frac{2}{3}(6x - 9) = 4x - 6$.
2. The $4x$ terms match, so $k - 10 = -6$.
3. Add 10: $k = 4$.

Simplify each side first: the left side is $4x - 6$. Since the coefficients of x already agree, the equation is an identity exactly when the constants agree, so $k - 10$ must equal -6.

17 **Answer: C**

1. Set the totals equal: $60 + 10c = 25c$.
2. Subtract $10c$: $60 = 15c$.
3. Divide by 15: $c = 4$.

Both totals are linear in the same variable, so the equal-cost point is found by setting the two expressions equal. The registration fee is paid once, so it is a constant, and the difference in the per-class rates is what eventually pays it off: \$15 per class against \$60.

18 **Answer: D**

1. Expand: $3x - 3a = 2x + 2a$.
2. Subtract $2x$ and add $3a$: $x = 5a$.

Treat a as a number and solve for x exactly as usual: expand, gather the x terms on one side and the a terms on the other. The two a terms are $-3a$ on the left and $+2a$ on the right, so moving $-3a$ across makes $5a$, not $-a$.

19 **Answer: A**

1. Let the integers be n, $n+2$, $n+4$.
2. Their sum: $3n+6=96$, so $3n=90$.
3. Divide by 3: $n=30$.

Consecutive even integers differ by 2, so naming them n, $n+2$, $n+4$ turns the sentence into $3n+6=96$. The variable was defined as the least integer, so n itself is the answer — dropping the $+6$ or answering with the largest are the two ways to lose the point.

20 **Answer: D**

1. Let n be the number of nickels; the number of dimes is $2n$.
2. Value in cents: $5n+10(2n)=250$, so $25n=250$.
3. Divide by 25: $n=10$.

Value problems need both a count relationship and a value equation. Writing the number of dimes as $2n$ keeps everything in one variable, and each coin must be counted at its own value: 5 cents for a nickel and 10 for a dime, with the total written in the same unit.

21 **Answer: 20**

1. Multiply every term by 8: $6x+4=5x+24$.
2. Subtract $5x$ and 4: $x=20$.

Multiplying every term by the least common denominator 8 clears all four fractions at once and leaves $6x+4=5x+24$. Clearing denominators before collecting like terms is what keeps the fractional coefficients from having to be subtracted.

22 **Answer: B**

1. Expand the left side: $2kx-6=6x+5-k$.
2. For no solution the coefficients must match: $2k=6$, so $k=3$.
3. Check: the constants become -6 and 2, which differ, so there is no solution.

No solution requires the variable terms to be identical while the constants differ. Expanding gives $2kx-6$, so $2k=6$ and $k=3$; with $k=3$ the constants read -6 and 2, which disagree, so the equation is indeed impossible. Matching the constants instead of the coefficients is the trap.

Topic 02 — Linear Functions

Module 1

1 **Answer: C**

1. Write $m=\frac{y_2-y_1}{x_2-x_1}$.
2. Substitute: $m=\frac{11-3}{6-2}$.
3. Simplify: $m=\frac{8}{4}=2$.

Slope is the change in y divided by the change in x, taken in the same order for both coordinates: $\frac{11-3}{6-2}=\frac{8}{4}=2$. Keeping the same point first in both differences is what guarantees the sign is right.

2 Answer: C

1. Substitute $x = 4$: $f(4) = 3(4) - 5$.
2. Multiply first: $3(4) = 12$.
3. Subtract: $12 - 5 = 7$.

Function notation means substitute: $f(4)$ asks for the output when the input is 4, so replace every x with 4 and evaluate $3(4) - 5 = 7$.

3 Answer: 4

1. Write $7x + 2 = 30$.
2. Subtract 2 from both sides: $7x = 28$.
3. Divide by 7: $x = 4$.

Here the output is known and the input is missing, so set the rule equal to 30 and undo the operations in reverse order: subtract 2, then divide by 7.

4 Answer: B

1. Compare $y = -3x + 8$ with $y = mx + b$.
2. The coefficient of x is -3.
3. So the slope is -3 (the 8 is the y-intercept).

An equation written as $y = mx + b$ shows the slope and the y-intercept directly: m is the coefficient of x, including its sign, and b is the constant. Here $m = -3$ and $b = 8$.

5 Answer: D

1. Evaluate the model at $h = 0$: $C(0) = 45(0) + 60 = 60$.
2. Zero hours of work still costs 60 dollars.
3. So 60 is the fixed charge, and 45 is the hourly rate.

The constant term is the output when the input is zero: $C(0) = 60$. Zero hours of work still costs 60 dollars, so 60 is the fixed fee, not a rate. The rate is the coefficient 45.

6 Answer: 1

1. Find the rate of change: $9 - 5 = 4$ for each increase of 1 in x.
2. Step backwards from $x = 1$ to $x = 0$: subtract 4.
3. $f(0) = 5 - 4 = 1$.

The inputs go up by 1 each time and the outputs go up by 4 each time, so the constant rate of change is 4. Stepping one unit backwards from $f(1) = 5$ subtracts 4, giving $f(0) = 1$.

7 Answer: A

1. Write the general form $f(x) = mx + b$.
2. $f(0) = b$, so $b = -2$.
3. The slope is $m = 5$, giving $f(x) = 5x - 2$.

For a linear function $f(x) = mx + b$, the value $f(0)$ is exactly b, because the mx term vanishes. So $b = -2$ and $m = 5$.

8 Answer: B

1. Apply $m = \frac{y_2-y_1}{x_2-x_1}$ with $(-4,9)$ first.
2. Numerator: $4-9=-5$. Denominator: $6-(-4)=10$.
3. Simplify: $m=-\frac{5}{10}=-\frac{1}{2}$.

Subtracting in a consistent order gives $\frac{4-9}{6-(-4)} = \frac{-5}{10} = -\frac{1}{2}$. The double negative in the denominator is where sign errors happen: $6-(-4)=10$, not 2.

9 Answer: 29

1. Rate of change: $\frac{17-5}{6-2}=3$.
2. From $x=6$ to $x=10$ is 4 more units.
3. $f(10)=17+3(4)=29$.

A linear function changes by the same amount for every equal step in the input. From $x=2$ to $x=6$ the output rises 12 over 4 units, a rate of 3 per unit, so the next four units add another 12.

10 Answer: A

1. Compare with $V(t)=V_0+mt$: the coefficient of t is -12.
2. A negative coefficient means the output decreases as t grows.
3. So the tank loses 12 gallons every minute.

The coefficient of the input variable is always a rate, and the minus sign in front of it makes that rate a decrease. Each additional minute subtracts 12 gallons.

11 Answer: A

1. The \$4 charge happens once per class: $4n$.
2. The \$25 fee happens once, no matter how many classes: $+25$.
3. Total cost is $4n+25$.

The quantity that repeats gets multiplied by the count, and the quantity charged once is added on its own. Only the per-class charge scales with n.

12 Answer: B

1. Read two lattice points from the graph: $(0,3)$ and $(4,0)$.
2. Slope $=\frac{0-3}{4-0}$.
3. Simplify: $-\frac{3}{4}$.

Pick the two points where the line meets the axes, $(0,3)$ and $(4,0)$. The line falls as it goes right, so the slope must be negative: $\frac{0-3}{4-0}=-\frac{3}{4}$.

13 Answer: 7

1. Rate of change: $\frac{8-14}{3-0}=-2$.
2. So $f(x)=14-2x$.
3. Set $14-2x=0$, giving $x=7$.

The outputs drop by 6 for every increase of 3 in x, a rate of -2 per unit. Starting from $f(6)=2$, one more unit of x removes 2 and lands exactly on zero.

14 **Answer: A**

1. Set $\frac{3}{4}x + 6 = 0$.
2. Subtract 6: $\frac{3}{4}x = -6$.
3. Multiply by $\frac{4}{3}$: $x = -8$.

Setting the output to zero locates the x-intercept. Undo the $+6$ first, then undo the multiplication by $\frac{3}{4}$ by multiplying by its reciprocal $\frac{4}{3}$.

15 **Answer: C**

1. Change in temperature: $56 - 68 = -12$ degrees.
2. Change in time: $8 - 2 = 6$ hours.
3. Rate: $\frac{-12}{6} = -2$ degrees per hour.

A rate of change is a difference of outputs divided by the matching difference of inputs, so the 12-degree drop must be spread over the 6 hours in which it happened.

16 **Answer: B**

1. Slope: $a = \frac{16-4}{5-1} = 3$.
2. Use $f(1) = 4$: $3(1) + b = 4$, so $b = 1$.
3. $f(0) = b = 1$.

Two points determine a and b. The slope a comes from the two given values, and then $b = f(0)$ is found by stepping back to $x = 0$; b is not the same as either given output.

17 **Answer: 5**

1. Write $\frac{17-k}{6-2} = 3$.
2. Multiply both sides by 4: $17 - k = 12$.
3. Solve: $k = 5$.

The slope formula still applies when one coordinate is unknown; it turns into an equation in k. Moving 4 units right raises y by $3(4) = 12$, so the earlier point must sit 12 lower.

18 **Answer: D**

1. Identify the coefficient of the input: 250.
2. The input m counts months, so the coefficient is a per-month rate.
3. The service gains 250 subscribers each month.

Because m is measured in months, the coefficient of m carries the units *subscribers per month*. The constant 3200 is the starting count, not a rate.

19 **Answer: C**

1. Subtract $4x$: $-6y = -4x + 18$.
2. Divide every term by -6: $y = \frac{2}{3}x - 3$.
3. The slope is $\frac{2}{3}$.

Standard form hides the slope, so solve for y. Dividing by -6 flips both signs on the right-hand side, which is exactly where the classic sign error occurs.

20 **Answer: 16**

1. Slope: $\frac{2-10}{7-3} = -2$.
2. Use $g(3) = 10$: $-2(3) + b = 10$, so $b = 16$.
3. $g(0) = b = 16$.

The rate of change is negative here, so moving left from $x = 3$ to $x = 0$ increases the output. Three steps of $+2$ are added to 10.

21 **Answer: C**

1. Substitute: $f(c) = 5c - 3$.
2. Set $5c - 3 = 2c + 9$.
3. Subtract $2c$ and add 3: $3c = 12$, so $c = 4$.

Substituting the unknown input into the rule gives $5c - 3$, and setting that equal to the given expression turns function notation into an ordinary linear equation.

22 **Answer: D**

1. $h(t)$ measures the candle's height in centimeters.
2. $h(t) = 0$ means the height has reached zero.
3. The solution is the time at which the candle is completely burned.

Setting the output of a model to zero asks when the quantity being measured runs out. Here the quantity is height, so $h(t) = 0$ is the moment the candle has no height left.

Module 2

1 **Answer: D**

1. Substitute $x = -3$: $f(-3) = -2(-3) + 11$.
2. Multiply: $-2(-3) = 6$.
3. Add: $6 + 11 = 17$.

A negative input multiplied by a negative coefficient gives a positive product: $-2(-3) = 6$. Losing either minus sign changes the answer.

2 **Answer: -2**

1. Numerator: $-7 - 5 = -12$.
2. Denominator: $4 - (-2) = 6$.
3. Slope: $\frac{-12}{6} = -2$.

Both differences must be taken in the same order. Going from the first point to the second, y falls by 12 while x rises by 6, so the line drops 2 units for every unit right.

3 **Answer: B**

1. Change in output: $12 - 3 = 9$.
2. Change in input: $5 - 2 = 3$.
3. Rate of change: $\frac{9}{3} = 3$.

The inputs are spaced 3 apart, not 1, so the jump of 9 in the outputs must be divided by 3 before it can be called a rate.

4 Answer: D

1. Substitute $t = 0$: $d(0) = 45(0) + 30 = 30$.
2. $t = 0$ is the moment the drive starts.
3. So the car began 30 miles from the rest stop.

Evaluating at $t = 0$ gives $d(0) = 30$, the distance at the instant the driving begins. The speed is the coefficient 45, in miles per hour.

5 Answer: D

1. Set $\frac{1}{2}x - 7 = -3$.
2. Add 7: $\frac{1}{2}x = 4$.
3. Multiply by 2: $x = 8$.

Adding 7 to both sides isolates $\frac{1}{2}x$; undoing a multiplication by $\frac{1}{2}$ means multiplying by 2, not dividing by 2.

6 Answer: B

1. Set $x = 0$: $2(0) + 5y = 20$.
2. So $5y = 20$.
3. Divide by 5: $y = 4$.

Every point on the y-axis has $x = 0$, so substituting $x = 0$ isolates the y-intercept. The constant 20 is not the intercept because the y-term has a coefficient of 5.

7 Answer: A

1. Write $f(x) = 4x + b$.
2. Substitute $(3, 5)$: $5 = 4(3) + b = 12 + b$.
3. So $b = -7$ and $f(x) = 4x - 7$.

The point $(3, 5)$ is not the y-intercept, so 5 cannot be used as b. Substituting the point into $f(x) = 4x + b$ gives $5 = 12 + b$, hence $b = -7$.

8 Answer: B

1. Substitute: $a(4) + 7 = -5$.
2. Subtract 7: $4a = -12$.
3. Divide by 4: $a = -3$.

Substituting the input and the output leaves a single equation in a. After subtracting 7 the product $4a$ equals -12, so a must still be divided by 4.

9 Answer: 4

1. Slope: $\frac{-5-13}{4-(-2)} = \frac{-18}{6} = -3$.
2. From $x = -2$ to $x = 1$ is 3 units to the right.
3. $f(1) = 13 - 3(3) = 4$.

The rate of change is -3 per unit, so each step to the right lowers the output by 3. From $x = -2$ to $x = 1$ is three steps, a total drop of 9.

10 **Answer: C**

1. Read two points: $(-2, 0)$ and $(0, 6)$.
2. Slope $= \frac{6-0}{0-(-2)} = 3$, and $f(0) = 6$, so $f(x) = 3x + 6$.
3. $f(3) = 3(3) + 6 = 15$.

Read the intercepts $(-2, 0)$ and $(0, 6)$ off the graph: the slope is $\frac{6-0}{0-(-2)} = 3$ and the y-intercept is 6, so $f(x) = 3x + 6$ and $f(3) = 15$.

11 **Answer: A**

1. The coefficient of t is 25, a rate in liters per minute.
2. The constant 150 is the amount already in the pool at $t = 0$.
3. So the pool gains 25 liters every minute.

The coefficient of t is the amount added per minute, and the constant is the amount already present at $t = 0$. Only the statement about 25 liters per minute describes the rate correctly.

12 **Answer: 6**

1. Write $\frac{0-9}{t-0} = -\frac{3}{2}$.
2. Cross-multiply: $-18 = -3t$.
3. Solve: $t = 6$.

The line starts at height 9 and must fall to 0; at $\frac{3}{2}$ units of drop per unit right, that takes $9 \div \frac{3}{2} = 6$ units.

13 **Answer: C**

1. Rate: $\frac{15-21}{5-2} = -2$ centimeters per hour.
2. Step back from $t = 2$ to $t = 0$: add $2(2) = 4$.
3. Initial height: $21 + 4 = 25$.

The height falls 6 centimeters in 3 hours, a rate of -2 per hour. Going backwards from $t = 2$ to $t = 0$ reverses the sign of the change, so 2 hours of burning are added back.

14 **Answer: B**

1. Substitute $x = 12$: $3(12) - 4y = 24$.
2. So $36 - 4y = 24$, giving $-4y = -12$.
3. Divide by -4: $y = 3$.

Substituting first keeps the arithmetic simple. Dividing 12 by 4 before subtracting the constant is the usual slip; the whole right-hand side must be divided.

15 **Answer: C**

1. Replace the input: $f(x + 3) = 2(x + 3) - 5$.
2. Distribute: $2x + 6 - 5$.
3. Combine: $2x + 1$.

The notation $f(x+3)$ means the entire quantity $x+3$ is used as the input, so it replaces x inside the rule before anything else happens: $2(x + 3) - 5 = 2x + 1$.

16 **Answer: A**

1. The stated difference means the slope is -4.
2. From $x = 3$ to $x = 6$ is 3 unit steps.
3. $f(6) = 8 + 3(-4) = -4$.

The condition $f(x+1) - f(x) = -4$ is a statement of the slope: each unit step right lowers the output by 4. Three steps from $x = 3$ to $x = 6$ lower it by 12.

17 **Answer: 1**

1. Slope: $m = \frac{5-(-1)}{6-(-3)} = \frac{6}{9} = \frac{2}{3}$.
2. Use $f(-3) = -1$: $\frac{2}{3}(-3) + b = -1$, so $-2 + b = -1$.
3. Therefore $b = 1$.

Once the slope $\frac{2}{3}$ is known, b follows from either point; b is the output at $x = 0$, which lies between the two given inputs, so its value must lie between -1 and 5.

18 **Answer: D**

1. At $t = 0$ (the year 2010), $P(0) = 1500$.
2. The coefficient of t is -45, a rate per year.
3. So the colony starts at 1500 birds and loses 45 birds per year.

The constant term is the value at $t = 0$, which the definition of t fixes as the year 2010, and the coefficient of t is a yearly rate whose minus sign makes it a decrease.

19 **Answer: C**

1. $f(4) = 12 + c$ and $f(1) = 3 + c$.
2. Set $12 + c = 2(3 + c) = 6 + 2c$.
3. Subtract c and 6: $c = 6$.

Both sides must be written in terms of c before comparing them; the factor of 2 multiplies the whole of $f(1)$, including the constant, which is the step most students skip.

20 **Answer: 32**

1. Set $35m + 80 = 1200$.
2. Subtract 80: $35m = 1120$.
3. Divide by 35: $m = 32$.

The one-time \$80 fee must be removed before dividing, because only the \$35 portion repeats each month; dividing 1200 by 35 directly would charge the join fee every month.

21 **Answer: C**

1. Slope: $\frac{8-23}{1-(-4)} = \frac{-15}{5} = -3$.
2. Use $f(1) = 8$: $-3(1) + b = 8$, so $b = 11$.
3. $f(0) = b = 11$.

The inputs are unevenly spaced, so the rate must be computed as a quotient rather than read off as a repeated difference: $\frac{8-23}{1-(-4)} = -3$. Then $f(0) = f(1) + 3 = 11$.

22 **Answer: B**

1. The slope is $\frac{3k-k}{4-0} = \frac{k}{2}$, so $f(x) = \frac{k}{2}x + k$.
2. Substitute $x = 2$: $f(2) = k + k = 2k$.
3. Set $2k = 10$, so $k = 5$.

Because f is linear and $x = 2$ is exactly halfway between 0 and 4, the output $f(2)$ is the average of $f(0)$ and $f(4)$, so $\frac{k+3k}{2} = 2k = 10$.

Topic 03 — Linear Equations in Two Variables

Module 1

1 **Answer: A**

1. Compare $y = -3x + 7$ with $y = mx + b$.
2. The coefficient of x is $m = -3$.

In slope-intercept form $y = mx + b$ the coefficient of x is the slope, so reading the equation is the whole job: $m = -3$. The number 7 is the y-intercept, not the slope.

2 **Answer: B**

1. Set $x = 0$ in $y = \frac{1}{2}x - 6$.
2. $y = \frac{1}{2}(0) - 6 = -6$.

A line crosses the y-axis where $x = 0$, and substituting $x = 0$ into $y = mx + b$ leaves $y = b$. Here $b = -6$, so the crossing point is $(0, -6)$.

3 **Answer: 19**

1. The point $(0, 4)$ gives $b = 4$, so $y = 3x + 4$.
2. Substitute $x = 5$: $y = 3(5) + 4 = 19$.

The point $(0, 4)$ is the y-intercept, so the line is $y = 3x + 4$ with no extra work. Substituting $x = 5$ gives $y = 19$.

4 **Answer: C**

1. Subtract $2x$ from both sides of $2x + y = 8$.
2. $y = -2x + 8$.

Solving for y means moving the x-term to the other side, which changes its sign: $y = -2x + 8$. Forgetting that sign change is the most common error on this conversion.

5 **Answer: D**

1. Set $y = 0$: $3x + 4(0) = 12$.
2. $3x = 12$, so $x = 4$.

Every point on the x-axis has $y = 0$, so set $y = 0$ and solve. The answer 3 belongs to the y-intercept; swapping the two intercepts is the classic slip here.

6 **Answer: 3**

1. Use $m = \frac{y_2-y_1}{x_2-x_1}$.
2. $m = \frac{16-4}{3-(-1)} = \frac{12}{4} = 3$.

Slope is the change in y divided by the change in x, taken in the same order for both coordinates: $\frac{16-4}{3-(-1)} = \frac{12}{4} = 3$. Subtracting the negative x-coordinate is what makes the denominator 4, not 2.

7 **Answer: B**

1. Parallel lines share the same slope, here $m = 4$.
2. $y = 4x + 5$ has slope 4 and a different intercept.

Parallel lines have equal slopes and different y-intercepts. Only $y = 4x + 5$ keeps the slope 4; the other choices change the slope, which tilts the line instead of shifting it.

8 **Answer: A**

1. $-5y = -3x + 15$.
2. Divide by -5: $y = \frac{3}{5}x - 3$, so $m = \frac{3}{5}$.

Rewrite as $y = \frac{3}{5}x - 3$. Dividing by the negative coefficient -5 flips both signs on the right, which is exactly where sign errors appear.

9 **Answer: C**

1. The point $(0, -2)$ gives $b = -2$.
2. With $m = \frac{5}{2}$, the line is $y = \frac{5}{2}x - 2$.

Because the given point has $x = 0$, it is the y-intercept, so $b = -2$ can be written down immediately and the equation is $y = \frac{5}{2}x - 2$.

10 **Answer: C**

1. $3y = -2x + 9$, so $y = -\frac{2}{3}x + 3$ and $m = -\frac{2}{3}$.
2. The negative reciprocal of $-\frac{2}{3}$ is $\frac{3}{2}$.

The given line has slope $-\frac{2}{3}$. A perpendicular slope is the negative reciprocal: flip to $-\frac{3}{2}$ and change the sign to $\frac{3}{2}$. Flipping without changing the sign is the standard trap.

11 **Answer: A**

1. $m = \frac{10-4}{3-1} = 3$.
2. $4 = 3(1) + b$, so $b = 1$ and $y = 3x + 1$.

Find the slope first, then use one point to pin down b. With $m = 3$ and the point $(1, 4)$, $4 = 3(1) + b$ gives $b = 1$, so the line is $y = 3x + 1$.

12 **Answer: 12**

1. Set $y = 0$: $0 = \frac{2}{3}x - 8$.
2. $\frac{2}{3}x = 8$, so $x = 12$.

Set $y = 0$ and solve. Multiplying by the reciprocal $\frac{3}{2}$ is faster than clearing the fraction first: $x = 8 \cdot \frac{3}{2} = 12$.

13 **Answer: D**

1. Multiply by 4: $4y = -3x + 8$.
2. Add $3x$: $3x + 4y = 8$.

Clear the fraction by multiplying every term by 4, then collect the variables on one side: $4y = -3x + 8$ becomes $3x + 4y = 8$. The x-term arrives on the left as $+3x$, not $-3x$.

14 **Answer: B**

1. Read the y-intercept from the graph: $b = -1$.
2. From $(0, -1)$ to $(2, 3)$, $m = \frac{3-(-1)}{2-0} = 2$.
3. So $y = 2x - 1$.

The line crosses the y-axis at -1, so $b = -1$. From $(0, -1)$ to $(2, 3)$ the line rises 4 while running 2, so $m = 2$ and the equation is $y = 2x - 1$.

15 **Answer: 15**

1. Line k has slope -2, so $y = -2x + b$.
2. $7 = -2(4) + b$, so $b = 15$.

Parallel means the slope is copied, so line k is $y = -2x + b$. Substituting the point gives $7 = -8 + b$, so $b = 15$.

16 **Answer: D**

1. The negative reciprocal of $\frac{3}{4}$ is $-\frac{4}{3}$.
2. $5 = -\frac{4}{3}(-3) + b = 4 + b$, so $b = 1$.
3. Line n is $y = -\frac{4}{3}x + 1$.

Perpendicular to slope $\frac{3}{4}$ means slope $-\frac{4}{3}$. Then $5 = -\frac{4}{3}(-3) + b = 4 + b$, so $b = 1$. Substituting a negative x into a negative slope produces a positive product, and losing that is what makes $b = 9$ look right.

17 **Answer: A**

1. $kx + 6y = 12$ has slope $-\frac{k}{6}$; $2x - 3y = 5$ has slope $\frac{2}{3}$.
2. Set the product equal to -1: $-\frac{2k}{18} = -1$.
3. $k = 9$.

The two slopes are $-\frac{k}{6}$ and $\frac{2}{3}$, and perpendicular lines have slopes whose product is -1. Solving $-\frac{k}{6} \cdot \frac{2}{3} = -1$ gives $k = 9$.

18 **Answer: 2**

1. $ky = -4x + 20$, so the slope is $-\frac{4}{k}$.
2. Set $-\frac{4}{k} = -2$.
3. $k = 2$.

Writing $4x + ky = 20$ as $y = -\frac{4}{k}x + \frac{20}{k}$ shows its slope is $-\frac{4}{k}$. Setting $-\frac{4}{k} = -2$ gives $k = 2$.

19 **Answer: C**

1. The intercepts are the points $(6, 0)$ and $(0, -4)$.
2. $m = \frac{-4-0}{0-6} = \frac{2}{3}$, so $y = \frac{2}{3}x - 4$.
3. Multiply by 3 and rearrange: $2x - 3y = 12$.

The intercepts are the points $(6, 0)$ and $(0, -4)$, so the slope is $\frac{-4-0}{0-6} = \frac{2}{3}$ and the line is $y = \frac{2}{3}x - 4$. Clearing the fraction gives $2x - 3y = 12$.

20 **Answer: B**

1. $m = \frac{1-3}{4-0} = -\frac{1}{2}$, so line ℓ has slope 2.
2. $-1 = 2(2) + b$ gives $b = -5$, so $y = 2x - 5$.
3. Set $y = 0$: $x = \frac{5}{2}$.

The line through $(0, 3)$ and $(4, 1)$ has slope $-\frac{1}{2}$, so ℓ has slope 2 and equation $y = 2x - 5$. Its x-intercept comes from $0 = 2x - 5$, giving $x = \frac{5}{2}$; the value -5 is the y-intercept, not the x-intercept.

21 **Answer: A**

1. $m = \frac{16-7}{5-2} = 3$.
2. $7 = 3(2) + b$, so $b = 1$.
3. $m + b = 3 + 1 = 4$.

The slope is $\frac{16-7}{5-2} = 3$, and $7 = 3(2) + b$ gives $b = 1$, so $m + b = 4$. Reversing the slope fraction to $\frac{1}{3}$ is what produces the value $\frac{20}{3}$.

22 **Answer: 11**

1. The negative reciprocal of $\frac{2}{5}$ is $-\frac{5}{2}$.
2. $1 = -\frac{5}{2}(4) + b = -10 + b$.
3. $b = 11$.

Line m has slope $-\frac{5}{2}$, the negative reciprocal of $\frac{2}{5}$. Then $1 = -\frac{5}{2}(4) + b = -10 + b$, so $b = 11$.

Module 2

1 **Answer: B**

1. $m = \frac{-4-4}{3-(-1)}$.
2. $m = \frac{-8}{4} = -2$.

Both differences must be taken in the same order: $\frac{-4-4}{3-(-1)} = \frac{-8}{4} = -2$. Subtracting in mixed order flips the sign of the answer.

2 **Answer: -4**

1. Set $x = 0$: $-5y = 20$.
2. $y = -4$.

Set $x = 0$, because every point on the y-axis has $x = 0$. Then $-5y = 20$ and $y = -4$.

3 Answer: C

1. The given slope is $-\frac{1}{2}$.
2. Its negative reciprocal is 2, so $y = 2x - 1$ works.

The negative reciprocal of $-\frac{1}{2}$ is 2: flip the fraction and change the sign. Choice A only changes the sign, and choice D only flips, so neither is perpendicular.

4 Answer: D

1. $-2y = -6x + 18$.
2. Divide by -2: $y = 3x - 9$.

Isolating y requires dividing by -2, which changes the sign of every term on the right: $-2y = -6x+18$ becomes $y = 3x - 9$.

5 Answer: A

1. $(0, 5)$ gives $b = 5$.
2. With $m = -\frac{3}{4}$, $y = -\frac{3}{4}x + 5$.

The point $(0, 5)$ is on the y-axis, so it is the y-intercept and $b = 5$. With $m = -\frac{3}{4}$ the equation is $y = -\frac{3}{4}x + 5$.

6 Answer: B

1. Substitute $(2, 3)$: $3 = 4(2) + c$.
2. $3 = 8 + c$, so $c = -5$.

A point on the graph makes the equation true, so $3 = 4(2) + c$. Then $c = 3 - 8 = -5$; adding instead of subtracting gives the trap value 11.

7 Answer: C

1. The y-intercept is 3, so $b = 3$.
2. From $(0, 3)$ to $(2, 0)$, $m = \frac{0-3}{2-0} = -\frac{3}{2}$.
3. The line is $y = -\frac{3}{2}x + 3$.

The line meets the y-axis at 3 and the x-axis at 2, so it falls 3 units while running 2 units: $m = -\frac{3}{2}$ and $b = 3$.

8 Answer: D

1. The x-intercept is the point $(8, 0)$.
2. $m = \frac{9-0}{2-8} = -\frac{3}{2}$.
3. $9 = -\frac{3}{2}(2) + b$, so $b = 12$.

The x-intercept is the point $(8, 0)$, so the slope is $\frac{9-0}{2-8} = -\frac{3}{2}$. Then $9 = -\frac{3}{2}(2) + b = -3 + b$, so $b = 12$.

9 Answer: $-\frac{2}{3}$

1. $3y = -2x + 12$, so $y = -\frac{2}{3}x + 4$.
2. Parallel lines share the slope, so $m = -\frac{2}{3}$.

Parallel graphs have equal slopes, and $2x + 3y = 12$ rearranges to $y = -\frac{2}{3}x + 4$, so $m = -\frac{2}{3}$. The different constant terms guarantee the lines really are distinct.

10 **Answer: A**

1. $2 = \frac{1}{2}(-4) + b = -2 + b$, so $b = 4$.
2. Set $y = 0$: $0 = \frac{1}{2}x + 4$.
3. $x = -8$.

From $2 = \frac{1}{2}(-4) + b$ we get $b = 4$, so ℓ is $y = \frac{1}{2}x + 4$. Setting $y = 0$ gives $x = -8$; the value 4 is the y-intercept, not the x-intercept.

11 **Answer: C**

1. Subtract 60: $C - 60 = 25m$.
2. Divide by 25: $m = \frac{C-60}{25}$.

Undo the operations in reverse order: subtract the joining fee first, then divide by the monthly rate. Dividing before subtracting produces the choice $m = \frac{C}{25} - 60$.

12 **Answer: B**

1. $\frac{9-(-3)}{k-1} = 4$, so $\frac{12}{k-1} = 4$.
2. $k - 1 = 3$, so $k = 4$.

The slope condition is $\frac{9-(-3)}{k-1} = 4$, so $12 = 4(k-1)$ and $k = 4$. Reading the rise as $9 - 3 = 6$ instead of $9 - (-3) = 12$ is the usual sign slip.

13 **Answer: D**

1. $-6y = -4x + 15$, so $y = \frac{2}{3}x - \frac{5}{2}$.
2. The negative reciprocal of $\frac{2}{3}$ is $-\frac{3}{2}$.
3. So $y = -\frac{3}{2}x + 1$ is perpendicular.

Rearranging gives $y = \frac{2}{3}x - \frac{5}{2}$, so the slope is $\frac{2}{3}$ and a perpendicular line has slope $-\frac{3}{2}$. Choice B keeps the size of the slope and only changes its sign, which is not enough.

14 **Answer: C**

1. From the graph the line passes through $(0, -6)$ and $(3, 0)$.
2. $m = \frac{0-(-6)}{3-0} = 2$, so $y = 2x - 6$.
3. Rearranged: $2x - y = 6$.

The line passes through $(0, -6)$ and $(3, 0)$, so $m = 2$ and $y = 2x - 6$. Moving the x-term left gives $-2x + y = -6$, or equivalently $2x - y = 6$.

15 **Answer: 6**

1. Set $y = 0$: $0 = -\frac{5}{3}x + 10$.
2. $\frac{5}{3}x = 10$, so $x = 6$.

Set $y = 0$: $\frac{5}{3}x = 10$, so $x = 10 \cdot \frac{3}{5} = 6$. A negative slope with a positive intercept must cross the x-axis to the right of the origin, so a positive answer is expected.

16 Answer: A

1. $m = \frac{8-2}{5-1} = \frac{3}{2}$.
2. $-4y = -ax + 12$, so $y = \frac{a}{4}x - 3$ and the slope is $\frac{a}{4}$.
3. $\frac{a}{4} = \frac{3}{2}$, so $a = 6$.

The two-point slope is $\frac{8-2}{5-1} = \frac{3}{2}$, and $ax - 4y = 12$ has slope $\frac{a}{4}$. Setting $\frac{a}{4} = \frac{3}{2}$ gives $a = 6$; note that dividing by -4 makes the slope positive, so a negative a cannot work.

17 Answer: -1

1. The negative reciprocal of $\frac{3}{5}$ is $-\frac{5}{3}$.
2. From $x = 3$ to $x = 6$ the run is 3, so the rise is $-\frac{5}{3}(3) = -5$.
3. $y = 4 - 5 = -1$.

Line m has slope $-\frac{5}{3}$. Moving from $x = 3$ to $x = 6$ is a run of 3, so y changes by $-\frac{5}{3} \cdot 3 = -5$, giving $y = 4 - 5 = -1$; no intercept is needed.

18 Answer: D

1. Midpoint: $(1, 1)$.
2. $m_\ell = \frac{-5-7}{4-(-2)} = -2$, so line m has slope $\frac{1}{2}$.
3. $1 = \frac{1}{2}(1) + b$, so $b = \frac{1}{2}$ and $y = \frac{1}{2}x + \frac{1}{2}$.

The midpoint is $\left(\frac{-2+4}{2}, \frac{7+(-5)}{2}\right) = (1, 1)$ and line ℓ has slope -2, so line m has slope $\frac{1}{2}$. Then $1 = \frac{1}{2}(1) + b$ gives $b = \frac{1}{2}$.

19 Answer: B

1. Set $y = 0$ in $2x + y = 16$: $x = 8$, so the point is $(8, 0)$.
2. Substitute into $y = mx + 4$: $0 = 8m + 4$.
3. $m = -\frac{1}{2}$.

The graph of $2x + y = 16$ meets the x-axis at $(8, 0)$. Substituting that point gives $0 = 8m + 4$, so $m = -\frac{1}{2}$; using the y-intercept $(0, 16)$ by mistake makes the equation unsolvable, which is a useful signal that the wrong intercept was taken.

20 Answer: C

1. $m = \frac{12-0}{0-a} = -\frac{12}{a}$.
2. Set $-\frac{12}{a} = -3$.
3. $a = 4$.

The slope between the two intercepts is $\frac{12-0}{0-a} = -\frac{12}{a}$. Setting $-\frac{12}{a} = -3$ gives $a = 4$, and the positive answer matches a falling line that crosses the x-axis to the right of the origin.

21 Answer: $\frac{9}{4}$

1. $ky = -3x + 12$, so the slope is $-\frac{3}{k}$.
2. The required slope is $-\frac{4}{3}$.
3. $-\frac{3}{k} = -\frac{4}{3}$ gives $k = \frac{9}{4}$.

The graph of $3x + ky = 12$ has slope $-\frac{3}{k}$, and perpendicularity to slope $\frac{3}{4}$ requires slope $-\frac{4}{3}$. Solving $-\frac{3}{k} = -\frac{4}{3}$ gives $k = \frac{9}{4}$.

22 **Answer: D**

1. $m = \frac{280-145}{5-2} = 45$ dollars per hour.
2. $145 = 45(2) + b$, so $b = 55$.
3. The one-time fee is the value at $h = 0$, namely 55 dollars.

The hourly rate is the slope, $\frac{280-145}{5-2} = 45$ dollars per hour, and the service fee is the value of C when $h = 0$, that is the C-intercept: $145 - 2(45) = 55$. Reporting 45 answers a different question, the hourly rate.

Topic 04 — Systems of Two Linear Equations in Two Variables

Module 1

1 **Answer: A**

1. Substitute $y = 2x$ into $3x + y = 20$: $3x + 2x = 20$.
2. Combine like terms: $5x = 20$, so $x = 4$.

The first equation already gives y in terms of x, so substitution is the fastest route: replacing y by $2x$ turns a two-variable system into a one-variable equation. Once $x = 4$ is known, $y = 8$ follows, but the question asked only for x.

2 **Answer: C**

1. Add the two equations: $(x + y) + (x - y) = 10 + 4$.
2. The y terms cancel: $2x = 14$, so $x = 7$.

When the same variable appears with opposite coefficients, adding the equations eliminates it in one line. The y terms cancel, leaving $2x = 14$. The value 3 is y, the variable the question did not ask about.

3 **Answer: 7**

1. Subtract: $(2x + y) - (x + y) = 17 - 12$, so $x = 5$.
2. Substitute into $x + y = 12$: $5 + y = 12$, so $y = 7$.

Subtracting the first equation from the second removes y immediately, because both equations carry exactly one y. That gives $x = 5$, and back-substitution gives $y = 7$.

4 **Answer: C**

1. Set the two expressions for y equal: $x + 3 = 2x - 1$.
2. Solve: $4 = x$.
3. Substitute into $y = x + 3$: $y = 7$.

Two expressions equal to the same y must be equal to each other. Setting them equal gives $x = 4$; substituting back into either equation gives $y = 7$. Stopping at $x = 4$ is the standard error here.

5 **Answer: D**

1. Substitute $y = x + 2$ into $4x + y = 17$: $4x + x + 2 = 17$.
2. Solve: $5x = 15$, so $x = 3$ and $y = 5$.
3. Therefore $x + y = 8$.

The SAT often asks for a combination rather than for a single variable. Solve the system fully, then build what was asked: $x = 3$ and $y = 5$ give $x + y = 8$. Answering 3 or 5 means answering a question that was not posed.

6 **Answer: B**

1. Add the equations: $5x = 15$, so $x = 3$.
2. Substitute into $2x + y = 7$: $6 + y = 7$, so $y = 1$.

The coefficients of y are $+1$ and -1, so adding the equations kills y at once and gives $x = 3$. The question, though, asks for y: substitute back to get $y = 7 - 6 = 1$.

7 **Answer: 10**

1. Let a be adult tickets and c child tickets: $a + c = 24$ and $12a + 8c = 232$.
2. Substitute $c = 24 - a$: $12a + 8(24 - a) = 232$.
3. Simplify: $4a + 192 = 232$, so $a = 10$.

Two unknowns need two equations: one counting tickets and one counting dollars. The count equation is the easy one to substitute with, since $c = 24 - a$ has no fractions.

8 **Answer: B**

1. Multiply the first equation by 2: $6x + 4y = 32$.
2. Add to $5x - 4y = -10$: $11x = 22$, so $x = 2$.
3. Substitute into $3x + 2y = 16$: $6 + 2y = 16$, so $y = 5$.

Neither variable cancels as written, so scale first: doubling the top equation makes the y coefficients $+4$ and -4. Adding then gives $x = 2$, and substituting back gives $y = 5$.

9 **Answer: 5**

1. Add the equations: $6x = 24$, so $x = 4$.
2. Substitute into $2x - 3y = 5$: $8 - 3y = 5$, so $y = 1$.
3. Therefore $x + y = 5$.

The y coefficients are already opposites, so no scaling is needed: adding gives $6x = 24$. Solve for both variables and then form the requested sum.

10 **Answer: A**

1. Add the equations: $4x = 12$, so $x = 3$.
2. Substitute into $x - 2y = -7$: $3 - 2y = -7$, so $y = 5$.
3. Therefore $2x - y = 6 - 5 = 1$.

Adding eliminates y and gives $x = 3$; back-substitution gives $y = 5$. The requested expression is $2x - y = 6 - 5 = 1$. Choice D is what you get if you compute $2x + y$ instead.

11 **Answer: C**

1. For one line to be a multiple of the other, compare y terms: $6 \div 3 = 2$.
2. Check the constants: $10 \div 5 = 2$, the same multiplier.
3. So the x terms must also match: $k = 2 \cdot 2 = 4$.

Infinitely many solutions means the two equations describe the same line, so one must be a constant multiple of the other. Comparing the y terms, $6 = 2 \cdot 3$, and the constants agree as well, $10 = 2 \cdot 5$; so the multiplier is 2 and $k = 2 \cdot 2 = 4$.

12 **Answer: D**

1. Compare the x terms: $6 \div 2 = 3$.
2. For parallel lines the y terms must scale the same way: $k = 3 \cdot 5 = 15$.
3. Check the constants: $3 \cdot 4 = 12 \neq 9$, so there is indeed no solution.

No solution means parallel but distinct lines: the coefficients are proportional while the constants are not. From $6 \div 2 = 3$ the multiplier is 3, so $k = 3 \cdot 5 = 15$; and since $9 \neq 3 \cdot 4$, the lines really are distinct.

13 **Answer: B**

1. Write the system: $p + n = 12$ and $2p + 5n = 39$.
2. Substitute $p = 12 - n$: $2(12 - n) + 5n = 39$.
3. Simplify: $24 + 3n = 39$, so $n = 5$.

Let p and n be the numbers of pens and notebooks. The count equation $p + n = 12$ substitutes cleanly into the cost equation. Notice the answer 7 is the number of pens: label your variables before you answer.

14 **Answer: C**

1. Combined speed: $300 \div 3 = 100$, so $r + s = 100$.
2. The faster car satisfies $r = s + 20$.
3. Substitute: $s + 20 + s = 100$, so $s = 40$ and $r = 60$.

In 3 hours the two cars together cover 300 miles, so the sum of their speeds is 100 mph. With the difference of the speeds given as 20, the sum-and-difference system solves in one step. Choice A is the slower car.

15 **Answer: 6**

1. Add the equations: $8x = 16$, so $x = 2$.
2. Substitute into $5x + 2y = 2$: $10 + 2y = 2$, so $y = -4$.
3. Therefore $x - y = 2 - (-4) = 6$.

The y terms are already opposites, so adding gives $8x = 16$ and $x = 2$. Then $y = -4$, and the requested difference is $2 - (-4) = 6$ – subtracting a negative is where this item is usually lost.

16 **Answer: A**

1. Multiply the first equation by 6: $3x + 2y = 24$.
2. From the second equation, $x = y + 3$.
3. Substitute: $3(y + 3) + 2y = 24$, so $5y = 15$ and $y = 3$.

Clear the fractions first: multiplying by the common denominator 6 turns the top equation into $3x + 2y = 24$. Then substitute $x = y + 3$ to get $5y + 9 = 24$, so $y = 3$ and $x = 6$. Choice B is x.

17 **Answer: B**

1. Add the equations: $11x + 11y = 44$.
2. Factor: $11(x + y) = 44$.
3. Divide by 11: $x + y = 4$.

When the coefficients are swapped between the two equations, adding them produces a multiple of $x+y$ directly – no need to find x and y separately. Here $11x + 11y = 44$, so $x + y = 4$. Choice D is the sum before dividing by 11.

18 **Answer: 2**

1. Set the coefficient ratios equal: $\frac{a}{3} = \frac{4}{6}$.
2. Solve: $a = 3 \cdot \frac{2}{3} = 2$.
3. Check: $\frac{10}{12} \neq \frac{2}{3}$, so the lines are parallel and distinct.

Parallel distinct lines have equal slopes. Matching the ratio of the x terms to the ratio of the y terms gives $a/3 = 4/6$, so $a = 2$; the constants 10 and 12 are not in that same ratio, which is exactly why no solution exists.

19 **Answer: D**

1. Let x and y be the liters of 10% and 30% solution: $x + y = 20$.
2. Acid: $0.10x + 0.30y = 0.25(20) = 5$.
3. Substitute $x = 20 - y$: $2 + 0.20y = 5$, so $y = 15$.

One equation tracks total volume and the other tracks total acid. The acid equation is $0.10x + 0.30y = 0.25(20) = 5$. Solving gives $x = 5$ and $y = 15$; choice A is the amount of the weak solution, the classic wrong-variable trap.

20 **Answer: D**

1. Multiply $2x - 5y = 8$ by -2: $-4x + 10y = -16$.
2. That is precisely the second equation.
3. The equations describe one line, so every point on it solves the system.

Multiplying the first equation by -2 reproduces the second equation exactly, so the two equations graph as the same line and every point on it is a solution. A system of two linear equations can never have exactly two solutions, which rules out choice C outright.

21 **Answer: 1**

1. Subtract: $(5x + 3y) - (3x + 5y) = 29 - 27$.
2. Simplify: $2x - 2y = 2$, so $2(x - y) = 2$.
3. Divide by 2: $x - y = 1$.

Subtracting the second equation from the first gives $2x - 2y = 2$, which is $2(x - y)$. The combination the question asks for appears without ever computing x or y on their own – always check whether the requested expression is a shortcut away.

22 **Answer: B**

1. Compare the x terms: $4 \div (-2) = -2$.
2. The same multiplier must work on the constants: $10 = -2c$.
3. Solve: $c = -5$.

The first equation is -2 times the left side of the second, so the two are the same line only when the constants obey the same factor: $10 = -2c$, giving $c = -5$. Forgetting the negative sign produces choice C.

Module 2

1 **Answer: C**

1. Set the expressions equal: $4x - 1 = 2x + 7$.
2. Solve: $2x = 8$, so $x = 4$.
3. Then $y = 2(4) + 7 = 15$, so $x + y = 19$.

Both equations are solved for y, so set the right sides equal: $4x - 1 = 2x + 7$ gives $x = 4$ and then $y = 15$. The question asks for the sum, 19; choices A and B are the individual coordinates.

2 **Answer: B**

1. Add the equations: $6x = 12$, so $x = 2$.
2. Substitute into $2x + 5y = 19$: $4 + 5y = 19$.
3. Solve: $5y = 15$, so $y = 3$.

The y coefficients are exact opposites, so adding the equations eliminates y and gives $x = 2$. Substituting back gives $5y = 15$, so $y = 3$; choice A is x.

3 **Answer: 16**

1. Let s and a be student and adult tickets: $s + a = 40$.
2. Money: $9s + 15a = 456$; substitute $s = 40 - a$.
3. $360 + 6a = 456$, so $a = 16$.

Substituting $s = 40 - a$ into the money equation is the efficient move: the \$9 baseline accounts for \$360, and every adult ticket adds \$6 more than a student ticket does.

4 **Answer: C**

1. Subtract the second equation from the first: $5x = 15$, so $x = 3$.
2. Substitute into $2x + 3y = 12$: $6 + 3y = 12$, so $y = 2$.
3. Therefore $2x - y = 6 - 2 = 4$.

Both equations carry $3y$, so subtracting removes y at once: $5x = 15$. With $x = 3$ and $y = 2$, the requested expression is $6 - 2 = 4$. Choice D is $2x + y$, the sign-slip answer.

5 **Answer: D**

1. Compare the x terms: $8 \div 2 = 4$.
2. Apply the same factor to the y terms: $k = 4 \cdot 5 = 20$.
3. Check the constants: $4 \cdot 7 = 28 \neq 12$, so the lines never meet.

No solution means the lines are parallel, so the coefficients scale by the same factor: $8 \div 2 = 4$ forces $k = 4 \cdot 5 = 20$. Since $12 \neq 4 \cdot 7$, the lines are distinct, confirming there is truly no solution.

6 **Answer: C**

1. Set $-2x + 9 = x - 3$.
2. Solve: $12 = 3x$, so $x = 4$ and $y = 1$.
3. Therefore $x + 2y = 4 + 2 = 6$.

An intersection point is exactly a solution of the system, so set the two expressions for y equal. That gives $x = 4$ and $y = 1$, so $x + 2y = 6$. Choice A is x alone and choice B is $x + y$.

7 Answer: D

1. Let m and s be muffins and scones: $m + s = 25$ and $3m + 4s = 87$.
2. Substitute $m = 25 - s$: $75 + s = 87$, so $s = 12$.
3. Therefore $m = 25 - 12 = 13$.

If all 25 items were muffins the total would be \$75; each scone adds \$1, so there were 12 scones and therefore 13 muffins. Choice C is the number of scones – the count the question did not ask for.

8 Answer: B

1. Multiply the first equation by 10: $5x + 2y = 26$.
2. From the second equation, $x = y + 1$.
3. Substitute: $5(y + 1) + 2y = 26$, so $7y = 21$ and $y = 3$.

Multiply the first equation by 10 to clear decimals: $5x + 2y = 26$. Substituting $x = y + 1$ gives $7y + 5 = 26$, so $y = 3$ and $x = 4$. Choice C is x; choice D is $x + y$.

9 Answer: 3

1. Double the second equation: $6x + 4y = 8$.
2. Subtract the first: $x = 2$.
3. Then $3(2) + 2y = 4$ gives $y = -1$, so $x - y = 3$.

Doubling the second equation matches its y term to the first, so subtracting eliminates y and leaves $x = 2$. Then $y = -1$, and $x - y = 2 - (-1) = 3$.

10 Answer: C

1. Note $6x + 9y = 3(2x + 3y)$.
2. Since $2x + 3y = 7$, the second equation reads $3(7) = c$.
3. So $c = 21$.

The second left side is exactly 3 times the first, so the equations agree only when the constants share that factor: $c = 3 \cdot 7 = 21$. For any other c the lines are parallel and there is no solution at all.

11 Answer: C

1. Let a and c be adult and child tickets: $a + c = 90$ and $8a + 5c = 600$.
2. Substitute $c = 90 - a$: $450 + 3a = 600$, so $a = 50$.
3. Therefore $c = 90 - 50 = 40$.

If all 90 tickets were child tickets the museum would collect \$450; each adult ticket adds \$3, so $150 \div 3 = 50$ adults and 40 children. Choice D is the number of adults.

12 Answer: 11

1. From the first equation, $y = 13 - 4x$.
2. Substitute: $2x + 3(13 - 4x) = 9$, so $-10x = -30$ and $x = 3$.
3. Then $y = 1$, so $3x + 2y = 9 + 2 = 11$.

Solve the system once, then assemble the requested expression. Substituting $y = 13 - 4x$ gives $x = 3$ and $y = 1$, so $3x + 2y = 9 + 2 = 11$.

13 **Answer: C**

1. Gym A: $C = 40 + 15m$. Gym B: $C = 10 + 20m$.
2. Set equal: $40 + 15m = 10 + 20m$.
3. Solve: $30 = 5m$, so $m = 6$.

Write a cost equation for each gym and set them equal – the point where two linear models agree is the solution of the system. The \$30 head start of Gym B is erased at \$5 per month, so it takes 6 months.

14 **Answer: A**

1. Divide the first equation by 3: $x + y = 5$.
2. Add $2x - y = 4$: $3x = 9$, so $x = 3$ and $y = 2$.
3. Therefore $2y - x = 4 - 3 = 1$.

Divide the first equation by 3 to get the far friendlier $x + y = 5$; adding it to $2x - y = 4$ gives $3x = 9$. With $x = 3$ and $y = 2$, the requested value is $4 - 3 = 1$. Choice D is $2y + x$.

15 **Answer: 2**

1. Downstream: $b + c = 24 \div 2 = 12$.
2. Upstream: $b - c = 24 \div 3 = 8$.
3. Subtract: $2c = 4$, so $c = 2$.

Downstream the current helps and upstream it hinders, so the effective speeds are $b + c$ and $b - c$. Those come to 12 and 8 mph, a sum-and-difference system whose solution is $b = 10$, $c = 2$.

16 **Answer: A**

1. Set the ratios equal: $\frac{a}{4} = \frac{5}{10} = \frac{1}{2}$.
2. Solve: $a = 2$.
3. Check: $\frac{12}{7} \neq \frac{1}{2}$, so the lines are parallel and distinct.

Parallel lines need proportional coefficients: $a/4 = 5/10$, so $a = 2$. The constants must not follow that ratio, and indeed $12/7 \neq 1/2$. Choice D comes from doubling the wrong equation.

17 **Answer: B**

1. Compare the x terms: $6 \div 3 = 2$.
2. Compare the constants: $16 \div 8 = 2$, the same factor.
3. So $10 = 2k$, giving $k = 5$.

Every part of the second equation is twice the corresponding part of the first: $6 = 2 \cdot 3$ and $16 = 2 \cdot 8$. For the equations to be identical, 10 must equal $2k$, so $k = 5$. Choice C is the coefficient copied without halving it.

18 **Answer: C**

1. Let x be the larger and y the smaller: $x + y = 34$.
2. Translate the second sentence: $3x - 2y = 62$.
3. Substitute $y = 34 - x$: $3x - 68 + 2x = 62$, so $5x = 130$ and $x = 26$.

Translate each sentence into an equation: $x + y = 34$ and $3x - 2y = 62$, where x is the larger. Substituting $y = 34 - x$ gives $5x = 130$, so $x = 26$ and $y = 8$. Choice A is the smaller number.

19 **Answer: B**

1. Subtract the second equation from the first: $2x - 2y = 14$.
2. Factor: $2(x - y) = 14$.
3. Divide by 2: $x - y = 7$.

With the coefficients swapped, subtracting the equations gives $2x - 2y = 14$, so $x - y = 7$ without ever isolating a variable. Choice A is $x + y$, which comes from adding instead; choice D is the difference before dividing by 2.

20 **Answer: 6**

1. Parallel lines have equal slopes: $k - 1 = 5$.
2. Solve: $k = 6$.
3. The intercepts 2 and -3 differ, so the system has no solution.

Two lines in slope-intercept form miss each other exactly when their slopes match but their intercepts do not. Setting $k - 1 = 5$ gives $k = 6$, and since $-3 \neq 2$ the lines really are parallel rather than identical.

21 **Answer: D**

1. Factor the second equation: $6x - 3y = 3(2x - y)$.
2. The first equation gives $2x - y = 4$.
3. Therefore $c = 3(4) = 12$.

Since $6x - 3y = 3(2x - y)$ and $2x - y = 4$, the second equation is forced to read $c = 12$. For any other value the two lines are parallel and the system has no solution at all, so 12 is the only value that permits a solution.

22 **Answer: D**

1. Write the system: $3s + 2h = 85$ and $2s + 5h = 75$.
2. Multiply the first by 5 and the second by 2: $15s + 10h = 425$ and $4s + 10h = 150$.
3. Subtract: $11s = 275$, so $s = 25$, $h = 5$ and $s + h = 30$.

Let s and h be the two prices. Elimination gives $s = 25$ and $h = 5$, so a shirt and a hat together cost \$30. Choices B and C are the individual prices; the question asks for their sum.

Topic 05 — Linear Inequalities in One or Two Variables

Module 1

1 **Answer: B**

1. Subtract 5 from both sides: $3x \leq 15$.
2. Divide both sides by 3: $x \leq 5$.

Because 3 is positive, dividing by it leaves the inequality symbol pointing the same way, so the solution set is every number at most 5. Testing $x = 0$ gives $5 \leq 20$, true, which confirms the direction.

2 **Answer: C**

1. "At most 1200" means the weight can equal 1200 but not exceed it.
2. Write $w \leq 1200$.

"At most" allows the limit itself: a load of exactly 1200 pounds is legal, so the symbol must include equality and point toward the smaller values.

3 **Answer: 7**

1. Add 7 to both sides: $2x > 12$.
2. Divide both sides by 2: $x > 6$.
3. The least integer greater than 6 is 7.

The inequality is strict, so $x = 6$ is not a solution even though it satisfies the related equation. The least integer strictly greater than 6 is 7.

4 **Answer: A**

1. Substitute $(0, 0)$: $0 > 2(0) - 1$, that is $0 > -1$, which is true.
2. The other three pairs give $1 > 1$, $3 > 3$ and $4 > 5$, all false.

A point is a solution exactly when substituting it makes the inequality true; the pairs that fail here all sit on or below the boundary line $y = 2x - 1$.

5 **Answer: D**

1. Cost of the notebooks: $3x$. Cost of the pens: $2y$.
2. Total cost is $3x + 2y$, and "no more than \$30" gives $3x + 2y \leq 30$.

Each notebook contributes 3 dollars and each pen contributes 2 dollars, so the total cost is $3x + 2y$; "no more than \$30" caps that total at 30 while still allowing exactly 30.

6 **Answer: B**

1. Subtract 3: $4x \leq 24$.
2. Divide by 4: $x \leq 6$, so the greatest value is 6.

The symbol $\leq$ includes equality, so the boundary value itself is a solution and is therefore the greatest one.

7 **Answer: C**

1. Subtract 5 from both sides: $-x \geq -3$.
2. Divide both sides by -1 and reverse the symbol: $x \leq 3$.

Isolating x requires dividing by -1, and that single step reverses the direction of the symbol; the check $x = 0$ gives $5 \geq 2$, true, so the solution set must contain 0.

8 **Answer: B**

1. Subtract 1 from all three parts: $-4 < 2x \leq 6$.
2. Divide all three parts by 2: $-2 < x \leq 3$.
3. Of the choices, only 3 lies in that interval.

Solving all three parts at once gives $-2 < x \leq 3$; the left end is excluded because that inequality is strict, while the right end is included.

9 Answer: 19

1. Subtract 4: $\frac{n}{3} > 6$.
2. Multiply both sides by 3: $n > 18$.
3. The least integer greater than 18 is 19.

Clearing the fraction first avoids decimals: the boundary is $n = 18$, and because the inequality is strict, 18 itself fails and the next integer wins.

10 Answer: C

1. Let t be the number of tickets: $9t + 6 \leq 60$.
2. Subtract 6: $9t \leq 54$.
3. Divide by 9: $t \leq 6$.

The fee is charged once, not once per ticket, so it is added outside the product; here the boundary comes out to a whole number, so 6 tickets cost exactly \$60 and are affordable.

11 Answer: D

1. Test $(2, 4)$: $2 + 4 = 6$, and $6 < 5$ is false.
2. Test $(3, 3)$: $3 + 3 = 6 < 5$ is false. Test $(4, 1)$: $1 \geq 2(4) - 3 = 5$ is false.
3. Test $(0, 0)$: $0 \geq -3$ is true and $0 + 0 = 0 < 5$ is true.

A solution must satisfy *both* inequalities; checking the simpler one first, $x + y < 5$, eliminates every pair whose coordinates sum to 5 or more.

12 Answer: 3

1. Subtract 12: $-5x \geq -15$.
2. Divide by -5 and reverse the symbol: $x \leq 3$, so $k = 3$.

The coefficient of x is negative, so the last step divides by -5 and turns $\geq$ into $\leq$ – which is exactly why the given solution has the form $x \leq k$ rather than $x \geq k$.

13 Answer: B

1. From $(0, -2)$ and $(1, 0)$ the slope is $\frac{0 - (-2)}{1 - 0} = 2$ and the y-intercept is -2, so the boundary is $y = 2x - 2$.
2. The line is dashed, so the symbol is strict.
3. The shading lies above the line, so $y > 2x - 2$.

The boundary has slope 2 and y-intercept -2, so the line is $y = 2x - 2$; a dashed line means points on it are excluded, and shading above a line means y is greater than the expression.

14 Answer: A

1. The average condition is $\frac{76 + 85 + 82 + s}{4} \geq 80$.
2. Multiply both sides by 4: $243 + s \geq 320$.
3. Subtract 243: $s \geq 77$.

"At least 80" means the average may equal 80, so the symbol is $\geq$; the requirement is on the average of all four scores, not on the fourth score by itself.

15 **Answer: 4**

1. Subtract 9: $-4x > -20$.
2. Divide by -4 and reverse the symbol: $x < 5$.
3. The greatest integer less than 5 is 4.

Dividing by -4 reverses the symbol, giving $x < 5$; because the inequality is strict, 5 is excluded and the greatest integer solution is one less.

16 **Answer: C**

1. Subtract 3 from all three parts: $-5 \le -2x < 4$.
2. Divide all three parts by -2 and reverse both symbols: $\frac{5}{2} \ge x > -2$.
3. Rewrite with the smaller number first: $-2 < x \le \frac{5}{2}$.

Every part is divided by -2, so both symbols reverse and the two endpoints trade places: the end that came from the strict inequality stays strict, but it is now the left end.

17 **Answer: B**

1. Pounds of fruit: $a + p$, and it must be at least 40, so $a + p \ge 40$.
2. Cost: apples contribute $2a$ dollars and pears contribute $3p$ dollars.
3. "At most \$90" gives $2a + 3p \le 90$.

One inequality counts pounds and one counts dollars. "At least 40 pounds" sets a minimum on $a + p$, and "at most \$90" sets a maximum on the cost $2a + 3p$; mixing the two symbols, or attaching \$3 to the apples, produces the other choices.

18 **Answer: B**

1. Let t be the number of texts: $25 + 0.10t \le 40$.
2. Subtract 25: $0.10t \le 15$.
3. Divide by 0.10: $t \le 150$.

Only the \$15 left after the fixed charge buys text messages, and each one costs a tenth of a dollar, so the count is $15 \div 0.10$ – ten times 15, not a fraction of it.

19 **Answer: A**

1. Both boundaries have slope -1, so the lines $y = -x + 4$ and $y = -x + 7$ are parallel.
2. Combining the two conditions would require $-x + 7 \le y \le -x + 4$, that is $7 \le 4$.
3. That is impossible, so the regions do not overlap.

The two boundary lines have the same slope, so they are parallel and never meet. The first inequality asks for points on or below the lower line and the second asks for points on or above the higher line, and no point can be both.

20 **Answer: 2**

1. The boundary line is $y = mx + 1$.
2. Substitute $(4, 9)$: $9 = 4m + 1$.
3. Solve: $4m = 8$, so $m = 2$.

The boundary line is the graph of $y = mx + 1$, so a point on the boundary satisfies that equation with equality; substituting it turns the geometry question into a one-step equation.

21 **Answer: C**

1. Multiply both sides by 6: $2(2x - 1) \geq 3(x + 4)$.
2. Distribute: $4x - 2 \geq 3x + 12$.
3. Subtract $3x$ and add 2: $x \geq 14$.

Multiplying by the positive number 6 clears both denominators at once and leaves the symbol alone; the only real risk is failing to distribute across the parentheses on each side.

22 **Answer: A**

1. Add $\frac{1}{2}x$ to both sides: $3 > \frac{3}{2}x - 6$.
2. Add 6: $9 > \frac{3}{2}x$.
3. Multiply by $\frac{2}{3}$: $6 > x$, that is $x < 6$.

Collecting the variable terms on the right keeps the coefficient of x positive, so no symbol has to be reversed; the result $6 > x$ is read from right to left as $x < 6$.

Module 2

1 **Answer: B**

1. Divide both sides by -3.
2. Because -3 is negative, reverse the symbol: $x \leq -4$.

Dividing both sides by -3 is the one operation that changes the direction of the symbol; a quick check settles it, since $x = -5$ gives $15 \geq 12$, true, and $x = 0$ gives $0 \geq 12$, false.

2 **Answer: 3**

1. Subtract 7: $-2x \geq -6$.
2. Divide by -2 and reverse the symbol: $x \leq 3$.

After dividing by -2 the symbol reverses to $\leq$, and because $\leq$ includes equality the boundary value 3 is itself the greatest solution.

3 **Answer: D**

1. Total riders must satisfy $3 + p \leq 8$.
2. Subtract 3: $p \leq 5$.

The capacity applies to everyone in the elevator, so the three people already inside use up part of it; the remaining room is $8 - 3 = 5$, and "at most" keeps equality.

4 **Answer: D**

1. $(0, 0)$ gives $0 \leq 12$; $(1, 1)$ gives $5 \leq 12$; $(3, 2)$ gives $6 + 6 = 12 \leq 12$.
2. $(6, 1)$ gives $12 + 3 = 15$, and $15 \leq 12$ is false.

Substituting each pair tests membership in the half-plane; three of the pairs give a value of at most 12, and the one that exceeds 12 lies on the far side of the boundary line.

5 Answer: A

1. Distribute: $5x - 10 < 3x$.
2. Subtract $3x$: $2x - 10 < 0$, so $2x < 10$.
3. Divide by 2: $x < 5$.

Distributing first turns the problem into an ordinary two-step inequality; every coefficient stays positive, so the strict symbol never changes direction.

6 Answer: 6

1. Subtract 5: $6x \geq 36$.
2. Divide by 6: $x \geq 6$.

The boundary is $x = 6$, and since $\geq$ allows equality that boundary value is itself the least solution – no rounding up is needed.

7 Answer: A

1. The total must satisfy $152 + p \geq 240$.
2. Subtract 152: $p \geq 88$.

"At least 240" constrains the three-game total, so the third score must make up the difference $240 - 152 = 88$; equality still qualifies, so the symbol is $\geq$.

8 Answer: C

1. Add 5 to all three parts: $9 \leq 3x \leq 21$.
2. Divide all three parts by 3: $3 \leq x \leq 7$.
3. The greatest possible value is 7.

Only the right-hand inequality controls the upper end, and because it is $\leq$ the boundary value belongs to the solution set.

9 Answer: 43

1. Let b be the number of boxes: $4200 + 65b \leq 7000$.
2. Subtract 4200: $65b \leq 2800$.
3. Divide by 65: $b \leq 43.07\ldots$, so at most 43 whole boxes.

The boundary works out to about 43.08 boxes, and a box cannot be split, so the answer must round *down* to stay under the limit: 43 boxes weigh 6995 pounds in total with the truck, while 44 would weigh 7060.

10 Answer: C

1. "At least 20 items" gives $x + y \geq 20$.
2. The cost is $4x + 7y$ dollars, and "at most \$100" gives $4x + 7y \leq 100$.

The item count and the cost are two different totals: $x+y$ counts items and $4x+7y$ counts dollars. A requirement uses $\geq$ and a budget uses $\leq$, and both allow the endpoint.

11 Answer: D

1. Add 2 to both sides: $\dfrac{3}{4}x \leq 9$.
2. Multiply both sides by $\dfrac{4}{3}$: $x \leq 12$.

Multiplying by $\frac{4}{3}$ – a positive number – leaves the symbol alone; the common error is to multiply by $\frac{3}{4}$ instead of dividing by it.

12 Answer: B

1. Slope: $\dfrac{0-3}{6-0} = -\dfrac{1}{2}$; y-intercept: 3.
2. The boundary line is solid, so use $\leq$ or $\geq$.
3. The region below the line is shaded, so $y \leq -\dfrac{1}{2}x + 3$.

The slope is $\frac{0-3}{6-0} = -\frac{1}{2}$ and the y-intercept is 3, so the boundary is $y = -\frac{1}{2}x + 3$. The line is solid, so the symbol includes equality, and the shading lies below the line, so y is less than the expression.

13 Answer: A

1. Subtract 1: $-\dfrac{x}{4} > 2$.
2. Multiply both sides by -4 and reverse the symbol: $x < -8$.

Multiplying by -4 reverses the symbol; students who multiply by 4 and forget the negative sign, or who keep the symbol as it was, land on the other choices.

14 Answer: C

1. Subtract 1 from all parts: $-8 < 2x \leq 8$.
2. Divide all parts by 2: $-4 < x \leq 4$.
3. The integers are $-3, -2, -1, 0, 1, 2, 3, 4$ – eight values.

The solution interval is $-4 < x \leq 4$: the left endpoint is excluded and the right endpoint is included, so the integers run from -3 to 4 and there are eight of them.

15 Answer: 5

1. Substitute $x = 3$ and $y = k$: $2(3) + k \geq 11$.
2. Simplify: $6 + k \geq 11$, so $k \geq 5$.

Substituting $x = 3$ turns the two-variable inequality into $k \geq 5$, and because $\geq$ allows equality the boundary value is attainable.

16 Answer: D

1. Isolating x: $-kx \geq 9$, so $x \leq -\dfrac{9}{k}$ when $k > 0$.
2. Match the endpoints: $-\dfrac{9}{k} = -3$.
3. Solve: $k = 3$.

The solution set has the form $x \leq$ something, so the division that isolated x must have reversed the symbol – which tells you $-k$ is negative, that is, k is positive. The endpoint $x = -3$ then satisfies the related equation.

17 **Answer: A**

1. Total study time is $m + s$, and "at most 12 hours" gives $m + s \leq 12$.
2. "At least 4 hours on science" gives $s \geq 4$.

The 12 hours cap the *combined* time, so that inequality involves both variables and uses $\leq$; the 4-hour requirement applies to science alone and uses $\geq$. Swapping either symbol or attaching the requirement to m gives one of the other choices.

18 **Answer: 16**

1. Multiply both sides by 15: $3(2x + 1) < 5(x + 4)$.
2. Distribute: $6x + 3 < 5x + 20$.
3. Subtract $5x$ and 3: $x < 17$, so the greatest integer is 16.

Multiplying by the positive number 15 clears both denominators without touching the symbol; the boundary $x = 17$ is excluded because the inequality is strict, so the greatest integer solution is 16.

19 **Answer: C**

1. Distribute the left side: $6 - 2x \leq 4x - 6$.
2. Add $2x$ and 6 to both sides: $12 \leq 6x$.
3. Divide by 6: $2 \leq x$, that is $x \geq 2$.

Moving the x-terms to the right keeps their coefficient positive, so nothing has to be reversed: $6 - 2x \leq 4x - 6$ becomes $12 \leq 6x$, and reading it backwards gives $x \geq 2$.

20 **Answer: 34**

1. Revenue from drinks: $2(40) = 80$ dollars.
2. Let h be the number of hot dogs: $3h + 80 \geq 180$.
3. Subtract 80 and divide by 3: $h \geq 33.3\ldots$, so $h = 34$.

The drinks have already brought in \$80, leaving \$100 to raise at \$3 each; that is $33\frac{1}{3}$ hot dogs, and since a partial hot dog cannot be sold the count must round *up* to 34 to clear the minimum.

21 **Answer: C**

1. $(0, 2)$: $3(0) - 2 = -2$, and $-2 > 6$ is false.
2. $(2, 0)$: $3(2) - 0 = 6$, and $6 > 6$ is false because the symbol is strict.
3. $(3, -2)$: $9 + 2 = 11 > 6$ is true, and $3 + 2(-2) = -1 \leq 4$ is true.

The first inequality is strict, so a pair that makes $3x - y$ exactly 6 fails; the surviving pair must then also satisfy the second inequality, where equality would be allowed.

22 Answer: A

1. $(2, 3)$: $3 \le -2 + 6 = 4$, $3 \ge 2$, and $2 \ge 1$ – all true.
2. $(0, 3)$ fails $x \ge 1$; $(2, 1)$ fails $y \ge 2$.
3. $(3, 4)$ fails $y \le -x + 6$, since $4 \le 3$ is false.

A point in the region must satisfy all three inequalities, boundaries included; each of the other three points violates exactly one of them, which is enough to rule it out.

Topic 06 — Equivalent Expressions

Module 1

1 Answer: A

1. Multiply: $x \cdot x = x^2$, $x \cdot 5 = 5x$, $3 \cdot x = 3x$, $3 \cdot 5 = 15$.
2. Combine like terms: $5x + 3x = 8x$.
3. The product is $x^2 + 8x + 15$.

Each term of the first factor multiplies each term of the second, so there are four products. The two middle products $5x$ and $3x$ are like terms and must be combined; a student who multiplies only the first terms and only the last terms gets $x^2 + 15$ and loses the entire middle term.

2 Answer: B

1. $3x \cdot 2x^2 = 6x^3$.
2. $3x \cdot (-4x) = -12x^2$.
3. $3x \cdot 5 = 15x$.

The distributive property sends the factor $3x$ to *every* term inside the parentheses, and the variable multiplies too: $3x \cdot 5 = 15x$, not 15. Stopping the distribution early is the mistake behind the other choices.

3 Answer: 1

1. Expand: $2x^2 - 2x + 3x - 3$.
2. Combine the x terms: $-2x + 3x = x$.
3. So the product is $2x^2 + x - 3$ and $b = 1$.

Two expressions are equivalent only when every matching coefficient agrees, so b is simply the coefficient of x in the expanded product. Expanding gives $-2x + 3x = x$, so $b = 1$ — not -2 or 3 alone.

4 Answer: D

1. Recognize $9 = 3^2$, so the expression is $x^2 - 3^2$.
2. Apply $a^2 - b^2 = (a + b)(a - b)$ with $a = x$ and $b = 3$.

A difference of two squares always factors as (first root + second root)(first root − second root). Since $x^2 - 9 = x^2 - 3^2$, the factors are $(x + 3)$ and $(x - 3)$. It is not $(x - 3)^2$, because that would expand to $x^2 - 6x + 9$.

5 Answer: A

1. $x^5 \cdot x^3 = x^{5+3} = x^8$.
2. $\dfrac{x^8}{x^2} = x^{8-2} = x^6$.

Multiplying powers of the same base *adds* exponents and dividing *subtracts* them. Here $5 + 3 = 8$ and then $8 - 2 = 6$. Multiplying the exponents instead of adding them is what produces the larger choices.

6 Answer: C

1. $\dfrac{12}{3} = 4$.
2. $\dfrac{x^5}{x^2} = x^{5-2} = x^3$.
3. The quotient is $4x^3$.

Divide the numerical coefficients and subtract the exponents separately: $12 \div 3 = 4$ and $x^{5-2} = x^3$. The coefficients are divided, never subtracted, and the exponents are subtracted, never divided.

7 Answer: D

1. The GCF of 6 and 9 is 3; both terms contain x, so the GCF is $3x$.
2. $6x^2 \div 3x = 2x$ and $9x \div 3x = 3$.
3. So $6x^2 + 9x = 3x(2x + 3)$.

Factoring out the greatest common factor is division: the GCF of $6x^2$ and $9x$ is $3x$, and dividing each term by $3x$ leaves $2x$ and 3. Checking by redistributing is the fastest way to reject the other choices.

8 Answer: D

1. Look for two numbers with product 12 and sum 7.
2. 3 and 4 work.
3. So $x^2 + 7x + 12 = (x + 3)(x + 4)$.

For a monic trinomial you need two numbers whose product is the constant term and whose sum is the middle coefficient. Here $3 \cdot 4 = 12$ and $3 + 4 = 7$. The pair $(2, 6)$ multiplies to 12 but sums to 8, which is why that choice fails.

9 Answer: C

1. Write $(2x - 5)^2 = (2x)^2 - 2(2x)(5) + 5^2$.
2. $(2x)^2 = 4x^2$ and $-2(2x)(5) = -20x$ and $5^2 = 25$.

Squaring a binomial is not squaring each term: $(a - b)^2 = a^2 - 2ab + b^2$ carries a middle term. With $a = 2x$ and $b = 5$ the middle term is $-2(2x)(5) = -20x$; halving it to $-10x$ is the other common slip.

10 Answer: 4

1. Factor the numerator: $x^2 - 16 = (x + 4)(x - 4)$.
2. Cancel the common factor $x - 4$ (legal because $x \neq 4$).
3. What is left is $x + 4$, so $c = 4$.

You may cancel only a common *factor*, so the numerator must be factored first. Since $x^2 - 16 = (x + 4)(x - 4)$, the factor $x - 4$ cancels and $x + 4$ remains, giving $c = 4$.

11 Answer: D

1. Factor: $x^2 + 5x + 6 = (x + 2)(x + 3)$.
2. Cancel the common factor $x + 2$.
3. The result is $x + 3$.

Factor before you cancel. The numerator is $(x + 2)(x + 3)$, so the whole factor $x + 2$ divides out and $x + 3$ remains. Cancelling the loose x in $x + 2$ against the x^2 is not allowed, because $x + 2$ is a sum, not a product.

12 Answer: A

1. $\left(3x^{-2}\right)^{-1} = 3^{-1}\left(x^{-2}\right)^{-1}$.
2. $3^{-1} = \dfrac{1}{3}$ and $\left(x^{-2}\right)^{-1} = x^{(-2)(-1)} = x^2$.
3. So the expression equals $\dfrac{x^2}{3}$.

An outer exponent hits every factor inside, including the coefficient: $3^{-1} = \frac{1}{3}$ and $\left(x^{-2}\right)^{-1} = x^2$. Leaving the 3 untouched, or leaving x in the denominator, are the two mistakes the distractors record.

13 Answer: 8

1. $16^{3/4} = \left(\sqrt[4]{16}\right)^3$.
2. $\sqrt[4]{16} = 2$ because $2^4 = 16$.
3. $2^3 = 8$.

A fractional exponent is a root and a power: the denominator gives the root, the numerator gives the power. Taking the fourth root first keeps the numbers small: $\sqrt[4]{16} = 2$, then $2^3 = 8$.

14 Answer: C

1. $\sqrt{72x^2} = \sqrt{36 \cdot 2 \cdot x^2}$.
2. $\sqrt{36} = 6$ and $\sqrt{x^2} = x$ for $x > 0$.
3. What is left under the radical is 2, so the value is $6x\sqrt{2}$.

Pull out the largest perfect-square factor. Here $72 = 36 \cdot 2$ and x^2 is already a perfect square, so $\sqrt{72x^2} = 6x\sqrt{2}$. The square root of x^2 is x, not x^2.

15 Answer: C

1. The common denominator is $2x$.
2. $\dfrac{1}{x} = \dfrac{2}{2x}$.
3. $\dfrac{2}{2x} + \dfrac{1}{2x} = \dfrac{3}{2x}$.

Fractions are added over a common denominator, never by adding denominators. Rewriting $\frac{1}{x}$ as $\frac{2}{2x}$ makes the sum $\frac{2}{2x} + \frac{1}{2x} = \frac{3}{2x}$.

16 **Answer: A**

1. The factors must have the form $(2x+m)(x+n)$ with $mn = 3$.
2. Try $m = 1$, $n = 3$: the middle term is $6x + x = 7x$.
3. So $2x^2 + 7x + 3 = (2x+1)(x+3)$.

When the leading coefficient is not 1, the middle term mixes the outer and inner products. For $(2x+1)(x+3)$ the outer and inner products are $6x$ and x, which total $7x$; for $(2x+3)(x+1)$ they total $5x$, so that pairing fails.

17 **Answer: D**

1. Factor: $x^2 - 4 = (x+2)(x-2)$.
2. $\dfrac{x+2}{(x+2)(x-2)}$ has the common factor $x+2$.
3. Cancelling it leaves $\dfrac{1}{x-2}$.

Factor the denominator as $(x+2)(x-2)$; the common factor $x+2$ cancels and leaves $\frac{1}{x-2}$. Cancelling the 2 in the numerator against the 4 in the denominator is illegal — both are inside sums, not factors.

18 **Answer: 2**

1. Factor the numerator: $x^2 - 9 = (x-3)(x+3)$.
2. Factor the denominator: $x^2 + x - 6 = (x+3)(x-2)$.
3. Cancel $x+3$ to get $\dfrac{x-3}{x-2}$, so $c = 2$.

Factor top and bottom completely before cancelling: $\frac{(x-3)(x+3)}{(x+3)(x-2)}$. Only the shared factor $x+3$ disappears, leaving $\frac{x-3}{x-2}$, so $c = 2$.

19 **Answer: C**

1. Common denominator: $(x-1)(x+1) = x^2 - 1$.
2. $\dfrac{x+1}{x^2-1} - \dfrac{x-1}{x^2-1}$.
3. $(x+1) - (x-1) = 2$, so the difference is $\dfrac{2}{x^2-1}$.

The common denominator is $(x-1)(x+1) = x^2 - 1$. The numerators become $(x+1) - (x-1) = 2$, and the minus sign must be distributed across both terms of the second numerator. Subtracting the numerators as $1 - 1 = 0$ ignores the rewriting step entirely.

20 **Answer: A**

1. $\left(8x^6\right)^{2/3} = 8^{2/3} \cdot x^{6\cdot(2/3)}$.
2. $8^{2/3} = \left(\sqrt[3]{8}\right)^2 = 2^2 = 4$.
3. $x^{6\cdot(2/3)} = x^4$, so the value is $4x^4$.

The outside exponent applies to the coefficient and to the variable power: $8^{2/3} = \left(\sqrt[3]{8}\right)^2 = 4$ and $x^{6\cdot 2/3} = x^4$. Multiplying 8 by $\frac{2}{3}$, or adding the exponents instead of multiplying them, gives the wrong choices.

21 **Answer: 8**

1. Expanding gives $6x^2 + (2a + 3b)x + ab$.
2. So $ab = 15$ and $2a + 3b = 19$.
3. The positive integer pair $a = 5$, $b = 3$ works, so $a + b = 8$.

Matching coefficients gives two conditions at once: $ab = 15$ for the constant term and $2a + 3b = 19$ for the x term. Only $a = 5$, $b = 3$ satisfies both, so a factor pair that gives the right constant is not enough — it must also give the right middle term.

22 **Answer: B**

1. Factor: $2x^2 + 7x - 15 = (x + 5)(2x - 3)$.
2. Cancel the common factor $x + 5$.
3. The result is $2x - 3$.

Factor the numerator as $(x+5)(2x-3)$ — the pair that produces the middle term $7x$ from $10x-3x$. Cancelling $x + 5$ leaves $2x - 3$. Choosing $2x + 3$ means the signs of the constant were never checked against -15.

Module 2

1 **Answer: A**

1. Expand: $x^2 + 4x - 4x - 16$.
2. The middle terms cancel, leaving $x^2 - 16$.

The outer and inner products are $4x$ and $-4x$, which cancel; that is exactly why a sum times a difference of the same two terms is a difference of squares.

2 **Answer: C**

1. $2x(3x - 7) = 6x^2 - 14x$.
2. Add $5x$: $6x^2 - 14x + 5x$.
3. Combine: $-14x + 5x = -9x$, giving $6x^2 - 9x$.

Distribute first, then combine like terms. After distributing you have $6x^2 - 14x + 5x$, and $-14x + 5x = -9x$. Only terms with the same power of x may be combined, so $6x^2$ stays alone.

3 **Answer: 5**

1. $(x + k)^2 = x^2 + 2kx + k^2$.
2. Match the middle term: $2k = 10$, so $k = 5$.
3. Check the constant: $5^2 = 25$.

A perfect-square trinomial has a constant equal to the square of k and a middle coefficient equal to $2k$. Both tests agree here: $5^2 = 25$ and $2(5) = 10$, so $k = 5$.

4 **Answer: C**

1. Write $4x^2 - 25 = (2x)^2 - 5^2$.
2. Apply $a^2 - b^2 = (a + b)(a - b)$ with $a = 2x$, $b = 5$.

Both terms are perfect squares: $4x^2 = (2x)^2$ and $25 = 5^2$. A difference of squares factors as $(2x + 5)(2x - 5)$; the squared forms are wrong because they would create a middle term.

5 Answer: D

1. $\dfrac{x^4}{x} = x^{4-1} = x^3$.
2. $\dfrac{y^3}{y^5} = y^{3-5} = y^{-2} = \dfrac{1}{y^2}$.
3. So the quotient is $\dfrac{x^3}{y^2}$.

Handle each base separately: $x^{4-1} = x^3$ and $y^{3-5} = y^{-2}$, and a negative exponent means the factor belongs in the denominator. Adding the y exponents instead of subtracting them gives y^8.

6 Answer: 9

1. $27^{2/3} = \left(\sqrt[3]{27}\right)^2$.
2. $\sqrt[3]{27} = 3$.
3. $3^2 = 9$.

The denominator of the exponent is the root and the numerator is the power. Taking the cube root first keeps the arithmetic small: $\sqrt[3]{27} = 3$, and $3^2 = 9$.

7 Answer: B

1. Factor out the GCF 3: $3(x^2 - 4)$.
2. $x^2 - 4 = (x+2)(x-2)$.
3. So $3x^2 - 12 = 3(x+2)(x-2)$.

Always take the greatest common factor out first, then look again: $3x^2 - 12 = 3(x^2 - 4)$, and $x^2 - 4$ is still a difference of squares. Stopping at $3(x^2 - 4)$ leaves the factoring unfinished.

8 Answer: B

1. $x\left(x^2 - 2x + 4\right) = x^3 - 2x^2 + 4x$.
2. $3\left(x^2 - 2x + 4\right) = 3x^2 - 6x + 12$.
3. Add: $x^3 + x^2 - 2x + 12$.

Every term of the trinomial must be multiplied by x and by 3, giving six products before like terms are collected: $-2x^2 + 3x^2 = x^2$ and $4x - 6x = -2x$. Multiplying only matching-degree terms is what produces the other answers.

9 Answer: 5

1. Expand the left side: $6x^2 + 2kx - 9x - 3k$.
2. Match constants: $-3k = -15$, so $k = 5$.
3. Check the x term: $2(5) - 9 = 1$.

Because the equation holds for every x, matching coefficients is legal. The constant term gives $-3k = -15$, so $k = 5$, and the middle term confirms it: $2k - 9 = 1$.

10 **Answer: A**

1. $3x^2 - 27 = 3\left(x^2 - 9\right) = 3(x+3)(x-3)$.
2. Cancel the common factor $x + 3$.
3. What remains is $3(x-3) = 3x - 9$.

Take out the common factor 3 first, then recognise the difference of squares: $3(x-3)(x+3)$. After cancelling $x + 3$ the 3 must stay attached to both terms, giving $3x - 9$ rather than $3x - 27$.

11 **Answer: B**

1. Common denominator: $x(x+1) = x^2 + x$.
2. $\dfrac{2x}{x^2+x} + \dfrac{3(x+1)}{x^2+x}$.
3. $2x + 3x + 3 = 5x + 3$, so the sum is $\dfrac{5x+3}{x^2+x}$.

The common denominator is the product $x(x+1) = x^2 + x$. Each numerator must be scaled by what its denominator was missing: $2x$ and $3(x+1)$, whose sum is $5x+3$. Adding numerators and denominators straight across is the classic error.

12 **Answer: 2**

1. $x^{1/2} \cdot x^{3/2} = x^{1/2+3/2}$.
2. $\dfrac{1}{2} + \dfrac{3}{2} = 2$.
3. So $k = 2$.

Powers of the same base multiply by adding exponents, and fractional exponents are added like ordinary fractions: $\frac{1}{2} + \frac{3}{2} = \frac{4}{2} = 2$. Multiplying the exponents would give $\frac{3}{4}$, which is not what the product rule says.

13 **Answer: B**

1. $\sqrt{48} = \sqrt{16 \cdot 3} = 4\sqrt{3}$.
2. $\sqrt{12} = \sqrt{4 \cdot 3} = 2\sqrt{3}$.
3. $4\sqrt{3} - 2\sqrt{3} = 2\sqrt{3}$.

Radicals combine only after they are simplified to the same radicand: $\sqrt{48} = 4\sqrt{3}$ and $\sqrt{12} = 2\sqrt{3}$, so the difference is $2\sqrt{3}$. Subtracting under one radical sign, $\sqrt{48 - 12} = 6$, is not a legal move.

14 **Answer: A**

1. $\left(2x^3y^{-2}\right)^3 = 2^3\left(x^3\right)^3\left(y^{-2}\right)^3$.
2. $2^3 = 8$, $\left(x^3\right)^3 = x^9$, $\left(y^{-2}\right)^3 = y^{-6}$.
3. $y^{-6} = \dfrac{1}{y^6}$, so the result is $\dfrac{8x^9}{y^6}$.

A power of a product raises each factor: $2^3 = 8$, $\left(x^3\right)^3 = x^9$, and $\left(y^{-2}\right)^3 = y^{-6}$, which moves to the denominator. Multiplying the coefficient by the exponent instead of raising it gives 6.

15 **Answer: C**

1. Factor: $x^2 - 5x + 6 = (x-2)(x-3)$.
2. Cancel the common factor $x - 3$.
3. The result is $x - 2$.

The numerator factors as $(x-2)(x-3)$, since $(-2)(-3) = 6$ and $-2-3 = -5$. Cancelling $x-3$ leaves $x-2$; the leftover factor is the one you did *not* cancel.

16 **Answer: B**

1. Look for two numbers with product -90 and sum -1: 9 and -10.
2. $6x^2 + 9x - 10x - 15 = 3x(2x+3) - 5(2x+3)$.
3. Factor out $2x+3$: $(2x+3)(3x-5)$.

Split the middle term using two numbers with product $6(-15) = -90$ and sum -1: those are 9 and -10. Grouping then gives $(2x+3)(3x-5)$; swapping the signs produces x instead of $-x$ in the middle.

17 **Answer: D**

1. $x^2 - 1 = (x-1)(x+1)$ and $x^2 + 3x = x(x+3)$.
2. Cancel $x-1$ and $x+3$.
3. What is left is $x(x+1) = x^2 + x$.

Factor every piece before multiplying: $\frac{(x-1)(x+1)}{x+3} \cdot \frac{x(x+3)}{x-1}$. Both $x-1$ and $x+3$ cancel diagonally, leaving $x(x+1) = x^2 + x$. Cancelling before factoring would strand the $x+3$ in the denominator.

18 **Answer: 3**

1. Factor: $2x^2 + 5x - 3 = (2x-1)(x+3)$.
2. Cancel the common factor $2x - 1$.
3. The quotient is $x+3$, so $c = 3$.

The numerator factors as $(2x-1)(x+3)$: the constant -3 comes from $(-1)(3)$ and the middle term from $6x - x = 5x$. Cancelling $2x-1$ leaves $x+3$, so $c = 3$.

19 **Answer: B**

1. $x^{-1} + y^{-1} = \dfrac{1}{x} + \dfrac{1}{y} = \dfrac{x+y}{xy}$.
2. Dividing 1 by that fraction flips it.
3. So the expression equals $\dfrac{xy}{x+y}$.

Rewrite the negative exponents as fractions first: $\frac{1}{x} + \frac{1}{y} = \frac{y+x}{xy}$. The whole expression is the reciprocal of that sum, which flips it to $\frac{xy}{x+y}$. Turning $\frac{1}{x^{-1}+y^{-1}}$ into $x+y$ treats a sum of reciprocals as the reciprocal of a sum.

20 **Answer: 24**

1. $(2x+3)^2 = 4x^2 + 12x + 9$ and $(2x-3)^2 = 4x^2 - 12x + 9$.
2. Subtract: $12x - (-12x) = 24x$.
3. So $k = 24$.

Expanding both squares gives $(4x^2 + 12x + 9) - (4x^2 - 12x + 9)$; the $4x^2$ and 9 cancel and the two middle terms add, leaving $24x$. Forgetting to distribute the minus sign over all three terms of the second square is what loses half of that.

21 **Answer: D**

1. $x^2 - x - 6 = (x-3)(x+2)$.
2. $x^2 - 4 = (x+2)(x-2)$.
3. Cancel $x+2$ to get $\dfrac{x-3}{x-2}$.

Factor both parts: $\frac{(x-3)(x+2)}{(x+2)(x-2)}$. Only $x+2$ is common, so it alone cancels and $\frac{x-3}{x-2}$ remains. Cancelling the x^2 terms first is the trap, because they are terms of a sum, not factors.

22 **Answer: C**

1. Common denominator: $(x-1)(x+1) = x^2 - 1$.
2. Numerators: $x(x+1) - 1(x-1) = x^2 + x - x + 1$.
3. That simplifies to $x^2 + 1$, so the difference is $\dfrac{x^2+1}{x^2-1}$.

Over the common denominator $x^2 - 1$ the numerator is $x(x+1) - (x-1) = x^2 + x - x + 1 = x^2 + 1$. The subtraction sign must reach both terms of $x - 1$; dropping it on the -1 leaves $x^2 + x - 1$ instead.

Topic 07 — Quadratic Equations

Module 1

1 **Answer: B**

1. Find two numbers with product 10 and sum -7: they are -2 and -5.
2. Factor: $(x-2)(x-5) = 0$.
3. Set each factor to zero: $x = 2$ or $x = 5$.
4. The larger solution is 5.

Two numbers multiplying to 10 and adding to -7 are -2 and -5, so the equation factors as $(x-2)(x-5) = 0$. The zero-product property gives $x = 2$ or $x = 5$, and the question asks for the larger one. The distractors 7 and 10 are the coefficients themselves, which are the sum and product of the roots, not the roots.

2 **Answer: C**

1. Take square roots of both sides, keeping $\pm$: $x - 3 = \pm 7$.
2. Solve each case: $x = 3 + 7 = 10$ or $x = 3 - 7 = -4$.
3. The greatest value is 10.

Taking the square root of both sides keeps both signs: $x - 3 = \pm 7$. That gives $x = 10$ and $x = -4$, and the greatest of these is 10. Choice A is the other solution, B forgets to add the 3 back, and D adds 49 instead of $\sqrt{49}$.

3 **Answer: 5**

1. Divide both sides by 2: $x^2 = 25$.
2. Take square roots: $x = \pm 5$.
3. The positive value is 5.

Divide by 2 before taking a root: $x^2 = 25$, so $x = \pm 5$. The question asks only for the positive value, which is why a single grid-in answer is possible; the equation itself still has two solutions.

4 Answer: D

1. Identify $a = 1$, $b = -9$.
2. The sum of the roots is $-b/a = 9$.
3. Check: the roots are 2 and 7, and $2 + 7 = 9$.

By Vieta's formulas the sum of the solutions of $ax^2 + bx + c = 0$ is $-b/a$, here $-(-9)/1 = 9$. No factoring is needed, though factoring confirms it: the roots are 2 and 7, whose sum is 9. Choice A drops the minus sign in $-b/a$, and B and C are the individual roots.

5 Answer: A

1. Set each factor to zero: $x = -4$ or $x = 6$.
2. Multiply the solutions: $(-4)(6) = -24$.

The zero-product property gives the solutions $x = -4$ and $x = 6$ directly, and their product is -24. Choice B is the sum of the roots and D forgets that one root is negative; a factor of $(x + 4)$ contributes the root -4, not $+4$.

6 Answer: B

1. Read off $a = 1$, $b = 6$, $c = 5$.
2. Compute $b^2 - 4ac = 36 - 20$.
3. The discriminant is 16.

The discriminant is $b^2 - 4ac = 6^2 - 4(1)(5) = 36 - 20 = 16$. Because it is positive, the equation has two distinct real solutions, which matches the factorisation $(x + 1)(x + 5) = 0$. Choice A is $\sqrt{16}$, C forgets the $-4ac$ term entirely, and D adds $4ac$ instead of subtracting it.

7 Answer: A

1. Find two numbers with product -12 and sum 1: they are 4 and -3.
2. Factor: $(x + 4)(x - 3) = 0$.
3. The solutions are $x = -4$ and $x = 3$; the listed one is 3.

Two numbers with product -12 and sum 1 are 4 and -3, so $x^2 + x - 12 = (x + 4)(x - 3) = 0$ and the solutions are -4 and 3. Only 3 appears among the choices; choice B is the sign-flipped version of the other root, and D is the constant term.

8 Answer: 5

1. Apply the quadratic formula with $a = 1$, $b = -6$, $c = 4$.
2. The discriminant is $36 - 16 = 20$, so $x = \frac{6 \pm 2\sqrt{5}}{2}$.
3. Divide every term by 2: $x = 3 \pm \sqrt{5}$, so $k = 5$.

The quadratic formula gives $x = \dfrac{6 \pm \sqrt{36 - 16}}{2} = \dfrac{6 \pm \sqrt{20}}{2} = 3 \pm \sqrt{5}$, since $\sqrt{20} = 2\sqrt{5}$ and every term of the numerator is divided by 2. So $k = 5$. Completing the square is just as quick: $(x - 3)^2 = 5$.

9 **Answer: D**

1. Write $x^2 + 8x = 3$.
2. Add $(8/2)^2 = 16$ to both sides: $x^2 + 8x + 16 = 19$.
3. The left side is a perfect square: $(x + 4)^2 = 19$, so $b = 19$.

Move the constant across first: $x^2 + 8x = 3$. Half of 8 is 4 and $4^2 = 16$, so adding 16 to both sides gives $(x + 4)^2 = 19$. Choice C adds 16 only on the left while leaving -3 in place, and A flips the sign of the result.

10 **Answer: C**

1. Identify $a = 2$, $b = -10$.
2. The sum of the roots is $-b/a = 10/2$.
3. The sum is 5.

The sum of the roots is $-b/a = -(-10)/2 = 5$. The leading coefficient matters here: forgetting to divide by $a = 2$ produces choice D. Choice B is the product c/a, and A has the sign of $-b/a$ reversed.

11 **Answer: 3**

1. Set $h = 0$: $-16t^2 + 48t = 0$.
2. Factor: $-16t(t - 3) = 0$, so $t = 0$ or $t = 3$.
3. Discard $t = 0$ (the kick itself); the ball lands after 3 seconds.

The ball is on the ground when $h = 0$, so $-16t(t - 3) = 0$ and $t = 0$ or $t = 3$. The root $t = 0$ is the instant of the kick, so the return happens at $t = 3$ seconds. Do not divide the equation by t: that would delete the launch time and hide the structure of the problem.

12 **Answer: B**

1. Set the discriminant to zero: $k^2 - 36 = 0$.
2. Solve: $k = \pm 6$.
3. Since $k > 0$, $k = 6$.

Exactly one real solution means the discriminant is zero: $k^2 - 4(1)(9) = 0$, so $k^2 = 36$ and $k = \pm 6$. The condition $k > 0$ selects 6, and indeed $x^2 + 6x + 9 = (x + 3)^2$. Choice A is the repeated root, not k, and C is the constant term.

13 **Answer: A**

1. Factor by grouping: $(2x - 1)(x - 3) = 0$.
2. Set each factor to zero: $x = \frac{1}{2}$ or $x = 3$.
3. The smaller solution is $\frac{1}{2}$.

Split $-7x$ into parts whose coefficients multiply to $2 \cdot 3 = 6$: $2x^2 - x - 6x + 3 = x(2x - 1) - 3(2x - 1) = (2x - 1)(x - 3)$. The solutions are $x = \frac{1}{2}$ and $x = 3$, and the smaller is $\frac{1}{2}$. Choice D is the other root, and C comes from mismatching the factor pair.

14 **Answer: B**

1. Take square roots: $2x - 1 = 5$ or $2x - 1 = -5$.
2. Solve each: $x = 3$ or $x = -2$.
3. Add them: $3 + (-2) = 1$.

Taking square roots gives $2x - 1 = \pm 5$, so $2x = 6$ or $2x = -4$, i.e. $x = 3$ or $x = -2$, and the sum is 1. Choices A and C are the individual solutions, offered to catch a student who stops after one branch of the $\pm$.

15 **Answer: 12**

1. Set $b^2 - 4(3)(12) = 0$.
2. So $b^2 = 144$ and $b = \pm 12$.
3. Since $b > 0$, $b = 12$.

One real solution means $b^2 - 4ac = 0$, so $b^2 = 4(3)(12) = 144$ and $b = \pm 12$. Because b is positive, $b = 12$; the equation is then $3(x + 2)^2 = 0$. The discriminant answers this without ever solving for x.

16 **Answer: C**

1. Write $2(x^2 + 6x) + 5 = 0$.
2. Complete the square inside: $x^2 + 6x = (x + 3)^2 - 9$.
3. Distribute the 2: $2(x + 3)^2 - 18 + 5 = 0$.
4. Therefore $2(x + 3)^2 = 13$ and $k = 13$.

Factor the 2 out of the first two terms: $2(x^2+6x)+5 = 0$. Completing the square inside gives $2\left[(x+3)^2-9\right]+5 = 0$, i.e. $2(x + 3)^2 - 18 + 5 = 0$, so $2(x + 3)^2 = 13$. The subtraction of 18, not 9, is the step most students miss — the 2 multiplies the correction term as well, which is what choice D ignores.

17 **Answer: A**

1. Identify $a = 3$, $c = -2$.
2. The product of the roots is $c/a = -\frac{2}{3}$.
3. Check by factoring: $(3x + 1)(x - 2) = 0$ gives $-\frac{1}{3}$ and 2.

The product of the roots is $c/a = -2/3$. Factoring confirms it: $(3x + 1)(x - 2) = 0$ gives $x = -\frac{1}{3}$ and $x = 2$, whose product is $-\frac{2}{3}$. Choice B is one of the roots, D is the sum $-b/a$, and C drops the sign of c.

18 **Answer: 9**

1. An x-intercept is a solution of $x^2 + 6x + c = 0$.
2. Exactly one intercept means $b^2 - 4ac = 0$: $36 - 4c = 0$.
3. So $c = 9$.

Touching the x-axis at exactly one point means $x^2 + 6x + c = 0$ has exactly one real solution, so $6^2 - 4(1)c = 0$ and $c = 9$. The parabola is then $y = (x + 3)^2$, whose vertex $(-3, 0)$ sits on the axis. Counting intersections with the x-axis is always a discriminant question in disguise.

19 **Answer: B**

1. Let the width be w; the length is $w+3$.
2. Area: $w(w+3)=88$, so $w^2+3w-88=0$.
3. Factor: $(w+11)(w-8)=0$.
4. Reject $w=-11$; the width is 8 meters.

If the width is w, the length is $w+3$ and $w(w+3)=88$, i.e. $w^2+3w-88=0$, which factors as $(w+11)(w-8)=0$. The roots are -11 and 8, and a width cannot be negative, so $w=8$. Choice C is the length, and A comes from mis-splitting 88.

20 **Answer: C**

1. Compute the discriminant: $(-4)^2-4(1)(-8)=16+32=48$.
2. Apply the formula: $x=\frac{4\pm\sqrt{48}}{2}=\frac{4\pm4\sqrt{3}}{2}$.
3. Divide every term by 2: $x=2\pm2\sqrt{3}$.
4. So $a=2$, $b=2$ and $a+b=4$.

The quadratic formula gives $x=\dfrac{4\pm\sqrt{16+32}}{2}=\dfrac{4\pm\sqrt{48}}{2}$. Since $\sqrt{48}=4\sqrt{3}$, this simplifies to $2\pm2\sqrt{3}$, so $a=2$, $b=2$ and $a+b=4$. Note that $-4ac=-4(1)(-8)=+32$: a negative c makes the discriminant larger, not smaller.

21 **Answer: B**

1. Product of the roots: $4r=40$, so $r=10$.
2. Sum of the roots: $4+10=-p$, so $p=-14$.
3. Then $p+r=-14+10=-4$.

The product of the roots is $c/a=40$, so $4r=40$ and $r=10$. The sum of the roots is $-p$, so $4+10=-p$ and $p=-14$. Therefore $p+r=-14+10=-4$. Choice D reverses the sign of p, the single most common slip in this Vieta setup.

22 **Answer: D**

1. Require $b^2-4ac<0$: $64-8k<0$.
2. Solve the inequality: $k>8$.
3. Among the choices only $k=10$ works.

No real solutions means a negative discriminant: $8^2-4(2)k<0$, so $64<8k$ and $k>8$. Only 10 satisfies this. The boundary value $k=8$ is deliberately offered: it gives a discriminant of exactly zero, which is one real solution, not none.

Module 2

1 **Answer: D**

1. Factor: $(x-5)(x+3)=0$.
2. The solutions are $x=5$ and $x=-3$.
3. The positive solution is 5.

Two numbers with product -15 and sum -2 are -5 and 3, so $(x-5)(x+3)=0$ and the solutions are 5 and -3. The positive one is 5. Choices A and C are those same roots with their signs flipped, the error you make by reading the roots straight out of the parentheses.

2 Answer: 3

1. Add 27 and divide by 3: $x^2 = 9$.
2. Take square roots: $x = \pm 3$.
3. The positive value is 3.

Isolate the square first: $3x^2 = 27$, so $x^2 = 9$ and $x = \pm 3$. The positive value is 3. Dividing by 3 before taking the root keeps the arithmetic clean and avoids a stray $\sqrt{27}$.

3 Answer: A

1. Compute $b^2 - 4ac = 16 - 28 = -12$.
2. The discriminant is negative.
3. Therefore the equation has 0 real solutions.

The discriminant is $4^2 - 4(1)(7) = 16 - 28 = -12$, which is negative, so there are no real solutions — the parabola $y = x^2 + 4x + 7$ sits entirely above the x-axis. A quadratic can never have three solutions, so choice D can be rejected on sight.

4 Answer: B

1. Write $x^2 + 10x = -18$.
2. Add $(10/2)^2 = 25$ to both sides.
3. $(x + 5)^2 = -18 + 25 = 7$, so $c = 7$.

Move the constant: $x^2 + 10x = -18$. Half of 10 is 5 and $5^2 = 25$, so adding 25 to both sides gives $(x + 5)^2 = -18 + 25 = 7$. Choice D adds 25 to $+18$ instead of -18, the sign slip that happens when the constant is not moved across first.

5 Answer: A

1. Identify $a = 1$, $c = -18$.
2. The product of the roots is $c/a = -18$.
3. Check: the roots are -9 and 2, and $(-9)(2) = -18$.

The product of the roots is $c/a = -18/1 = -18$; factoring as $(x+9)(x-2) = 0$ confirms it, since $(-9)(2) = -18$. Choice D drops the sign of c, and B is the sum of the roots, $-b/a = -7$.

6 Answer: 5

1. Set $-16t^2 + 64t + 80 = 0$.
2. Divide by -16: $t^2 - 4t - 5 = 0$.
3. Factor: $(t - 5)(t + 1) = 0$, so $t = 5$ or $t = -1$.
4. Time cannot be negative, so $t = 5$ seconds.

Set $h = 0$ and divide by -16 to simplify: $t^2 - 4t - 5 = 0$, which factors as $(t - 5)(t + 1) = 0$. The roots are 5 and -1, and only a nonnegative time makes sense, so the stone lands after 5 seconds. The value 80 is the cliff's height, not an answer.

7 **Answer: A**

1. Compute the discriminant: $16 - 4 = 12$.
2. So $x = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm 2\sqrt{3}}{2}$.
3. Divide the whole numerator by 2: $x = 2 \pm \sqrt{3}$.

The quadratic formula gives $x = \dfrac{4 \pm \sqrt{16-4}}{2} = \dfrac{4 \pm 2\sqrt{3}}{2} = 2 \pm \sqrt{3}$. Choice D is what you get by dividing only the first term of the numerator by $2a$, choice C flips the sign of $-b$, and choice B computes the discriminant as $16 + 4$.

8 **Answer: C**

1. Let the width be w, so the length is $w + 5$.
2. Solve $w(w+5) = 84$, i.e. $w^2 + 5w - 84 = 0$.
3. Factor: $(w+12)(w-7) = 0$.
4. Reject $w = -12$; the width is 7 feet.

With width w the area is $w(w+5) = 84$, so $w^2 + 5w - 84 = 0$ and $(w+12)(w-7) = 0$. The roots are -12 and 7; a width must be positive, so $w = 7$ and the length is 12. Choices A and D are the rejected root and the length, both offered because they appear in correct work.

9 **Answer: -15**

1. The product of the roots equals $c/a = c$.
2. Compute $(-3)(5) = -15$.
3. So $c = -15$.

A quadratic with those roots is $(x+3)(x-5) = x^2 - 2x - 15$, so $c = -15$ (and $b = -2$). Vieta gives the same thing immediately: c/a is the product of the roots, $(-3)(5) = -15$. The sign of c is negative because the roots have opposite signs.

10 **Answer: C**

1. Write $3(x^2 - 4x) + 5 = 0$.
2. Inside: $x^2 - 4x = (x-2)^2 - 4$.
3. Distribute: $3(x-2)^2 - 12 + 5 = 0$.
4. Therefore $3(x-2)^2 = 7$ and $k = 7$.

Factor 3 from the first two terms: $3(x^2 - 4x) + 5 = 0$. Completing the square inside gives $3\left[(x-2)^2 - 4\right] + 5 = 0$, so $3(x-2)^2 - 12 + 5 = 0$ and $3(x-2)^2 = 7$. The correction term is $3 \cdot 4 = 12$, not 4; using 4 produces choice A.

11 **Answer: B**

1. Set the discriminant to zero: $144 - 36k = 0$.
2. Solve: $36k = 144$.
3. So $k = 4$.

Exactly one real solution means $b^2 - 4ac = 0$, here $12^2 - 4k(9) = 144 - 36k = 0$, so $k = 4$. The equation becomes $4x^2 + 12x + 9 = (2x+3)^2 = 0$. Choice C comes from halving 12 and choice A from the perfect-square constant.

12 **Answer: 2**

1. Set $-16t^2 + 96t = 128$.
2. Divide by -16: $t^2 - 6t + 8 = 0$.
3. Factor: $(t-2)(t-4) = 0$.
4. The smaller value is $t = 2$ seconds.

Set $-16t^2 + 96t = 128$ and divide by -16: $t^2 - 6t + 8 = 0$, so $(t-2)(t-4) = 0$ and $t = 2$ or $t = 4$. Both times are physically real — the rocket passes 128 feet on the way up at $t = 2$ and again on the way down at $t = 4$ — and the smaller is 2.

13 **Answer: B**

1. Compute the discriminant: $36 - 8 = 28$.
2. So $x = \frac{6 \pm 2\sqrt{7}}{2} = 3 \pm \sqrt{7}$.
3. Subtract: $(3+\sqrt{7}) - (3-\sqrt{7}) = 2\sqrt{7}$.

The formula gives $x = \dfrac{6 \pm \sqrt{36-8}}{2} = \dfrac{6 \pm 2\sqrt{7}}{2} = 3 \pm \sqrt{7}$, so the difference is $(3+\sqrt{7}) - (3-\sqrt{7}) = 2\sqrt{7}$. The distance between the roots is always twice the square-root part, which is why choice A — the square-root part alone — is wrong.

14 **Answer: D**

1. Require $b^2 - 4ac > 0$: $k^2 - 16 > 0$.
2. So $k > 4$ or $k < -4$.
3. Among the choices only $k = 5$ works.

Two distinct real solutions require $k^2 - 16 > 0$, that is $|k| > 4$. Only $k = 5$ qualifies. The value $k = 4$ gives a discriminant of exactly zero — a single repeated root, $(x+2)^2 = 0$ — which is the trap in this item.

15 **Answer: C**

1. Identify $a = 4$, $b = -20$.
2. The sum of the roots is $-b/a = 20/4$.
3. The sum is 5.

The sum of the roots is $-b/a = 20/4 = 5$; factoring as $(2x-1)(2x-9) = 0$ gives $\frac{1}{2} + \frac{9}{2} = 5$. Choice A is the product c/a, choice B is $-b/(4a)$ from dividing by $2a$ twice, and D forgets to divide by a at all.

16 **Answer: C**

1. Move everything to one side: $x^2 + 6x - 16 = 0$.
2. The sum of the roots is $-b/a = -6$.
3. Check: the roots are -8 and 2, and $-8 + 2 = -6$.

Vieta applies only in standard form, so rewrite as $x^2 + 6x - 16 = 0$; the sum of the roots is then $-b/a = -6$. Factoring confirms it: $(x+8)(x-2) = 0$ gives -8 and 2, whose sum is -6. Choices A and B are pieces of that work mistaken for the whole answer.

17 **Answer: C**

1. Multiply both sides by x: $12 = x^2 - x$.
2. Rearrange: $x^2 - x - 12 = 0$.
3. Factor: $(x-4)(x+3) = 0$, so $x = 4$ or $x = -3$.
4. The greatest value is 4.

Multiply both sides by x: $12 = x^2 - x$, so $x^2 - x - 12 = 0$ and $(x-4)(x+3) = 0$. The solutions are 4 and -3, both valid since neither is 0, and the greatest is 4. Multiplying through by the variable is the standard move that turns a rational equation into a quadratic one.

18 **Answer: 4**

1. Set the two expressions equal: $x^2 + k = 2x + 3$.
2. Rearrange: $x^2 - 2x + (k-3) = 0$.
3. One solution means $(-2)^2 - 4(k-3) = 0$.
4. So $k - 3 = 1$ and $k = 4$.

Setting the expressions equal gives $x^2 + k = 2x + 3$, i.e. $x^2 - 2x + (k-3) = 0$. One point of intersection means one real solution, so the discriminant vanishes: $4 - 4(k-3) = 0$, giving $k - 3 = 1$ and $k = 4$. The line is then tangent to the parabola at $(1, 5)$.

19 **Answer: 45**

1. The roots sum to 14 and differ by 4.
2. Solve: the roots are 9 and 5.
3. Their product is k: $9 \cdot 5 = 45$.

The two roots add to $-b/a = 14$ and differ by 4, so they are 9 and 5. Their product is $c/a = k$, giving $k = 45$. Setting up sum and difference together is faster than guessing factor pairs of an unknown constant.

20 **Answer: C**

1. Set $-16t^2 + 40t + 24 = 0$.
2. Divide by -8: $2t^2 - 5t - 3 = 0$.
3. Factor: $(2t+1)(t-3) = 0$.
4. Reject $t = -\frac{1}{2}$; the ball lands at $t = 3$ seconds.

Set $h = 0$ and divide by -8: $2t^2 - 5t - 3 = 0$, which factors as $(2t+1)(t-3) = 0$. The roots are $-\frac{1}{2}$ and 3; a negative time is impossible, so the ball lands after 3 seconds. Choice A is the time at the vertex, $-b/(2a)$, which is when the ball is highest, not when it lands.

21 **Answer: A**

1. Compute the discriminant: $64 - 24 = 40$.
2. So $x = \frac{8 \pm \sqrt{40}}{4} = \frac{8 \pm 2\sqrt{10}}{4}$.
3. Divide numerator and denominator by 2: $x = \frac{4 \pm \sqrt{10}}{2}$.
4. Then $a = 4$, $b = 10$ and $a + b = 14$.

The formula gives $x = \dfrac{8 \pm \sqrt{64-24}}{4} = \dfrac{8 \pm \sqrt{40}}{4} = \dfrac{8 \pm 2\sqrt{10}}{4} = \dfrac{4 \pm \sqrt{10}}{2}$, so $a = 4$, $b = 10$ and $a + b = 14$. Choice B stops at the unsimplified $\dfrac{8 \pm \sqrt{40}}{4}$ and reads off 8 and 10; choice C reads off 4 and 40.

22 **Answer: C**

1. Set the discriminant to zero: $k^2 - 4(k+3) = 0$.
2. Simplify: $k^2 - 4k - 12 = 0$.
3. Factor: $(k-6)(k+2) = 0$, so $k = 6$ or $k = -2$.
4. The sum is $6 + (-2) = 4$.

One real solution means the discriminant is zero: $k^2 - 4(1)(k+3) = 0$, i.e. $k^2 - 4k - 12 = 0$, which factors as $(k-6)(k+2) = 0$. So $k = 6$ or $k = -2$, and their sum is 4. Vieta on the equation in k gives the same total in one step: the sum is $-(-4)/1 = 4$. Choice D is only one of the two values.

Topic 08 — Quadratic and Polynomial Functions

Module 1

1 **Answer: C**

1. Compare $f(x) = (x-4)^2 + 7$ with $f(x) = a(x-h)^2 + k$.
2. $x - h = x - 4$, so $h = 4$ (and $k = 7$).
3. The vertex is $(4, 7)$, so $h = 4$.

Vertex form is $f(x) = a(x-h)^2 + k$, whose vertex is (h, k). The parabola turns where the squared part is zero, at $x = 4$, so $h = 4$. Because the form subtracts h, the number written in the parentheses is already the x-coordinate, not its opposite; the constant 7 is k, the other coordinate.

2 **Answer: 3**

1. Identify $a = 1$ and $b = -6$.
2. Compute $x = -\frac{b}{2a} = -\frac{-6}{2} = 3$.

A parabola is symmetric about the vertical line through its vertex, and the vertex of ax^2+bx+c sits at $x = -\frac{b}{2a}$. With $a = 1$ and $b = -6$ that line is $x = 3$.

3 **Answer: D**

1. Set $g(x) = 0$, so $(x-2)(x+5) = 0$.
2. $x - 2 = 0$ gives $x = 2$; $x + 5 = 0$ gives $x = -5$.

A product is zero exactly when one of its factors is zero, so set each factor equal to 0. The intercepts are the values that make the factors vanish, which are the opposites of the numbers you see written: $x - 2 = 0$ gives $x = 2$ and $x + 5 = 0$ gives $x = -5$.

4 **Answer: A**

1. The y-intercept occurs where $x = 0$.
2. Substitute: $y = (0-3)(0+8) = (-3)(8) = -24$.

Every y-intercept is found the same way, in any form: substitute $x = 0$. Here $(0-3)(0+8) = -24$. The choice 24 comes from multiplying 3 and 8 without tracking the signs.

5 **Answer: D**

1. $(x-1)^2 \geq 0$ for all x, so $-(x-1)^2 \leq 0$.
2. The greatest value of f occurs when $(x-1)^2 = 0$, that is at $x = 1$.
3. $f(1) = -0 + 6 = 6$.

Because $(x-1)^2$ is never negative, $-(x-1)^2$ is never positive, so the largest possible output happens when the squared term is 0, at $x = 1$. The value there is 6. The number 1 is where the maximum occurs, not the maximum itself.

6 **Answer: 5**

1. Dividing by $x - 2$ means $a = 2$.
2. By the remainder theorem the remainder is $p(2)$.
3. $p(2) = 2^2 + 3(2) - 5 = 4 + 6 - 5 = 5$.

The remainder theorem replaces division entirely: dividing by $x - a$ leaves the remainder $p(a)$. Here $a = 2$, so evaluate $p(2) = 4 + 6 - 5 = 5$.

7 **Answer: C**

1. Vertex $(h, k) = (5, -2)$ means $h = 5$ and $k = -2$.
2. Substitute into $y = (x-h)^2 + k$ to get $y = (x-5)^2 - 2$.

Vertex form $y = (x-h)^2 + k$ has vertex (h, k), so a vertex of $(5, -2)$ requires $h = 5$ and $k = -2$; that is, x minus 5, then minus 2 at the end. Writing $x + 5$ would place the vertex at $x = -5$, and swapping the two numbers gives the vertex $(2, 5)$ instead.

8 **Answer: D**

1. Vertex form displays the minimum as the constant k.
2. $x^2 + 8x = (x+4)^2 - 16$.
3. So $x^2 + 8x + 3 = (x+4)^2 - 16 + 3 = (x+4)^2 - 13$.

The minimum value is displayed by vertex form, so complete the square: half of 8 is 4, and $(x+4)^2 = x^2+8x+16$, which overshoots the constant by 16. Subtracting that gives $(x+4)^2 - 13$, whose expansion returns the original expression, so the minimum value is -13.

9 **Answer: 2**

1. Identify $a = 3$ and $b = -12$.
2. Compute $x = -\frac{b}{2a} = \frac{12}{6} = 2$.

With a leading coefficient other than 1 the formula $x = -\frac{b}{2a}$ still applies, but the $2a$ in the denominator matters: $-\frac{-12}{2(3)} = 2$. Forgetting the a would give 6.

10 **Answer: D**

1. The parabola opens downward, so the vertex is the maximum.
2. $t = -\frac{b}{2a} = -\frac{64}{-32} = 2$ seconds.
3. $h(2) = -16(4) + 64(2) + 5 = -64 + 128 + 5 = 69$ feet.

The greatest height is the y-value of the vertex, so first find when it occurs, $t = -\frac{64}{2(-16)} = 2$ seconds, then substitute back to get the height itself. The number 2 is the time, and 5 is the height at the moment of release, so neither answers the question asked.

11 **Answer: B**

1. Look for two numbers with product -15 and sum -2.
2. They are -5 and 3, so $x^2 - 2x - 15 = (x-5)(x+3)$.
3. The intercepts $x = 5$ and $x = -3$ are now visible.

The x-intercepts are displayed by factored form, so look for two numbers multiplying to -15 and adding to -2: those are -5 and $+3$. Expanding $(x+5)(x-3)$ instead gives $x^2 + 2x - 15$, the right constant but the wrong middle term, which is why checking the linear coefficient is the fast way to eliminate.

12 **Answer: D**

1. The zeros come from $x = 0$, $x - 3 = 0$, and $x + 4 = 0$.
2. The factor $(x-3)$ appears with exponent 2.
3. So the zero $x = 3$ has multiplicity 2.

The multiplicity of a zero is the exponent on its factor. Only $(x-3)$ is squared, so the zero $x = 3$ has multiplicity 2; the exponent 2 is not itself a zero, which is the trap in choice C.

13 **Answer: B**

1. The leading term is $-2x^3$.
2. Odd degree means the ends behave oppositely.
3. The negative coefficient sends the graph down on the right and up on the left.

End behaviour is decided entirely by the leading term $-2x^3$. The degree 3 is odd, so the two ends go opposite ways, and the leading coefficient -2 is negative, so the right-hand end goes down. The lower-degree terms $5x-1$ are irrelevant once $|x|$ is large.

14 **Answer: C**

1. For each choice $x - a$, evaluate $p(a)$.
2. $p(-1) = 12$, $p(-2) = 12$, $p(3) = 12$: none is zero.
3. $p(1) = 1 - 7 + 6 = 0$, so $x - 1$ is a factor.

By the factor theorem, $x - a$ is a factor exactly when $p(a) = 0$, so test the candidates rather than dividing. $p(1) = 1 - 7 + 6 = 0$, so $x - 1$ is a factor; by contrast $p(-1) = -1 + 7 + 6 = 12$ and $p(3) = 27 - 21 + 6 = 12$, so those are not.

15 **Answer: B**

1. The vertex lies on the axis of symmetry, midway between the x-intercepts.
2. $x = \frac{1+7}{2} = 4$.

A parabola is symmetric, so its two x-intercepts are the same distance from the axis of symmetry, and the vertex sits exactly halfway between them. The midpoint of 1 and 7 is $\frac{1+7}{2} = 4$. Choice C is their difference and choice D their sum, neither of which is a midpoint.

16 **Answer: B**

1. Substitute $x = 4$, $f(x) = 5$ into $f(x) = a(x-2)^2 - 3$.
2. $a(4-2)^2 - 3 = 5$, so $4a - 3 = 5$.
3. $4a = 8$, hence $a = 2$.

The vertex fixes h and k; one more point is exactly enough to pin down a. Substituting $(4, 5)$ gives $a(2)^2 - 3 = 5$, so $4a = 8$ and $a = 2$. Choice D is the value of $4a$, the number you get if you forget to divide.

17 **Answer: -3**

1. By the remainder theorem, $p(2) = 4$.
2. $2(2)^3 + k(2) - 6 = 16 + 2k - 6 = 2k + 10$.
3. Set $2k + 10 = 4$, so $2k = -6$ and $k = -3$.

The remainder theorem turns a division statement into a single equation: dividing by $x - 2$ leaves $p(2)$, so $p(2) = 4$. Expanding gives $16 + 2k - 6 = 4$, and solving yields $k = -3$.

18 **Answer: D**

1. The lowest point of the graph is $(1, -4)$, so $h = 1$ and $k = -4$.
2. The parabola opens upward, so $a > 0$.
3. Vertex form gives $f(x) = (x-1)^2 - 4$.

Read the vertex off the picture: it is at $(1, -4)$, so vertex form must be $(x-1)^2 - 4$. The graph opens upward, which rules out the negative leading coefficient, and the x-intercepts at -1 and 3 confirm the choice, since $(x-1)^2 - 4 = (x-3)(x+1)$.

19 **Answer: C**

1. Expand the square: $(x-3)^2 = x^2 - 6x + 9$.
2. Subtract 16: $x^2 - 6x + 9 - 16 = x^2 - 6x - 7$.
3. The constant -7 is the y-coordinate of the y-intercept.

The y-intercept is displayed by standard form, whose constant term is $f(0)$. Expanding, $(x-3)^2 - 16 = x^2 - 6x + 9 - 16 = x^2 - 6x - 7$, so the graph crosses the y-axis at $(0, -7)$. Choice A is what you get by expanding the square but forgetting the -16.

20 **Answer: 2**

1. Write $x + 2$ as $x - (-2)$, so the factor theorem gives $p(-2) = 0$.
2. $(-2)^3 + a(-2)^2 - 9(-2) - 18 = -8 + 4a + 18 - 18 = 4a - 8$.
3. Set $4a - 8 = 0$, so $a = 2$.

"$x+2$ is a factor" means $p(-2) = 0$ – note the sign, since $x+2 = x-(-2)$. Substituting gives $-8+4a+18-18 = 0$, so $4a = 8$ and $a = 2$.

21 **Answer: C**

1. $R(p) = p(120 - 4p)$ is zero at $p = 0$ and $p = 30$.
2. The vertex lies midway: $p = \frac{0+30}{2} = 15$.
3. $R(15) = 15(120 - 60) = 15(60) = 900$ dollars.

In factored form the zeros of R are $p = 0$ and $p = 30$, so by symmetry the maximum occurs halfway between, at $p = 15$; the revenue there is $15(120 - 60) = 900$. Choice A is the price that produces the maximum, not the revenue, and choice B is the second factor's value.

22 **Answer: B**

1. Vertex form gives vertex $(3, 5)$ and axis of symmetry $x = 3$.
2. $a < 0$ means the parabola opens downward, so the vertex is the highest point.
3. Therefore the maximum value of f is 5.

Since $a < 0$ the parabola opens downward, so the vertex $(3, 5)$ is a maximum and the greatest output is 5. The axis of symmetry is $x = 3$, not $x = 5$, and because the vertex lies above the x-axis on a downward parabola the graph must in fact cross the x-axis twice.

Module 2

1 **Answer: B**

1. $f(x) = -(x + 6)^2 + 11$ is greatest when $(x + 6)^2$ is smallest.
2. $(x + 6)^2 = 0$ when $x = -6$.

The squared term is subtracted, so f is largest when $(x + 6)^2 = 0$, which happens at $x = -6$. Writing vertex form as $a(x - h)^2 + k$ shows $x + 6 = x - (-6)$, so $h = -6$; the value 11 is the maximum itself, not where it occurs.

2 **Answer: 4**

1. Identify $a = 2$ and $b = -16$.
2. $x = -\frac{b}{2a} = \frac{16}{2(2)} = 4$.

The line of symmetry passes through the vertex, at $x = -\frac{b}{2a}$. With $a = 2$ and $b = -16$ this is $\frac{16}{4} = 4$; dividing by a alone rather than $2a$ would wrongly give 8.

3 **Answer: A**

1. Set each factor equal to zero.
2. $x = 0$; $x - 7 = 0$ gives $x = 7$; $x + 2 = 0$ gives $x = -2$.

Each factor contributes one zero: $x = 0$ from the bare x, $x = 7$ from $x - 7$, and $x = -2$ from $x + 2$. Dropping the factor x itself, as in choice C, is the most common slip, since it looks like a coefficient rather than a factor.

4 **Answer: B**

1. Write $x + 1 = x - (-1)$, so $a = -1$.
2. $p(-1) = (-1)^3 - 2(-1)^2 + 5 = -1 - 2 + 5 = 2$.

Dividing by $x + 1$ means dividing by $x - (-1)$, so the remainder is $p(-1) = -1 - 2 + 5 = 2$. Evaluating $p(1) = 4$ instead – the sign slip – is choice C.

5 Answer: 7

1. Substitute $x = 0$: $h(0) = 3(0-2)^2 - 5$.
2. $(0-2)^2 = 4$, so $h(0) = 12 - 5 = 7$.

The y-intercept is always $h(0)$, whatever form the rule is in: $3(0-2)^2 - 5 = 3(4) - 5 = 7$. Vertex form does not display the y-intercept, so the constant -5 is the minimum value, not the intercept.

6 Answer: A

1. The leading term is $4x^4$.
2. Even degree means both ends go the same direction.
3. The positive leading coefficient sends both ends upward.

Only the leading term $4x^4$ matters far from the origin. The degree 4 is even, so both ends behave the same way, and the coefficient 4 is positive, so both ends rise. The $-x^3$ term changes the middle of the graph but not its ends.

7 Answer: C

1. An x-intercept at $(6, 0)$ needs the factor $x - 6$.
2. An x-intercept at $(-1, 0)$ needs the factor $x + 1$.
3. So $y = (x-6)(x+1)$.

A zero at $x = r$ comes from the factor $x - r$, so intercepts at 6 and -1 require the factors $x - 6$ and $x - (-1) = x + 1$. Choice A reverses both signs and would give intercepts at -6 and 1.

8 Answer: 1

1. $h = -\frac{b}{2a} = 5$.
2. $k = f(5) = 25 - 50 + 21 = -4$.
3. $h + k = 5 + (-4) = 1$.

Half of -10 is -5, and $(x-5)^2 = x^2 - 10x + 25$, which is 4 too large, so $f(x) = (x-5)^2 - 4$. That gives $h = 5$ and $k = -4$, so $h + k = 1$.

9 Answer: B

1. The parabola opens downward, so the vertex gives the greatest height.
2. $t = -\frac{b}{2a} = -\frac{40}{-10} = 4$ seconds.

The question asks *when*, so it wants the x-coordinate of the vertex: $t = -\frac{40}{2(-5)} = 4$ seconds. Choice D, 82, is the greatest height in metres, and choice A is the launch height – both are numbers in the problem, but neither is a time.

10 Answer: C

1. $h = -\frac{b}{2a} = \frac{12}{4} = 3$.
2. $k = f(3) = 2(9) - 12(3) + 13 = -5$.
3. So $2x^2 - 12x + 13 = 2(x-3)^2 - 5$.

Vertex form displays the minimum. Here $h = -\frac{-12}{2(2)} = 3$ and $k = f(3) = 18 - 36 + 13 = -5$, so the expression is $2(x-3)^2 - 5$; expanding confirms $2x^2 - 12x + 18 - 5 = 2x^2 - 12x + 13$. Choice B uses b itself instead of $-\frac{b}{2a}$ for h.

11 **Answer: D**

1. $p(2) = (2-1)(2+3)(2+4)$.
2. $= (1)(5)(6) = 30$.

Substitute directly into the factored form rather than expanding: $p(2) = (2-1)(2+3)(2+4) = (1)(5)(6) = 30$. Choice A comes from mishandling the sign of the first factor as $1-2=-1$.

12 **Answer: A**

1. The zero $x=-1$ has multiplicity 3, which is odd, so the graph crosses there.
2. The zero $x=2$ has multiplicity 2, which is even, so the graph touches there.

Multiplicity decides the behaviour at a zero. The factor $(x+1)$ has odd multiplicity 3, so the graph passes through the axis at $x=-1$; the factor $(x-2)$ has even multiplicity 2, so near $x=2$ the sign of p cannot change and the graph touches and turns back.

13 **Answer: A**

1. Read the vertex from the graph: $(-2,-9)$.
2. $h=-2$ gives the factor $(x-(-2))^2 = (x+2)^2$, and $k=-9$.
3. The graph opens upward, so $y=(x+2)^2-9$.

The lowest point of the graph is $(-2,-9)$, so vertex form needs $h=-2$ and $k=-9$, and $h=-2$ appears in the equation as $x+2$. The curve opens upward, so the leading coefficient is positive; as a check, the x-intercepts -5 and 1 satisfy $(x+2)^2-9=(x+5)(x-1)$.

14 **Answer: 2**

1. The axis of symmetry is $x=\frac{1+9}{2}=5$.
2. $f(5)=a(5-1)(5-9)=-16a$.
3. Set $-16a=-32$, so $a=2$.

The minimum occurs on the axis of symmetry, midway between the intercepts, at $x=5$. Then $f(5)=a(4)(-4)=-16a$, and setting $-16a=-32$ gives $a=2$.

15 **Answer: D**

1. For a factor $x-a$, the factor theorem requires $p(a)=0$.
2. $p(2)=-18$, $p(1)=-20$, $p(-3)=12$: none works.
3. $p(3)=27+18-33-12=0$, so $x-3$ is a factor.

Test each candidate with the factor theorem instead of dividing. $p(3)=27+18-33-12=0$, so $x-3$ is a factor. The other candidates give $p(2)=-18$, $p(1)=-20$ and $p(-3)=12$, none of which is zero.

16 **Answer: B**

1. Factor out the common factor: $2x^2-2x-12=2(x^2-x-6)$.
2. $x^2-x-6=(x-3)(x+2)$.
3. So the expression is $2(x-3)(x+2)$, showing intercepts at $x=3$ and $x=-2$.

Factor the common 2 out first: $2x^2-2x-12=2(x^2-x-6)=2(x-3)(x+2)$. Choice A factors correctly but drops the leading 2, so it is not equivalent to the original expression – and equivalence is part of what the question demands, even though both would suggest the same intercepts.

17 Answer: -6

1. The factor theorem gives $p(2) = 0$.
2. $3(8) - 5(4) + 2k + 8 = 24 - 20 + 2k + 8 = 2k + 12$.
3. Set $2k + 12 = 0$, so $k = -6$.

By the factor theorem, $x - 2$ being a factor means $p(2) = 0$. Evaluating gives $24 - 20 + 2k + 8 = 2k + 12$, so $2k + 12 = 0$ and $k = -6$.

18 Answer: B

1. $x = -\frac{b}{2a} = -\frac{5}{2}$.
2. $f\left(-\frac{5}{2}\right) = \frac{25}{4} - \frac{25}{2} + 3$.
3. $= \frac{25-50+12}{4} = -\frac{13}{4}$.

The minimum value is the output at the vertex, so first find $x = -\frac{5}{2}$, then substitute: $\frac{25}{4} - \frac{25}{2} + 3 = -\frac{13}{4}$. Choice C is the x-coordinate of the vertex, the classic wrong-coordinate answer on this topic.

19 Answer: C

1. Vertex $(3, -4)$ with leading coefficient 1 gives $y = (x-3)^2 - 4$.
2. Expand: $x^2 - 6x + 9 - 4 = x^2 - 6x + 5$.
3. So $b = -6$ and $c = 5$.

Start from the form that shows the vertex, $y = (x-3)^2 - 4$, then expand into standard form to read off c: $x^2 - 6x + 9 - 4 = x^2 - 6x + 5$. Equivalently, $c = f(0) = 9 - 4 = 5$. Choice A is b, and choice B is the y-coordinate of the vertex.

20 Answer: 212

1. $x = -\frac{b}{2a} = -\frac{56}{-4} = 14$ units.
2. $P(14) = -2(196) + 56(14) - 180$.
3. $= -392 + 784 - 180 = 212$ dollars.

The parabola opens downward, so the maximum profit is the value at the vertex. It occurs at $x = -\frac{56}{2(-2)} = 14$ units, and $P(14) = -392 + 784 - 180 = 212$ dollars. Stopping at 14 answers "how many units", not "how much profit".

21 Answer: A

1. $(x-1)^2$ has even multiplicity, so the graph touches at $x = 1$.
2. The degree is $2 + 1 = 3$ and the leading coefficient is -1.
3. Odd degree with a negative lead means $p(x)$ falls without bound as x grows.

Two independent readings are needed. The factor $(x-1)^2$ has even multiplicity, so the graph touches the axis at $x = 1$, while $(x+3)$ has multiplicity 1, so it crosses at $x = -3$. Expanding mentally, the leading term is $-x^3$: odd degree with a negative coefficient, so the right-hand end falls.

22 **Answer: A**

1. Confirm $p(3) = 27 - 54 + 33 - 6 = 0$, so $x - 3$ is a factor.
2. The quotient is quadratic with leading coefficient 1 and constant $\frac{-6}{-3} = 2$.
3. Expanding $(x-3)(x^2 - 3x + 2) = x^3 - 6x^2 + 11x - 6$ confirms $q(x) = x^2 - 3x + 2$.

Because $p(3) = 27 - 54 + 33 - 6 = 0$, the factor $x - 3$ does divide p, and the quotient must be a quadratic $x^2 + mx + 2$: its leading coefficient is 1 to reproduce x^3, and its constant term must satisfy $(-3)(\text{constant}) = -6$, so the constant is 2. Matching the x^2 term, $m - 3 = -6$ gives $m = -3$, so $q(x) = x^2 - 3x + 2$.

Topic 09 — Exponential Functions, Radicals and Rational Expressions

Module 1

1 **Answer: B**

1. The year 2015 corresponds to $x = 0$.
2. Evaluate $P(0) = 750(1.06)^0$.
3. Any nonzero base raised to the 0 power is 1, so $P(0) = 750$.

In $a \cdot b^x$ the number a is the value when $x = 0$, because $b^0 = 1$. Setting $x = 0$ gives $P(0) = 750(1.06)^0 = 750$, so the initial population is 750. The number 1.06 is the growth factor, not a population.

2 **Answer: D**

1. Each year the car keeps $100\% - 15\% = 85\%$ of its value.
2. Write 85% as the decimal 0.85; this is the decay factor.
3. Multiply the starting value by the decay factor once per year: $V(t) = 24000(0.85)^t$.

A loss of 15% leaves 85% of the value, so each year the value is multiplied by $1 - 0.15 = 0.85$. The decay factor is 0.85, not 0.15: multiplying by 0.15 would leave only 15% of the car's value after one year.

3 **Answer: 24**

1. Square both sides: $2x + 1 = 49$.
2. Subtract 1: $2x = 48$.
3. Divide by 2: $x = 24$, and $\sqrt{49} = 7$ checks.

Squaring both sides removes the radical: $2x + 1 = 49$, so $x = 24$. Checking, $\sqrt{2(24)+1} = \sqrt{49} = 7$, so the solution is genuine and not extraneous.

4 **Answer: C**

1. Multiply both sides by $x - 2$: $12 = 4(x - 2)$.
2. Divide by 4: $3 = x - 2$.
3. Add 2: $x = 5$, and $x = 5$ does not make the denominator zero.

A rational equation with a single fraction is fastest to clear by multiplying both sides by the denominator: $12 = 4(x - 2)$. Then $3 = x - 2$ and $x = 5$. Answer 3 comes from stopping at $x - 2 = 3$.

5 Answer: C

1. Find the common ratio: $15 \div 5 = 3$.
2. So $f(x) = 5(3)^x$.
3. Evaluate $f(4) = 5(3)^4 = 5(81) = 405$.

Each step in x multiplies the output by the same factor: $15/5 = 3$ and $45/15 = 3$, so $f(x) = 5(3)^x$. Then $f(4) = 5 \cdot 81 = 405$. A common slip is to answer $f(3) = 135$.

6 Answer: 10

1. Substitute $x = 3$: $A(3) = 80(0.5)^3$.
2. Compute $(0.5)^3 = 0.125$.
3. Multiply: $80(0.125) = 10$ grams.

Each half-life multiplies the amount by 0.5, so after three of them the sample has been halved three times: $80 \to 40 \to 20 \to 10$. Equivalently $A(3) = 80(0.5)^3 = 80/8 = 10$.

7 Answer: C

1. An increase of 7% gives $100\% + 7\% = 107\%$ of the original.
2. Convert 107% to a decimal: 1.07.
3. The growth factor is $1 + r = 1.07$.

"Increases by 7%" means the new amount is the old amount plus 7% of it, that is $100\% + 7\% = 107\%$ of the old amount. As a decimal that is 1.07. Using 0.07 instead of 1.07 models keeping only 7% of the quantity, which is the single most common error in this topic.

8 Answer: B

1. Square both sides: $x + 7 = x^2 + 2x + 1$.
2. Rearrange: $x^2 + x - 6 = 0$, so $(x + 3)(x - 2) = 0$.
3. Check both: $x = -3$ gives $2 = -2$, false; $x = 2$ gives $3 = 3$, true.

Squaring gives $x + 7 = x^2 + 2x + 1$, so $x^2 + x - 6 = 0$ and $x = 2$ or $x = -3$. Squaring can create solutions the original equation never had: at $x = -3$ the left side is $\sqrt{4} = 2$ while the right side is -2. Only $x = 2$ survives the check.

9 Answer: 2

1. Rewrite with denominator $2x$: $\frac{2}{2x} + \frac{1}{2x} = \frac{3}{2x}$.
2. Set $\frac{3}{2x} = \frac{3}{4}$; equal numerators force equal denominators.
3. So $2x = 4$ and $x = 2$.

The two fractions on the left have a common denominator of $2x$: $\frac{2}{2x} + \frac{1}{2x} = \frac{3}{2x}$. Setting $\frac{3}{2x} = \frac{3}{4}$ forces $2x = 4$, so $x = 2$, which is allowed since it does not make a denominator zero.

10 Answer: C

1. The base of the exponential is the yearly multiplier.
2. Multiplying by 1.03 is the same as adding 3%.
3. So each year's population is 1.03 times the previous year's.

In $a \cdot b^t$ the base b is a multiplier, not an amount added. Because $b = 1.03 = 1 + 0.03$, the population is 1.03 times as large each year, an increase of 3% per year — not 103% and not 1.03 people.

11 **Answer: D**

1. Monthly rate: $0.06 \div 12 = 0.005$, so the monthly factor is 1.005.
2. Number of compounding periods in t years: $12t$.
3. Balance: $B(t) = 1500(1.005)^{12t}$.

Compounding monthly means the annual rate is split into 12 equal pieces, $0.06/12 = 0.005$ per month, and there are $12t$ months in t years. So the factor is $(1.005)^{12t}$. Using 1.06 with the exponent $12t$ would compound a full 6% twelve times a year.

12 **Answer: A**

1. Cross multiply: $5(x-1) = 2(x+1)$.
2. Expand: $5x - 5 = 2x + 2$.
3. Solve: $3x = 7$, so $x = \frac{7}{3}$.

Two equal fractions can be cross multiplied: $5(x-1) = 2(x+1)$, giving $5x-5 = 2x+2$ and $x = \frac{7}{3}$. Since $\frac{7}{3}$ is neither 1 nor -1, no denominator is zero and the solution is valid.

13 **Answer: B**

1. Each year the machine keeps $100\% - 20\% = 80\%$ of its value.
2. After 3 years the value is $30000(0.8)^3$.
3. Compute $(0.8)^3 = 0.512$, so the value is \$15,360.

Percent decay is repeated multiplication, not repeated subtraction: the machine keeps 80% of its value each year, so its value is $30000(0.8)^3 = 30000(0.512) = 15360$. Subtracting 20% of the original three times would wrongly give \$12,000.

14 **Answer: A**

1. Square both sides: $3x + 4 = x^2$.
2. Solve $x^2 - 3x - 4 = 0$: $x = -1$ or $x = 4$.
3. Test each in the original: $x = -1$ gives $1 = -1$, false, so -1 is extraneous.

Squaring gives $3x+4 = x^2$, so $x^2 - 3x - 4 = 0$ and $x = -1$ or $x = 4$. A square root is never negative, so the right side must satisfy $x \geq 0$; $x = -1$ fails that test ($\sqrt{1} = 1 \neq -1$) and is extraneous, while $x = 4$ checks.

15 **Answer: 4800**

1. Count doublings: $20 \div 5 = 4$.
2. Multiply by 2 four times: $300(2)^4$.
3. $300 \cdot 16 = 4800$ bacteria.

The number of doublings is $20 \div 5 = 4$, so the population is multiplied by 2 four times: $300 \cdot 2^4 = 300 \cdot 16 = 4800$. Writing $300(2)^{t/5}$ and substituting $t = 20$ gives the same thing.

16 **Answer: B**

1. The factor for one 4-year period is $1 + 0.12 = 1.12$.
2. In t years there are $t/4$ such periods.
3. So $V(t) = 5000(1.12)^{t/4}$.

The growth factor 1.12 belongs to a 4-year period, so the exponent must count 4-year periods: after t years there are $t/4$ of them. Writing $4t$ in the exponent would apply the 12% growth four times per year instead of once every four years.

17 **Answer: A**

1. Multiply both sides by $x - 3$: $x^2 = 9$.
2. So $x = 3$ or $x = -3$.
3. Reject $x = 3$: it makes $x - 3 = 0$, so the original expression is undefined.

Multiplying by $x - 3$ gives $x^2 = 9$, so $x = 3$ or $x = -3$. But $x = 3$ makes the original denominator zero, so it is not in the domain and must be thrown out. Only $x = -3$ is a solution.

18 **Answer: 4**

1. Isolate a radical: $\sqrt{x+5} = 1 + \sqrt{x}$.
2. Square: $x + 5 = 1 + 2\sqrt{x} + x$, so $2\sqrt{x} = 4$.
3. Then $\sqrt{x} = 2$ and $x = 4$, which checks in the original.

Isolate one radical before squaring: $\sqrt{x+5} = 1 + \sqrt{x}$. Squaring gives $x + 5 = 1 + 2\sqrt{x} + x$, so $4 = 2\sqrt{x}$ and $\sqrt{x} = 2$, hence $x = 4$. Checking, $\sqrt{9} - \sqrt{4} = 3 - 2 = 1$.

19 **Answer: D**

1. The base is 2 because the value doubles.
2. One doubling period is 9 years, so t years contain $t/9$ periods.
3. Therefore $V(t) = 2500(2)^{t/9}$.

"Doubles every 9 years" means the base is 2 and one full period is 9 years, so the exponent counts periods: $t/9$. The exponent $9t$ would double the money nine times a year.

20 **Answer: C**

1. Write $8 = 2^3$, so $8^{x+1} = 2^{3(x+1)} = 2^{3x+3}$.
2. Divide by 2^x by subtracting exponents: $2^{3x+3-x} = 2^{2x+3}$.
3. So $a = 2$ and $b = 3$, giving $a + b = 5$.

Rewrite every power with the same base before dividing: $8^{x+1} = (2^3)^{x+1} = 2^{3x+3}$. Dividing powers of the same base subtracts exponents, so $\frac{2^{3x+3}}{2^x} = 2^{3x+3-x} = 2^{2x+3}$. Then $a = 2$, $b = 3$ and $a + b = 5$.

21 **Answer: 11**

1. Square both sides: $2x + 3 = x^2 - 12x + 36$.
2. Rearrange: $x^2 - 14x + 33 = 0$, so $(x - 3)(x - 11) = 0$.
3. Check: $x = 3$ gives $3 = -3$, false; $x = 11$ gives $5 = 5$, true.

Squaring gives $2x + 3 = x^2 - 12x + 36$, so $x^2 - 14x + 33 = 0$ and $x = 3$ or $x = 11$. Because the left side is never negative, any solution must satisfy $x - 6 \geq 0$; $x = 3$ fails and is extraneous, and $x = 11$ checks: $\sqrt{25} = 5 = 11 - 6$.

22 Answer: A

1. Find the multiplier over 10 years: $12000 \div 8000 = 1.5$.
2. That factor applies once per 10 years, so use the exponent $t/10$.
3. Therefore $P(t) = 8000(1.5)^{t/10}$.

Over the 10 years the population was multiplied by $12000/8000 = 1.5$, so 1.5 is the factor for a 10-year period and the exponent must be $t/10$. The model is $P(t) = 8000(1.5)^{t/10}$; a linear model $8000 + 400t$ would not keep the ratio constant.

Module 2

1 Answer: B

1. Write the base as $1 + r$: $1.15 = 1 + 0.15$.
2. The rate of increase is $r = 0.15$.
3. As a percent, $r = 15\%$, so $p = 15$.

The base 1.15 equals $1 + 0.15$, and the 0.15 is the part added, so the percent increase is $0.15 \cdot 100 = 15$. Reading the whole base as the percent gives 115, a classic misread.

2 Answer: 1

1. Multiply both sides by $x + 3$: $8 = 2(x + 3)$.
2. Divide by 2: $4 = x + 3$.
3. Subtract 3: $x = 1$.

Multiplying by $x+3$ gives $8 = 2(x+3)$, so $4 = x+3$ and $x = 1$. The value $x = 1$ does not make the denominator zero, so it is a genuine solution.

3 Answer: D

1. Subtract 4: $\sqrt{x-1} = 5$.
2. Square both sides: $x - 1 = 25$.
3. Add 1: $x = 26$, and $\sqrt{25} + 4 = 9$ checks.

Isolate the radical first: $\sqrt{x-1} = 5$. Only then square, giving $x - 1 = 25$ and $x = 26$. Squaring before isolating, or forgetting to undo the -1, produces 24.

4 Answer: D

1. After the decrease, $100\% - 35\% = 65\%$ of the value remains.
2. Write 65% as a decimal: 0.65.
3. The decay factor is $1 - r = 0.65$.

A decrease of 35% leaves $100\% - 35\% = 65\%$, so the factor is 0.65. The decay factor is what remains, not what is lost, so 0.35 is the trap answer.

5 Answer: C

1. Substitute $x = -1$: $g(-1) = 6(3)^{-1}$.
2. Rewrite $3^{-1} = \frac{1}{3}$.
3. Multiply: $6 \cdot \frac{1}{3} = 2$.

A negative exponent means a reciprocal: $3^{-1} = \frac{1}{3}$, so $g(-1) = 6 \cdot \frac{1}{3} = 2$. A negative exponent never makes the output negative, so -18 is wrong.

6 Answer: B

1. The base is 0.5 because the amount halves.
2. A period is 12 years long, so t years contain $t/12$ periods.
3. Therefore $A(t) = 400(0.5)^{t/12}$.

Half-life means the base is $\frac{1}{2}$ and one period lasts 12 years, so the exponent must count periods: $t/12$. The exponent $12t$ would halve the sample twelve times a year.

7 Answer: 882

1. Substitute $t = 2$: $V(2) = 800(1.05)^2$.
2. Compute $(1.05)^2 = 1.1025$.
3. Multiply: $800(1.1025) = 882$ dollars.

Substituting $t = 2$ gives $800(1.05)^2 = 800(1.1025) = 882$. Adding 5% of 800 twice would give 880; compound growth is applied to the new balance each year, not the original.

8 Answer: A

1. Multiply each term by $x - 1$: $x = 2 + 3(x - 1)$.
2. Expand and collect: $x = 3x - 1$, so $-2x = -1$.
3. Then $x = \frac{1}{2}$, which does not make $x - 1 = 0$.

Multiply every term by $x - 1$: $x = 2 + 3(x - 1)$, so $x = 3x - 1$ and $x = \frac{1}{2}$. Since $\frac{1}{2} \neq 1$ the denominator is not zero, so this solution is genuine.

9 Answer: 5

1. Square both sides: $5x - 4 = x^2$.
2. Solve $x^2 - 5x + 4 = 0$: $x = 1$ or $x = 4$.
3. Both check in the original equation, so the sum is $1 + 4 = 5$.

Squaring gives $5x - 4 = x^2$, so $x^2 - 5x + 4 = 0$ and $x = 1$ or $x = 4$. Both make the right side nonnegative and both check ($\sqrt{1} = 1$ and $\sqrt{16} = 4$), so neither is extraneous and the sum is 5.

10 Answer: D

1. Convert months to years: $t = \frac{m}{12}$.
2. Substitute into the model: $3000(1.08)^{m/12}$.
3. The base stays 1.08 because the yearly growth has not changed.

Changing the time unit changes the exponent, not the base: m months is $m/12$ years, so replace t by $m/12$. The yearly factor 1.08 stays; dividing the base by 12 is not the same as dividing the exponent.

11 **Answer: D**

1. Count periods: $12 \div 4 = 3$.
2. Multiply by 3 three times: $90(3)^3$.
3. $90 \cdot 27 = 2430$ bacteria.

There are $12 \div 4 = 3$ tripling periods, so the count is multiplied by 3 three times: $90 \cdot 3^3 = 90 \cdot 27 = 2430$. Tripling only once gives 270, and multiplying by 12 gives 1080.

12 **Answer: B**

1. Multiply through by $4x(x+2)$: $4(x+2) + 4x = 3x(x+2)$.
2. Simplify: $8x + 8 = 3x^2 + 6x$, so $3x^2 - 2x - 8 = 0$.
3. Factor $(3x+4)(x-2) = 0$ and keep the positive root, $x = 2$.

Multiplying by the common denominator $4x(x+2)$ gives $4(x+2) + 4x = 3x(x+2)$, so $8x + 8 = 3x^2 + 6x$ and $3x^2 - 2x - 8 = 0$. Factoring, $(3x+4)(x-2) = 0$, so $x = -\frac{4}{3}$ or $x = 2$; the positive solution is 2.

13 **Answer: C**

1. The base of an exponential model multiplies the previous value.
2. Here that multiplier is 0.88, so 88% of the value remains.
3. Equivalently, the car loses 12% of its value each year.

The base is the yearly multiplier, so each year the car is worth 0.88 times its previous value — a loss of 12%, not 88%. Confusing what remains (88%) with what is lost (12%) is the standard decay-model error.

14 **Answer: A**

1. Factor the numerator: $x^2 - 9 = (x+3)(x-3)$.
2. Write the quotient as $\frac{(x+3)(x-3)}{x+3}$.
3. Cancel $x + 3$ to get $x - 3$.

The numerator is a difference of squares: $x^2 - 9 = (x+3)(x-3)$. Cancelling the common factor $x + 3$ leaves $x - 3$. Cancelling is legal only because $x \neq -3$, which is why the restriction is stated.

15 **Answer: C**

1. Square both sides: $4x = x^2 - 6x + 9$.
2. Solve $x^2 - 10x + 9 = 0$: $x = 1$ or $x = 9$.
3. Check: $x = 1$ gives $2 = -2$, false; $x = 9$ gives $6 = 6$, true.

Squaring gives $4x = x^2 - 6x + 9$, so $x^2 - 10x + 9 = 0$ and $x = 1$ or $x = 9$. At $x = 1$ the equation reads $2 = -2$, which is false, so 1 is extraneous; $x = 9$ gives $6 = 6$ and is the only solution.

16 **Answer: C**

1. Set $1600\left(\frac{1}{2}\right)^{t/8} = 100$, so $\left(\frac{1}{2}\right)^{t/8} = \frac{1}{16}$.
2. Since $\frac{1}{16} = \left(\frac{1}{2}\right)^4$, the exponents match: $\frac{t}{8} = 4$.
3. Therefore $t = 32$ years.

Divide to find the number of halvings: $\frac{100}{1600} = \frac{1}{16} = \left(\frac{1}{2}\right)^4$, so $t/8 = 4$ and $t = 32$. Reading the 8 as the answer, or halving once too few times, gives 8 or 24.

17 **Answer: B**

1. Square both sides: $2x + 7 = x + 4 + 2\sqrt{x+3}$, so $x + 3 = 2\sqrt{x+3}$.
2. Square again: $(x+3)^2 = 4(x+3)$, giving $x = -3$ or $x = 1$.
3. Check both: $x = -3$ gives $1 = 1$ and $x = 1$ gives $3 = 3$, so the sum is $-3 + 1 = -2$.

Squaring gives $2x + 7 = 1 + 2\sqrt{x+3} + x + 3$, so $x + 3 = 2\sqrt{x+3}$. Squaring again gives $(x+3)^2 = 4(x+3)$, so $x + 3 = 0$ or $x + 3 = 4$, that is $x = -3$ or $x = 1$. Both check here, so the sum is -2; the lesson is that a candidate is extraneous only if the check says so.

18 **Answer: A**

1. Find the two-year multiplier: $720 \div 500 = 1.44$.
2. The annual factor satisfies $b^2 = 1.44$, so $b = \sqrt{1.44} = 1.2$.
3. Since $b = 1 + 0.20$, the annual increase is 20%, so $p = 20$.

Two years of growth multiply the population by $\frac{720}{500} = 1.44$, so the one-year factor b satisfies $b^2 = 1.44$ and $b = 1.2$. Then $p = 20$. Halving the total percent increase of 44% would give 22, which ignores compounding.

19 **Answer: 9/5**

1. Factor $x^2 - 1 = (x-1)(x+1)$ to find the common denominator.
2. Multiply through: $3(x+1) + 2(x-1) = 10$.
3. Simplify: $5x + 1 = 10$, so $x = \frac{9}{5}$; check that $x \neq \pm 1$.

Since $x^2 - 1 = (x-1)(x+1)$, the common denominator is $(x-1)(x+1)$. Multiplying through gives $3(x+1) + 2(x-1) = 10$, so $5x + 1 = 10$ and $x = \frac{9}{5}$. That value is neither 1 nor -1, so it does not make a denominator zero and is a genuine solution.

20 **Answer: 5**

1. Rewrite 27 as 3^3: $27^{x-2} = 3^{3(x-2)} = 3^{3x-6}$.
2. With equal bases, set the exponents equal: $2x - 1 = 3x - 6$.
3. Solve: $x = 5$.

Write both sides with the same base: $27 = 3^3$, so $27^{x-2} = 3^{3x-6}$. When two equal powers share a base, their exponents are equal: $2x - 1 = 3x - 6$, so $x = 5$.

21 **Answer: B**

1. Convert years to decades: $d = \frac{y}{10}$.
2. Substitute into $P(d) = 8(2)^d$.
3. This gives $P(y) = 8(2)^{y/10}$.

One decade is 10 years, so $d = \frac{y}{10}$. Substituting changes only the exponent: $P(y) = 8(2)^{y/10}$. Multiplying the exponent by 10 instead would double the population every year.

22 Answer: A

1. Factor $x^2 - 4 = (x-2)(x+2)$ and multiply every term by it.
2. This gives $(x+2) + (x-2) = 4$, so $2x = 4$ and $x = 2$.
3. Check: $x = 2$ makes $x - 2 = 0$, so it is extraneous and there are no solutions.

With $x^2 - 4 = (x-2)(x+2)$, multiplying through gives $(x+2) + (x-2) = 4$, so $2x = 4$ and $x = 2$. But $x = 2$ makes two of the original denominators zero, so it is extraneous and must be discarded — the equation has no solution at all.

Topic 10 — Nonlinear Systems and Function Transformations

Module 1

1 Answer: A

1. A shift up adds to the output: replace y by $y - 5$, i.e. $y = x^2 + 5$.
2. Check the vertex: $(0, 0)$ moves to $(0, 5)$, and $y = 0^2 + 5 = 5$.

Adding a constant *outside* the function changes the output, so every point moves straight up: $y = x^2 + 5$. The choices with 5 inside the parentheses move the graph sideways, not up.

2 Answer: B

1. The change $x - 3$ is inside the function, so the shift is horizontal.
2. $g(3) = f(0)$: the point that f had at $x = 0$ now sits at $x = 3$.
3. So the graph moves 3 units to the right.

A change inside the parentheses moves the graph horizontally, and it moves it the *opposite* way from the sign you see. To get the same output that f gave at some input, g needs an x that is 3 larger, so every point slides 3 units right.

3 Answer: 3

1. Set $2x = x^2 - 3$.
2. Rearrange: $x^2 - 2x - 3 = 0$.
3. Factor: $(x-3)(x+1) = 0$, so $x = 3$ or $x = -1$.
4. The greatest is $x = 3$.

Two graphs meet where their y-values agree, so setting the two expressions equal turns the system into a single quadratic equation in x.

4 Answer: A

1. A reflection across the x-axis sends (x, y) to $(x, -y)$.
2. The x-values are untouched, so the negative sign goes outside: $y = -f(x)$.

Reflecting across the x-axis flips the sign of every output, which is a change on the outside: $y = -f(x)$. Changing the sign of the input instead, $f(-x)$, would flip the graph left-to-right.

5 **Answer: C**

1. The factor 3 multiplies the output, not the input.
2. A point (a, b) becomes $(a, 3b)$.
3. That is a vertical stretch by a factor of 3.

Multiplying the whole function by 3 triples every output while leaving every input alone, so the graph is pulled away from the x-axis: a vertical stretch by a factor of 3.

6 **Answer: D**

1. Set $x^2 = x + 6$, so $x^2 - x - 6 = 0$.
2. Factor: $(x - 3)(x + 2) = 0$, giving $x = 3$ and $x = -2$.
3. Then $y = 3^2 = 9$ and $y = (-2)^2 = 4$.
4. The sum is $9 + 4 = 13$.

Solve for the x-coordinates first, then feed each one back into either equation. A common slip is to stop after finding the x-values, or to report only one of the two points.

7 **Answer: A**

1. Set $x^2 + 1 = x - 2$, so $x^2 - x + 3 = 0$.
2. Discriminant: $(-1)^2 - 4(1)(3) = 1 - 12 = -11$.
3. A negative discriminant means no real solutions, so the graphs never meet.

The number of intersection points is the number of real solutions of the combined equation, and the discriminant settles that without solving anything.

8 **Answer: B**

1. Substitute: $x^2 - 4x + 7 = 2x - 1$.
2. Rearrange: $x^2 - 6x + 8 = 0$, so $(x - 2)(x - 4) = 0$.
3. Then $x = 2$ gives $y = 2(2) - 1 = 3$, and $x = 4$ gives $y = 7$.
4. $(2, 3)$ is one of the two solutions.

Substituting the line into the parabola is the whole method: it removes y and leaves a quadratic in x. A point must satisfy *both* equations, so always finish by computing the matching y.

9 **Answer: -4**

1. Substitute: $x^2 = 4x + c$, so $x^2 - 4x - c = 0$.
2. One solution means the discriminant is 0: $(-4)^2 - 4(1)(-c) = 0$.
3. $16 + 4c = 0$, so $c = -4$.

Tangent means the line touches the curve exactly once, so the quadratic you get after substituting must have exactly one real solution — its discriminant is zero.

10 Answer: B

1. Substitute $y = x + 1$: $x^2 + (x + 1)^2 = 25$.
2. Expand: $2x^2 + 2x - 24 = 0$, so $x^2 + x - 12 = 0$.
3. Factor: $(x + 4)(x - 3) = 0$, giving $x = -4$ and $x = 3$.
4. Then $y = -3$ and $y = 4$; the greatest is 4.

A circle and a line are handled exactly like a parabola and a line: substitute the linear equation into the other one. The quadratic that appears has two roots, and each root produces its own y.

11 Answer: D

1. $g(1) = f(1 + 2) = f(3)$.
2. From the table, $f(3) = 8$.
3. So $g(1) = 8$.

Evaluating a transformed function is a substitution, not a guess: $g(1)$ means f evaluated at $1 + 2 = 3$. Reading $f(1)$ or $f(-1)$ instead is the usual error, and the table is built so both wrong readings are available.

12 Answer: C

1. $x + 4$ inside the square shifts the graph 4 units *left*, since the vertex needs $x + 4 = 0$, i.e. $x = -4$.
2. -3 outside subtracts 3 from every output: 3 units down.
3. The new vertex is $(-4, -3)$.

Read the two changes separately. The $+4$ sits inside the square, so it is horizontal and it moves the graph the opposite way: 4 units left. The -3 sits outside, so it means exactly what it says: 3 units down.

13 Answer: C

1. $y = 3f(x)$ means the output is tripled.
2. At $x = 2$: $3f(2) = 3(5) = 15$.
3. So $k = 15$.

A vertical stretch multiplies the y-coordinate and leaves the x-coordinate alone, so $(2, 5)$ becomes $(2, 15)$. Adding 3 instead of multiplying, or dividing by 3, are the two mistakes this question is built to catch.

14 Answer: B

1. $g(x) = f(-x) = (-x)^2 - 6(-x)$.
2. $(-x)^2 = x^2$ and $-6(-x) = 6x$.
3. So $g(x) = x^2 + 6x$.

Replacing x by $-x$ reflects the graph across the y-axis. Do it algebraically: every x in the formula becomes $-x$, so the even-power term is unchanged and the odd-power term flips sign.

15 Answer: 4

1. Set $-x^2 + 6x = 5$.
2. Rearrange: $x^2 - 6x + 5 = 0$.
3. Factor: $(x - 1)(x - 5) = 0$, so $x = 1$ and $x = 5$.
4. The positive difference is $5 - 1 = 4$.

A horizontal line makes the substitution trivial: set the quadratic equal to 5 and solve. The two roots are the x-coordinates, and their difference is what the question asks for.

16 **Answer: B**

1. Substitute: $x^2 = mx - 4$, so $x^2 - mx + 4 = 0$.
2. Exactly one solution means $(-m)^2 - 4(1)(4) = 0$.
3. $m^2 - 16 = 0$, so $m = 4$ or $m = -4$.
4. Since $m > 0$, $m = 4$.

Tangency is a discriminant question in disguise. Substituting gives a quadratic whose discriminant must be zero; that produces $m^2 = 16$, and the condition $m > 0$ picks the root.

17 **Answer: 12**

1. Set $x^2 - 2x - 3 = 2x + 2$.
2. Rearrange: $x^2 - 4x - 5 = 0$, so $(x-5)(x+1) = 0$.
3. $x = 5$ gives $y = 2(5) + 2 = 12$; $x = -1$ gives $y = 2(-1) + 2 = 0$.
4. $y_1 + y_2 = 12 + 0 = 12$.

Find both x-values, then convert each to a y-value with the easier of the two equations — the line. Adding the x-values instead of the y-values is the trap here.

18 **Answer: A**

1. From the graph, the vertex of f is $(0, -4)$.
2. $g(x) = f(x+3)$ shifts the graph 3 units left.
3. Check: $g(-3) = f(0) = -4$.
4. The vertex is $(-3, -4)$.

The vertex of f is at $(0, -4)$. Since the $+3$ is inside the function, the graph slides horizontally, and it slides 3 units *left*, carrying the vertex to $(-3, -4)$. The height is untouched because nothing was added outside.

19 **Answer: D**

1. Multiplying by -1 turns the minimum of f into a maximum of g.
2. That extreme output is $-2 \cdot (-3) = 6$.
3. So the maximum value of g is 6, still at $x = 1$.

The factor -2 does two things at once: the negative sign flips the graph across the x-axis, so a minimum becomes a maximum, and the 2 doubles the distance from the axis. So the least output -3 becomes the greatest output 6.

20 **Answer: 15**

1. Substitute: $x^2 + (2x + c)^2 = 45$.
2. Expand: $5x^2 + 4cx + (c^2 - 45) = 0$.
3. Set the discriminant to 0: $(4c)^2 - 4(5)(c^2 - 45) = 0$.
4. $16c^2 - 20c^2 + 900 = 0$, so $c^2 = 225$ and $c = 15$ (as $c > 0$).

Tangency to a circle works the same way as tangency to a parabola: substitute, collect a quadratic in x, and force the discriminant to zero so the two intersection points collapse into one.

21 **Answer: C**

1. Shift right 2: $y = f(x-2)$.
2. Reflect across the x-axis: negate the output.
3. $y = -f(x-2)$.

Handle the two steps in order. The shift right 2 replaces x by $x-2$, giving $y = f(x-2)$; reflecting that across the x-axis negates the whole output, giving $y = -f(x-2)$. The negative sign never reaches inside the parentheses.

22 **Answer: A**

1. $g(x) = 0$ when $f(2x) = 0$.
2. f is zero when its input is 6, so $2x = 6$.
3. Therefore $x = 3$.

A factor multiplying x inside the function compresses the graph horizontally: to make the inside equal 6, the new input only has to be half as large. Multiplying by 2 instead of dividing is the standard error.

Module 2

1 **Answer: A**

1. $x + 6$ is inside the function, so the shift is horizontal.
2. $h(-6) = f(0)$, so points move to smaller x-values.
3. The graph shifts 6 units to the left.

A constant added inside the parentheses shifts the graph horizontally in the direction opposite to its sign. Here $h(-6) = f(0)$, so the point f had at $x = 0$ now appears at $x = -6$: the graph moves 6 units left.

2 **Answer: 2**

1. Set $x^2 + 3x = x + 8$.
2. Rearrange: $x^2 + 2x - 8 = 0$.
3. Factor: $(x+4)(x-2) = 0$, so $x = -4$ or $x = 2$.
4. The greater value is 2.

Setting the two expressions equal eliminates y and leaves a quadratic. Both roots are x-coordinates of genuine intersection points, so the question is only asking which one is larger.

3 **Answer: B**

1. Set $x^2 - 4 = -4$.
2. Simplify: $x^2 = 0$.
3. The only solution is $x = 0$, so the graphs meet at exactly one point.

The line $y = -4$ passes exactly through the vertex of the parabola, so it touches the curve once instead of cutting it twice. Algebraically, the combined equation $x^2 = 0$ has a single (repeated) root.

4 Answer: D

1. The -7 is applied to the output, not the input.
2. Every point (a, b) becomes $(a, b-7)$.
3. The graph shifts 7 units down.

Subtracting 7 outside the function lowers every output by 7, so the whole graph slides straight down. Only changes inside the parentheses move a graph sideways.

5 Answer: 13

1. $f(-3) = 8$.
2. The new function value at $x = -3$ is $f(-3) + 5$.
3. $k = 8 + 5 = 13$.

Adding 5 outside the function raises the output at every input, and it never touches the x-coordinate. So the point at $x = -3$ keeps its position horizontally and climbs 5 units.

6 Answer: C

1. $h(x) = f(-x) = (-x)^3 - 2(-x)$.
2. $(-x)^3 = -x^3$ and $-2(-x) = 2x$.
3. So $h(x) = -x^3 + 2x$.

Substitute $-x$ everywhere: the odd power flips sign and so does the $-2x$ term. The result, $-x^3 + 2x$, is the mirror image of f across the y-axis.

7 Answer: D

1. $\frac{1}{2}$ is outside the function, so the effect is vertical.
2. A point (a, b) becomes $\left(a, \frac{b}{2}\right)$.
3. That is a vertical shrink by a factor of $\frac{1}{2}$.

The factor $\frac{1}{2}$ multiplies the output, so every point moves halfway toward the x-axis while keeping its x-coordinate. That is a vertical shrink, not a horizontal one.

8 Answer: 7

1. Substitute: $x^2 + 6x + 11 = 2x + c$.
2. Rearrange: $x^2 + 4x + (11 - c) = 0$.
3. Set the discriminant to 0: $4^2 - 4(1)(11 - c) = 0$.
4. $16 - 44 + 4c = 0$, so $c = 7$.

Exactly one intersection point means the quadratic left after substitution has a repeated root, so its discriminant is zero. Solving that equation for c is the whole problem.

9 Answer: C

1. Set $x^2 - 6x + 9 = x - 3$.
2. Rearrange: $x^2 - 7x + 12 = 0$.
3. Discriminant: $49 - 48 = 1 > 0$, so there are two real solutions ($x = 3$ and $x = 4$).

Even though the parabola is a perfect square and the line passes through its vertex region, the combined quadratic has a positive discriminant, so the line cuts the curve twice.

10 **Answer: D**

1. $g(5) = f(5-2) + 3 = f(3) + 3$.
2. From the table, $f(3) = 6$.
3. $g(5) = 6 + 3 = 9$.

Do the inside first: $g(5)$ needs $f(5-2) = f(3)$, and only then add the 3 that sits outside. Reading $f(7)$ (shifting the wrong way) or forgetting the $+3$ are the two failures this item checks.

11 **Answer: B**

1. $-(\cdot)$ outside negates every output: reflection across the x-axis.
2. $x - 2$ inside shifts the graph 2 units right (the vertex needs $x - 2 = 0$).
3. The new vertex is $(2, 0)$ and the parabola opens downward.

The minus sign is outside the square, so it reflects the parabola across the x-axis. The -2 is inside, so it is horizontal and moves the graph in the opposite direction from its sign: 2 units right.

12 **Answer: C**

1. A right shift of 3 means evaluating $f(x-3)$.
2. $f(x-3) = ((x-3)+1)^2 = (x-2)^2$.
3. The vertex moves from $(-1, 0)$ to $(2, 0)$.

A shift 3 units right replaces x by $x - 3$ in the formula: $f(x-3) = ((x-3)+1)^2 = (x-2)^2$. Replacing x by $x + 3$ — shifting the wrong way — gives $(x+4)^2$, which is the distractor to avoid.

13 **Answer: 8**

1. $g(x) = \left(\frac{x}{2}\right)^2 - 16 = \frac{x^2}{4} - 16$.
2. Set $g(x) = 0$: $\frac{x^2}{4} = 16$, so $x^2 = 64$.
3. $x = \pm 8$; the positive intercept is 8.

Dividing the input by 2 stretches the graph horizontally by a factor of 2, so each x-intercept moves twice as far from the y-axis. The stretch acts on x, never on the outputs.

14 **Answer: B**

1. Set $x^2 + 5x + 4 = 3x + 7$.
2. Rearrange: $x^2 + 2x - 3 = 0$, so $(x+3)(x-1) = 0$.
3. $x = 1$ gives $y = 10$; $x = -3$ gives $y = -2$.
4. The sum is $10 + (-2) = 8$.

Substituting the line into the parabola gives the x-values; the line then converts each one to a y-value in a single step. Reporting only one point's y-value, or the sum of the x-values, are the two traps.

15 **Answer: A**

1. $f(-x)$ reflects the graph across the y-axis.
2. The leading minus sign then reflects that result across the x-axis.
3. A point (a, b) on f ends up at $(-a, -b)$.

There are two independent sign changes: the inner $-x$ reflects the graph across the y-axis, and the outer minus sign reflects it across the x-axis. Applying both is the same as rotating the graph 180° about the origin.

16 **Answer: C**

1. From $y = x^2 - 4$, $x^2 = y + 4$.
2. Substitute into the circle: $(y + 4) + y^2 = 16$, so $y^2 + y - 12 = 0$.
3. $(y + 4)(y - 3) = 0$, so $y = -4$ or $y = 3$.
4. $y = 3$ gives $x^2 = 7$, two points; $y = -4$ gives $x^2 = 0$, one point.
5. In total there are 3 intersection points.

Here it is cleaner to eliminate x^2 rather than y: the parabola gives $x^2 = y + 4$, which turns the circle equation into a quadratic in y. Each y-value must then be checked for how many x-values it produces — one of them yields a single point, not two.

17 **Answer: A**

1. Set $x^2 - 2x + 4 = kx$, so $x^2 - (2 + k)x + 4 = 0$.
2. Tangency: $(2 + k)^2 - 4(1)(4) = 0$.
3. $(2 + k)^2 = 16$, so $2 + k = \pm 4$ and $k = 2$ or $k = -6$.
4. Since $k > 0$, $k = 2$.

Substituting gives $x^2 - (2+k)x+4 = 0$, and tangency forces its discriminant to zero. That yields $(2+k)^2 = 16$, whose two roots are $k = 2$ and $k = -6$; only the positive one is wanted.

18 **Answer: D**

1. From the graph, the vertex of f is $(2, -1)$.
2. $f(x + 1)$ shifts the graph 1 unit left: the vertex x-coordinate becomes $2 - 1 = 1$.
3. Subtracting 2 lowers it: $-1 - 2 = -3$.
4. The vertex is $(1, -3)$.

Two transformations, read separately: $x + 1$ inside moves the graph 1 unit *left*, and -2 outside moves it 2 units down. The vertex $(2, -1)$ therefore lands at $(1, -3)$.

19 **Answer: 12**

1. $h(2) = 2f(2 + 1) = 2f(3)$.
2. From the table, $f(3) = 6$.
3. $h(2) = 2(6) = 12$.

Work from the inside out: the input 2 becomes $2 + 1 = 3$ before f is applied, and only the result is doubled. The factor 2 multiplies the output, so it can never change which row of the table you read.

20 **Answer: B**

1. $2 \cdot (\,\cdot\,)$ outside: vertical stretch by a factor of 2.
2. $x + 3$ inside: horizontal shift 3 units left, since the vertex needs $x = -3$.
3. The vertex of the image is $(-3, 0)$ and the parabola is twice as steep.

The 2 multiplies the output, so it is a vertical stretch; the $+3$ lives inside the square, so it is a horizontal shift in the opposite direction — 3 units left. Mixing up which variable the 2 acts on, or shifting right, are the intended errors.

21 **Answer: 4**

1. From $y = x^2 - 13$, $x^2 = y + 13$.
2. Substitute: $(y + 13) + y^2 = 25$, so $y^2 + y - 12 = 0$.
3. $(y + 4)(y - 3) = 0$, so $y = 3$ or $y = -4$.
4. $y = 3$ gives $x^2 = 16$, i.e. $x = \pm 4$; $y = -4$ gives $x^2 = 9$, i.e. $x = \pm 3$.
5. That is 4 intersection points.

Eliminating x^2 turns the system into a quadratic in y. Each admissible y must then be traced back: here both y-values give a positive x^2, so each contributes two points.

22 **Answer: A**

1. Set $x^2 - 2x + c = 2x - 5$.
2. Rearrange: $x^2 - 4x + (c + 5) = 0$.
3. One solution requires $(-4)^2 - 4(1)(c + 5) = 0$.
4. $16 - 4c - 20 = 0$, so $-4c = 4$ and $c = -1$.

Substituting leaves $x^2 - 4x + (c + 5) = 0$, and *exactly one* real solution means a discriminant of zero. Note the $+5$: moving -5 across the equals sign is where the arithmetic usually goes wrong.

Topic 11 — Ratios, Rates, Proportional Relationships and Units

Module 1

1 **Answer: D**

1. Write the counts as $2k$ boys and $3k$ girls.
2. Solve $2k = 18$, so $k = 9$.
3. The number of girls is $3k = 3(9) = 27$.

The ratio $2 : 3$ means the choir is built from blocks of 2 boys and 3 girls. Since 18 boys is 9 such blocks, there are 9 blocks of girls as well, giving 27. Choice A comes from multiplying by $\frac{2}{3}$ instead of $\frac{3}{2}$, which inverts the ratio.

2 **Answer: 7**

1. A unit rate is $\dfrac{\text{total distance}}{\text{number of litres}}$.
2. Compute $\dfrac{245}{35} = 7$ kilometres per litre.

"Per litre" means a denominator of 1 litre, so divide the distance by the number of litres. The unit rate 7 km/L is the constant of proportionality between distance and fuel used.

3 **Answer: B**

1. Multiply by $\dfrac{1000\text{ g}}{1\text{ kg}}$ so that kilograms cancel.
2. $3.5 \times 1000 = 3500$ grams.

A gram is a much smaller unit than a kilogram, so the number of grams must be much larger than 3.5. Multiplying by the conversion factor $\frac{1000\text{ g}}{1\text{ kg}}$ cancels the kilograms and leaves 3500 grams. Choice A divides by 10 somewhere along the way.

4 Answer: D

1. The ratio has $2+3=5$ parts, so one part is $\dfrac{60}{5}=12$ dollars.
2. The larger share is 3 parts: $3(12)=36$ dollars.

The profit is given as a total, so use part-to-whole fractions: the parts add to 5, and the larger share is $\frac{3}{5}$ of the whole. Choice C is what you get by splitting the profit evenly instead of in the given ratio.

5 Answer: A

1. One notebook costs $\dfrac{6}{4}=1.50$ dollars.
2. Ten notebooks cost $10(1.50)=15$ dollars.

Cost is proportional to the number of notebooks, so find the unit price and multiply. Choice D multiplies the price of four notebooks by 10, which prices 40 notebooks rather than 10.

6 Answer: 150

1. Actual distance = (map distance) × (kilometres per centimetre).
2. $6\times 25=150$ kilometres.

This scale already carries its units, so no unit conversion is needed: each map centimetre is worth 25 real kilometres, and the actual distance is proportional to the map distance.

7 Answer: B

1. Use density $=\dfrac{\text{mass}}{\text{volume}}$.
2. $\dfrac{480\text{ g}}{60\text{ cm}^3}=8$ grams per cubic centimetre.

Density is mass per unit volume, so divide mass by volume. Choice A is the reciprocal, $\frac{60}{480}$, which gives the volume per gram instead.

8 Answer: B

1. Write $3k$ red and $4k$ blue marbles, so $3k+4k=84$.
2. Then $7k=84$, so $k=12$.
3. The number of red marbles is $3k=36$.

Because a total is given, the reds are $\frac{3}{7}$ of the bag, not $\frac{3}{4}$ of it. Choice D is exactly the trap: it treats the part-to-part ratio $\frac{3}{4}$ as a part-to-whole fraction. Choice C counts the blue marbles.

9 Answer: 20

1. $\dfrac{72\text{ km}}{1\text{ h}}\times\dfrac{1000\text{ m}}{1\text{ km}}\times\dfrac{1\text{ h}}{3600\text{ s}}$.
2. $\dfrac{72\,000}{3600}=20$ metres per second.

Chain the two conversion factors so kilometres and hours both cancel. A metre per second is a bigger unit than a kilometre per hour, so the number must come out smaller than 72 — a useful check against inverting a factor.

10 **Answer: B**

1. Set up $\dfrac{3}{24} = \dfrac{f}{60}$.
2. Cross-multiply: $24f = 180$, so $f = 7.5$ cups.

Flour is proportional to cookies, so the ratio $\frac{\text{cups}}{\text{cookies}}$ is constant. Choice D comes from dividing 60 by 3, which uses the wrong pair of quantities entirely.

11 **Answer: A**

1. Multiply by $\dfrac{1 \text{ euro}}{1.25 \text{ dollars}}$ so that dollars cancel.
2. $\dfrac{400}{1.25} = 320$ euros.

A euro is worth more than a dollar, so the number of euros must be smaller than 400; that alone rules out choices C and D. Dividing by 1.25 cancels the dollars, whereas choice D multiplies by 1.25 — the classic inverted conversion factor.

12 **Answer: C**

1. The scale factor is $r = \dfrac{10}{4} = 2.5$.
2. The new length is $6(2.5) = 15$ inches.

An enlargement multiplies every length by the same scale factor $r = \frac{10}{4} = 2.5$. Choice B adds 6 inches to the length because the width gained 6 inches, but proportional scaling is multiplicative, not additive.

13 **Answer: D**

1. $1 \text{ m}^2 = 100^2 = 10\,000 \text{ cm}^2$.
2. $2 \times 10\,000 = 20\,000$ square centimetres.

The conversion factor must be squared along with the unit: $1 \text{ m}^2 = (100 \text{ cm})^2 = 10\,000 \text{ cm}^2$. Choice B uses the plain factor 100 and is the answer for anyone who forgot the square.

14 **Answer: C**

1. Use mass = density × volume.
2. $2.7 \times 50 = 135$ grams.

Rearranging density $= \frac{\text{mass}}{\text{volume}}$ gives mass = density × volume. Choice B adds the two given numbers, which the units immediately forbid: you cannot add grams per cubic centimetre to cubic centimetres.

15 **Answer: B**

1. Find the constant: $k = \dfrac{45}{12} = \dfrac{15}{4}$.
2. Then $y = \dfrac{15}{4}(20) = 75$.

Direct proportion means $y = kx$ with k constant, so $\frac{y}{x}$ never changes. Choice A adds 8 to y because x increased by 8; that is additive reasoning, which a proportional relationship does not obey.

16 **Answer: D**

1. Bottles filled: $250 \times 8 = 2000$.
2. Volume: $2000 \times 0.75 = 1500$ litres.
3. Convert: $\dfrac{1500}{1000} = 1.5$ cubic metres.

Build one chain in which bottles, hours and litres all cancel: $250 \cdot 8 \cdot 0.75 = 1500$ litres, and dividing by 1000 gives 1.5 cubic metres. Choice C divides by 1000 a second time; choices A and B divide by 1000 three and two times over.

17 **Answer: 2000**

1. The ratio has $5 + 2 + 3 = 10$ parts.
2. Sugar is 2 parts, so one part is $\dfrac{400}{2} = 200$ grams.
3. The total is 10 parts: $10(200) = 2000$ grams.

With a three-term ratio the method is unchanged: one part is the size of one share, and the total is the sum of all parts. The 400 grams matches the 2 in the ratio, not the 10 of the total, so divide by 2 before multiplying by 10.

18 **Answer: C**

1. Actual distance $= 8 \times 25\,000 = 200\,000$ centimetres.
2. 1 km $= 100\,000$ cm, so $\dfrac{200\,000}{100\,000} = 2$ kilometres.

A unitless scale means one map centimetre stands for 25 000 real centimetres, so the actual distance is 200 000 cm and must then be converted to kilometres by dividing by 100 000. Choice D stops at metres divided by 100; choice B divides by an extra factor of 10.

19 **Answer: C**

1. $1\ \text{cm}^2$ represents $4^2 = 16$ square kilometres.
2. $12 \times 16 = 192$ square kilometres.

Areas scale by the square of the length scale factor: $1\ \text{cm}^2$ on the map represents $4^2 = 16\ \text{km}^2$. Choice A multiplies by 4 instead of 16, the standard error of forgetting to square the scale factor.

20 **Answer: A**

1. Fuel used: $\dfrac{8\ \text{L}}{100\ \text{km}} \times 250\ \text{km} = 20$ litres.
2. Cost: $20 \times 1.60 = 32$ dollars.

Two rates chain together: litres per kilometre, then dollars per litre. The trip uses $\frac{8}{100} \times 250 = 20$ litres, and 20 litres cost \$32. Choice D forgets that the 8 litres covers 100 kilometres, not one.

21 **Answer: 1.05**

1. Volume $= 5^3 = 125$ cubic centimetres.
2. Mass $= 8.4 \times 125 = 1050$ grams.
3. Convert: $\dfrac{1050}{1000} = 1.05$ kilograms.

Find the volume from the edge length first, then convert mass to the unit the question asks for. The final division by 1000 is what the question is really testing: the density is in grams, but the answer is wanted in kilograms.

22 **Answer: C**

1. Running time: $3 \times 60 = 180$ minutes.
2. Sheets used: $15 \times 180 = 2700$.
3. Cost: $\dfrac{2700}{500} \times 6 = 32.40$ dollars.

Convert hours to minutes before applying the printing rate, then convert sheets to packages before applying the price. Choice D uses 3 hours as 300 minutes; choice B prices the paper at a whole number of packages by rounding down instead of using the exact rate.

Module 2

1 **Answer: D**

1. One part is $\dfrac{28}{4} = 7$ litres.
2. Blue paint is 7 parts: $7 \times 7 = 49$ litres.

The 28 litres corresponds to the 4 in the ratio, so one part is 7 litres and the blue paint is 7 parts. Choice A multiplies by $\frac{4}{7}$ rather than $\frac{7}{4}$, inverting the ratio.

2 **Answer: 14**

1. Unit rate $= \dfrac{84 \text{ pages}}{6 \text{ min}}$.
2. $\dfrac{84}{6} = 14$ pages per minute.

The phrase "per minute" asks for the rate with a denominator of 1 minute, so divide the number of pages by the number of minutes.

3 **Answer: B**

1. Multiply by $\dfrac{1 \text{ h}}{3600 \text{ s}}$ so that seconds cancel.
2. $\dfrac{5400}{3600} = 1.5$ hours.

An hour is a much larger unit than a second, so the number of hours must be far smaller than 5400: divide by 3600. Choice A divides by 60 twice but keeps only one of the divisions in the right place.

4 **Answer: C**

1. The ratio has $1 + 2 + 5 = 8$ parts, so one part is $\dfrac{96}{8} = 12$ dollars.
2. The largest share is 5 parts: $5(12) = 60$ dollars.

A total is given, so the largest share is $\frac{5}{8}$ of the prize. Choice D uses $\frac{5}{6}$, treating the ratio of the largest share to the other two shares as if it were a share of the whole.

5 **Answer: A**

1. One book has mass $\dfrac{2.8}{7} = 0.4$ kilograms.
2. Twelve books have mass $12(0.4) = 4.8$ kilograms.

Mass is proportional to the number of books, so find the mass of one book and multiply. Choice B doubles the mass of seven books, which would be the mass of 14 books, not 12.

6 Answer: 450

1. Actual length = model length $\times 18$.
2. $25 \times 18 = 450$ centimetres.

A scale of $1 : 18$ means every length on the actual car is 18 times the corresponding length on the model, so the actual car must be longer than the model — multiply rather than divide.

7 Answer: C

1. Population = density $\times$ area.
2. $250 \times 36 = 9000$ people.

Population density is population per unit area, so population = density $\times$ area. Choice A divides the area by the density, which produces a quantity in square kilometres per person.

8 Answer: B

1. Write $5k$ for tea and $7k$ for coffee.
2. $7k - 5k = 96$, so $2k = 96$ and $k = 48$.
3. The total surveyed is $12k = 12(48) = 576$ people.

The given number is a difference, and the difference between $7k$ and $5k$ is $2k$ — two parts, not twelve. Once $k = 48$, the total is $12k = 576$. Choice A comes from matching the 96 to the wrong number of parts.

9 Answer: 54

1. $\dfrac{15\text{ m}}{1\text{ s}} \times \dfrac{3600\text{ s}}{1\text{ h}} = 54\,000$ metres per hour.
2. $\dfrac{54\,000}{1000} = 54$ kilometres per hour.

Going from metres per second to kilometres per hour multiplies by 3600 and divides by 1000, a net factor of 3.6; because a kilometre per hour is a smaller unit than a metre per second, the number must come out larger than 15.

10 Answer: D

1. $1\text{ m}^3 = 100^3 = 1\,000\,000\text{ cm}^3$.
2. $2.5 \times 1\,000\,000 = 2\,500\,000$ cubic centimetres.

For a volume the conversion factor is cubed: $1\text{ m}^3 = 100^3 = 1\,000\,000\text{ cm}^3$. Choice B uses 100^2 and choice A uses 100 alone, the two ways of under-applying the factor.

11 Answer: C

1. The scale factor is $r = \dfrac{25}{10} = 2.5$.
2. The first perimeter is $6 + 8 + 10 = 24$ centimetres.
3. The second perimeter is $24(2.5) = 60$ centimetres.

Similar figures have a single scale factor, here $r = \frac{25}{10} = 2.5$, and a perimeter is a sum of lengths, so it scales by r as well. Choice B is the perimeter of the original triangle doubled, not scaled by 2.5.

12 Answer: D

1. Cost in euros: $1.40 \times 40 = 56$ euros.
2. Cost in dollars: $56 \times 1.10 = 61.60$ dollars.

Two conversion factors chain: litres to euros, then euros to dollars. Because a euro is worth more than a dollar, the dollar figure must exceed the euro figure of 56; choice B divides by 1.10 instead of multiplying, inverting the exchange factor.

13 Answer: 24

1. 12 km $= 12 \times 100\,000 = 1\,200\,000$ centimetres.
2. Map distance $= \dfrac{1\,200\,000}{50\,000} = 24$ centimetres.

The scale is unitless, so both distances must be in the same unit before dividing: convert 12 kilometres to 1 200 000 centimetres, then divide by 50 000 because the map distance is the smaller of the two.

14 Answer: A

1. One part is $\dfrac{40}{10} = 4$ litres, so there are 28 litres of water and 12 of syrup.
2. Adding only syrup leaves the water at 28 litres.
3. Syrup needed: $28 - 12 = 16$ litres.

Break the mixture into parts first: one part is 4 litres, so there are 28 litres of water and 12 of syrup. Only syrup is added, so the water stays at 28 and the syrup must reach 28 as well. Choice D is the target amount of syrup rather than the amount added.

15 Answer: C

1. 1.2 kg $= 1200$ grams.
2. Volume $= \dfrac{1200}{0.8} = 1500$ cubic centimetres.

Convert the mass to grams so that it matches the units of the density, then divide: volume $= \frac{\text{mass}}{\text{density}}$. Choice B multiplies by the density instead of dividing, and because the density is less than 1 the volume must exceed the number of grams.

16 Answer: B

1. Area $= 250 \times 160 = 40\,000$ square metres.
2. $\dfrac{40\,000}{10\,000} = 4$ hectares.
3. Fertiliser $= 4 \times 3 = 12$ kilograms.

The rate is per hectare, so the area must be converted from square metres to hectares before it is used: $40\,000$ m^2 is 4 hectares, needing 12 kilograms. Choice D applies the rate directly to the 40 000 square metres without converting.

17 Answer: 300

1. $\dfrac{90 \text{ km}}{1 \text{ h}} \times \dfrac{1000 \text{ m}}{1 \text{ km}} \times \dfrac{1 \text{ h}}{3600 \text{ s}} = 25$ metres per second.
2. Distance $= 25 \times 12 = 300$ metres.

Convert the speed to metres per second before applying the time, so that the seconds cancel: 90 km/h is 25 m/s, and 25 metres each second for 12 seconds is 300 metres.

18 Answer: B

1. The length scale factor is $\frac{5}{3}$.
2. Areas scale by $\left(\frac{5}{3}\right)^2 = \frac{25}{9}$.
3. $27 \times \frac{25}{9} = 75$ square centimetres.

Lengths scale by $\frac{5}{3}$, so areas scale by $\left(\frac{5}{3}\right)^2 = \frac{25}{9}$, giving $27 \cdot \frac{25}{9} = 75$. Choice A multiplies the area by $\frac{5}{3}$ and is the answer for anyone who forgot to square the scale factor.

19 Answer: 2.8

1. Volume $= 20 \times 15 \times 8 = 2400$ cubic centimetres.
2. Mass $= 6.72$ kg $= 6720$ grams.
3. Density $= \frac{6720}{2400} = 2.8$ grams per cubic centimetre.

The requested unit fixes the plan: express the mass in grams and the volume in cubic centimetres, then divide. Leaving the mass in kilograms would give a density 1000 times too small.

20 Answer: A

1. $\frac{30 \text{ mi}}{1 \text{ gal}} \times \frac{1.6 \text{ km}}{1 \text{ mi}} = 48$ kilometres per gallon.
2. $\frac{48 \text{ km}}{4 \text{ L}} = 12$ kilometres per litre.

Convert the numerator from miles to kilometres and the denominator from gallons to litres in the same chain: $\frac{30 \cdot 1.6}{4} = 12$. Choice C converts only the miles, and choice D divides by 1.6 instead of multiplying, inverting one factor.

21 Answer: C

1. Actual length: $4.5 \times 200 = 900$ cm $= 9$ metres.
2. Actual width: $3 \times 200 = 600$ cm $= 6$ metres.
3. Area $= 9 \times 6 = 54$ square metres.

Scale each length by 200 first — 9 metres by 6 metres — and only then multiply. Choice A is the area on the plan converted as if it were already in square metres; choice B scales the plan area by 200 once instead of by 200^2.

22 Answer: B

1. Capacity $= 4.5 \times 1000 = 4500$ litres.
2. Rate $= \frac{15 \text{ L}}{10 \text{ s}} \times \frac{60 \text{ s}}{1 \text{ min}} = 90$ litres per minute.
3. Time $= \frac{4500}{90} = 50$ minutes.

Put both quantities in matching units before dividing: the tank holds 4500 litres and the hose delivers 90 litres per minute, so the fill takes 50 minutes. Choice D reports the answer in seconds rather than minutes.

Topic 12 — Percentages

Module 1

1 Answer: B

1. Write 30% as $\frac{30}{100} = 0.30$.
2. The word *of* means multiply: $0.30 \times 250 = 75$.

A percent is a fraction over 100, so 30% of a quantity is 0.30 times that quantity. Choice A is 3% of 250 and choice C is what is left after removing 30%; the question asks for the part itself, not the remainder.

2 Answer: 25

1. Translate: $\frac{p}{100} \cdot 180 = 45$.
2. Divide the part by the whole: $\frac{45}{180} = 0.25$.
3. Multiply by 100 to express the ratio as a percent: 25.

"What percent of 180" means 180 is the whole and goes in the denominator. The ratio $\frac{45}{180} = \frac{1}{4}$ is the fraction of the whole, and multiplying a fraction by 100 converts it to a percent.

3 Answer: C

1. A 25% discount leaves $100\% - 25\% = 75\%$ of the price.
2. Multiply by the multiplier 0.75: $80 \times 0.75 = 60$.

The question asks for the price paid, not the money saved. Choice A is the discount itself ($20) and choice B subtracts 25 dollars rather than 25 percent; the multiplier 0.75 gets you to the amount the customer actually hands over in one step.

4 Answer: B

1. Find the change: $250 - 200 = 50$ members.
2. Divide by the *original* number: $\frac{50}{200} = 0.25$.
3. Convert to a percent: 25%.

Percent change is always measured against the value you started from. Choice A comes from dividing the change by the new total, $\frac{50}{250}$, which answers a different question: it says the increase is 20% of the final membership, not 20% of the starting membership.

5 Answer: C

1. Adding 15% means multiplying by $1 + 0.15 = 1.15$.
2. Compute $60 \times 1.15 = 69$.

Because the tip is added to the bill rather than replacing it, the multiplier is 1.15, not 0.15. Choice A is the tip alone and choice D adds $15 instead of 15% of $60.

6 Answer: A

1. The part is 24 and the whole is 150.
2. Compute $\frac{24}{150} = 0.16$.
3. Multiply by 100: 16%.

A count and a percent are different units: 24 bicycles out of 150 is not 24%, which is what choice B assumes. Choice D is the percent that are *not* electric, $\frac{126}{150}$.

7 **Answer: D**

1. Divide the number correct by the number of questions: $\frac{34}{40} = 0.85$.
2. Convert to a percent: 85%.

The whole is the 40 questions on the test, so it belongs in the denominator. Choice A is the number of questions missed and choice B is the percent *missed*, $\frac{6}{40}$; choice C mistakes the raw count for a percent.

8 **Answer: D**

1. The original quantity is 100% of itself, or 1.
2. Adding 8% gives $100\% + 8\% = 108\%$.
3. As a decimal, $108\% = 1.08$.

An increase keeps the whole quantity and adds to it, so the multiplier must be greater than 1. Choice A is the increase alone and choice C is the multiplier for an 8% *decrease*; only 1.08 preserves the original and adds 8% of it.

9 **Answer: 20**

1. Find the change: $850 - 680 = 170$ dollars.
2. Divide by the original price: $\frac{170}{850} = 0.2$.
3. Convert to a percent: 20.

The reduction is measured against the price before the sale, so 850 is the denominator. A useful check: a 20% decrease has multiplier 0.80, and $850 \times 0.80 = 680$, exactly the new price.

10 **Answer: B**

1. Let the number be x and translate: $0.40x = 26$.
2. Divide both sides by 0.40: $x = \frac{26}{0.40} = 65$.

Here 26 is the part, not the whole, so the whole must be larger: dividing by the multiplier undoes it. Choice A multiplies by 0.40 instead of dividing, and choice D divides by 0.25, the multiplier for 25%.

11 **Answer: A**

1. Car owners: $0.45 \times 800 = 360$.
2. Of those, bicycle owners: $0.25 \times 360 = 90$.

The second percent is taken of the car owners, not of all 800 residents, so the two multipliers are applied one after the other: $800 \times 0.45 \times 0.25 = 90$. Choice B takes 25% of the whole survey, choice C stops after the first step, and choice D adds the percents to get 70% of 800.

12 **Answer: C**

1. Add the two categories: $96 + 84 = 180$ students.
2. Divide by the total: $\frac{180}{400} = 0.45$.
3. Convert to a percent: 45%.

"Mathematics or science" means the two counts are combined before comparing with the total of 400. Choices A and B are the individual percentages, $\frac{84}{400}$ and $\frac{96}{400}$, and choice D is the percent who chose neither.

13 **Answer: A**

1. Find the positive difference: $|480 - 500| = 20$ centimeters.
2. Divide by the actual value: $\frac{20}{500} = 0.04$.
3. Convert to a percent: 4%.

Percent error compares the size of the mistake with the true measurement, so the actual value 500 is the denominator. Choice B divides by the estimate 480 instead, and choice C reports the error in centimeters as though it were already a percent.

14 **Answer: D**

1. A 20% decrease has multiplier $1 - 0.20 = 0.80$.
2. Let x be the old rent: $0.80x = 960$.
3. Divide: $x = \frac{960}{0.80} = 1200$.

The \$960 is the result of the decrease, so it must be divided by 0.80 to get back to the start; after a decrease the original is always the larger number. Choice A multiplies by 0.80 a second time and choice C adds 20% of the *new* rent, 960×1.20, which is too small because 20% of 960 is less than 20% of 1200.

15 **Answer: C**

1. Apply the increase: $200 \times 1.10 = 220$.
2. Apply the decrease to the *new* price: $220 \times 0.90 = 198$.

The two 10% changes are not equal amounts of money: the rise is 10% of \$200 but the fall is 10% of \$220, so the price ends \$2 below where it started. Choice D is the trap that a rise and a fall of the same percent cancel; the combined multiplier is $1.10 \times 0.90 = 0.99$, never 1.

16 **Answer: B**

1. A 30% discount has multiplier 0.70 and a 10% discount has multiplier 0.90.
2. Multiply in one product: $120 \times 0.70 \times 0.90 = 75.60$.

The second discount applies to the already reduced price of \$84, not to the original \$120, so the two percents cannot be added. Choice A is what you get by treating the discounts as a single 40% off, and choice D stops after the first discount; the true combined multiplier is $0.70 \times 0.90 = 0.63$, a 37% discount.

17 **Answer: 250**

1. Adding a 6% tax has multiplier 1.06.
2. Let x be the pre-tax price: $1.06x = 265$.
3. Divide: $x = \frac{265}{1.06} = 250$.

The \$265 already contains the tax, so it is the output of the multiplication and must be divided by 1.06. Taking 6% of \$265 and subtracting gives \$249.10, which is wrong because the tax was charged on the smaller pre-tax price, not on the total.

18 **Answer: A**

1. Write the multipliers: 1.20 for the increase and 0.75 for the decrease.
2. Multiply: $1.20 \times 0.75 = 0.90$.
3. As a percent, $0.90 = 90\%$.

Suppose the investment started at \$100: it grows to \$120, then loses 25% of \$120, which is \$30, ending at \$90. Choice B comes from combining the percents as $+20 - 25 = -5$, which fails because the 25% loss is taken from the larger, grown value.

19 **Answer: 100**

1. Combine the multipliers: $1.30 \times 0.70 = 0.91$.
2. Let x be the original: $0.91x = 91$.
3. Divide: $x = \frac{91}{0.91} = 100$.

Two opposite changes of the same percent always leave a net decrease, here to 91% of the start, so the original must be larger than 91. Collapsing both changes into the single multiplier 0.91 turns a two-step story into one division.

20 **Answer: D**

1. Translate the sentence: $0.60x = 0.45 \times 80$.
2. Compute the right side: $0.45 \times 80 = 36$.
3. Divide: $x = \frac{36}{0.60} = 60$.

Both sides are percents of different wholes, so evaluate the side you can and then undo the multiplier on the other. Choice A is 45% of 80 — the value of the common quantity, not of x — and choice C is 60% of 80, which answers the question backwards.

21 **Answer: C**

1. Combine the multipliers: $1.25 \times 0.75 = 0.9375$.
2. Let x be the original price: $0.9375x = 60$.
3. Divide: $x = \frac{60}{0.9375} = 64$.

The 25% reduction is taken from the raised price, so it removes more money than the increase added and the lamp ends below its starting price — meaning the original is above \$60. Choice B is the trap that the two changes cancel, and choice D undoes only the discount.

22 **Answer: 20**

1. The increase multiplies the price by 1.25.
2. The discount multiplier m must satisfy $1.25m = 1$, so $m = \frac{1}{1.25} = 0.80$.
3. A multiplier of 0.80 is a discount of $1 - 0.80 = 0.20$, or 20 percent.

Undoing a percent increase requires a *smaller* percent decrease, because the decrease is taken from the larger, raised price. Concretely, \$100 becomes \$125, and returning to \$100 means cutting \$25 from \$125, which is 20% of \$125 — not 25%.

Module 2

1 Answer: C

1. Write 65% as 0.65.
2. Multiply: $0.65 \times 400 = 260$.

Taking a percent of a number means multiplying by that percent written as a decimal. Choice B is the remaining 35% hidden as a subtraction of 165, and choice D adds 65 to the number instead of taking 65% of it.

2 Answer: 40

1. Set up $\frac{p}{100} \cdot 45 = 18$.
2. Divide: $\frac{18}{45} = 0.4$.
3. Convert to a percent: 40.

The number following *of* is the whole, so 45 is the denominator. Checking forward, 40% of 45 is $0.4 \times 45 = 18$, which confirms the direction of the division.

3 Answer: B

1. Write 35% as 0.35.
2. Multiply by the total sold: $0.35 \times 240 = 84$.

The percent describes a share of the 240 phones, so it must be multiplied by 240. Choice C is the number that were *not* refurbished (65% of 240), and choice A repeats the percent as though it were a count.

4 Answer: A

1. Find the change: $250 - 200 = 50$ dollars.
2. Divide by Monday's sales: $\frac{50}{250} = 0.2$.
3. Convert to a percent: 20%.

A decrease is measured against the value before the drop, so 250 belongs in the denominator. Choice B divides by the smaller Tuesday figure, and choice D is the percent of Monday's sales that *remained*, not the percent lost.

5 Answer: D

1. Let the number be x: $0.30x = 72$.
2. Divide by 0.30: $x = \frac{72}{0.30} = 240$.

Since 72 is only 30% of the number, the number must be more than three times as large. Choice A multiplies by 0.30 rather than dividing, and choice C divides by $\frac{1}{3}$ instead of 0.30, treating 30% as one third.

6 Answer: B

1. The full quantity is 100%.
2. Removing 35% leaves $100\% - 35\% = 65\%$.
3. As a decimal, $65\% = 0.65$.

A decrease keeps what is left over, so the multiplier is 1 minus the percent as a decimal. Choice A is the part removed rather than the part kept, and choice C is the multiplier for a 35% increase.

7 **Answer: C**

1. Find the change: $75 - 60 = 15$ points.
2. Divide by the original score: $\frac{15}{60} = 0.25$.
3. Convert to a percent: 25%.

The increase is compared with the score the student started from, 60. Choice B divides by the new score 75, and choice A reports the raw point gain as a percent.

8 **Answer: 128**

1. A 25% discount has multiplier 0.75.
2. Let x be the original price: $0.75x = 96$.
3. Divide: $x = \frac{96}{0.75} = 128$.

The \$96 is what remains after the discount, so it represents 75% of the original price and must be divided by 0.75. Adding 25% to \$96 gives \$120, which is wrong because that adds a quarter of the sale price rather than restoring a quarter of the larger original.

9 **Answer: C**

1. After 2021: $500 \times 1.10 = 550$.
2. After 2022: $550 \times 1.20 = 660$.

The second year's growth applies to the 550 people alive at the start of that year, not to the original 500. Choice B comes from adding the percents to get a single 30% increase; the correct combined multiplier is $1.10 \times 1.20 = 1.32$, a 32% increase.

10 **Answer: B**

1. Find the excess: $265 - 250 = 15$ grams.
2. Divide by the amount called for: $\frac{15}{250} = 0.06$.
3. Convert to a percent: 6%.

"Percent greater than" compares the excess with the reference amount named after *than*, which is the 250 grams in the recipe. Choice A divides by the 265 grams actually used, and choice D is the amount used as a percent *of* the recipe amount, which answers a different question.

11 **Answer: 48**

1. Count the tickets that are not student tickets: $180 + 60 = 240$.
2. Divide by the total: $\frac{240}{500} = 0.48$.
3. Convert to a percent: 48.

The word *not* redirects the numerator to every other row of the table while the denominator stays the overall total of 500. Equivalently, student tickets are $\frac{260}{500} = 52\%$ of the sales, and the rest is $100\% - 52\% = 48\%$.

12 **Answer: D**

1. Set up $\frac{p}{100} \cdot 60 = 96$.
2. Divide: $\frac{96}{60} = 1.6$.
3. Convert to a percent: 160%.

Because 96 is larger than the whole it is compared with, the answer must exceed 100% — percents above 100 are perfectly legal. Choice C reverses the fraction to $\frac{60}{96}$, and choice A reports the difference $96 - 60$ as a percent.

13 **Answer: A**

1. Write the multipliers: 0.60 for the decrease and 1.40 for the increase.
2. Multiply: $0.60 \times 1.40 = 0.84$.
3. As a percent, $0.84 = 84\%$.

Start at 100: the quantity falls to 60, and the 40% increase then adds 40% of 60, which is only 24, giving 84. Choice C is the trap that opposite changes of equal percent undo one another; they never do, because the second percent is applied to a different amount.

14 **Answer: C**

1. A 20% discount has multiplier 0.80.
2. Let x be the original price: $0.80x = 52$.
3. Divide: $x = \frac{52}{0.80} = 65$.

The sale price is 80% of the original, so dividing by 0.80 recovers the full price. Choice A multiplies by 0.80 again, and choice B adds 20% of the sale price, which is smaller than 20% of the original and therefore falls short.

15 **Answer: 104**

1. Let the number be x and translate: $0.40x = 0.15x + 26$.
2. Subtract $0.15x$ from both sides: $0.25x = 26$.
3. Divide: $x = \frac{26}{0.25} = 104$.

Both percents refer to the same unknown, so the difference between them is itself a percent of that unknown: $40\% - 15\% = 25\%$ of the number equals 26. Recognising that lets you replace a two-term equation with a single reverse-percent division.

16 **Answer: B**

1. After 2022: $8000 \times 0.85 = 6800$.
2. After 2023: $6800 \times 1.20 = 8160$.

The 20% gain is taken on the reduced population of 6800, so it does not fully offset the earlier loss taken on 8000. Choice C treats the two changes as a net +5% on the original, which overstates the recovery; the true combined multiplier is $0.85 \times 1.20 = 1.02$.

17 **Answer: B**

1. Combine the multipliers: $1.20 \times 0.90 = 1.08$.
2. Let x be the January starting price: $1.08x = 216$.
3. Divide: $x = \frac{216}{1.08} = 200$.

Two changes collapse into one multiplier, and because that multiplier is greater than 1 the starting price must be below \$216. Choice D undoes only the February decrease, and choice A applies the decrease a second time instead of reversing it.

18 Answer: D

1. Markup multiplier: 1.40. Discount multiplier: 0.75.
2. Multiply: $1.40 \times 0.75 = 1.05$.
3. As a percent, $1.05 = 105\%$.

Let the store pay \$100: the listed price is \$140, and a 25% discount removes \$35, leaving \$105. Choice A comes from subtracting the percents, $40\% - 25\%$, and choice B from applying 25% off to the original cost rather than to the higher listed price.

19 Answer: 25

1. The fall multiplies the price by 0.80.
2. The rise multiplier m must satisfy $0.80m = 1$, so $m = \frac{1}{0.80} = 1.25$.
3. A multiplier of 1.25 is an increase of 0.25, or 25 percent.

Recovering from a percent loss always takes a *larger* percent gain, because the gain is computed on the smaller reduced price. A share worth \$100 falls to \$80, and getting the missing \$20 back is $\frac{20}{80} = 25\%$ of \$80, not 20%.

20 Answer: C

1. Read the two revenues: 250 in 2021 and 324 in 2024.
2. Find the change: $324 - 250 = 74$.
3. Divide by the 2021 revenue: $\frac{74}{250} = 0.296$, or 29.6%.

"Percent greater than its revenue in 2021" fixes 250 as the reference value, so the intervening years are irrelevant. Choice A divides the change by the 2024 revenue instead, and choice D states the change in thousands of dollars as though it were a percent.

21 Answer: D

1. Combine the multipliers: $0.75 \times 0.80 = 0.60$.
2. Let x be the original price: $0.60x = 180$.
3. Divide: $x = \frac{180}{0.60} = 300$.

Two discounts multiply, so the printer sold for 60% of its original price — a 40% total discount, not the 45% you would get by adding. Choice A reverses only the first discount, and choice C divides by 0.625, treating the combined discount as 37.5%.

22 Answer: 100000

1. Write the multipliers: 1.25 from 2022 to 2023 and 0.92 from 2023 to 2024.
2. Combine them: $1.25 \times 0.92 = 1.15$.
3. Let x be the 2022 revenue: $1.15x = 115000$, so $x = \frac{115000}{1.15} = 100000$.

Each percent is taken of the year immediately before it, so the two multipliers chain into one factor of 1.15 connecting 2022 directly to 2024. Note that the net effect is a 15% increase, not the 17% you would get by subtracting 8 from 25, because the 8% loss is measured against the larger 2023 revenue.

Topic 13 — One-Variable Data: Distributions and Measures of Center and Spread

Module 1

1 Answer: C

1. Add the five values: $61 + 70 + 72 + 78 + 84 = 365$.
2. There are 5 values, so divide: $365 \div 5 = 73$.

The mean is the total divided by the number of values, so it depends on every value in the list. Choice B, 72, is the middle value of the ordered list — that is the median, not the mean, and the two agree only by coincidence.

2 Answer: B

1. Write the values in increasing order: $2, 3, 4, 5, 6, 7, 9$.
2. There are 7 values, so the median is the 4th one, which is 5.

The median is the middle value *of the ordered list*. Choice A is the value sitting in the middle of the list as it was printed, which is the single most common error on this topic; choice D is simply the largest value.

3 Answer: 15

1. The largest mass is 22 and the smallest is 7.
2. The range is $22 - 7 = 15$ grams.

The range measures spread, not centre: it is one subtraction, largest minus smallest. Nothing between the extremes affects it.

4 Answer: C

1. Count each size: 7 appears once, 8 twice, 9 three times, 10 once, 11 once.
2. The size with the greatest count is 9.

The mode is the *value* that occurs most often, not how often it occurs. Choice A is the number of times size 9 appeared, and choice B is a size that appeared only twice.

5 Answer: B

1. There are 15 dots, so the median is the 8th value counting from the left.
2. Running totals: 2 dots at 0, then 5 at 1 (through the 7th value), then 4 at 2.
3. The 8th value is therefore 2.

Each dot is one family, so the dot plot is a list written vertically. With 15 values the median is the 8th, which you find by counting dots from the left. Choice A is the mode (the tallest stack) and choice C is the midpoint of the axis, neither of which is the median.

6 Answer: 17

1. Add the eight distances: $14 + 18 + 12 + 20 + 16 + 22 + 19 + 15 = 136$.
2. Divide by 8: $136 \div 8 = 17$ miles per week.

"Per week" signals a mean: pool all the miles, then share them equally across the eight weeks.

7 Answer: C

1. The line drawn inside the box marks the median.
2. That line sits at 24 on the axis.

A box plot shows five numbers: the two whisker ends are the minimum and maximum, the two ends of the box are Q_1 and Q_3, and the line inside the box is the median. Choices A, B and D are the minimum, Q_1 and Q_3.

8 **Answer: B**

1. Total siblings: $0(4)+1(7)+2(5)+3(3)+4(1)=0+7+10+9+4=30$.
2. Total students: $4+7+5+3+1=20$.
3. Mean $=\frac{30}{20}=1.5$.

A frequency table is a compressed list: the value 1 really appears seven times. Multiply value by frequency before adding. Choice D is the mean of the frequencies, $\frac{20}{5}=4$, and choice C is the mean of the five listed values — both ignore how often each value actually occurred.

9 **Answer: B**

1. Sort: $28, 31, 33, 36, 39, 41, 45, 47, 50, 52$.
2. With 10 values the median is the mean of the 5th and 6th: $\frac{39+41}{2}=40$.

With an even number of values there is no single middle value, so the median is the mean of the two middle ones *after sorting*. Choice D is what you get by averaging the fifth and sixth entries of the unsorted list, 39 and 47.

10 **Answer: C**

1. Five scores with mean 84 must total $5\times 84=420$.
2. The four known scores total $78+91+85+80=334$.
3. The fifth score is $420-334=86$.

Convert the mean into a total. Knowing that the five scores must add to 420 turns the question into one subtraction. Choice A is the mean of the four given scores and choice B assumes the missing score equals the mean.

11 **Answer: D**

1. Original: total $=162$, so the mean is $\frac{162}{9}=18$; the median is the 5th value, 18.
2. After adding 90: total $=252$, so the mean is $\frac{252}{10}=25.2$.
3. With 10 values the median is $\frac{18+19}{2}=18.5$.
4. The mean rose by 7.2; the median rose by 0.5.

The mean is built from the total, so one huge value drags it a long way; the median only counts positions, so it slides by half a place. Here the mean goes from 18 to 25.2 while the median goes from 18 to 18.5.

12 **Answer: 8.5**

1. Total score: $6(1)+7(3)+8(6)+9(5)+10(5)=6+21+48+45+50=170$.
2. There are $1+3+6+5+5=20$ dots.
3. Mean $=\frac{170}{20}=8.5$.

A dot plot is a frequency table drawn with dots, so the mean is found the same way: value times frequency, added, then divided by the number of dots — not by the number of columns.

13 **Answer: D**

1. Both data sets have mean 22, so compare distances from 22.
2. Data set X: distances $2, 1, 0, 1, 2$. Data set Y: distances $10, 5, 0, 5, 10$.
3. The values of Y sit farther from the mean, so Y has the larger standard deviation.

Standard deviation measures typical distance from the mean, so compare how tightly the values cluster. The values of X are within 2 of 22; the values of Y reach 10 away. Equal means say nothing at all about spread, which rules out choices B and C.

14 **Answer: A**

1. From the box plot, $Q_1 = 10$ and $Q_3 = 22$.
2. $\text{IQR} = Q_3 - Q_1 = 22 - 10 = 12$ minutes.

The interquartile range is the width of the *box*, $Q_3 - Q_1$; the range is the distance between the whisker ends. Choice D, $30 - 4 = 26$, is the range — the distractor this question is built around.

15 **Answer: 10**

1. Six numbers with mean 15 total $6 \times 15 = 90$.
2. Five numbers with mean 16 total $5 \times 16 = 80$.
3. The number removed is $90 - 80 = 10$.

Two means, two totals. The removed number is the difference between them — and notice that removing a number *raised* the mean, so the number removed had to be below the original mean.

16 **Answer: A**

1. Total goals: $0(3) + 1(5) + 2(7) + 3(4) + 4(1) = 0 + 5 + 14 + 12 + 4 = 35$.
2. Total games: $3 + 5 + 7 + 4 + 1 = 20$.
3. Mean $= \frac{35}{20} = 1.75$ goals per game.

Read each bar as a value with a frequency, then take a weighted mean. Choice C, 4, is the mean of the five bar heights, $\frac{20}{5}$; choice D is the tallest bar. Neither uses the goal values at all.

17 **Answer: 9**

1. Total of the values: $2(5) + 3(8) + 4x + 5(3) = 10 + 24 + 4x + 15 = 49 + 4x$.
2. Number of values: $5 + 8 + x + 3 = 16 + x$.
3. Set $\frac{49+4x}{16+x} = 3.4$, so $49 + 4x = 54.4 + 3.4x$.
4. Then $0.6x = 5.4$, so $x = 9$.

Write the mean formula with x in both the numerator and the denominator, then clear the fraction. The unknown frequency changes the total *and* the count, which is why it appears twice.

18 **Answer: D**

1. Both data sets have mean 6, so compare distances from 6.
2. In P most values are at 6 itself, and none is more than 2 away with only one at each end.
3. In Q eight of the ten values are 2 away from the mean.
4. Q has the larger typical distance from the mean, so it has the larger standard deviation.

With the means equal, the comparison is entirely about clustering. Data set P piles up at 6; data set Q piles up at 4 and 8, so a typical value of Q sits 2 away from the mean while a typical value of P sits well under 1 away.

19 **Answer: A**

1. Original mean: $\frac{32+35+36+38+40+42+45+180}{8} = \frac{448}{8} = 56$.
2. Original median: $\frac{38+40}{2} = 39$.
3. After removing 180: mean $= \frac{268}{7} \approx 38.3$ and median $= 38$.
4. The mean falls by about 18; the median falls by 1.

This is the outlier rule in numbers: the mean falls from 56 to about 38.3, a drop of roughly 18, while the median falls only from 39 to 38. The median is resistant to outliers precisely because it depends on position rather than on size.

20 **Answer: 82**

1. First class total: $24 \times 78 = 1872$.
2. Second class total: $16 \times 88 = 1408$.
3. Combined mean: $\frac{1872+1408}{40} = \frac{3280}{40} = 82$.

You cannot average two means directly unless the groups are the same size — $\frac{78+88}{2} = 83$ is wrong here. Rebuild both totals, add them, and divide by the combined count so the larger class carries more weight.

21 **Answer: C**

1. Total students: $6 + 9 + 8 + 7 + 5 = 35$, so the median is the 18th value in order.
2. Cumulative totals: 6 students at 0 day, 15 through 1 day, 23 through 2 days.
3. The 18th value falls in the 2-day group, so the median is 2.

Find the total frequency first: with 35 values the median is the 18th, so run a cumulative total until you pass 18. Choice B is the mode — the value with the largest frequency — and choice D is the median of the frequency column, both of which answer a question that was not asked.

22 **Answer: C**

1. Twelve numbers with mean 7 total $12 \times 7 = 84$.
2. Thirteen numbers with mean 8 total $13 \times 8 = 104$.
3. The new number is $104 - 84 = 20$.

Raising the mean of 12 numbers by 1 requires enough extra to lift all thirteen values, not just one: 13 extra units on top of the new value's own share. Choice A assumes the new number simply equals the new mean, which would have left the mean unchanged at 7.

Module 2

1 Answer: B

1. Sort the prices: $9, 11, 12, 14, 16, 18, 21, 25$.
2. With 8 values the median is the mean of the 4th and 5th: $\frac{14+16}{2} = 15$.

An even count means the median is the average of the two middle values of the sorted list. Choices A and D are those two middle values on their own, and choice C is the mean.

2 Answer: A

1. The largest frequency in the table is 19.
2. That frequency belongs to the value 2, so the mode is 2 children.

In a frequency table the mode is the value in the left column that carries the largest frequency, not the frequency itself. Choice D, 19, is how many households had that number of children.

3 Answer: A

1. The leftmost dot is above 3 and the rightmost dot is above 9.
2. The range is $9 - 3 = 6$.

The range is a subtraction of the two extreme *values*, and on a dot plot those are the leftmost and rightmost columns that actually contain dots. Choice B counts the columns, choice C reports the largest value alone, and choice D counts the dots.

4 Answer: D

1. The box runs from $Q_1 = 34$ to $Q_3 = 58$.
2. The right-hand edge of the box is at 58, so $Q_3 = 58$.

Q_3 is the right-hand edge of the box: a quarter of the data lies above it. Choice A is Q_1 (the left edge), choice B is the median (the line inside), and choice C is the midpoint of the two whisker ends, which no box plot marks.

5 Answer: 3

1. Add: $2.4 + 3.1 + 1.8 + 2.9 + 3.6 + 4.2 = 18$.
2. Divide by 6: $18 \div 6 = 3$ inches.

Decimals change nothing about the method: add every value, then divide by how many values there are.

6 Answer: D

1. Both data sets have a mean of 50.
2. In data set A every value equals the mean, so the standard deviation is 0.
3. Data set B has values up to 4 away from the mean, so its standard deviation is greater than 0.

Standard deviation is the typical distance from the mean. In data set A every value *is* the mean, so that distance is 0 — the smallest a standard deviation can ever be. Any variation at all beats it.

7 Answer: A

1. Total students: $2 + 3 + 4 + 6 = 15$, so the median is the 8th value.
2. Cumulative totals: 2 through score 1, 5 through score 2, 9 through score 3.
3. The 8th value falls in the score-3 group, so the median is 3.

With 15 values the median is the 8th in order; a cumulative count of the frequencies locates it. Choice B is the mode, choice C is the largest frequency, and choice D is the number of students.

8 **Answer: B**

1. Total cars: $0(5) + 1(8) + 2(6) + 3(4) + 4(2) = 0 + 8 + 12 + 12 + 8 = 40$.
2. Total families: $5 + 8 + 6 + 4 + 2 = 25$.
3. Mean $= \frac{40}{25} = 1.6$ cars per family.

Weight each value by its frequency. Choice D is the mean of the frequencies ($\frac{25}{5} = 5$), choice C is the mean of the five listed values, and choice A is the median. Only choice B uses both columns of the table.

9 **Answer: 15**

1. With 6 values the median is the mean of the 3rd and 4th values: $\frac{11+x}{2}$.
2. Set $\frac{11+x}{2} = 13$, so $11 + x = 26$.
3. Therefore $x = 15$.

For six values the median is the average of the third and fourth, so the stated median gives an equation in x rather than a value you can read off. The answer, 15, does lie between 11 and 17, so the list really is in increasing order.

10 **Answer: C**

1. The original total is $10 \times 46 = 460$.
2. Replacing 23 by 63 increases the total by $63 - 23 = 40$, giving 500.
3. There are still 10 values, so the new mean is $\frac{500}{10} = 50$.

Only the total changes, and it changes by $63 - 23 = 40$. That extra 40 is then shared among all ten values, raising the mean by 4. Choice D adds the whole 40 to the mean without dividing.

11 **Answer: B**

1. Total students: $4 + 9 + 7 + 5 + 3 + 2 = 30$, so the median averages the 15th and 16th values.
2. Cumulative totals: 4 through 0 siblings, 13 through 1, 20 through 2.
3. Both the 15th and the 16th value are 2, so the median is 2.

Convert the histogram to a running count. With 30 values the median is the average of the 15th and 16th, and both land in the same bar, so no averaging is actually needed. Choice A is the mode (the tallest bar) and choice D is the largest value on the axis.

12 **Answer: A**

1. Class A: minimum 55, $Q_1 = 68$, median 76, $Q_3 = 84$, maximum 95.
2. Class B: minimum 62, $Q_1 = 72$, median 76, $Q_3 = 80$, maximum 90.
3. The medians are equal at 76.
4. IQR of A is $84 - 68 = 16$; IQR of B is $80 - 72 = 8$.

Read all five marks off each plot before judging. The two medians coincide at 76, but class A's box is twice as wide ($84 - 68 = 16$ against $80 - 72 = 8$), and its whiskers are wider too ($95 - 55 = 40$ against $90 - 62 = 28$), so class A is the more spread out of the two.

13 **Answer: B**

1. The original total height is $18 \times 62 = 1116$ inches.
2. Adding the two new students gives $1116 + 70 + 74 = 1260$ inches.
3. The new mean is $\frac{1260}{20} = 63$ inches.

Averaging the old mean with the mean of the two newcomers gives 67, choice D — but that treats two students as though they outweighed eighteen. Work with totals so the group sizes are respected.

14 **Answer: 9**

1. The eight values are already in order, so split them into two halves of four.
2. Lower half $3, 7, 8, 12$: $Q_1 = \frac{7+8}{2} = 7.5$.
3. Upper half $13, 15, 18, 25$: $Q_3 = \frac{15+18}{2} = 16.5$.
4. $\text{IQR} = 16.5 - 7.5 = 9$.

Split the ordered list in half and take the median of each half: Q_1 from the lower four values, Q_3 from the upper four. The IQR describes the middle half of the data, so the extreme value 25 has no effect on it — unlike the range, which would be 22.

15 **Answer: D**

1. Original mean: $\frac{88}{7} \approx 12.6$; original median: the 4th value, 9.
2. After removal the six values total 48, so the mean is $\frac{48}{6} = 8$.
3. The new median is $\frac{7+9}{2} = 8$.
4. The mean fell by about 4.6; the median fell by 1.

Removing a high outlier pulls the total down hard, so the mean falls from about 12.6 to 8; the median only shifts from 9 to 8 because it merely moves half a place in the ordered list.

16 **Answer: 25**

1. Seven numbers with mean 24 total $7 \times 24 = 168$.
2. Removing 15 and 28 leaves a total of $168 - 43 = 125$.
3. Five numbers remain, so the mean is $\frac{125}{5} = 25$.

Totals again: subtract the two removed values from the original total, then divide by the new count of five. Because the two removed numbers averaged only 21.5 — below 24 — the mean of what is left has to rise.

17 **Answer: D**

1. Both data sets have the same mean, 50, so compare distances from 50.
2. In R, most values are within 2 of the mean.
3. In S, most values are more than 10 from the mean.
4. A larger typical distance from the mean means a larger standard deviation, so S is larger.

You are never asked to compute a standard deviation on this test, and here you do not need to: with equal means, the data set whose values sit farther from that mean has the larger standard deviation. Choice C states the opposite of how the SAT treats this topic.

18 Answer: C

1. Adding 4 to every value adds 4 to the mean: $15 + 4 = 19$.
2. Doubling every value doubles the mean: $2 \times 19 = 38$.

Doing the same operations to every value does the same operations to the mean, but the *order* matters: increase first, then double. Choice B doubles before adding, which reverses the instructions.

19 Answer: C

1. Total students: $6 + 9 + 11 + 9 + 5 = 40$, so the median averages the 20th and 21st scores.
2. Cumulative totals: 6 through score 70, 15 through 75, 26 through 80.
3. Both the 20th and the 21st score are 80, so the median is 80.

The median is a position, not an average: with 40 scores it lies between the 20th and the 21st, and both of those fall in the 80 group. Choice B is the mean, 79.75, which is pulled below the median by the six low scores of 70.

20 Answer: A

1. The new value is larger than every existing value, so the total rises and the mean rises.
2. The new value lies far from the mean, so it increases the typical distance from the mean.
3. Therefore both the mean and the standard deviation are larger.

A value far above every other value raises the total, so the mean rises; and it sits an enormous distance from the mean, so the typical distance from the mean — the standard deviation — rises as well. The median, by contrast, moves at most half a place, so it cannot rise by more than the mean does.

21 Answer: 75

1. Class total: $30 \times 79 = 2370$.
2. Group A total: $12 \times 85 = 1020$.
3. The other 18 students total $2370 - 1020 = 1350$.
4. Their mean is $\frac{1350}{18} = 75$.

The whole-class total splits into the two group totals. Subtracting group A's total leaves the other group's total, which then divides by 18. The answer must fall below 79, since group A scored above the class mean.

22 Answer: D

1. The original total is $20 \times 42 = 840$.
2. The remaining 18 values total $18 \times 40 = 720$.
3. The two removed values total $840 - 720 = 120$.
4. Their mean is $\frac{120}{2} = 60$.

Removing two values dropped the mean of eighteen numbers by 2, which means those two values carried $18 \times 2 = 36$ units of surplus on top of their own share. Choice A, the average of the two stated means, ignores the group sizes entirely.

Topic 14 — Two-Variable Data: Models and Scatterplots

Module 1

1 **Answer: A**

1. Follow the points from left to right.
2. The scores rise as the hours rise, so the association is positive.
3. The points hug a straight path, so the form is linear.

As the hours studied increase, the scores increase, and the points lie close to a straight line. That is a positive, approximately linear association. Nothing in the picture doubles, so it is not exponential.

2 **Answer: C**

1. Find $x = 5$ on the horizontal axis.
2. Move up to the line of best fit, not to a plotted point.
3. Read the height: $5(5) + 55 = 80$.

A prediction from a line of best fit is read off the line, not off a data point. Go up from $x = 5$ until you meet the line; the line has equation $\hat{y} = 5x + 55$, so $\hat{y} = 5(5) + 55 = 80$.

3 **Answer: 4**

1. Compute the rise: $27 - 11 = 16$.
2. Compute the run: $6 - 2 = 4$.
3. Divide: $16/4 = 4$.

Slope is the change in y divided by the change in x between any two points that lie on the line itself. Data points would not do; these two points are on the line.

4 **Answer: A**

1. Identify the slope: 24.
2. One unit of x is one month; y is measured in members.
3. So each extra month adds a predicted 24 members.

The slope of 24 attaches to x, and one unit of x is one month, so 24 is the predicted change in members per month. The number 150 is the intercept and belongs to the moment the gym opened, not to each month.

5 **Answer: D**

1. Set $x = 0$ in the model: $\hat{y} = 4$.
2. Translate $x = 0$ into the story: the moment the seedling was potted.
3. Attach the units of y: centimeters.

The y-intercept is the predicted y when $x = 0$, and $x = 0$ is the week the seedling was potted. So the model predicts a height of 4 centimeters at that moment. The 1.8 is the weekly growth, a different question.

6 **Answer: 85**

1. Substitute $x = 12$.
2. Compute $6(12) = 72$.
3. Add the intercept: $72 + 13 = 85$.

A prediction is a substitution: put the given x into the model and evaluate. The answer is a prediction, not a guarantee about any one café.

7 **Answer: A**

1. Check the differences: 200, 400, 800 — not constant.
2. Check the ratios: 2, 2, 2 — constant.
3. A constant ratio identifies an exponential model.

Going from one y to the next multiplies by 2 each time ($400/200 = 2$, $800/400 = 2$, $1600/800 = 2$), while the differences $200, 400, 800$ are not constant. A constant ratio means exponential.

8 **Answer: B**

1. Write residual = actual − predicted.
2. Substitute: $64 - 58$.
3. The residual is 6 gallons.

A residual is actual minus predicted, in that order. Here $64 - 58 = 6$, and the positive sign says the point lies above the line: the model underpredicted that day's sales.

9 **Answer: D**

1. Compute residual = actual − predicted = $197 - 212 = -15$.
2. A negative residual means the prediction was too high.
3. So the model overpredicts by \$15.

Residual $= 197 - 212 = -15$. A negative residual means the predicted value was too big, so the model overpredicts, and the data point lies below the line. Sign and wording must agree.

10 **Answer: D**

1. The slope is -3.4.
2. One unit of x is one year.
3. A negative slope means the predicted y falls by 3.4 each year.

The slope -3.4 is the predicted change in subscriptions per 100 households for each additional year, and the negative sign makes that change a decrease. The 61 describes the year 2010 only.

11 **Answer: 84**

1. Substitute $x = 16$: $4.5(16) = 72$.
2. Add the intercept: $72 + 12 = 84$.
3. The predicted height is 84 feet.

Substituting $x = 16$ extends the pattern past most of the data, which is why the value is only a prediction; the arithmetic itself is unchanged.

12 **Answer: A**

1. The value 3.5 is the y-intercept.
2. Set $x = 0$: a ride of zero miles.
3. The model predicts a fare of \$3.50 for that ride.

The number 3.5 is added on regardless of distance, so it is the predicted fare at $x = 0$ miles — the flat charge for getting in. The per-mile rate is the slope 2.4.

13 Answer: 5

1. Predicted value: $7(5) + 20 = 55$.
2. Actual value from the table at $x = 5$: 60.
3. Residual $= 60 - 55 = 5$.

Find what the model says at $x = 5$, then subtract that from what actually happened at $x = 5$. The residual is positive, so this point sits above the line of best fit.

14 Answer: C

1. Compute the ratios of consecutive values.
2. Each ratio equals 1.5.
3. A constant multiplier means exponential, with multiplier 1.5.

The differences 40, 60, 90 are not constant, but each value is 1.5 times the one before it ($120/80 = 1.5$, $180/120 = 1.5$, $270/180 = 1.5$). A constant multiplier is the signature of an exponential model.

15 Answer: D

1. The slope is 1.6, in units of $\hat{P}$ per year.
2. $\hat{P}$ is measured in thousands of people.
3. So the yearly increase is 1.6 thousand $= 1600$ people.

The slope is 1.6 thousand people per year. Because $\hat{P}$ is measured in thousands, 1.6 must be converted: $1.6 \times 1000 = 1600$ people per year.

16 Answer: C

1. The difference in x is $18 - 11 = 7$ minutes.
2. Each minute adds a predicted 13 words.
3. So the difference is $13 \times 7 = 91$ words.

A difference of two predictions from the same line depends only on the slope: the intercept cancels. Each extra minute adds 13 words, and there are 7 extra minutes.

17 Answer: B

1. Start from actual = predicted + residual.
2. Substitute: $138 + (-24)$.
3. The actual attendance was 114 visitors.

Rearranging residual = actual − predicted gives actual = predicted + residual. The residual is negative, so the true attendance must be *below* the prediction.

18 Answer: D

1. The slope 17 pairs a 1-unit increase in x with a change in y.
2. One unit of x is one practice test; y is measured in points.
3. Word it as a prediction on average, not as a cause.

A line of best fit describes an average, predicted change and an association, never a guaranteed effect and never causation. The slope predicts the change in y for one more unit of x, not the reverse.

19 **Answer: 20**

1. Set $2.5x + 18 = 68$.
2. Subtract: $2.5x = 50$.
3. Divide: $x = 20$ weeks.

Here the output is given and the input is unknown, so the prediction becomes an equation to solve rather than an expression to evaluate.

20 **Answer: C**

1. Compute residual = actual − predicted for each day.
2. The residuals are -1, 2, -2, 2, -1.
3. Three residuals are negative, so the model overpredicts on 3 days.

The model overpredicts exactly when predicted exceeds actual, that is when the residual is negative. Days 1, 3 and 5 have residuals -1, -2 and -1; days 2 and 4 have residual $+2$.

21 **Answer: B**

1. Slope = 34: the predicted monthly increase in fish.
2. Intercept = 210: the predicted number of fish at $x = 0$.
3. $x = 0$ is the moment of restocking.

Two numbers, two jobs: the slope 34 is a rate per month, and the intercept 210 is the predicted count at $x = 0$, the moment of restocking. Every wrong choice assigns at least one of them the other's job.

22 **Answer: C**

1. Look at how much the height changes from one week to the next.
2. The changes get larger and larger, so the graph curves upward.
3. Increasing at an increasing rate is exponential growth, not linear.

The membership climbs by 2, then 3, then 5, then 8, then 13, then 21 hundred: the increases themselves are growing, so the points bend upward rather than following a straight line. That is exponential growth.

Module 2

1 **Answer: A**

1. Read the plot left to right.
2. The number of tickets falls as the days increase, so the association is negative.
3. The points lie close to a straight line, so the form is linear.

The points fall steadily from left to right along a nearly straight path, so the association is negative and approximately linear. A negative association does not mean the relationship is weak.

2 **Answer: C**

1. Set $x = 0$: $\hat{y} = 15$.
2. $x = 0$ is the moment the candle was lit.
3. So the predicted starting height is 15 centimeters.

The number 15 is the constant term, so it is the predicted height when $x = 0$ minutes — the instant the candle was lit. The rate at which the candle shortens is the other number, 0.2 centimeters per minute.

3 Answer: 40

1. Substitute $x = 35$.
2. Compute $0.8(35) = 28$.
3. Add: $28 + 12 = 40$ hundred ballots.

Substitute and evaluate, keeping both variables in the same units (hundreds) so no conversion is needed.

4 Answer: B

1. The slope is 0.05.
2. One unit of x is one text message; y is in dollars.
3. So each additional text raises the predicted bill by \$0.05.

The slope 0.05 has units of dollars per text message, so each extra text adds a predicted 5 cents to the bill. The 30 is the predicted bill when no texts are sent.

5 Answer: -2.5

1. Rise: $5 - 30 = -25$.
2. Run: $14 - 4 = 10$.
3. Slope: $-25/10 = -2.5$.

The y-values drop as the x-values rise, so the slope must come out negative; a positive answer means the subtraction was done in mismatched orders.

6 Answer: A

1. Ask whether the change per unit is a fixed amount or a fixed percent.
2. A fixed amount per unit gives a constant difference, which is linear.
3. A fixed percent of the current value gives a constant ratio, which is exponential.

A change of a fixed *percent* of a changing amount is exponential, because the amount added or removed changes as the quantity changes. Fixed amounts per unit of time or per item are linear.

7 Answer: C

1. Below the line means actual < predicted.
2. Residual = actual − predicted, so the residual is negative.
3. A prediction that is too high is an over-prediction.

Below the line means the actual value is smaller than the predicted value, so actual − predicted is negative and the model guessed too high. Negative residual and over-prediction always travel together.

8 Answer: -4

1. Predicted: $5.5(8) + 30 = 44 + 30 = 74$.
2. Residual = actual − predicted $= 70 - 74$.
3. The residual is -4.

Predict first, then subtract in the order actual minus predicted. The negative result says the point lies below the line and the model overpredicts there.

9 **Answer: D**

1. At $t = 0$ the model gives $P = 4800$, the population in 2015.
2. The base 0.85 multiplies the population each year.
3. Losing 15% leaves 85%, so the decay rate is 15% per year.

In $a(b)^t$, a is the value at $t = 0$ and b is the yearly multiplier. Since $0.85 = 1 - 0.15$, the population keeps 85% of itself and therefore loses 15% each year, starting from 4800 in 2015.

10 **Answer: D**

1. Note the range of the data: x from 1 to 12.
2. The prediction uses $x = 90$, far outside that range.
3. Outside the observed range the pattern is not supported by evidence.

Extrapolation is the issue: a line of best fit only summarizes the range of x that was actually observed. Beyond it the relationship may bend, level off, or stop existing, and the model has no evidence either way.

11 **Answer: 50**

1. Set $-0.6x + 45 = 15$.
2. Subtract 45: $-0.6x = -30$.
3. Divide: $x = 50$.

Because the slope is negative, reaching a smaller predicted y requires a larger x; setting up the equation avoids guessing at that direction.

12 **Answer: A**

1. The slope is 0.19, measured in dollars per year.
2. Convert: $\$0.19 = 19$ cents.
3. So the predicted price rose about 19 cents per year.

The slope is 0.19 dollars per year, and 0.19 dollars is 19 cents. Reading the slope as 19 dollars per year, or reading the intercept as the yearly change, are the two standard errors here.

13 **Answer: B**

1. Compute $0.8^3 = 0.512$.
2. Multiply: $25000(0.512)$.
3. The predicted value is \$12800.

Each year the machine keeps 80% of its value, so three years leave $0.8^3 = 0.512$ of the original. Subtracting 20% three times as a fixed dollar amount would be the linear mistake.

14 **Answer: 3**

1. A positive residual means the point lies above the line.
2. The line is $\hat{y} = 2x + 1$, giving predictions $3, 5, 7, 9, 11, 13, 15$.
3. The points $(1, 5)$, $(3, 10)$ and $(5, 15)$ lie above it, so the count is 3.

A positive residual means the actual value is greater than the predicted value, so the point sits strictly above the line. Counting points above the line answers the question directly.

15 Answer: D

1. Set $x = 0$: $\hat{y} = 68$.
2. $x = 0$ means 0 hours after noon, that is, noon itself.
3. So the model predicts 68 degrees Fahrenheit at noon.

Setting $x = 0$ puts the clock at noon, so 68 is the predicted temperature at noon. The hourly rise is the slope 2.3; swapping the two numbers produces two of the wrong choices.

16 Answer: C

1. Ratios: $18/6 = 3$, $54/18 = 3$, $162/54 = 3$, so the base is 3.
2. At $t = 0$ the count is 6, so the initial value is 6.
3. The model is $N = 6(3)^t$.

The values triple each hour, so the model is exponential with base 3, and the value at $t = 0$ is 6, so the coefficient in front is 6. Writing $3(6)^t$ swaps the starting value with the growth factor.

17 Answer: B

1. Compute residual = actual − predicted for each day: $6, -7, 9, -12$.
2. Over-prediction corresponds to a negative residual: days 2 and 4.
3. The larger over-prediction is 12 visitors on day 4.

Over-prediction happens where predicted exceeds actual, that is, where the residual is negative. The residuals are $6, -7, 9$ and -12, so the largest over-prediction is 12 visitors, on day 4. Days 1 and 3 are under-predictions and cannot be the answer at all.

18 Answer: D

1. The slope is -0.006, in hours of sleep per minute of social media.
2. The negative sign makes the predicted change a decrease.
3. So one more minute per day predicts 0.006 fewer hours of sleep.

The slope pairs one extra minute of social media with a predicted change in sleep, and the negative sign makes it a decrease. Reversing the roles of the two variables, or reading 8.4 as the rate, gives the trap choices.

19 Answer: 108

1. The change in x is $25 - 10 = 15$.
2. Each unit of x changes the prediction by 7.2.
3. So the difference is $7.2(15) = 108$.

Differences of predictions depend only on the slope: the constant 15 appears in both predictions and cancels when they are subtracted.

20 Answer: C

1. Track the signs in order: $+, +, -, -, -, +, +$.
2. That is a systematic U-shaped pattern, not random scatter.
3. A systematic pattern in the residuals means the linear form is wrong.

Residuals from a good linear fit should scatter randomly around zero. Here they march from positive to negative and back, a U-shaped pattern that says the data curve and a straight line is the wrong shape — no matter how small the residuals are.

21 Answer: B

1. Predicted: $3.4(15) + 12 = 51 + 12 = 63$.
2. Actual = predicted + residual $= 63 + (-7)$.
3. The actual y-value is 56.

Predict first, then apply the residual with its sign: actual = predicted + residual. Stopping at the prediction, 63, answers a question the problem did not ask.

22 Answer: A

1. Slope 2.8 thousand dollars per year = \$2800 per year of experience.
2. Intercept 41 thousand dollars = \$41000 predicted at $x = 0$ years.
3. $x = 0$ describes an employee with no experience.

Both numbers are measured in thousands of dollars, so 2.8 means \$2800 per year of experience and 41 means \$41000 at zero years. Skipping the unit conversion, swapping the two roles, or reversing the variables produces the other three choices.

Topic 15 — Probability and Conditional Probability

Module 1

1 Answer: A

1. Count all the marbles: $5 + 7 + 8 = 20$.
2. Count the favourable outcomes: 5 are red.
3. $P(\text{red}) = \frac{5}{20} = \frac{1}{4}$.

The denominator of a simple probability is the total number of marbles, and that total includes the red marbles themselves: $5 + 7 + 8 = 20$. Choice B is what you get by dividing 5 by the 15 marbles that are not red, which is not a probability at all.

2 Answer: B

1. The basket holds $12 + 18 = 30$ candies.
2. The candies that are not lemon are the 18 cherry candies.
3. $P(\text{not lemon}) = \frac{18}{30} = \frac{3}{5}$.

"Not lemon" means cherry, so the numerator is 18 and the denominator is still the whole basket, 30. Equivalently $1 - \frac{12}{30} = \frac{3}{5}$. Choice C compares lemon to cherry rather than lemon to the total, which is a ratio, not a probability.

3 Answer: C

1. The multiples of 5 from 1 to 20 are $5, 10, 15, 20$ — four values.
2. There are 20 equally likely integers in all.
3. $P = \frac{4}{20} = \frac{1}{5}$.

List the favourable outcomes rather than guessing at them: 5, 10, 15 and 20 are the only multiples of 5 in the list, so 4 of the 20 equally likely integers work. Choice D comes from counting 0 as a fifth multiple even though 0 is not in the range.

4 **Answer: 2/5**

1. Boys: $25 - 15 = 10$.
2. $P(\text{boy}) = \frac{10}{25} = \frac{2}{5}$.

The number of boys is not given directly, so subtract first: $25 - 15 = 10$. The denominator remains the whole class, 25, because the student is chosen from the class and not from some smaller group.

5 **Answer: B**

1. Robotics members: the column total is 50.
2. The pool is all 100 students.
3. $P(\text{robotics}) = \frac{50}{100} = \frac{1}{2}$.

The student is drawn from all 100 members, so the denominator is the grand total. The robotics column total is $24 + 26 = 50$, giving $\frac{50}{100} = \frac{1}{2}$. Choices C and D use the robotics column or the senior row as the denominator, which no phrase in this question authorises.

6 **Answer: C**

1. The condition restricts the pool to the 60 juniors.
2. Juniors in debate: 36.
3. $P(\text{debate} \mid \text{junior}) = \frac{36}{60} = \frac{3}{5}$.

"Given that the member is a junior" shrinks the pool to the junior row, whose total is 60. Of those juniors, 36 are in debate, so the probability is $\frac{36}{60} = \frac{3}{5}$. Choice A divides by the grand total 100 and choice D reverses the condition by dividing by the debate column total 50.

7 **Answer: A**

1. The junior-and-robotics cell contains 24 students.
2. The student is chosen from all 100 members.
3. $P = \frac{24}{100} = \frac{6}{25}$.

"A junior in the robotics club" names one cell of the table, the overlap of the junior row and the robotics column, which holds 24 students. Nothing in the sentence restricts the pool, so the denominator is the grand total 100: $\frac{24}{100} = \frac{6}{25}$. Choices B and C divide the same cell by a row or column total, and choice D counts the union of juniors and robotics members.

8 **Answer: 13/25**

1. Restrict to the robotics column: 50 students.
2. Seniors in that column: 26.
3. $P(\text{senior} \mid \text{robotics}) = \frac{26}{50} = \frac{13}{25}$.

"A member of the robotics club is selected" is a condition even though the word *given* never appears: the pool is the robotics column, total 50. Of those, 26 are seniors, so the probability is $\frac{26}{50} = \frac{13}{25} = 0.52$. Dividing by the senior row total 40 would answer the reversed question instead.

9 **Answer: D**

1. $P(\text{Maria wins}) = \frac{10}{250} = \frac{1}{25}$.
2. $P(\text{does not win}) = 1 - \frac{1}{25} = \frac{24}{25}$.

Maria wins with probability $\frac{10}{250} = \frac{1}{25}$, so she does not win with probability $1 - \frac{1}{25} = \frac{24}{25}$; counting the 240 tickets she does not own gives the same fraction. Choice B is the ratio of Maria's tickets to everyone else's, which is not a probability.

10 **Answer: B**

1. Students who walk: the column total is 50.
2. All students: 160.
3. $P(\text{walk}) = \frac{50}{160} = \frac{5}{16}$.

No grade is specified, so use the "Walk" column total, 50, over the grand total, 160: $\frac{50}{160} = \frac{5}{16}$. Choice C uses only the grade 9 row ($\frac{30}{90}$) and choice A uses only the grade 9 walkers over the grand total.

11 **Answer: D**

1. Restrict to the grade 10 row: 70 students.
2. Grade 10 students who take the bus: 35.
3. $P(\text{bus} \mid \text{grade } 10) = \frac{35}{70} = \frac{1}{2}$.

The condition names grade 10, so the pool is that row and the denominator is 70, not 160. Of those 70 students, 35 take the bus: $\frac{35}{70} = \frac{1}{2}$. Choice A divides by the grand total and choice C reverses the condition by dividing by the bus column total 80.

12 **Answer: 9/16**

1. Bus riders: 80.
2. Bus riders in grade 9: 45.
3. $P(\text{grade } 9 \mid \text{bus}) = \frac{45}{80} = \frac{9}{16}$.

The student comes from the bus column, so the denominator is 80. Of those bus riders, 45 are in grade 9, giving $\frac{45}{80} = \frac{9}{16}$. Using the grade 9 row total 90 instead would answer "given grade 9, what is the probability of taking the bus?" — the reversed question, whose answer is $\frac{1}{2}$.

13 **Answer: D**

1. $P(\text{car}) = \frac{30}{160} = \frac{3}{16}$.
2. $P(\text{not car}) = 1 - \frac{3}{16} = \frac{13}{16}$.

The complement is faster than adding two columns: $P(\text{car}) = \frac{30}{160} = \frac{3}{16}$, so $P(\text{not car}) = 1 - \frac{3}{16} = \frac{13}{16}$. Adding the bus and walk totals, $\frac{80+50}{160}$, gives the same value. Choice A is the probability of the opposite event and choice B conditions on grade 10 for no reason.

14 **Answer: B**

1. Let t be the total: $\frac{15}{t} = \frac{3}{8}$.
2. Cross-multiply: $3t = 120$, so $t = 40$.
3. Green marbles: $40 - 15 = 25$.

Run the probability formula backwards: if $\frac{15}{t} = \frac{3}{8}$ then $3t = 120$ and $t = 40$ marbles in all. The question asks for green marbles, not the total, so subtract: $40 - 15 = 25$. Choice C is the total, the answer to a question that was not asked.

15 **Answer: A**

1. The grade 10 and walk cell contains 20 students.
2. The pool is all 160 students.
3. $P = \frac{20}{160} = \frac{1}{8}$.

The word *and* points at a single cell: 20 students are both in grade 10 and walkers. The selection is from all 160 students, so $\frac{20}{160} = \frac{1}{8}$. Choice B divides that cell by the grade 10 row total and choice D divides it by the walk column total — both answer conditional questions this stem never asked.

16 **Answer: 4**

1. Let x be the number of red marbles added; the bag then holds $8 + x$ red out of $20 + x$ marbles.
2. $\frac{8+x}{20+x} = \frac{1}{2}$ gives $2(8 + x) = 20 + x$.
3. $16 + 2x = 20 + x$, so $x = 4$.

Each marble added is red, so it raises the red count *and* the total by one; the classic error is to hold the total at 20. Solving $\frac{8+x}{20+x} = \frac{1}{2}$ gives $16 + 2x = 20 + x$, so $x = 4$. Check: 12 red out of 24 marbles is indeed one half.

17 **Answer: A**

1. Restrict to the debate column: 50 members.
2. Juniors among them: 36.
3. $P(\text{junior} \mid \text{debate}) = \frac{36}{50} = \frac{18}{25}$.

The pool is the debate column, total 50, and 36 of those members are juniors: $\frac{36}{50} = \frac{18}{25}$. Choice B, $\frac{36}{60}$, answers the reversed question "a junior is selected; what is the probability of debate?", choice C divides by the grand total, and choice D is the probability that a debate member is a senior.

18 **Answer: C**

1. Debate members: 50. Seniors: 40. Seniors in debate: 14.
2. $n(\text{debate or senior}) = 50 + 40 - 14 = 76$.
3. $P = \frac{76}{100} = \frac{19}{25}$.

The 14 students who are seniors in debate belong to both groups, so adding the 50 debate members to the 40 seniors counts them twice. Subtract the overlap once: $50 + 40 - 14 = 76$, giving $\frac{76}{100} = \frac{19}{25}$. Choice D is the double-counted sum, choice A is the intersection, and choice B drops the overlap entirely instead of counting it once.

19 **Answer: D**

1. After x students join, robotics has $50 + x$ members out of $100 + x$.
2. $\frac{50+x}{100+x} = \frac{3}{5}$ gives $5(50 + x) = 3(100 + x)$.
3. $250 + 5x = 300 + 3x$, so $2x = 50$ and $x = 25$.

Every new member increases the robotics count and the club total together, so the equation is $\frac{50+x}{100+x} = \frac{3}{5}$, not $\frac{50+x}{100} = \frac{3}{5}$ (which would give choice A). Cross-multiplying, $250 + 5x = 300 + 3x$, so $x = 25$; the check is $\frac{75}{125} = \frac{3}{5}$.

20 **Answer: 2**

1. After adding x blue shirts there are $6+x$ blue shirts out of $30+x$.
2. $\frac{6+x}{30+x} = \frac{1}{4}$ gives $4(6+x) = 30+x$.
3. $24+4x = 30+x$, so $3x = 6$ and $x = 2$.

Adding x blue shirts makes the display $30+x$ shirts of which $6+x$ are blue, so $\frac{6+x}{30+x} = \frac{1}{4}$ gives $24+4x = 30+x$ and $x = 2$. Check: 8 blue out of 32 shirts is $\frac{1}{4}$. Answers larger than a few shirts are a sign that the total was left unchanged.

21 **Answer: C**

1. Restrict to grade 9: 90 students.
2. Grade 9 students who do not take the bus: $90-45 = 45$ (or $30+15$).
3. $P = \frac{45}{90} = \frac{1}{2}$.

Two ideas at once: the condition fixes the denominator at the grade 9 row total, 90, and the complement supplies the numerator, $90-45 = 45$ students who walk or ride in a car. So the probability is $\frac{45}{90} = \frac{1}{2}$. Choice D uses the bus column total 80, the reversed condition, and choice A abandons the condition altogether.

22 **Answer: B**

1. $n(\text{tea or coffee}) = 35+30-12 = 53$.
2. $n(\text{neither}) = 60-53 = 7$.
3. $P(\text{neither}) = \frac{7}{60}$.

Count the people who like at least one drink first, subtracting the overlap so the 12 people who like both are counted once: $35+30-12 = 53$. The remaining $60-53 = 7$ people like neither, so the probability is $\frac{7}{60}$. Choice D is the probability of the opposite event and choice C is just the overlap.

Module 2

1 **Answer: B**

1. There are 8 equally likely sections.
2. 3 of them are shaded.
3. $P(\text{shaded}) = \frac{3}{8}$.

Equal areas make the eight sections equally likely, so the denominator is 8 and the numerator is the 3 shaded sections. Choice C compares shaded sections to unshaded ones, a ratio of 3 to 5 rather than a probability, and choice D is the complement.

2 **Answer: A**

1. Supporters: $120+80 = 200$.
2. All respondents: 400.
3. $P(\text{support}) = \frac{200}{400} = \frac{1}{2}$.

The resident is drawn from all 400 respondents, so the denominator is the grand total and the numerator is the support column total: $\frac{200}{400} = \frac{1}{2}$. Choices B, C and D are all correct answers to *conditional* questions about this table, which is exactly why they are offered here.

3 Answer: D

1. Restrict to the under-40 row: 160 residents.
2. Supporters in that row: 120.
3. $P(\text{support} \mid \text{under } 40) = \frac{120}{160} = \frac{3}{4}$.

"Given that the resident is under 40" fixes the denominator at the under-40 row total, 160. That row contains 120 supporters, so the probability is $\frac{120}{160} = \frac{3}{4}$. Choice A keeps the grand total, and choice C divides by the support column total 200, which reverses the condition.

4 Answer: 2/5

1. The 40-or-older and oppose cell contains 160 residents.
2. The pool is all 400 residents.
3. $P = \frac{160}{400} = \frac{2}{5}$.

The word *and* identifies the single cell where the "40 or older" row meets the "Oppose" column: 160 residents. Because no condition restricts the pool, divide by the grand total: $\frac{160}{400} = \frac{2}{5}$. Dividing by 240 or by 200 would answer a conditional question instead.

5 Answer: C

1. Restrict to the support column: 200 residents.
2. Supporters under 40: 120.
3. $P(\text{under } 40 \mid \text{support}) = \frac{120}{200} = \frac{3}{5}$.

The resident is drawn from the supporters, so the denominator is the support column total, 200, and the numerator is the 120 supporters who are under 40: $\frac{120}{200} = \frac{3}{5}$. Choice B is the same cell over the under-40 row total, the reversed condition asked in the previous question.

6 Answer: C

1. Primes from 1 to 12: $2, 3, 5, 7, 11$.
2. That is 5 favourable cards out of 12.
3. $P(\text{prime}) = \frac{5}{12}$.

The primes at most 12 are 2, 3, 5, 7 and 11 — five of them, so the probability is $\frac{5}{12}$. Choice D comes from wrongly counting 1 as prime, and choice B from forgetting 11.

7 Answer: A

1. Defective items: $15 + 35 = 50$.
2. All items: 500.
3. $P(\text{defective}) = \frac{50}{500} = \frac{1}{10}$.

No machine is named, so the pool is the whole shift: $\frac{50}{500} = \frac{1}{10}$. Choice B conditions on machine B, choice C conditions on the defective items, and choice D is the probability that an item passed.

8 **Answer: 7/40**

1. Restrict to the machine B row: 200 items.
2. Defective items from machine B: 35.
3. $P(\text{defective} \mid B) = \frac{35}{200} = \frac{7}{40}$.

The item comes from machine B, so the denominator is that row's total, 200, and not the 500 items of the whole shift. Machine B produced 35 defective items, giving $\frac{35}{200} = \frac{7}{40} = 0.175$. Using 500 would produce 0.07, a number that describes the whole shift rather than machine B.

9 **Answer: A**

1. Restrict to the defective column: 50 items.
2. Defective items from machine A: 15.
3. $P(A \mid \text{defective}) = \frac{15}{50} = \frac{3}{10}$.

Selecting from the defective items makes the defective column total, 50, the denominator; 15 of those came from machine A, so the probability is $\frac{15}{50} = \frac{3}{10}$. Choice C, $\frac{15}{300}$, reverses the condition (the probability that a machine A item is defective) and choice D divides by the grand total.

10 **Answer: B**

1. Restrict to the passed column: 450 items.
2. Passed items from machine B: 165.
3. $P(B \mid \text{passed}) = \frac{165}{450} = \frac{11}{30}$.

The condition is "passed", so the denominator is the passed column total, 450, and the numerator is the 165 items machine B passed: $\frac{165}{450} = \frac{11}{30}$. Choice D, $\frac{165}{200}$, is the reversed condition and choice A divides by all 500 items.

11 **Answer: D**

1. Restrict to the oppose column: 200 residents.
2. Opposers who are 40 or older: 160.
3. $P(40+ \mid \text{oppose}) = \frac{160}{200} = \frac{4}{5}$.

The pool is the oppose column, total 200, of whom 160 are 40 or older: $\frac{160}{200} = \frac{4}{5}$. Choice C is $\frac{160}{240}$, the reversed condition (given 40 or older, the probability of opposing), and choice A divides the same cell by all 400 residents.

12 **Answer: 48**

1. Green marbles: $\frac{1}{5} \cdot 60 = 12$.
2. Marbles that are not green: $60 - 12 = 48$.

Work from the probability back to a count: $\frac{1}{5}$ of 60 is 12 green marbles, so $60 - 12 = 48$ are not green. Equivalently, $P(\text{not green}) = 1 - \frac{1}{5} = \frac{4}{5}$ and $\frac{4}{5} \cdot 60 = 48$. The trap is stopping at 12, the count the question did not ask for.

13 **Answer: D**

1. Restrict to the machine A row: 300 items.
2. Machine A items that passed: 285.
3. $P(\text{passed} \mid A) = \frac{285}{300} = \frac{19}{20}$.

The item comes from machine A, so the denominator is that row's total, 300: $\frac{285}{300} = \frac{19}{20}$. Choice B divides the same cell by the 500 items of the whole shift and choice C is the pass rate of the shift as a whole, $\frac{450}{500}$ — close to the right answer, but describing the wrong group.

14 **Answer: C**

1. Multiples of 3 from 1 to 30: $\frac{30}{3} = 10$.
2. $P(\text{multiple of } 3) = \frac{10}{30} = \frac{1}{3}$.
3. $P(\text{not a multiple of } 3) = 1 - \frac{1}{3} = \frac{2}{3}$.

Counting the multiples of 3 is far quicker than counting everything else: there are $30 \div 3 = 10$ of them, so $P(\text{multiple of } 3) = \frac{10}{30} = \frac{1}{3}$ and the complement is $1 - \frac{1}{3} = \frac{2}{3}$. Choice A answers the opposite event.

15 **Answer: B**

1. Males who answered Yes: $70 - 30 = 40$.
2. Total who answered Yes: $40 + 20 = 60$.
3. $P(\text{female} \mid \text{Yes}) = \frac{20}{60} = \frac{1}{3}$.

Fill the blanks before choosing a denominator: the male row gives $70 - 30 = 40$ men who answered Yes, so the Yes column totals $40 + 20 = 60$ (which also equals $120 - 60$). The condition restricts the pool to those 60 respondents, of whom 20 are female: $\frac{20}{60} = \frac{1}{3}$. Choice C uses the female row total and choice A uses all 120 respondents.

16 **Answer: 40**

1. After x defective items are added there are $50 + x$ defective out of $500 + x$ items.
2. $\frac{50+x}{500+x} = \frac{1}{6}$ gives $6(50 + x) = 500 + x$.
3. $300 + 6x = 500 + x$, so $5x = 200$ and $x = 40$.

The new items increase both the defective count and the group size, so the equation is $\frac{50+x}{500+x} = \frac{1}{6}$. Then $300 + 6x = 500 + x$, giving $x = 40$; the check is $\frac{90}{540} = \frac{1}{6}$. Holding the total at 500 would give the much smaller (and wrong) value $x = \frac{100}{3}$, which is not even a whole number of items.

17 **Answer: B**

1. After x supporters are added, $200 + x$ of the $400 + x$ respondents support the proposal.
2. $\frac{200+x}{400+x} = \frac{5}{9}$ gives $9(200 + x) = 5(400 + x)$.
3. $1800 + 9x = 2000 + 5x$, so $4x = 200$ and $x = 50$.

Each additional resident is a supporter, so both the support total and the survey total grow by x: $\frac{200+x}{400+x} = \frac{5}{9}$. Cross-multiplying gives $1800 + 9x = 2000 + 5x$, so $x = 50$, and $\frac{250}{450} = \frac{5}{9}$ confirms it. Choice D is what you get if the denominator is frozen at 400.

18 **Answer: C**

1. Residents who do not support: the oppose column, 200 residents.
2. Of those, the under-40 cell holds 40.
3. $P(\text{under } 40 \mid \text{oppose}) = \frac{40}{200} = \frac{1}{5}$.

Because every resident either supports or opposes, "do not support" is the oppose column, total 200. Of those, 40 are under 40 years old, so the fraction is $\frac{40}{200} = \frac{1}{5}$. Choice B, $\frac{40}{160}$, reverses the condition by dividing by the under-40 row, and choice D uses the grand total.

19 **Answer: 12**

1. After adding x red marbles there are $6 + x$ red out of $24 + x$ marbles.
2. $\frac{6+x}{24+x} = \frac{1}{2}$ gives $2(6 + x) = 24 + x$.
3. $12 + 2x = 24 + x$, so $x = 12$.

Adding x red marbles gives $6+x$ red marbles out of $24+x$, and $\frac{6+x}{24+x} = \frac{1}{2}$ leads to $12+2x = 24+x$, so $x = 12$. Check: 18 red out of 36 marbles. A useful shortcut for the target $\frac{1}{2}$: the reds must equal the non-reds, and there are 18 non-red marbles, so $18 - 6 = 12$ reds are needed.

20 **Answer: C**

1. $n(\text{sport or instrument}) = 70 + 45 - 25 = 90$.
2. $n(\text{neither}) = 120 - 90 = 30$.
3. $P(\text{neither}) = \frac{30}{120} = \frac{1}{4}$.

The 25 students who do both are inside the 70 and inside the 45, so the number who do at least one is $70 + 45 - 25 = 90$, not 115. That leaves $120 - 90 = 30$ students who do neither, and $\frac{30}{120} = \frac{1}{4}$. Choice A is the probability of the opposite event and choice D is the overlap.

21 **Answer: B**

1. Take 1000 lamps: 600 from X and 400 from Y.
2. Defectives: $0.05(600) = 30$ from X and $0.10(400) = 40$ from Y.
3. $P(\text{defective}) = \frac{30+40}{1000} = \frac{7}{100}$.

The two percentages have different denominators, so they cannot be added or averaged directly. Imagine 1000 lamps: 600 from plant X give 30 defectives and 400 from plant Y give 40, so 70 of the 1000 lamps are defective, a probability of $\frac{7}{100}$. Choice C is the plain average of 5% and 10%, which would only be right if the plants supplied equal numbers of lamps.

22 **Answer: 80**

1. Work inside the 40-or-older row only: 160 opposers out of 240.
2. After x more opposers in that row, $\frac{160+x}{240+x} = \frac{3}{4}$.
3. $4(160 + x) = 3(240 + x)$ gives $640 + 4x = 720 + 3x$, so $x = 80$.

The probability in question is conditional on being 40 or older, so only that row matters: the grand total 400 never enters the equation. The row currently holds 160 opposers out of 240 people, and each new respondent adds one to both, giving $\frac{160+x}{240+x} = \frac{3}{4}$, hence $640 + 4x = 720 + 3x$ and $x = 80$. Check: $\frac{240}{320} = \frac{3}{4}$.

Topic 16 — Inference, Margin of Error and Evaluating Statistical Claims

Module 1

1 Answer: C

1. The sample proportion is $\frac{60}{200} = 0.3$.
2. Apply that proportion to the whole population: 0.3×1500.
3. $0.3 \times 1500 = 450$.

A random sample is used as a miniature of the population, so the proportion found in the sample is the best available estimate of the proportion in the population; multiplying it by the population size converts that proportion into a count.

2 Answer: 39

1. A plausible range runs from estimate − margin of error to estimate + margin of error.
2. The lower endpoint is $42 - 3$.
3. $42 - 3 = 39$.

The margin of error measures how far the population value may reasonably sit from the sample estimate, so the smallest plausible value is the estimate minus the margin of error.

3 Answer: B

1. Identify the list from which the sample was drawn: the list of all registered members.
2. The population is that whole list, not the sample and not a wider group.

The population is the group the sample was randomly drawn from, because that is the only group the random selection makes the sample representative of.

4 Answer: A

1. A larger sample carries more information about the population.
2. More information means less uncertainty, so the margin of error shrinks.

The margin of error measures uncertainty about the population value, and taking a larger random sample reduces that uncertainty, so the margin of error decreases.

5 Answer: B

1. The sample proportion is $\frac{28}{80} = 0.35$.
2. Estimate the population count: 0.35×1200.
3. $0.35 \times 1200 = 420$.

Scaling a sample proportion up to the population size is the standard way to turn a sample count into an estimate for the whole group.

6 Answer: A

1. The estimate sits at the center of the interval: $\frac{12.4 + 13.6}{2} = 13.0$.
2. The margin of error is the distance from the center to an endpoint: $13.6 - 13.0 = 0.6$.

An interval built as estimate ± margin of error is symmetric about the estimate, so the margin of error is half the width of the interval.

7 **Answer: C**

1. Ask whether every customer has a chance of being chosen.
2. Sampling only during the busiest hour systematically favors long waits.

A sample is biased when the method of choosing it systematically favors one kind of member; restricting the sample to the busiest hour guarantees waits longer than those of a typical customer.

8 **Answer: D**

1. Form the interval: $15.2 - 1.3 = 13.9$ and $15.2 + 1.3 = 16.5$.
2. The interval describes plausible values of the population mean, not of individual students.

A margin of error attaches to the estimate of the population mean, so the interval it creates lists plausible values for that one mean; it says nothing about how the individual students are spread out.

9 **Answer: 28.7**

1. The greatest plausible value is estimate + margin of error.
2. $27.5 + 1.2 = 28.7$.

The margin of error bounds how far the population mean may plausibly lie from the sample mean in either direction, so the largest plausible value is the sample mean plus the margin of error.

10 **Answer: A**

1. The sample was drawn at random from the 400 club members, so conclusions apply to those members.
2. 34 out of 50 is 68%, a clear majority of the sample.

A random sample supports a general statement about the population it was drawn from, and about nothing wider; it also gives an estimate rather than an exact count.

11 **Answer: B**

1. The sample proportion is $\dfrac{208}{320} = 0.65$.
2. Estimate the population count: 0.65×12000.
3. $0.65 \times 12000 = 7800$.

The random sample estimates the proportion of the whole county that plans to vote, and multiplying that proportion by the number of registered voters converts it into an estimated count.

12 **Answer: D**

1. Hold the confidence level fixed and compare sample sizes.
2. The larger sample, Poll B, produces the smaller margin of error.

With the confidence level fixed, the margin of error depends on the sample size: a larger random sample pins down the population value more tightly.

13 **Answer: 0.42**

1. The estimate is the center of the interval.
2. $\dfrac{0.38 + 0.46}{2} = 0.42$.

An interval of plausible values is built as estimate $\pm$ margin of error, so the estimate sits exactly halfway between the endpoints.

14 **Answer: A**

1. Check for random assignment: each volunteer was randomly assigned to a group.
2. Random assignment balances the two groups, so the difference is attributed to the treatment.
3. The volunteers were not randomly selected, so the result cannot be extended beyond them.

Random assignment is what licenses a cause-and-effect conclusion; random selection is a separate question that controls only how widely the conclusion may be generalized.

15 **Answer: C**

1. The interval gives plausible values for the population mean.
2. 2.5 lies between 2.1 and 2.7, so it is a plausible value of that mean.

A confidence interval is a statement about one number, the population mean; any value inside it is plausible, no value is certain, and individual fish are not described at all.

16 **Answer: D**

1. The nutritionist observed habits; no treatment was assigned, so no cause can be established.
2. The sample was random from this university only, so the conclusion is limited to that university.
3. What remains is an association within that population.

An observational study with a random sample supports generalizing an association to the sampled population, but without random assignment it cannot separate the treatment from the other ways the two groups differ.

17 **Answer: 2560**

1. The greatest plausible proportion is $0.36 + 0.04 = 0.40$.
2. Convert to a count: 0.40×6400.
3. $0.40 \times 6400 = 2560$.

First push the proportion to the far end of its plausible range, then scale that proportion up to the population; the sample size of 500 has already been used to produce the margin of error and plays no further role.

18 **Answer: A**

1. Raising the confidence level means the procedure must succeed more often.
2. Succeeding more often from the same data requires casting a wider net.

Confidence and precision trade off: with the sample fixed, the only way to be right more often is to state a wider interval, so a higher confidence level always brings a larger margin of error.

19 **Answer: D**

1. Note that the respondents selected themselves.
2. Self-selected respondents differ systematically from the population, which biases the estimate.
3. A large sample does not repair that bias.

Bias comes from how a sample is chosen, not from how big it is; a self-selected sample of magazine readers over-represents people motivated to respond, so no sample size can rescue the estimate.

20 **Answer: 2.7**

1. The width of the interval is $148.3 - 142.9 = 5.4$.
2. The margin of error is half the width: $\frac{5.4}{2} = 2.7$.

The interval is centered at the sample mean and reaches one margin of error in each direction, so the margin of error is half the total width.

21 **Answer: B**

1. The confidence level describes the long-run behavior of the method, not this one interval.
2. The population mean is a fixed number; it either is or is not inside 48 to 62.
3. The interval also describes a mean, not individual students.

The 95% is a property of the procedure: repeated random samples give many different intervals, and about 95% of them capture the fixed population mean. This particular interval is not assigned a probability.

22 **Answer: C**

1. The study is observational: the adults chose whether to own a dog.
2. Only random assignment to the groups can rule out other differences between dog owners and others.
3. More data or a margin of error improves precision, not the type of conclusion.

No amount of extra data turns an observational study into an experiment; cause requires that the researcher, not the participant, decide who receives the treatment, and that the decision be made at random.

Module 2

1 **Answer: C**

1. The sample proportion is $\frac{45}{150} = 0.3$.
2. Estimate the population count: $0.3 \times 900 = 270$.

The proportion observed in a random sample is the best estimate of the proportion in the whole collection, so multiplying it by 900 estimates the count.

2 **Answer: 0.55**

1. The least plausible value is estimate $-$ margin of error.
2. $0.58 - 0.03 = 0.55$.

The plausible range is built symmetrically around the estimate, so its lower endpoint is the estimate minus the margin of error.

3 **Answer: A**

1. Both studies used the same confidence level, so that is not the difference.
2. Study B's sample is nine times as large, which reduces the uncertainty about the population mean.

The margin of error reflects how much the sample mean might miss the population mean by; averaging over more randomly chosen days makes that miss smaller.

4 Answer: A

1. The sample was randomly selected from the employees at one branch.
2. Random selection allows generalizing to exactly that group.

Random selection is what makes a sample stand in for a larger group, and the group it stands in for is the one the sample was drawn from.

5 Answer: 440

1. The sample proportion is $\frac{24}{300} = 0.08$.
2. Estimate the population count: 0.08×5500.
3. $0.08 \times 5500 = 440$.

A random sample estimates the proportion of car owners in the whole university, and multiplying that proportion by 5,500 turns it into an estimated number of students.

6 Answer: B

1. The plausible range is $7.4 - 0.5 = 6.9$ to $7.4 + 0.5 = 7.9$.
2. Only 7.1 falls inside that range.

Any value inside estimate $\pm$ margin of error is consistent with the sample; values outside it are not, so the test reduces to checking which choice lies between 6.9 and 7.9.

7 Answer: D

1. Ask which method systematically favors one kind of resident.
2. People leaving a fitness center exercise far more than a typical resident.

A sample taken where the behavior being measured is concentrated is biased upward; the three random methods choose residents without regard to how much they exercise, so none of them tilts the estimate high.

8 Answer: C

1. No treatment was assigned, so this is an observational study.
2. Observational data can show that two variables move together.
3. It cannot say which one drives the other, or whether a third factor drives both.

Without random assignment the two groups may differ in many ways besides coffee, so the study is limited to reporting the association it observed.

9 Answer: 23.1

1. The sample estimate is the midpoint of the interval.
2. $\frac{21.4 + 24.8}{2} = \frac{46.2}{2} = 23.1$.

The interval is the sample estimate plus or minus one margin of error, so the estimate is recovered by averaging the endpoints.

10 **Answer: D**

1. The sample was random from the organization's membership only.
2. A sample yields an estimate, not an exact count, and nothing about cause.

Random selection from the membership list supports one thing: an estimate, with uncertainty, of a proportion for that membership.

11 **Answer: A**

1. The center is the sample estimate, which does not depend on the confidence level.
2. Raising the confidence level enlarges the margin of error, widening the interval.

Changing the confidence level changes only how far the interval reaches from the sample estimate; the estimate itself comes from the data and stays put.

12 **Answer: D**

1. The sample proportion is $\frac{84}{240} = 0.35$.
2. Estimate the population count: 0.35×8000.
3. $0.35 \times 8000 = 2800$.

The random sample estimates the fraction of the park's trees that are maples, and that fraction applied to all 8,000 trees gives the estimated count.

13 **Answer: D**

1. Plausible values are exactly those between 28.6 and 31.2.
2. 32.4 lies outside the interval.

A confidence interval is a list of plausible values for the population mean, so a value fails to be plausible precisely when it falls outside the endpoints.

14 **Answer: B**

1. Everyone received the survey, but only volunteers answered.
2. That makes the respondents a self-selected group rather than a random sample.

Sending a survey to everyone does not make the responses random; the students who bother to reply may hold different opinions, and nothing in the data reveals how the silent 880 would have answered.

15 **Answer: 4.05**

1. The greatest plausible value is estimate + margin of error.
2. $3.6 + 0.45 = 4.05$.

The plausible range reaches one margin of error above the estimate, so its upper endpoint is $3.6 + 0.45$.

16 **Answer: C**

1. Random selection was used, so the finding extends to the city's residents.
2. No random assignment was used, so cause is out of reach.
3. Combine the two: a generalizable association.

Random selection and random assignment answer different questions: the first settles who the result applies to, the second settles whether the result is causal. This study has only the first.

17 **Answer: B**

1. Study 1: random selection, no random assignment, so a generalizable association.
2. Study 2: random assignment, no random selection, so cause for the volunteers only.
3. Match each study to the conclusion its design supports.

Each design buys exactly one thing. Random selection buys generalization, random assignment buys causation, and a study earns only what its own design provides.

18 **Answer: 2025**

1. The greatest plausible proportion is $0.42 + 0.03 = 0.45$.
2. Convert to a count: 0.45×4500.
3. $0.45 \times 4500 = 2025$.

Push the proportion to the top of its plausible range first, then scale to the population; the sample size of 400 is already accounted for inside the margin of error.

19 **Answer: C**

1. Halving the margin of error requires doubling $\sqrt{n}$.
2. Doubling $\sqrt{n}$ requires multiplying n by $2^2 = 4$.
3. $4 \times 400 = 1600$.

Because the margin of error falls like $1/\sqrt{n}$, precision is expensive: each halving of the margin of error costs four times the sample.

20 **Answer: C**

1. The percent of all voters who support the measure is a fixed unknown number, not a random one.
2. The randomness lives in the sampling, so the 95 percent describes the method.
3. The sample percent is the center, 49 percent, not the whole interval.

The confidence level is a track record of the procedure across all possible samples; once a particular interval is written down, the population value is either in it or not, so no probability attaches to this one interval.

21 **Answer: 8100**

1. The margin of error must shrink by a factor of $\dfrac{2.4}{0.8} = 3$.
2. So $\sqrt{n}$ must grow by a factor of 3, and n must grow by a factor of $3^2 = 9$.
3. $9 \times 900 = 8100$.

Since the margin of error behaves like $1/\sqrt{n}$, dividing it by 3 costs nine times as many people; this is why very precise polls are so large.

22 **Answer: B**

1. Random assignment was used, so a cause-and-effect conclusion is available.
2. The volunteers answered an advertisement, so they are not a random sample of adults.
3. Cause holds for the participants; generalization does not.

This is the standard split the test asks about: random assignment earns the word “caused,” while the absence of random selection keeps that causal claim confined to the people who actually took part.

Topic 17 — Area and Volume

Module 1

1 **Answer: D**

1. Area of a rectangle is $A = \ell w$.
2. $A = 12 \cdot 7 = 84$.

Area multiplies the two dimensions; adding them (19) or doubling that sum (38, the perimeter) answers a different question. The answer is 84 square inches.

2 **Answer: B**

1. Use $A = \frac{1}{2}bh$.
2. $A = \frac{1}{2}(10)(6) = 30$.

A triangle is half of the parallelogram with the same base and height, so the $\frac{1}{2}$ is not optional: $10 \cdot 6 = 60$ is the parallelogram’s area, and the triangle’s is 30 square centimeters.

3 **Answer: C**

1. Use $A = \pi r^2$.
2. $A = \pi(5)^2 = 25\pi$.

Squaring the radius is what makes the answer an area. Here 10π is the circumference and 100π is what you get by using the diameter 10 in place of the radius, so the area is 25π square inches.

4 **Answer: 180**

1. Use $V = \ell wh$.
2. $V = 9 \cdot 5 \cdot 4 = 180$.

The volume of any prism is the area of its base times its height: the base here is $9 \cdot 5 = 45$ square inches, and stacking that base through 4 inches of height gives 180 cubic inches.

5 **Answer: B**

1. The circumference is $C = \pi d$.
2. $C = \pi(14) = 14\pi$.

Because the diameter is given directly, use $C = \pi d$ rather than $C = 2\pi r$; treating 14 as the radius doubles the answer to 28π, and squaring the radius gives the area 49π. The circumference is 14π feet.

6 Answer: D

1. Use $A = bh$ with the perpendicular height.
2. $A = 9 \cdot 4 = 36$.

A parallelogram is a rectangle with one corner triangle slid across, so no factor of $\frac{1}{2}$ appears; halving would give 18, which is the area of a triangle with this base and height. The area is 36.

7 Answer: 96

1. A cube has 6 congruent square faces.
2. Each face has area $4^2 = 16$.
3. $SA = 6 \cdot 16 = 96$.

Surface area counts material, so add up all six faces rather than multiplying the three dimensions (that would give the volume, 64). The surface area is 96 square centimeters.

8 Answer: B

1. Use $A = \frac{1}{2}(b_1 + b_2)h$.
2. $A = \frac{1}{2}(8 + 12)(5) = \frac{1}{2}(20)(5) = 50$.

A trapezoid behaves like a parallelogram whose base is the average of the two parallel sides, $\frac{8+12}{2} = 10$, so the area is $10 \cdot 5 = 50$. Using one base alone gives 40 or 60, and forgetting the $\frac{1}{2}$ gives 100.

9 Answer: C

1. The base is a circle of area $\pi(3)^2 = 9\pi$.
2. $V = Bh = 9\pi \cdot 10 = 90\pi$.

Volume of a cylinder is base area times height, so the radius must be squared: 30π comes from forgetting to square it, 60π is the lateral surface area $2\pi rh$, and 360π comes from using the diameter 6 as the radius. The volume is 90π cubic inches.

10 Answer: A

1. Use $V = \frac{1}{3}\pi r^2 h$.
2. $V = \frac{1}{3}\pi(6)^2(10) = \frac{1}{3}(360\pi) = 120\pi$.

A cone fills exactly one third of the cylinder with the same base and height, and that cylinder holds 360π; dropping the $\frac{1}{3}$ gives 360π and using $\frac{1}{2}$ gives 180π. The volume is 120π cubic centimeters.

11 Answer: 7

1. Write $\frac{1}{2}(24)h = 84$.
2. $12h = 84$.
3. $h = 7$.

When the area is the number you are given, put the unknown into the same formula and solve rather than looking for a new one: $\frac{1}{2}bh = 84$ with $b = 24$ gives $h = 7$ units.

12 Answer: B

1. Complete the shape to a 10 by 6 rectangle, area 60.
2. The missing corner is 4 by 3, area 12.
3. $60 - 12 = 48$.

A composite figure should be cut into rectangles or completed to one; here the whole rectangle is 60 and the notch removes 12, so the region is 48. Splitting instead into a 6×6 square and a 4×3 rectangle gives $36 + 12 = 48$ as well.

13 Answer: C

1. If A has area ℓw, then B has area $(3\ell)(3w)$.
2. $(3\ell)(3w) = 9\ell w$.

Area multiplies two lengths, so a scale factor of 3 appears twice: the area is multiplied by $3^2 = 9$, not by 3. The factor $27 = 3^3$ would apply to a volume, where three lengths are multiplied.

14 Answer: B

1. Use $V = \frac{4}{3}\pi r^3$.
2. $r^3 = 27$.
3. $V = \frac{4}{3}\pi(27) = 36\pi$.

Cube the radius first, then take $\frac{4}{3}$ of it: $4\pi(27) = 108\pi$ drops the division by 3, and 288π comes from using the diameter 6 as the radius. The volume is 36π cubic inches.

15 Answer: 5

1. Write $\pi r^2(4) = 100\pi$.
2. Divide both sides by 4π: $r^2 = 25$.
3. $r = 5$.

Set the volume formula equal to the given volume and solve for the letter; dividing by π first clears it from the problem entirely, leaving $r^2 = 25$, so the radius is 5 centimeters.

16 Answer: B

1. The volume is $\frac{1}{3}\pi r^2 h$.
2. Replacing r with $2r$ and h with $3h$ gives $\frac{1}{3}\pi(2r)^2(3h) = \frac{1}{3}\pi(4r^2)(3h)$.
3. That is 12 times the original.

This is not the uniform-scaling law, because the two dimensions grow by different factors, so substitute directly: the radius is squared in the formula and so contributes $2^2 = 4$, while the height contributes 3, for a total factor of 12. Multiplying the factors as $2 \cdot 3 = 6$ ignores the square.

17 Answer: A

1. The radius, the height and the slant height form a right triangle: $5^2 + h^2 = 13^2$.
2. $h^2 = 144$, so $h = 12$.
3. $V = \frac{1}{3}\pi(5)^2(12) = 100\pi$.

The slant height runs up the sloped surface, so it is the hypotenuse, never the height in the volume formula; using 13 directly produces $\frac{325\pi}{3}$. Finding $h = 12$ from the 5-12-13 triangle first gives the volume 100π.

18 **Answer: C**

1. Let the base edge be s: $s^2(5) = 500$, so $s^2 = 100$ and $s = 10$.
2. The two square bases contribute $2(10)^2 = 200$.
3. The four side faces contribute $4(10)(5) = 200$.
4. $SA = 200 + 200 = 400$.

Work backwards from the volume to the missing length, then reread the question, which asks for surface area rather than that length. Counting only the four sides gives 200 and adding just one base gives 300; the full surface area is 400 square inches.

19 **Answer: 400**

1. Lengths scale by $k = 4$.
2. Surface area scales by $k^2 = 16$.
3. $25 \cdot 16 = 400$.

Surface area is an area, so the scale factor enters squared: multiplying by 4 instead of 16 is the classic error. The statue's surface area is 400 square inches.

20 **Answer: B**

1. The full tank holds $\pi(4)^2(9) = 144\pi$.
2. The water is $\frac{2}{3}$ of that.
3. $\frac{2}{3}(144\pi) = 96\pi$.

Find the whole first and take the fraction of it at the end; 144π is the full tank and 48π is the empty third. Using the diameter 8 as the radius would give 384π. The water occupies 96π cubic feet.

21 **Answer: C**

1. Let the shorter side be x, so the longer is $3x$.
2. $\frac{1}{2}(x + 3x)(6) = 60$, so $12x = 60$.
3. $x = 5$, so the longer side is $3(5) = 15$.

Naming the shorter side x turns the sentence into one equation in one unknown; the trap is stopping at $x = 5$, which is the shorter side. The longer parallel side is 15 units.

22 **Answer: B**

1. Volumes scale by k^3, so $k^3 = 27$ and $k = 3$.
2. Surface areas scale by k^2.
3. $k^2 = 9$.

Run the scaling law backwards: the volume ratio gives the length ratio only after a cube root, and that length ratio is then squared for the surface area. Reusing 27 for the area is the intended error; the answer is 9.

Module 2

1 **Answer: C**

1. The side length satisfies $s^2 = 144$, so $s = 12$.
2. $P = 4s = 4(12) = 48$.

The square root of the area is the side, and the question asks for the distance around, which is four of those sides. Stopping at $s = 12$ gives the first choice; the perimeter is 48 meters.

2 **Answer: B**

1. From $\pi r^2 = 49\pi$, $r^2 = 49$ and $r = 7$.
2. $C = 2\pi r = 14\pi$.

Cancel the π on both sides to recover the radius, then use it in the circumference formula; using 49 itself as the radius produces 98π. The circumference is 14π centimeters.

3 **Answer: A**

1. The base area is $8 \cdot 6 = 48$ square inches.
2. $48h = 240$.
3. $h = 5$.

Volume is base area times height, so divide by the whole base area, not by a single edge: dividing by 8 gives 30 and by 6 gives 40. The height is 5 inches.

4 **Answer: 150**

1. $s^3 = 125$, so $s = 5$.
2. Each of the 6 faces has area 25.
3. $SA = 6(25) = 150$.

Take the cube root of the volume to get the edge, then build the surface area from that edge; the volume and the surface area are different measurements of the same cube. The surface area is 150 square centimeters.

5 **Answer: C**

1. The radius is $\frac{16}{2} = 8$.
2. $A = \pi(8)^2 = 64\pi$.

The area formula needs the radius, so halve the diameter before squaring; using 16 as the radius gives 256π, four times too large. The area is 64π square inches.

6 **Answer: D**

1. The volume is s^3.
2. Replacing s with $4s$ gives $(4s)^3 = 64s^3$.

Volume multiplies three lengths, so the scale factor appears three times: $4^3 = 64$. The choice 16 is the factor for surface area and 12 comes from multiplying 4 by the three dimensions instead of cubing.

7 **Answer: C**

1. The distance between the parallel sides is the height, $h = 4$.
2. $A = \frac{1}{2}(6 + 10)(4)$.
3. $A = \frac{1}{2}(16)(4) = 32$.

"Distance between them" is the trapezoid's height, and both parallel sides must appear in the sum; using only one base gives 24 or 20, and forgetting the $\frac{1}{2}$ gives 64. The area is 32 square feet.

8 **Answer: A**

1. The rectangle contributes $20 \cdot 10 = 200$.
2. Each shorter side is 10, so each semicircle has diameter 10 and radius 5.
3. The two semicircles together form one circle: $\pi(5)^2 = 25\pi$.
4. Total: $200 + 25\pi$.

Two semicircles of equal radius combine into one full circle, and the side they sit on is a diameter, not a radius. Calling the radius 10 gives $200 + 100\pi$ and counting two full circles gives $200 + 50\pi$; the area is $200 + 25\pi$ square meters.

9 **Answer: 6**

1. Write $\frac{1}{3}\pi r^2(8) = 96\pi$.
2. Divide by π and multiply by 3: $8r^2 = 288$.
3. $r^2 = 36$, so $r = 6$.

Clearing the $\frac{1}{3}$ and the π before touching r keeps the arithmetic clean; forgetting the $\frac{1}{3}$ would give $r^2 = 12$ and no clean answer, which is a signal you dropped it. The radius is 6 inches.

10 **Answer: C**

1. The two circular ends give $2\pi(5)^2 = 50\pi$.
2. The curved side unrolls into a rectangle of area $2\pi rh = 2\pi(5)(6) = 60\pi$.
3. $SA = 50\pi + 60\pi = 110\pi$.

A closed cylinder has three surfaces: two circles and one rolled-up rectangle whose width is the circumference. Using only the side gives 60π and counting one end gives 85π; the total is 110π square centimeters.

11 **Answer: D**

1. $2(12 + w) = 34$, so $12 + w = 17$.
2. $w = 5$.
3. $A = 12 \cdot 5 = 60$.

Half the perimeter is the sum of a length and a width, which is 17, so the other side is 5; the question then asks for the area, not that side. The area is 60 square inches.

12 **Answer: 300**

1. Lengths scale by $k = 2.5$.
2. Area scales by $k^2 = 6.25$.
3. $48(6.25) = 300$.

Both dimensions grow, so the area grows by the square of the factor; multiplying 48 by 2.5 alone would answer for a single length. The enlarged area is 300 square inches.

13 **Answer: B**

1. Cylinder: $\pi(3)^2(10) = 90\pi$.
2. Hemisphere: $\frac{1}{2} \cdot \frac{4}{3}\pi(3)^3 = \frac{2}{3}\pi(27) = 18\pi$.
3. $90\pi + 18\pi = 108\pi$.

A composite solid is the sum of pieces you have formulas for, and a hemisphere is half a sphere: using a whole sphere gives 126π and using the diameter 6 as the hemisphere's radius gives 234π. The volume is 108π cubic inches.

14 **Answer: D**

1. The three distinct faces have areas $3 \cdot 4 = 12$, $4 \cdot 5 = 20$ and $3 \cdot 5 = 15$.
2. Each occurs twice: $SA = 2(12 + 20 + 15)$.
3. $SA = 2(47) = 94$.

Covering the outside asks for surface area, and a box has six faces in three matching pairs, so the sum 47 must be doubled; 60 is the volume, which is measured in cubic inches and cannot answer a question about paper. The answer is 94 square inches.

15 **Answer: C**

1. The walkway adds 2 feet on each side, so the outer rectangle is $15 + 4 = 19$ by $8 + 4 = 12$.
2. Outer area: $19 \cdot 12 = 228$.
3. Garden area: $15 \cdot 8 = 120$.
4. Walkway: $228 - 120 = 108$.

A border of uniform width adds twice its width to each dimension, not once, and the walkway is the difference of the two rectangles. Treating it as perimeter times width gives $46 \cdot 2 = 92$, which double-counts nothing and misses the four corners; the walkway covers 108 square feet.

16 **Answer: A**

1. The slant height, the vertical height, and half a base edge form a right triangle: $h^2 + 3^2 = 5^2$.
2. $h^2 = 16$, so $h = 4$.
3. $V = \frac{1}{3}(6^2)(4) = \frac{1}{3}(144) = 48$.

The slant height is the hypotenuse of a right triangle whose other legs are the vertical height and half a base edge, so it can never go straight into the volume formula; using 5 as the height gives 60 and dropping the $\frac{1}{3}$ gives 144. The volume is 48 cubic centimeters.

17 **Answer: 540**

1. Volumes of similar solids scale by k^3.
2. $k^3 = 3^3 = 27$.
3. $20(27) = 540$.

Similar means both the radius and the height are multiplied by 3, and the volume formula multiplies three lengths, so the volume grows by 27. Cylinder B has volume 540 cubic centimeters.

18 **Answer: B**

1. $\frac{4}{3}\pi r^3 = 288\pi$, so $r^3 = 216$.
2. $r = 6$.
3. $SA = 4\pi(6)^2 = 144\pi$.

The radius is the bridge between the two formulas: recover it from the volume, then feed it into the surface-area formula. Using the diameter 12 as the radius gives 576π, and the surface area is 144π square units.

19 **Answer: A**

1. $V_P = \pi r^2 h$.
2. $V_Q = \pi\left(\frac{r}{2}\right)^2(2h) = \pi \cdot \frac{r^2}{4} \cdot 2h = \frac{1}{2}\pi r^2 h$.
3. $\frac{V_Q}{V_P} = \frac{1}{2}$.

Halving and doubling do not cancel here, because the radius is squared and the height is not: the radius contributes $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$ against the height's 2. The volume of Q is $\frac{1}{2}$ of the volume of P.

20 **Answer: 5**

1. Let the base edge be s; the height is $4s$.
2. $V = s^2(4s) = 4s^3 = 500$.
3. $s^3 = 125$, so $s = 5$.

Writing every dimension in terms of one letter turns the volume into a single cubic equation; the height being a multiple of the base edge is what makes s^3 appear rather than s^2. The base edge is 5 centimeters.

21 **Answer: D**

1. Bottom disk: $\pi(3)^2 = 9\pi$.
2. Curved side of the cylinder: $2\pi(3)(8) = 48\pi$.
3. Curved part of the hemisphere: $\frac{1}{2} \cdot 4\pi(3)^2 = 18\pi$.
4. $9\pi + 48\pi + 18\pi = 75\pi$.

Count only the surfaces you could actually paint: the top circle of the cylinder is covered by the hemisphere, so it is not part of the outside, while the hemisphere adds half of a sphere's curved surface. Stopping at the cylinder alone gives 57π or 66π; the total is 75π square inches.

22 **Answer: 10**

1. Volumes scale by k^3: $k^3 = \frac{125}{27}$.
2. $k = \frac{5}{3}$.
3. A radius is a length, so it scales by k: $6 \cdot \frac{5}{3} = 10$.

The ratio of volumes must be cube-rooted before it can be applied to a length; scaling the radius by $\frac{125}{27}$ instead would be the classic error. The larger cone's base radius is 10 centimeters.

Topic 18 — Lines, Angles and Triangles

Module 1

1 Answer: B

1. The two marked angles together make a straight line, so they are supplementary.
2. $x + 118 = 180$.
3. $x = 62$.

A straight line is a $180°$ angle, so the ray splits it into two angles whose measures add to $180°$; nothing else in the figure is needed. Subtracting gives $x = 62$. Choice A is the complement, $90 - 62$, and choice C simply repeats the given angle, which would be correct only for a vertical angle.

2 Answer: 63

1. Complementary angles add to $90°$.
2. $m\angle B = 90 - 27 = 63$.

Complementary means the two angles together form a right angle, so their measures add to $90°$, not $180°$. That gives $m\angle B = 63°$. Answering 153 means using the supplementary rule by mistake.

3 Answer: B

1. Vertical angles are equal, so $3x + 8 = x + 40$.
2. $2x = 32$.
3. $x = 16$.

The two marked angles sit opposite each other where the lines cross, so they are vertical angles and therefore equal; that is what turns the figure into an equation. Solving $3x + 8 = x + 40$ gives $x = 16$. Choice A comes from adding x to both sides instead of subtracting, and choice D comes from treating the pair as supplementary.

4 Answer: D

1. Co-interior (same-side interior) angles between parallel lines are supplementary.
2. $x + 50 = 180$.
3. $x = 130$.

Both marked angles lie between the parallel lines and on the same side of the transversal, which makes them co-interior: they add to $180°$ rather than being equal. Choice B is the answer you get by treating them as alternate interior angles, and the figure rules that out because one marked angle is clearly obtuse and the other acute.

5 Answer: A

1. The three interior angles of a triangle add to $180°$.
2. $47 + 68 = 115$.
3. $180 - 115 = 65$.

Every triangle's interior angles total $180°$, so the third angle is whatever is left after the two given angles are removed. Choice C is the sum of the two given angles rather than the remainder, and choice D subtracts only one of them.

6 **Answer: 70**

1. Since $AB = AC$, the angles opposite them are equal: $\angle B = \angle C$.
2. $\angle B + \angle C = 180 - 40 = 140$.
3. $\angle B = 140/2 = 70$.

Equal sides face equal angles, so the two base angles share whatever is left of the 180° after the vertex angle is taken out. Half of 140° is 70°. Answering 140 gives the pair of base angles together rather than one of them.

7 **Answer: B**

1. $\angle ACD$ is an exterior angle of the triangle at C.
2. An exterior angle equals the sum of the two remote interior angles.
3. $\angle ACD = 35 + 65 = 100$.

The exterior angle at C does not touch $\angle A$ or $\angle B$, so those two are its remote interior angles and it equals their sum. You can check it the long way: $\angle ACB = 180 - 35 - 65 = 80$, and $\angle ACD = 180 - 80 = 100$. Choice A is that interior angle $\angle ACB$, the supplement of what was asked.

8 **Answer: C**

1. Corresponding angles between parallel lines are equal: $3x + 20 = x + 60$.
2. $2x = 40$, so $x = 20$.
3. Each angle measures $3(20) + 20 = 80$.

Corresponding angles sit in matching positions at the two crossings, so setting the expressions equal is legitimate; the work then has two stages, because the question asks for the angle and not for x. Choice A stops at $x = 20$, choice B comes from adding the expressions to 180°, and choice D is the supplement of the correct angle.

9 **Answer: C**

1. The third side c satisfies $12 - 7 < c < 12 + 7$.
2. So $5 < c < 19$.
3. Of the choices, only 17 lies strictly between 5 and 19.

The triangle inequality says each side must be shorter than the sum of the other two and longer than their difference, which pins the third side strictly between 5 and 19. Both endpoints are offered as choices on purpose: at $c = 5$ or $c = 19$ the three lengths collapse onto a straight line and no triangle exists.

10 **Answer: 13**

1. The third side c satisfies $9 - 5 < c < 9 + 5$, so $4 < c < 14$.
2. The greatest integer strictly less than 14 is 13.

The upper bound is the sum of the two known sides, and it is a strict bound, so 14 itself is impossible. The largest integer that works is 13; answering 14 is the classic boundary error.

11 **Answer: A**

1. The correspondence gives $\dfrac{AB}{DE} = \dfrac{BC}{EF}$.
2. $\dfrac{8}{6} = \dfrac{12}{EF}$.
3. $EF = 12 \cdot \dfrac{6}{8} = 9$.

The order of the letters in $\triangle ABC \sim \triangle DEF$ pairs $\overline{AB}$ with $\overline{DE}$ and $\overline{BC}$ with $\overline{EF}$, so the scale factor from the first triangle to the second is $6/8 = 3/4$ and every length shrinks by that factor. Choice C multiplies by $8/6$ instead, and choice B subtracts the difference $8 - 6 = 2$ from 12, treating a proportional relationship as an additive one.

12 **Answer: A**

1. Same-side interior angles between parallel lines are supplementary.
2. $(3x + 10) + (2x + 40) = 180$.
3. $5x + 50 = 180$, so $5x = 130$ and $x = 26$.

Because the two angles are on the same side of the transversal and between the parallel lines, they add to $180°$; setting them equal instead gives choice B. Choices C and D are the two angle measures themselves, $88°$ and $92°$, which do add to 180 but are not what the question asked for.

13 **Answer: 25**

1. $\angle CBD$ is the exterior angle at B, so it equals the sum of the remote interior angles $\angle A$ and $\angle C$.
2. $5x = 40 + (3x + 10)$.
3. $2x = 50$, so $x = 25$.

The exterior angle at B is remote from $\angle A$ and $\angle C$, so it equals their sum; that single equation carries all three pieces of information. Solving gives $x = 25$, which makes the exterior angle $125°$ and the two remote angles $40°$ and $85°$ – a valid check, since $40 + 85 = 125$.

14 **Answer: B**

1. Since $AB = AC$, $\angle B = \angle C = (2x)°$.
2. $(x + 20) + 2x + 2x = 180$.
3. $5x = 160$, so $x = 32$.

The equal sides $\overline{AB}$ and $\overline{AC}$ force the two angles opposite them, $\angle C$ and $\angle B$, to be equal, so the base angle expression must be used twice in the angle sum. Choice A comes from setting the vertex angle equal to a base angle, choice C is the resulting vertex angle $52°$, and choice D is a base angle $64°$.

15 **Answer: C**

1. The ratio of corresponding lengths is $k = \dfrac{5}{3}$.
2. Areas of similar figures are in the ratio $k^2 = \dfrac{25}{9}$.
3. Area $= 18 \cdot \dfrac{25}{9} = 50$.

Scaling a figure by k multiplies every length by k but every area by k^2, because area depends on two perpendicular dimensions at once. Choice B applies the length ratio $5/3$ straight to the area, the single most common error in this topic, and choice A applies it upside down.

16 **Answer: B**

1. Because $\overline{DE} \parallel \overline{BC}$, the parallel line cuts the two sides proportionally: $\dfrac{AD}{DB} = \dfrac{AE}{EC}$.
2. $\dfrac{6}{4} = \dfrac{9}{EC}$.
3. $6 \cdot EC = 36$, so $EC = 6$.

The parallel segment creates $\triangle ADE \sim \triangle ABC$, and a consequence is that the pieces of the two sides are in the same ratio, so $AD : DB$ must be matched with $AE : EC$ – piece to piece. Choice D is $AC = 15$, the whole side rather than the piece asked for, and choice C comes from inverting the ratio.

17 **Answer: 25**

1. $\overline{ST} \parallel \overline{QR}$ gives $\triangle PST \sim \triangle PQR$, with S matching Q and T matching R.
2. $PQ = PS + SQ = 8 + 12 = 20$.
3. $\dfrac{PS}{PQ} = \dfrac{ST}{QR}$, so $\dfrac{8}{20} = \dfrac{10}{QR}$ and $QR = 25$.

Here $\overline{ST}$ and $\overline{QR}$ are sides of the two similar triangles, so they must be compared with a pair of full sides, $\overline{PS}$ and $\overline{PQ}$ – not with the piece $\overline{SQ}$. Adding $8+12$ first is the whole trick; using $8/12$ instead would give 15, which is why the sum must be taken before the proportion is written.

18 **Answer: C**

1. Draw the line through P parallel to ℓ and m.
2. That line splits $\angle APB$ into two parts, each an alternate interior angle with one of the given angles.
3. $x = 35 + 48 = 83$.

A point trapped between two parallel lines is always handled the same way: run a third parallel through it. The upper part of $\angle APB$ is then alternate interior with the 35° angle and the lower part is alternate interior with the 48° angle, so the whole angle is their sum. Choice A subtracts them and choice D is the supplement of the answer.

19 **Answer: C**

1. The triangle inequality gives $11 - 5 < x < 11 + 5$, so $6 < x < 16$.
2. The greatest integer value is $x = 15$.
3. Perimeter $= 5 + 11 + 15 = 31$.

The perimeter is largest when the unknown side is as long as the triangle inequality allows, and since the bound $x < 16$ is strict, the largest legal integer is 15. Choice D uses the forbidden boundary value $x = 16$, and choice B is the smallest possible perimeter, from $x = 7$.

20 **Answer: 54**

1. Perimeter is a length, so the ratio of corresponding lengths is also $2 : 3$; the scale factor is $k = \dfrac{3}{2}$.
2. Areas are in the ratio $k^2 = \dfrac{9}{4}$.
3. Area $= 24 \cdot \dfrac{9}{4} = 54$.

Perimeter is built from lengths, so a perimeter ratio is the same as the side ratio and can be squared to get the area ratio. Multiplying 24 by $3/2$ instead would give 36, the error of applying a length ratio to an area.

21 **Answer: A**

1. The sun's rays make equal angles, so the person and shadow form a triangle similar to the one formed by the flagpole and its shadow.
2. $\dfrac{\text{height}}{\text{shadow}}$ is the same for both: $\dfrac{6}{9} = \dfrac{h}{24}$.
3. $h = 24 \cdot \dfrac{6}{9} = 16$.

Both triangles have a right angle at the ground and the same angle of elevation to the sun, so they are similar by AA and height-to-shadow is a constant ratio. Choice D pairs the sides the wrong way round, computing $24 \cdot \frac{9}{6}$, and choices B and C add or subtract the difference $9 - 6 = 3$ instead of scaling.

22 **Answer: D**

1. The two triangles share $\angle A$ and have $\angle ADE = \angle ABC$, so $\triangle ADE \sim \triangle ABC$ with D matching B and E matching C.
2. $AB = AD + DB = 4 + 6 = 10$.
3. $\dfrac{AD}{AB} = \dfrac{DE}{BC}$, so $\dfrac{4}{10} = \dfrac{6}{BC}$ and $BC = 15$.

Two equal angles are enough for similarity (AA), and the correspondence $\triangle ADE \sim \triangle ABC$ pairs $\overline{AD}$ with the *whole* side $\overline{AB}$, not with the piece $\overline{DB}$. Choice B is what you get from $\frac{4}{6} = \frac{6}{BC}$, using the piece $DB = 6$ in place of $AB = 10$, and choice A inverts the ratio.

Module 2

1 **Answer: B**

1. The three angles together make a straight angle: $x + 2x + 3x = 180$.
2. $6x = 180$.
3. $x = 30$.

The angles on one side of a straight line always total 180°, however many rays split it, so the six equal parts of x share that 180°. Choice D comes from using 360°, the full turn around a point, which would apply only if the rays split the whole plane around O.

2 **Answer: C**

1. The angle corresponding to the 72° angle sits above line m on the right of t, so it also measures 72°.
2. The marked angle $x°$ is on a straight line with that 72° angle.
3. $x = 180 - 72 = 108$.

Carry the given angle down to the second line with the corresponding-angles rule, then take a supplement, because the marked angle is below line m rather than above it. Choice B stops one step early with the corresponding angle itself, and choice A takes the complement instead of the supplement.

3 **Answer: 80**

1. Write the angles as $2k$, $3k$, and $4k$.
2. $2k + 3k + 4k = 180$, so $9k = 180$ and $k = 20$.
3. The largest angle is $4k = 80$.

A ratio only fixes the relative sizes, so introduce a common multiplier k and let the 180° angle sum determine it. The angles are 40°, 60°, and 80°, which do add to 180.

4 Answer: D

1. $AB = AC$ makes $\angle B$ and $\angle C$ the base angles, so $\angle C = 37°$.
2. $\angle A = 180 - 37 - 37$.
3. $\angle A = 106$.

The equal sides face the two base angles, so the given 37° occurs twice and only the vertex angle is unknown. Choice C is the pair of base angles combined, and choice B is the complement of 37°, which has no role here.

5 Answer: B

1. $\angle CBD$ is the exterior angle at B, so it equals $\angle A + \angle C$.
2. $130 = 55 + \angle C$.
3. $\angle C = 75$.

The exterior angle at B equals the sum of the two angles it does not touch, so one subtraction finishes the problem. Choice A is the interior angle $\angle ABC = 180 - 130 = 50$, and choice C is the supplement of the required angle.

6 Answer: B

1. The correspondence pairs $\overline{RS}$ with $\overline{XY}$ and $\overline{ST}$ with $\overline{YZ}$.
2. $\dfrac{RS}{XY} = \dfrac{ST}{YZ}$, so $\dfrac{12}{8} = \dfrac{15}{YZ}$.
3. $YZ = 15 \cdot \dfrac{8}{12} = 10$.

The scale factor from $\triangle RST$ to $\triangle XYZ$ is $8/12 = 2/3$, so every side of the second triangle is two thirds of its partner in the first. Choice D multiplies by $12/8$ instead, and choice C subtracts the difference $12 - 8 = 4$ from 15.

7 Answer: 75

1. The scale factor from the smaller to the larger is $k = \dfrac{5}{2}$.
2. Areas scale by $k^2 = \dfrac{25}{4}$.
3. Area $= 12 \cdot \dfrac{25}{4} = 75$.

Doubling every length of a figure quadruples its area, and in general lengths scale by k while areas scale by k^2. Multiplying 12 by $5/2$ would give 30, which is the answer to a question about a length, not an area.

8 Answer: D

1. Same-side interior angles are supplementary: $(7x - 20) + 3x = 180$.
2. $10x = 200$, so $x = 20$.
3. The angles measure $7(20) - 20 = 120$ and $3(20) = 60$; the larger is 120.

The two angles lie between the parallel lines on one side of the transversal, so they add to 180°; solving for x is only the first half of the work, since the question asks for an angle. Choice A stops at x, and choice C is the smaller angle – the supplement of the one requested.

9 **Answer: 100**

1. $(x + 10) + (2x - 20) + (3x + 10) = 180$.
2. $6x = 180$, so $x = 30$.
3. The angles are $40°$, $40°$, and $100°$; the largest is 100.

Collect the three expressions, set the sum to $180°$, and only then substitute back – the question asks for an angle, not for x. The triangle turns out to be isosceles, and $40 + 40 + 100 = 180$ confirms the arithmetic.

10 **Answer: C**

1. The third side c satisfies $15 - 8 < c < 15 + 8$, so $7 < c < 23$.
2. The integers strictly between are $8, 9, \ldots, 22$.
3. That is $22 - 8 + 1 = 15$ values.

Both bounds are strict, so 7 and 23 are excluded and the list runs from 8 to 22; counting a run of consecutive integers means subtracting and adding one. Choice A comes from forgetting to add that one, and choice D from including one of the forbidden endpoints.

11 **Answer: A**

1. $\overline{DE} \parallel \overline{BC}$ gives $\triangle ADE \sim \triangle ABC$, with D matching B and E matching C.
2. $\dfrac{AD}{AB} = \dfrac{DE}{BC}$, so $\dfrac{9}{15} = \dfrac{DE}{20}$.
3. $DE = 20 \cdot \dfrac{9}{15} = 12$.

Because $\overline{DE}$ and $\overline{BC}$ are corresponding sides of the two similar triangles, they must be compared using two corresponding full sides, $\overline{AD}$ and $\overline{AB}$. Choice D uses $AD : DB = 9 : 6$ by mistake, a ratio that belongs only to the pieces of the sides, and choice B subtracts the difference $15 - 9 = 6$ from 20.

12 **Answer: A**

1. $\angle ACB = 180 - 118 = 62$, so the base angles are $\angle B = \angle C = 62°$.
2. $\angle A = 180 - 62 - 62$.
3. $\angle A = 56$.

The exterior angle gives the base angle by a supplement, and the isosceles condition duplicates it, leaving only $\angle A$ unknown. Faster: the exterior angle equals $\angle A + \angle B = \angle A + 62$, so $\angle A = 118 - 62 = 56$. Choice B is the base angle rather than the vertex angle.

13 **Answer: A**

1. The $42°$ angle at A and $\angle ABC$ are alternate interior angles for $\ell \parallel m$ cut by $\overline{AB}$, so $\angle ABC = 42°$.
2. $\angle BAC = 180 - 42 - 63$.
3. $\angle BAC = 75$.

The parallel lines are the only bridge between the angle given at ℓ and the triangle sitting on m; once $42°$ has been carried down to B, the angle sum finishes it. Choice B is the supplement of the answer and choices C and D are supplements of the two given angles.

14 **Answer: C**

1. The ratio of areas is $\frac{48}{27} = \frac{16}{9}$.
2. The ratio of lengths is the square root of that: $\sqrt{\frac{16}{9}} = \frac{4}{3}$.
3. $PQ = 9 \cdot \frac{4}{3} = 12$.

Areas scale by the square of the length ratio, so going from areas back to lengths means taking a square root. Choice D applies the area ratio 48/27 directly to the side, and choice A uses the length ratio upside down.

15 **Answer: C**

1. Congruent triangles have equal corresponding sides, so $AB = XY$ and $BC = YZ$.
2. $3x - 4 = 20$ gives $x = 8$; $2y + 1 = 15$ gives $y = 7$.
3. $x + y = 8 + 7 = 15$.

Congruence is similarity with a scale factor of 1, so corresponding sides are simply equal and each equation can be solved on its own. The correspondence matters: $\overline{AB}$ must be matched with $\overline{XY}$, not with $\overline{YZ}$. Choices A and B are the individual values of y and x.

16 **Answer: 4**

1. A line parallel to one side cuts the other two sides proportionally: $\frac{AD}{DB} = \frac{AE}{EC}$.
2. $\frac{AD}{6} = \frac{8}{12} = \frac{2}{3}$.
3. $AD = 6 \cdot \frac{2}{3} = 4$.

Here every given length is a *piece* of a side, so the piece-to-piece proportion is the right one; matching AD with AE and DB with EC keeps the two sides of the triangle in step. As a check, $AB = 10$ and $AC = 20$, and $AD : AB = 4 : 10 = 8 : 20 = AE : AC$.

17 **Answer: D**

1. $\angle AEB$ and $\angle CED$ are vertical angles, and $\angle BAE = \angle DCE$ are alternate interior angles because $\overline{AB} \parallel \overline{CD}$.
2. So $\triangle AEB \sim \triangle CED$, with A matching C, E matching E, and B matching D.
3. $\frac{AE}{CE} = \frac{BE}{DE}$, so $\frac{4}{10} = \frac{6}{DE}$ and $DE = 15$.

The two triangles point away from each other at E, so the correspondence runs $A \leftrightarrow C$ and $B \leftrightarrow D$ – across the intersection, never within the same triangle. The scale factor from $\triangle AEB$ to $\triangle CED$ is $10/4$, so $ED = 6 \cdot \frac{10}{4} = 15$. Choice A inverts that ratio; choices B and C assume the triangles are congruent and copy BE or EC.

18 **Answer: C**

1. The $34°$ angle and $\angle ABC$ are alternate interior angles, so $\angle ABC = 34°$.
2. $AB = AC$ makes $\angle ACB = \angle ABC = 34°$.
3. $\angle BAC = 180 - 34 - 34 = 112$.

Two separate facts are needed: the parallel lines transfer the $34°$ down to B, and the isosceles condition copies it across to C. Choice B is the two base angles combined rather than the vertex angle, choice D is the supplement of $34°$, and choice A is its complement.

19 **Answer: B**

1. The scale factor is $k = 1.5 = \dfrac{3}{2}$.
2. Areas scale by $k^2 = \dfrac{9}{4}$.
3. Area $= 40 \cdot \dfrac{9}{4} = 90$.

Sides that are 50% longer do not give an area that is 50% larger: the area is multiplied by $1.5^2 = 2.25$. Choice A multiplies the area by 1.5, and choice D uses a scale factor of 2 by mistake.

20 **Answer: 12**

1. The area of $\triangle ABC$ is $16 + 48 = 64$.
2. $\triangle ADE \sim \triangle ABC$, and the ratio of their areas is $\dfrac{16}{64} = \dfrac{1}{4}$.
3. The ratio of corresponding lengths is $\sqrt{\dfrac{1}{4}} = \dfrac{1}{2}$, so $AB = 2 \cdot AD = 12$.

The quadrilateral is not a similar figure, so the first move is to rebuild the whole triangle: $\triangle ABC$ has area 64. From there the area ratio $1:4$ gives a length ratio $1:2$ by taking a square root, and $\overline{AD}$ and $\overline{AB}$ are corresponding sides. Using the area ratio 4 as a length ratio would give 24.

21 **Answer: D**

1. The third side must equal 5 or 12, since the triangle is isosceles.
2. Sides 5, 5, 12 fail the triangle inequality, because $5 + 5 = 10 < 12$.
3. So the sides are 5, 12, 12, and the perimeter is 29.

There are only two candidate triangles, and the triangle inequality eliminates one of them: two sides of length 5 cannot reach across a side of length 12. Choice B is the perimeter of that impossible triangle, choice C forgets the third side entirely, and choice A adds only the two given lengths.

22 **Answer: D**

1. Triangles BAD and BCA share $\angle B$, and $\angle BAD = \angle BCA$ is given, so $\triangle BAD \sim \triangle BCA$.
2. The correspondence is $B \leftrightarrow B$, $A \leftrightarrow C$, $D \leftrightarrow A$, so $\dfrac{BA}{BC} = \dfrac{BD}{BA}$.
3. $BA^2 = BC \cdot BD$, so $64 = 4 \cdot BC$ and $BC = 16$.

The two triangles overlap, which is exactly why the correspondence has to be written down before any proportion: $\overline{BA}$ in the small triangle matches $\overline{BC}$ in the large one, while $\overline{BD}$ in the small matches $\overline{BA}$ in the large. That makes $\overline{AB}$ the geometric mean of $\overline{BD}$ and $\overline{BC}$. Choice C is $DC = BC - BD = 12$, the remaining piece rather than the whole side, and choice B assumes the two triangles are congruent.

Topic 19 — Right Triangles and Trigonometry

Module 1

1 Answer: B

1. Write $6^2 + 8^2 = c^2$.
2. Then $36 + 64 = 100$, so $c^2 = 100$.
3. Take the positive square root: $c = 10$.

The hypotenuse is the side across from the right angle, so it is the c in $a^2 + b^2 = c^2$ and the legs are a and b. Adding the squares of the legs gives c^2, and the last step—the one students skip—is taking the square root.

2 Answer: B

1. Write $5^2 + b^2 = 13^2$.
2. Then $b^2 = 169 - 25 = 144$.
3. So $b = 12$.

Running the theorem backwards means subtracting, not adding: the square of a leg equals the square of the hypotenuse minus the square of the other leg. The 5-12-13 triple appears constantly on the test.

3 Answer: 15

1. $9 = 3 \cdot 3$ and $12 = 3 \cdot 4$, so the triangle is a 3-4-5 triangle scaled by 3.
2. The hypotenuse is $3 \cdot 5 = 15$.
3. Check: $81 + 144 = 225 = 15^2$.

This is the 3-4-5 triple multiplied by 3. Recognising a scaled triple lets you write the answer down without squaring anything, which is exactly the time saving the test is measuring.

4 Answer: B

1. The side ratio is $x : x : x\sqrt{2}$ with $x = 7$.
2. The hypotenuse is the $x\sqrt{2}$ side.
3. So the hypotenuse is $7\sqrt{2}$.

In a 45°-45°-90° triangle the sides are in the ratio $1 : 1 : \sqrt{2}$, so the hypotenuse is a leg times $\sqrt{2}$. Multiplying, never dividing, is what takes you from leg to hypotenuse.

5 Answer: A

1. The side ratio is $x : x\sqrt{3} : 2x$ with $2x = 10$.
2. So $x = 5$.
3. The side opposite 30° is the x side, so it is 5.

In a 30°-60°-90° triangle the sides are in the ratio $1 : \sqrt{3} : 2$, in the order short leg, long leg, hypotenuse. The shortest side always sits opposite the smallest angle, so it is half the hypotenuse.

6 Answer: A

1. Identify the side opposite A: it has length 8.
2. Identify the hypotenuse: it has length 17.
3. $\sin A = \dfrac{\text{opposite}}{\text{hypotenuse}} = \dfrac{8}{17}$.

SOH: sine is opposite over hypotenuse. The side opposite angle A is the one that does not touch A, which here is the side of length 8, and the hypotenuse is always the side across from the right angle.

7 Answer: A

1. The two angles x° and $(90-x)^\circ$ are complementary.
2. For complementary angles, sine and cosine swap: $\cos\left((90-x)^\circ\right) = \sin(x^\circ)$.
3. So the value is $\frac{3}{5}$.

The sine of an angle and the cosine of its complement are the same number, because in a right triangle the side opposite one acute angle is the side adjacent to the other. No angle measure is needed—the identity $\sin(x^\circ) = \cos\left((90-x)^\circ\right)$ hands you the answer.

8 Answer: 24

1. The wall and the ground meet at a right angle, so the ladder is the hypotenuse.
2. Write $10^2 + h^2 = 26^2$.
3. Then $h^2 = 676 - 100 = 576$, so $h = 24$.

The ladder is the hypotenuse because it is the slanted side, opposite the right angle formed by the wall and the ground. Putting the ladder in as a leg is the single most common error in this problem type.

9 Answer: B

1. $\tan A = \dfrac{BC}{AC}$.
2. Substitute: $\dfrac{3}{4} = \dfrac{BC}{20}$.
3. So $BC = 20 \cdot \dfrac{3}{4} = 15$.

TOA: tangent is opposite over adjacent, and both of those sides are legs. For angle A the opposite leg is $\overline{BC}$ and the adjacent leg is $\overline{AC}$, so the tangent equation links exactly the two lengths in the question.

10 Answer: C

1. The short leg is the side opposite 30°, so $x = 6$.
2. The side opposite 60° is $x\sqrt{3}$.
3. So it has length $6\sqrt{3}$.

The long leg sits opposite the 60° angle and equals the short leg times $\sqrt{3}$; the hypotenuse is the short leg times 2. Mixing those two multipliers is the classic slip here.

11 Answer: C

1. $\overline{AC}$ and $\overline{DF}$ correspond, so the scale factor is $\dfrac{9}{6} = \dfrac{3}{2}$.
2. $\overline{BC}$ and $\overline{EF}$ correspond.
3. So $EF = 8 \cdot \dfrac{3}{2} = 12$.

Two right triangles with a pair of congruent acute angles are similar, so corresponding sides are in a constant ratio. Find the scale factor from the pair of sides you know completely, then apply it once.

12 **Answer: A**

1. $P + Q = 90^\circ$ because angle R takes the other 90°.
2. For complementary angles $\sin Q = \cos P$.
3. So $\sin Q = \frac{7}{25}$.

Angles P and Q are the two acute angles of a right triangle, so they add to 90°. The side adjacent to P is the side opposite Q, which makes $\sin Q$ and $\cos P$ the same ratio of the same two sides.

13 **Answer: C**

1. The pole and the shadow are the two legs of a right triangle.
2. The angle of elevation is 45°, so the other acute angle is also 45° and the legs are equal.
3. The height equals the shadow: 18 feet.

The pole, its shadow and the sun's ray form a right triangle whose acute angles are both 45°, so it is isosceles: the height equals the shadow length. The $18\sqrt{2}$ answer is the sun's ray, not the pole.

14 **Answer: D**

1. The hypotenuse is $\sqrt{8^2 + 15^2} = \sqrt{289} = 17$.
2. Add the three sides: $8 + 15 + 17$.
3. The perimeter is 40.

Find the missing side first, then add all three. The 8-15-17 triple is worth memorising: it turns a two-step problem into a one-step one.

15 **Answer:** $\frac{4}{5}$

1. $\sin A = \frac{3}{5}$ means opposite $= 3k$ and hypotenuse $= 5k$.
2. The adjacent leg is $\sqrt{(5k)^2 - (3k)^2} = 4k$.
3. $\cos A = \dfrac{4k}{5k} = \dfrac{4}{5}$.

A ratio alone is enough to rebuild the whole triangle. Read $\sin A = \frac{3}{5}$ as opposite $= 3$ and hypotenuse $= 5$, get the third side from the Pythagorean theorem, then read off the cosine.

16 **Answer: B**

1. $AC = \sqrt{9^2 + 12^2} = 15$.
2. Area $= \frac{1}{2}(9)(12) = 54$ and also $\frac{1}{2}(15)(BD)$.
3. So $BD = \dfrac{2 \cdot 54}{15} = \dfrac{36}{5}$.

The area of the triangle can be computed two ways: from the two legs, or from the hypotenuse and the altitude drawn to it. Setting those equal is faster than chasing the similar triangles.

17 **Answer: B**

1. Since $a + b = 90$, $\cos(b^\circ) = \sin(a^\circ) = \frac{5}{13}$.
2. From $\sin(a^\circ) = \frac{5}{13}$ the legs are 5 and 12, so $\tan(a^\circ) = \frac{5}{12}$.
3. Multiply: $\frac{5}{13} \cdot \frac{5}{12} = \frac{25}{156}$.

Two ideas stack here. First $a + b = 90$, so $\cos(b^\circ) = \sin(a^\circ) = \frac{5}{13}$. Second, a sine of $\frac{5}{13}$ forces a 5-12-13 triangle, so the tangent is $\frac{5}{12}$.

18 **Answer:** 15

1. $\sin\theta = \cos\varphi$ when $\theta + \varphi = 90$.
2. So $(3x+10)+(x+20)=90$, giving $4x+30=90$.
3. Then $4x=60$, so $x=15$.

When a sine equals a cosine, the two angles are complements. Turning the trigonometric equation into the linear equation $(3x+10)+(x+20)=90$ is the whole problem.

19 **Answer:** C

1. The altitude splits the base into two segments of length 6.
2. That creates a 30°-60°-90° triangle with short leg 6 and hypotenuse 12.
3. The altitude is the long leg, $6\sqrt{3}$.

An altitude of an equilateral triangle cuts it into two 30°-60°-90° triangles whose hypotenuse is a full side and whose short leg is half a side. The altitude is the long leg, so it is the short leg times $\sqrt{3}$.

20 **Answer:** A

1. $\tan\theta = \frac{20}{21}$ means opposite $= 20$ and adjacent $= 21$.
2. The hypotenuse is $\sqrt{20^2+21^2} = \sqrt{841} = 29$.
3. $\sin\theta = \dfrac{20}{29}$.

A tangent gives you both legs, so the Pythagorean theorem gives the hypotenuse and every other ratio follows. Here 20-21-29 is the triangle; picking $\frac{21}{29}$ means the opposite and adjacent legs were swapped.

21 **Answer:** 10

1. $\sin\theta = \dfrac{\text{height}}{\text{ramp}}$.
2. So $\dfrac{5}{13} = \dfrac{h}{26}$.
3. Then $h = 26 \cdot \dfrac{5}{13} = 10$.

The ramp is the hypotenuse and the height is the side opposite θ, so sine is exactly the ratio that connects them. Multiplying the hypotenuse by the sine is the one step required.

22 **Answer:** B

1. Let $BC = 1$, so $AB = 2$ and $AC = \sqrt{2^2-1^2} = \sqrt{3}$.
2. For angle A, the opposite leg is $BC = 1$ and the adjacent leg is $AC = \sqrt{3}$.
3. $\tan A = \dfrac{1}{\sqrt{3}} = \frac{1}{3}\sqrt{3}$.

Set $BC = 1$ and $AB = 2$; the triangle is forced to be a 30°-60°-90° triangle, with $AC = \sqrt{3}$. Then $\tan A$ is opposite over adjacent, which is $\frac{1}{\sqrt{3}}$, and rationalising gives $\frac{1}{3}\sqrt{3}$. Choosing $\sqrt{3}$ means the ratio was inverted.

Module 2

1 Answer: B

1. Write $20^2 + b^2 = 25^2$.
2. Then $b^2 = 625 - 400 = 225$.
3. So $b = 15$.

The hypotenuse is the longest side, so it must be the c in $a^2 + b^2 = c^2$; the missing side is found by subtracting. This is the 3-4-5 triple scaled by 5.

2 Answer: D

1. The ratio is leg : leg : hypotenuse $= x : x : x\sqrt{2}$.
2. So $x\sqrt{2} = 8\sqrt{2}$.
3. Then $x = 8$.

Going from hypotenuse to leg reverses the ratio, so you divide by $\sqrt{2}$ instead of multiplying. Because the hypotenuse was already written with a $\sqrt{2}$, the division is clean.

3 Answer: $\frac{7}{24}$

1. The leg opposite A is $\overline{BC} = 7$.
2. The leg adjacent to A is $\overline{AB} = 24$.
3. $\tan A = \dfrac{7}{24}$.

Tangent uses the two legs only, and in a right triangle at B both $\overline{AB}$ and $\overline{BC}$ are legs. The side opposite A is $\overline{BC}$, so the hypotenuse never enters the calculation.

4 Answer: D

1. $\tan A = \frac{5}{12}$ means the leg opposite A is 5 and the leg adjacent is 12.
2. Those roles swap for angle B: opposite 12, adjacent 5.
3. So $\tan B = \frac{12}{5}$.

What is opposite angle A is adjacent to angle B and vice versa, so the tangents of complementary angles are reciprocals. Answering $\frac{5}{12}$ means the identity was applied as if tangent behaved like sine and cosine.

5 Answer: D

1. The long leg is $x\sqrt{3}$, so $x\sqrt{3} = 9\sqrt{3}$ and $x = 9$.
2. The hypotenuse is $2x$.
3. So the hypotenuse is 18.

Work back to the short leg first: the long leg is the short leg times $\sqrt{3}$, so the short leg is 9. The hypotenuse is then twice the short leg.

6 Answer: C

1. The pole and ground form the legs: 24 and 7.
2. The wire is the hypotenuse: $w^2 = 24^2 + 7^2 = 576 + 49 = 625$.
3. So $w = 25$.

The pole and the ground are perpendicular, so the wire is the hypotenuse—the longest side. Any answer smaller than 24 can be rejected immediately.

7 **Answer:** $\frac{8}{17}$

1. $D + E = 90°$.
2. For complementary angles, $\cos E = \sin D$.
3. So $\cos E = \frac{8}{17}$.

Angles D and E are complementary, and the leg opposite D is the leg adjacent to E. Both ratios are therefore the same leg over the same hypotenuse, so no computation is needed at all.

8 **Answer: D**

1. $\sin A = \dfrac{BC}{AB}$, so $\dfrac{3}{5} = \dfrac{9}{AB}$.
2. Cross-multiply: $3 \cdot AB = 45$.
3. So $AB = 15$.

$\overline{BC}$ is opposite angle A and $\overline{AB}$ is the hypotenuse, so those two sides are exactly the ones sine relates. Since the unknown is in the denominator, you divide rather than multiply.

9 **Answer:** 16

1. $\cos\theta = \sin\varphi$ when $\theta + \varphi = 90$.
2. So $(4x - 6) + (x + 16) = 90$, giving $5x + 10 = 90$.
3. Then $5x = 80$, so $x = 16$.

A cosine equals a sine exactly when the two angles are complementary, so the trigonometry collapses into one linear equation. The angle expressions themselves never need to be evaluated.

10 **Answer: D**

1. $AC = \sqrt{8^2 + 6^2} = 10$.
2. The scale factor is $\dfrac{XY}{AB} = \dfrac{20}{8} = 2.5$.
3. So $XZ = 10 \cdot 2.5 = 25$.

Similar triangles share every angle, so every trigonometric ratio matches; the sides just scale. Find the scale factor from the matched pair $\overline{AB}$ and $\overline{XY}$, then scale the hypotenuse.

11 **Answer: A**

1. $\tan(\text{elevation}) = \dfrac{40}{40\sqrt{3}} = \dfrac{1}{\sqrt{3}}$.
2. The short leg is opposite the smallest angle, and $\tan 30° = \dfrac{1}{\sqrt{3}}$.
3. So the angle of elevation is $30°$.

The tangent of the elevation angle is height over shadow, which here is $\frac{1}{\sqrt{3}}$—the tangent of $30°$. Answering 60 means the two legs were used in the wrong order.

12 **Answer:** 60

1. $10 = 2 \cdot 5$ and $24 = 2 \cdot 12$, so the hypotenuse is $2 \cdot 13 = 26$.
2. Add the three sides: $10 + 24 + 26$.
3. The perimeter is 60.

This is the 5-12-13 triple doubled, so the hypotenuse is 26 without any squaring. The question then only asks for a sum, but you cannot answer it until the third side exists.

13 **Answer: C**

1. Adjacent $= 5$ and hypotenuse $= 13$.
2. The opposite leg is $\sqrt{13^2 - 5^2} = 12$.
3. $\tan A = \dfrac{12}{5}$.

One ratio determines the triangle: $\cos A = \frac{5}{13}$ gives adjacent 5 and hypotenuse 13, so the opposite leg is 12. Tangent is then opposite over adjacent, and putting them the other way round produces $\frac{5}{12}$.

14 **Answer: B**

1. The diagonal is side $\times\sqrt{2}$, so $s\sqrt{2} = 10$ and $s = 5\sqrt{2}$.
2. Area $= s^2 = (5\sqrt{2})^2$.
3. So the area is 50.

A diagonal cuts the square into two 45°-45°-90° triangles, so the side is $\frac{10}{\sqrt{2}} = 5\sqrt{2}$ and the area is the side squared. Squaring the diagonal instead gives 100.

15 **Answer: B**

1. Angle $B = 60^\circ$, so $\overline{AC}$ is the long leg and $\overline{BC}$ is the short leg.
2. Long leg $=$ short leg $\times\sqrt{3}$, so $12 = BC\sqrt{3}$.
3. $BC = \dfrac{12}{\sqrt{3}} = 4\sqrt{3}$.

$\overline{AC}$ is the long leg, because it lies opposite the 60° angle at B, and $\overline{BC}$ is the short leg. So $\overline{BC}$ is $\overline{AC}$ divided by $\sqrt{3}$, not multiplied by it.

16 **Answer: C**

1. $\sin A = \frac{2}{3}$ gives opposite $= 2$, hypotenuse $= 3$.
2. The adjacent leg is $\sqrt{9-4} = \sqrt{5}$.
3. $\cos A = \frac{1}{3}\sqrt{5}$.

The legs of a right triangle need not be integers. With opposite $= 2$ and hypotenuse $= 3$, the adjacent leg is $\sqrt{3^2 - 2^2} = \sqrt{5}$, so the cosine carries a radical. Subtracting the numerators to get $\frac{1}{3}$ is the trap.

17 **Answer:** 15

1. $\tan A = \frac{3}{4}$ makes the triangle a 3-4-5 triangle: legs $3k$ and $4k$, hypotenuse $5k$.
2. The perimeter is $3k + 4k + 5k = 12k = 36$, so $k = 3$.
3. The hypotenuse is $5k = 15$.

The tangent fixes the shape but not the size, so write the sides as $3k$, $4k$ and $5k$ and let the perimeter pin down k. $\overline{AB}$ is the hypotenuse because it is the side opposite the right angle at C.

18 **Answer: C**

1. $\sin(b^\circ) = \cos(a^\circ) = \frac{4}{5}$.
2. From the 3-4-5 triangle, $\sin(a^\circ) = \frac{3}{5}$, so $\cos(b^\circ) = \frac{3}{5}$.
3. The quotient is $\dfrac{4/5}{3/5} = \dfrac{4}{3}$.

Because $a + b = 90$, the identities give $\sin(b^\circ) = \cos(a^\circ) = \frac{4}{5}$ and $\cos(b^\circ) = \sin(a^\circ) = \frac{3}{5}$. The quotient of sine over cosine is the tangent, so the answer is $\tan(b^\circ) = \frac{4}{3}$; forgetting to swap gives $\frac{3}{4}$.

19 **Answer: A**

1. Triangle ACD is similar to triangle CBD.
2. So $\dfrac{AD}{CD} = \dfrac{CD}{DB}$, giving $CD^2 = 4 \cdot 9 = 36$.
3. So $CD = 6$.

The altitude to the hypotenuse creates two triangles similar to the original and to each other, so $\frac{AD}{CD} = \frac{CD}{DB}$. That proportion makes CD the geometric mean of the two pieces, not their average—which is why $\frac{13}{2}$ is offered.

20 **Answer:** 100

1. $\sin B = \cos B$ means $B = 90° - B$, so $B = 45°$.
2. Each leg is $\dfrac{20}{\sqrt{2}} = 10\sqrt{2}$.
3. Area $= \frac{1}{2}(10\sqrt{2})(10\sqrt{2}) = 100$.

A sine equal to a cosine at the same angle forces that angle to be $45°$, since $\sin B = \cos(90° - B)$ means $B = 90° - B$. The triangle is therefore isosceles with legs $\frac{20}{\sqrt{2}} = 10\sqrt{2}$.

21 **Answer: C**

1. $\sin\theta = \frac{8}{17}$ gives an 8-15-17 triangle, so $\cos\theta = \frac{15}{17}$.
2. The horizontal run is $34\cos\theta$.
3. So the run is $34 \cdot \dfrac{15}{17} = 30$.

Sine gives the rise, but the question asks for the run, so you need cosine. The 8-15-17 triple turns $\sin\theta = \frac{8}{17}$ into $\cos\theta = \frac{15}{17}$ instantly; 16 is the rise, offered for anyone who answers the wrong question.

22 **Answer: A**

1. $A + B = 90°$ and $A = 2B$, so $3B = 90°$ and $B = 30°$.
2. $\overline{AC}$ is opposite angle B and $\overline{AB}$ is the hypotenuse, so $\dfrac{AC}{AB} = \sin B$.
3. $\sin 30° = \dfrac{1}{2}$.

The acute angles are complementary, so $2B + B = 90$ and $B = 30°$. $\overline{AC}$ is the leg opposite B and $\overline{AB}$ is the hypotenuse, so the ratio is $\sin B = \sin 30° = \frac{1}{2}$; reading it as $\sin A$ instead gives $\frac{1}{2}\sqrt{3}$.

Topic 20 — Circles

Module 1

1 **Answer: B**

1. Use $C = 2\pi r$.
2. Substitute $r = 7$: $C = 2\pi(7) = 14\pi$.

Circumference is $2\pi r$, not πr and not πr^2. With $r = 7$ the circumference is 14π; the choice 49π is the area, and 28π comes from treating 7 as the diameter and then doubling again.

2 Answer: C

1. The radius is half the diameter: $r = \frac{10}{2} = 5$.
2. Use $A = \pi r^2 = \pi(5)^2 = 25\pi$.

The area formula wants the radius, so halve the diameter first. Substituting 10 directly into πr^2 gives 100π, which is the classic diameter-for-radius error and is offered as choice D.

3 Answer: 7

1. Compare with $(x-h)^2 + (y-k)^2 = r^2$.
2. Here $r^2 = 49$, so $r = \sqrt{49} = 7$.

The number on the right side of the standard form is r^2, not r. Taking the square root gives a radius of 7; reporting 49 would be reporting the square of the radius.

4 Answer: C

1. Multiply by $\frac{\pi}{180}$ to convert degrees to radians.
2. $60 \cdot \frac{\pi}{180} = \frac{\pi}{3}$.

A half turn is $180° = \pi$ radians, so one degree is $\frac{\pi}{180}$ radians. Since 60 is a third of 180, the answer is a third of π.

5 Answer: A

1. The standard form is $(x-h)^2 + (y-k)^2 = r^2$.
2. Write $(x+4)^2$ as $(x-(-4))^2$, so $h = -4$.
3. $(y-1)^2$ gives $k = 1$, so the center is $(-4, 1)$.

The standard form subtracts the coordinates of the center, so the sign you read inside the parentheses is the opposite of the coordinate. A plus sign means a negative coordinate: $(x+4)^2$ puts the center at $x = -4$.

6 Answer: A

1. Set $\pi r^2 = 36\pi$.
2. Divide by π: $r^2 = 36$.
3. Take the positive square root: $r = 6$.

Dividing by π first keeps the arithmetic exact. What remains is $r^2 = 36$, so the radius is 6; the value 36 is r^2 and 18 would be half of that number, neither of which is a length here.

7 Answer: B

1. The central angle AOB and the inscribed angle ACB intercept the same arc AB.
2. An inscribed angle is half of the central angle on the same arc.
3. $\frac{1}{2}(80°) = 40°$.

Both angles open onto the same arc, so the inscribed angle is half the central angle. The value 160° comes from doubling instead of halving, and 80° from assuming the two angles are equal.

8 Answer: A

1. The angle is $\frac{40}{360} = \frac{1}{9}$ of the full circle.
2. The full circumference is $2\pi(9) = 18\pi$.
3. The arc is $\frac{1}{9}(18\pi) = 2\pi$.

An arc is the same fraction of the circumference that its central angle is of $360°$. Choice D is the entire circumference, choice B comes from using 9 as the diameter's half again, and choice C is the sector's area, not a length.

9 Answer: B

1. The sector is $\frac{60}{360} = \frac{1}{6}$ of the circle.
2. The full area is $\pi(6)^2 = 36\pi$.
3. The sector area is $\frac{1}{6}(36\pi) = 6\pi$.

Take the same fraction of the area that the angle is of the whole turn. Choice A, 2π, is the arc length of that sector, and choice D is the area of the entire circle.

10 Answer: 150

1. Multiply by $\frac{180}{\pi}$ to convert radians to degrees.
2. $\frac{5\pi}{6} \cdot \frac{180}{\pi} = 5 \cdot 30 = 150$.

Because π radians is $180°$, the factor $\frac{180}{\pi}$ cancels the π and leaves a plain number. Five sixths of 180 is 150.

11 Answer: D

1. Use $(x-h)^2 + (y-k)^2 = r^2$ with $h = 0$, $k = 3$, $r = 5$.
2. $(x-0)^2 + (y-3)^2 = 5^2$.
3. Simplify: $x^2 + (y-3)^2 = 25$.

The center is subtracted, so a center of $y = 3$ produces $(y-3)^2$, and the right side is $r^2 = 25$, not $r = 5$. Choice A flips the sign of k and choice B leaves the radius unsquared.

12 Answer: 14

1. From $(x+2)^2$, $h = -2$; from $(y-6)^2$, $k = 6$.
2. From $r^2 = 100$, $r = 10$.
3. $h + k + r = -2 + 6 + 10 = 14$.

Each piece is read off separately, and the only two places to slip are the sign of h and taking the square root for r. With $h = -2$, $k = 6$ and $r = 10$, the sum is 14.

13 Answer: C

1. $\overline{AB}$ is a diameter, so angle ACB is inscribed in a semicircle and measures $90°$.
2. The angles of triangle ABC sum to $180°$.
3. $\angle ABC = 180° - 90° - 35° = 55°$.

An inscribed angle that intercepts a semicircle is half of $180°$, so it is a right angle. That makes triangle ABC right-angled at C, and the two acute angles must be complementary, giving $55°$.

14 Answer: C

1. A tangent is perpendicular to the radius at the point of tangency, so angle $OPQ = 90°$.
2. Triangle OPQ is a right triangle with legs $OP = 5$ and PQ, and hypotenuse $OQ = 13$.
3. $PQ = \sqrt{13^2 - 5^2} = \sqrt{144} = 12$.

The tangent-radius right angle is what turns this into a Pythagorean triple: 5-12-13. Choices B and D come from subtracting or adding the two given lengths instead of using the Pythagorean theorem.

15 Answer: C

1. With the angle in radians, use $s = r\theta$.
2. $s = 10 \cdot \frac{3\pi}{4} = \frac{30\pi}{4}$.
3. Simplify: $s = \frac{15\pi}{2}$.

In radians the arc length is a single product, $r\theta$, with no fraction of 360 involved. Choice B is what you get using $r = 5$, half the radius, and choice D uses $\theta = 3\pi$ instead of $\frac{3\pi}{4}$.

16 Answer: C

1. Group and move the constant: $(x^2 - 6x) + (y^2 + 8y) = 11$.
2. Complete the squares by adding 9 and 16 to both sides.
3. $(x - 3)^2 + (y + 4)^2 = 11 + 9 + 16 = 36$.
4. So $r = \sqrt{36} = 6$.

Whatever you add on the left to complete a square must also be added on the right, which is why 11 grows to 36. Choice A comes from forgetting to add 9 and 16 on the right, choice B from subtracting the constant instead of adding it, and choice D reports r^2 rather than r.

17 Answer: 4

1. Rearrange: $(x^2 + 10x) + (y^2 - 4y) = -13$.
2. Add 25 and 4 to both sides.
3. $(x + 5)^2 + (y - 2)^2 = -13 + 25 + 4 = 16$.
4. $r = \sqrt{16} = 4$.

Moving the $+13$ across the equal sign makes it -13; the completing-the-square constants 25 and 4 then push the right side up to 16. The radius is the square root of that, namely 4.

18 Answer: D

1. Group: $(x^2 - 12x) + (y^2 + 2y) = -12$.
2. Complete the squares: add 36 and 1 to both sides.
3. $(x - 6)^2 + (y + 1)^2 = 25$.
4. The center is $(6, -1)$.

Half of -12 is -6, which lands in the equation as $(x - 6)^2$ and therefore as $h = 6$; half of 2 is 1, giving $(y + 1)^2$ and $k = -1$. The signs of the centre are opposite to the signs inside the parentheses, and the three distractors are exactly the three ways to get one or both signs wrong.

19 **Answer: B**

1. The sector is $\frac{120}{360} = \frac{1}{3}$ of the circle, so $\frac{1}{3}\pi r^2 = 27\pi$.
2. Multiply by 3: $\pi r^2 = 81\pi$.
3. Divide by π and take the square root: $r^2 = 81$, so $r = 9$.

Reverse the sector formula: if a third of the area is 27π, the whole area is 81π. Choice D reports r^2 instead of r, and choice A comes from dividing 27 by the angle fraction in the wrong direction.

20 **Answer: D**

1. An inscribed angle is half its intercepted arc, so arc AB measures $2(30°) = 60°$.
2. The circumference is $2\pi(12) = 24\pi$.
3. The arc is $\frac{60}{360}(24\pi) = 4\pi$.

The inscribed angle must be doubled before it can be used as a fraction of $360°$. Using $30°$ directly gives 2π, which is choice B and the mistake this question is built around.

21 **Answer: 13**

1. The radius is the distance from the center to any point on the circle.
2. $r = \sqrt{(9-4)^2 + (9-(-3))^2} = \sqrt{25+144}$.
3. $r = \sqrt{169} = 13$.

Every point on a circle is exactly one radius from the center, so the distance formula gives the radius directly. The legs 5 and 12 form a Pythagorean triple with hypotenuse 13.

22 **Answer: A**

1. Group: $(x^2 + 4x) + (y^2 - 10y) = -20$.
2. Add 4 and 25 to both sides.
3. $(x+2)^2 + (y-5)^2 = -20 + 4 + 25 = 9$.

Completing both squares turns the expanded equation into standard form, showing a circle of centre $(-2, 5)$ and radius 3. Choice C flips both signs inside the parentheses, choice B adds 20 instead of subtracting it, and choice D writes r where r^2 belongs.

Module 2

1 **Answer: C**

1. Use $C = \pi d$.
2. $\pi d = 36\pi$, so $d = 36$.

Written as $C = \pi d$, the circumference formula gives the diameter in one step. Choice B is the radius, and choice D doubles the diameter a second time.

2 **Answer: B**

1. From $\pi r^2 = 49\pi$, $r^2 = 49$ and $r = 7$.
2. Then $C = 2\pi r = 2\pi(7) = 14\pi$.

There is no shortcut from area to circumference; you must pass through the radius. Once $r = 7$, the circumference is 14π. Choice D comes from doubling the area instead.

3 Answer: 135

1. Multiply by $\frac{180}{\pi}$.
2. $\frac{3\pi}{4} \cdot \frac{180}{\pi} = 3 \cdot 45 = 135$.

Since π radians is a straight angle of $180°$, three quarters of π is three quarters of 180, which is 135.

4 Answer: A

1. The right side is r^2, so $r = \sqrt{45}$.
2. Simplify: $\sqrt{45} = \sqrt{9 \cdot 5} = 3\sqrt{5}$.

The radius is the square root of the right-hand side, and 45 has the perfect-square factor 9, so the radical simplifies to $3\sqrt{5}$. Choice D reports r^2 as if it were r, and choice B mixes up which factor comes out of the radical.

5 Answer: C

1. The sector is $\frac{45}{360} = \frac{1}{8}$ of the circle.
2. The full area is $\pi(8)^2 = 64\pi$.
3. The sector area is $\frac{1}{8}(64\pi) = 8\pi$.

One eighth of the circle's area is 8π. Choice A is the arc length of the same sector, a length rather than an area, and choice D would be the area of a $90°$ sector.

6 Answer: B

1. Angle BAC is inscribed in a semicircle, so it measures $90°$.
2. The angles of triangle ABC sum to $180°$.
3. $\angle BCA = 180° - 90° - 28° = 62°$.

Because $\overline{BC}$ is a diameter, the inscribed angle at A intercepts a semicircle and is right. The remaining two angles are then complementary, so $28°$ and $62°$ pair up. Choice D is $180° - 28°$, which forgets the right angle.

7 Answer: D

1. Multiply by $\frac{\pi}{180}$: $210 \cdot \frac{\pi}{180} = \frac{210\pi}{180}$.
2. Reduce by 30: $\frac{7\pi}{6}$.

Since $210°$ is more than a straight angle, the answer must be more than π, which already eliminates the first three choices. Reducing $\frac{210}{180}$ gives $\frac{7}{6}$, and choice C is that fraction accidentally inverted.

8 Answer: 5

1. Use the distance formula from the center to the given point.
2. $r = \sqrt{(6-2)^2 + (1-(-2))^2} = \sqrt{16+9}$.
3. $r = \sqrt{25} = 5$.

The radius is the distance from the centre to any point of the circle, and here the horizontal and vertical gaps are 4 and 3, the legs of a 3-4-5 right triangle. Subtracting the y-coordinates carefully matters: $1 - (-2) = 3$, not -1.

9 **Answer: C**

1. The whole circle has area $\pi(12)^2 = 144\pi$.
2. The sector is $\frac{24\pi}{144\pi} = \frac{1}{6}$ of the circle.
3. So the angle is $\frac{1}{6}(360°) = 60°$.

Compare the sector's area with the whole circle's area to find what fraction of the circle it is, then take that fraction of 360°. Choice D doubles the correct fraction, which is what happens if you compare with the area of a semicircle instead.

10 **Answer: 120**

1. The circumference is $2\pi(15) = 30\pi$.
2. The arc is $\frac{10\pi}{30\pi} = \frac{1}{3}$ of the circle.
3. The angle is $\frac{1}{3}(360°) = 120°$.

The ratio of the arc to the circumference is the same as the ratio of the central angle to 360°, and the factors of π cancel in that ratio. One third of a full turn is 120°.

11 **Answer: A**

1. The center is the midpoint: $\left(\frac{1+7}{2}, \frac{2+10}{2}\right) = (4, 6)$.
2. The diameter has length $\sqrt{6^2 + 8^2} = 10$, so $r = 5$.
3. The equation is $(x-4)^2 + (y-6)^2 = 25$.

The centre of a circle is the midpoint of any diameter, and the radius is half the diameter's length. Choice B uses the full diameter 10 as the radius, and choice D centres the circle at an endpoint instead of the midpoint.

12 **Answer: D**

1. Use $s = r\theta$ with $s = 12\pi$ and $\theta = \frac{2\pi}{3}$.
2. $r = \frac{12\pi}{2\pi/3} = 12\pi \cdot \frac{3}{2\pi}$.
3. $r = 18$.

Dividing by a fraction multiplies by its reciprocal, and the π's cancel, leaving $r = 18$. Choice A comes from dividing by 2π and forgetting the $\frac{1}{3}$, and choice B is half the correct radius.

13 **Answer: 6**

1. Group: $(x^2 - 8x) + (y^2 + 6y) = 11$.
2. Add 16 and 9 to both sides.
3. $(x-4)^2 + (y+3)^2 = 11 + 16 + 9 = 36$.
4. $r = \sqrt{36} = 6$.

Half of -8 squared is 16, and half of 6 squared is 9; both must be added on the right as well, taking 11 up to 36. The radius is the square root, 6, not 36.

14 **Answer: A**

1. The central angle is twice the inscribed angle on the same arc.
2. $x + 40 = 2(3x)$.
3. $x + 40 = 6x$, so $40 = 5x$ and $x = 8$.

Set the central angle equal to twice the inscribed angle, not equal to it. Solving gives $x = 8$, so the inscribed angle is $24°$ and the central angle is $48°$. Choice B is what you get from setting the two angles equal, and choice D from doubling the central angle instead of the inscribed one.

15 **Answer: C**

1. From $2\pi r = 24\pi$, the radius is $r = 12$.
2. The full area is $\pi(12)^2 = 144\pi$.
3. The sector is $\frac{30}{360} = \frac{1}{12}$ of that: $\frac{1}{12}(144\pi) = 12\pi$.

The circumference is only there to hand you the radius. Choice A is the arc length of the sector, and choice D is the area of the entire circle rather than of one twelfth of it.

16 **Answer: C**

1. The circumference is $2\pi(12) = 24\pi$, so the sector is $\frac{8\pi}{24\pi} = \frac{1}{3}$ of the circle.
2. The full area is $\pi(12)^2 = 144\pi$.
3. The sector area is $\frac{1}{3}(144\pi) = 48\pi$.

The same fraction that produced the arc produces the sector, so find the fraction once and reuse it. Equivalently, the sector's area is $\frac{1}{2}rs = \frac{1}{2}(12)(8\pi) = 48\pi$. Choice D is that product without the factor of one half.

17 **Answer: A**

1. Group: $(x^2 + 6x) + (y^2 - 14y) = -22$.
2. Add 9 and 49 to both sides.
3. $(x + 3)^2 + (y - 7)^2 = -22 + 9 + 49 = 36$.

Half of 6 is 3 and half of -14 is -7, so the completed squares are $(x + 3)^2$ and $(y - 7)^2$, and the constants 9 and 49 must be added to the right of -22. Choice D keeps $+22$ on the right, and choice C writes the radius where r^2 belongs.

18 **Answer: B**

1. Divide the entire equation by 2: $x^2 + y^2 - 6x + 2y - 3 = 0$.
2. Group and move the constant: $(x^2 - 6x) + (y^2 + 2y) = 3$.
3. Add 9 and 1 to both sides: $(x - 3)^2 + (y + 1)^2 = 13$.
4. $r = \sqrt{13}$.

Completing the square only works when the coefficients of x^2 and y^2 are 1, so divide by 2 before doing anything else. Skipping that division doubles the right side and gives $\sqrt{26}$; choice D reports r^2 as the radius.

19 **Answer: 6**

1. Group: $(x^2 - 10x) + (y^2 + 4y) = -20$.
2. Add 25 and 4 to both sides: $(x-5)^2 + (y+2)^2 = 9$.
3. So $h = 5$, $k = -2$, $r = 3$.
4. $h + k + r = 5 - 2 + 3 = 6$.

Completing both squares gives centre $(5, -2)$ and $r^2 = 9$, so the radius is 3. Every one of the three values is a place to slip a sign or forget a square root, and the sum 6 only comes out right if all three are read correctly.

20 **Answer: D**

1. A circle tangent to the x-axis touches it at exactly one point, so the radius equals the distance from the center to the x-axis.
2. That distance is $|4| = 4$, so $r = 4$.
3. The equation is $(x-3)^2 + (y-4)^2 = 16$.

Tangency to a horizontal line means the vertical distance from the centre to that line is exactly the radius, so $r = 4$ and $r^2 = 16$. Choice A uses the x-coordinate as the radius, and choice C uses the distance from the origin to the centre.

21 **Answer: 25**

1. Group: $(x^2 - 4x) + (y^2 - 2y) = 20$.
2. Add 4 and 1 to both sides: $(x-2)^2 + (y-1)^2 = 25$.
3. So $r^2 = 25$ and the area is $\pi r^2 = 25\pi$.
4. Therefore $k = 25$.

Because the area formula uses r^2 and completing the square delivers r^2 directly, there is no need to take a square root at all here. The right side, 25, is already the value of k.

22 **Answer: B**

1. Group: $(x^2 - 2x) + (y^2 + 8y) = 8$.
2. Add 1 and 16 to both sides: $(x-1)^2 + (y+4)^2 = 25$.
3. So $r = 5$.
4. $C = 2\pi r = 10\pi$.

Completing the square gives $r^2 = 25$, so the radius is 5 and the circumference is 10π. Choice A uses πr instead of $2\pi r$, and choice C is the area of the circle rather than its circumference.

Answer Key

Topic 01 — Linear Equations in One Variable

Module 1

1 A	2 7	3 B	4 C	5 B	6 10	7 C	8 C
9 A	10 3	11 A	12 D	13 B	14 4	15 B	16 C
17 D	18 12	19 A	20 D	21 41	22 B		

Module 2

1 B	2 30	3 C	4 C	5 D	6 C	7 7	8 C
9 B	10 A	11 14	12 A	13 D	14 B	15 A	16 4
17 C	18 D	19 A	20 D	21 20	22 B		

Topic 02 — Linear Functions

Module 1

1 C	2 C	3 4	4 B	5 D	6 1	7 A	8 B
9 29	10 A	11 A	12 B	13 7	14 A	15 C	16 B
17 5	18 D	19 C	20 16	21 C	22 D		

Module 2

1 D	2 -2	3 B	4 D	5 D	6 B	7 A	8 B
9 4	10 C	11 A	12 6	13 C	14 B	15 C	16 A
17 1	18 D	19 C	20 32	21 C	22 B		

Topic 03 — Linear Equations in Two Variables

Module 1

1 A	2 B	3 19	4 C	5 D	6 3	7 B	8 A
9 C	10 C	11 A	12 12	13 D	14 B	15 15	16 D
17 A	18 2	19 C	20 B	21 A	22 11		

Module 2

1 B	2 -4	3 C	4 D	5 A	6 B	7 C	8 D
9 $-\frac{2}{3}$	10 A	11 C	12 B	13 D	14 C	15 6	16 A
17 -1	18 D	19 B	20 C	21 $\frac{9}{4}$	22 D		

Topic 04 — Systems of Two Linear Equations in Two Variables

Module 1

1 A	2 C	3 7	4 C	5 D	6 B	7 10	8 B
9 5	10 A	11 C	12 D	13 B	14 C	15 6	16 A
17 B	18 2	19 D	20 D	21 1	22 B		

Module 2

1 C	2 B	3 16	4 C	5 D	6 C	7 D	8 B
9 3	10 C	11 C	12 11	13 C	14 A	15 2	16 A
17 B	18 C	19 B	20 6	21 D	22 D		

Topic 05 — Linear Inequalities in One or Two Variables

Module 1

1 B	2 C	3 7	4 A	5 D	6 B	7 C	8 B
9 19	10 C	11 D	12 3	13 B	14 A	15 4	16 C
17 B	18 B	19 A	20 2	21 C	22 A		

Module 2

1 B	2 3	3 D	4 D	5 A	6 6	7 A	8 C
9 43	10 C	11 D	12 B	13 A	14 C	15 5	16 D
17 A	18 16	19 C	20 34	21 C	22 A		

Topic 06 — Equivalent Expressions

Module 1

1 A	2 B	3 1	4 D	5 A	6 C	7 D	8 D
9 C	10 4	11 D	12 A	13 8	14 C	15 C	16 A
17 D	18 2	19 C	20 A	21 8	22 B		

Module 2

1 A	2 C	3 5	4 C	5 D	6 9	7 B	8 B
9 5	10 A	11 B	12 2	13 B	14 A	15 C	16 B
17 D	18 3	19 B	20 24	21 D	22 C		

Topic 07 — Quadratic Equations

Module 1

1 B	2 C	3 5	4 D	5 A	6 B	7 A	8 5
9 D	10 C	11 3	12 B	13 A	14 B	15 12	16 C
17 A	18 9	19 B	20 C	21 B	22 D		

Module 2

1 D	2 3	3 A	4 B	5 A	6 5	7 A	8 C
9 -15	10 C	11 B	12 2	13 B	14 D	15 C	16 C
17 C	18 4	19 45	20 C	21 A	22 C		

Topic 08 — Quadratic and Polynomial Functions

Module 1

1 C	2 3	3 D	4 A	5 D	6 5	7 C	8 D
9 2	10 D	11 B	12 D	13 B	14 C	15 B	16 B
17 -3	18 D	19 C	20 2	21 C	22 B		

Module 2

1 B	2 4	3 A	4 B	5 7	6 A	7 C	8 1
9 B	10 C	11 D	12 A	13 A	14 2	15 D	16 B
17 -6	18 B	19 C	20 212	21 A	22 A		

Topic 09 — Exponential Functions, Radicals and Rational Expressions

Module 1

1 B	2 D	3 24	4 C	5 C	6 10	7 C	8 B
9 2	10 C	11 D	12 A	13 B	14 A	15 4800	16 B
17 A	18 4	19 D	20 C	21 11	22 A		

Module 2

1 B	2 1	3 D	4 D	5 C	6 B	7 882	8 A
9 5	10 D	11 D	12 B	13 C	14 A	15 C	16 C
17 B	18 A	19 9/5	20 5	21 B	22 A		

Topic 10 — Nonlinear Systems and Function Transformations

Module 1

1 A	2 B	3 3	4 A	5 C	6 D	7 A	8 B
9 -4	10 B	11 D	12 C	13 C	14 B	15 4	16 B
17 12	18 A	19 D	20 15	21 C	22 A		

Module 2

1 A	2 2	3 B	4 D	5 13	6 C	7 D	8 7
9 C	10 D	11 B	12 C	13 8	14 B	15 A	16 C
17 A	18 D	19 12	20 B	21 4	22 A		

Topic 11 — Ratios, Rates, Proportional Relationships and Units

Module 1

1 D	2 7	3 B	4 D	5 A	6 150	7 B	8 B
9 20	10 B	11 A	12 C	13 D	14 C	15 B	16 D
17 2000	18 C	19 C	20 A	21 1.05	22 C		

Module 2

1 D	2 14	3 B	4 C	5 A	6 450	7 C	8 B
9 54	10 D	11 C	12 D	13 24	14 A	15 C	16 B
17 300	18 B	19 2.8	20 A	21 C	22 B		

Topic 12 — Percentages

Module 1

1 B	2 25	3 C	4 B	5 C	6 A	7 D	8 D
9 20	10 B	11 A	12 C	13 A	14 D	15 C	16 B
17 250	18 A	19 100	20 D	21 C	22 20		

Module 2

1 C	2 40	3 B	4 A	5 D	6 B	7 C	8 128
9 C	10 B	11 48	12 D	13 A	14 C	15 104	16 B
17 B	18 D	19 25	20 C	21 D	22 100000		

Topic 13 — One-Variable Data: Distributions and Measures of Center and Spread

Module 1

1 C	2 B	3 15	4 C	5 B	6 17	7 C	8 B
9 B	10 C	11 D	12 8.5	13 D	14 A	15 10	16 A
17 9	18 D	19 A	20 82	21 C	22 C		

Module 2

1 B	2 A	3 A	4 D	5 3	6 D	7 A	8 B
9 15	10 C	11 B	12 A	13 B	14 9	15 D	16 25
17 D	18 C	19 C	20 A	21 75	22 D		

Topic 14 — Two-Variable Data: Models and Scatterplots

Module 1

1 A	2 C	3 4	4 A	5 D	6 85	7 A	8 B
9 D	10 D	11 84	12 A	13 5	14 C	15 D	16 C
17 B	18 D	19 20	20 C	21 B	22 C		

Module 2

1 A	2 C	3 40	4 B	5 -2.5	6 A	7 C	8 -4
9 D	10 D	11 50	12 A	13 B	14 3	15 D	16 C
17 B	18 D	19 108	20 C	21 B	22 A		

Topic 15 — Probability and Conditional Probability

Module 1

1 A	2 B	3 C	4 2/5	5 B	6 C	7 A	8 13/25
9 D	10 B	11 D	12 9/16	13 D	14 B	15 A	16 4
17 A	18 C	19 D	20 2	21 C	22 B		

Module 2

1 B	2 A	3 D	4 2/5	5 C	6 C	7 A	8 7/40
9 A	10 B	11 D	12 48	13 D	14 C	15 B	16 40
17 B	18 C	19 12	20 C	21 B	22 80		

Topic 16 — Inference, Margin of Error and Evaluating Statistical Claims

Module 1

1 C	2 39	3 B	4 A	5 B	6 A	7 C	8 D
9 28.7	10 A	11 B	12 D	13 0.42	14 A	15 C	16 D
17 2560	18 A	19 D	20 2.7	21 B	22 C		

Module 2

1 C	2 0.55	3 A	4 A	5 440	6 B	7 D	8 C
9 23.1	10 D	11 A	12 D	13 D	14 B	15 4.05	16 C
17 B	18 2025	19 C	20 C	21 8100	22 B		

Topic 17 — Area and Volume

Module 1

1 D	2 B	3 C	4 180	5 B	6 D	7 96	8 B
9 C	10 A	11 7	12 B	13 C	14 B	15 5	16 B
17 A	18 C	19 400	20 B	21 C	22 B		

Module 2

1 C	2 B	3 A	4 150	5 C	6 D	7 C	8 A
9 6	10 C	11 D	12 300	13 B	14 D	15 C	16 A
17 540	18 B	19 A	20 5	21 D	22 10		

Topic 18 — Lines, Angles and Triangles

Module 1

1 B	2 63	3 B	4 D	5 A	6 70	7 B	8 C
9 C	10 13	11 A	12 A	13 25	14 B	15 C	16 B
17 25	18 C	19 C	20 54	21 A	22 D		

Module 2

1 B	2 C	3 80	4 D	5 B	6 B	7 75	8 D
9 100	10 C	11 A	12 A	13 A	14 C	15 C	16 4
17 D	18 C	19 B	20 12	21 D	22 D		

Topic 19 — Right Triangles and Trigonometry

Module 1

1 B	2 B	3 15	4 B	5 A	6 A	7 A	8 24
9 B	10 C	11 C	12 A	13 C	14 D	15 $\frac{4}{5}$	16 B
17 B	18 15	19 C	20 A	21 10	22 B		

Module 2

1 B	2 D	3 $\frac{7}{24}$	4 D	5 D	6 C	7 $\frac{8}{17}$	8 D
9 16	10 D	11 A	12 60	13 C	14 B	15 B	16 C
17 15	18 C	19 A	20 100	21 C	22 A		

Topic 20 — Circles

Module 1

1 B	**2** C	**3** 7	**4** C	**5** A	**6** A	**7** B	**8** A
9 B	**10** 150	**11** D	**12** 14	**13** C	**14** C	**15** C	**16** C
17 4	**18** D	**19** B	**20** D	**21** 13	**22** A		

Module 2

1 C	**2** B	**3** 135	**4** A	**5** C	**6** B	**7** D	**8** 5
9 C	**10** 120	**11** A	**12** D	**13** 6	**14** A	**15** C	**16** C
17 A	**18** B	**19** 6	**20** D	**21** 25	**22** B		